DATA SUPPLEMENT

Appendix A. Search strategies

Table S1. Search strategies for systematic review topics

Search dates: January 1, 2013 through January to June 2025, depending on topic.
[Selected searches will be updated during public review.]

Research Question (RQ): Topic	Database	Search strategy
RQ 1: KDIGO AKI definition	PubMed	("Acute Kidney Injury"[Mesh] OR "Cardio-Renal Syndrome"[Mesh] OR "Hepatorenal Syndrome"[Mesh] OR (("Acute Kidney"[tiab] OR "Acute Renal"[tiab]) AND (Injury[tiab] OR failure[tiab] OR disease[tiab] OR impairment[tiab] OR insufficiency[tiab])) OR hepatorenal[tiab] OR hepato-renal[tiab] OR cardiorenal[tiab] OR "cardio-renal"[tiab]) AND ("risk factor" OR "risk prediction" OR severity OR staging OR stage OR progress* OR prognosis OR diagnos* OR Oliguria OR "serum creatinine" OR "urine output" OR RIFLE OR AKIN OR AWARE OR AWAKEN) AND (KDIGO or "kidney disease improving global outcomes") NOT ("addresses" OR "autobiography"[pt] OR "bibliography"[pt] OR "biography"[pt] OR "case reports" OR "comment"[pt] OR "congresses" OR "dictionary"[pt] OR "directory"[pt] OR "festschrift"[pt] OR "government publications" OR "historical article"[pt] OR "interview"[pt] OR "lectures" OR "legal cases" OR "legislation"[pt] OR "news"[pt] OR "newspaper article"[pt] OR "patient education handout"[pt] OR "periodical index"[pt] OR "comment on" OR ("review"[pt] NOT "systematic review"[pt]) OR ("Animals"[Mesh] NOT "Humans"[Mesh]) OR rats[tw] OR cow[tw] OR cows[tw] OR chicken*[tw] OR horse[tw] OR horses[tw] OR mice[tw] OR mouse[tw] OR bovine[tw] OR sheep OR ovine OR murinae OR "animal model")
	Embase	#1 'acute kidney failure'/exp OR 'acute kidney failure' #2 'acute kidney injury' #3 'cardiorenal syndrome'

		#4 'cardio-renal syndrome' #5 'hepatorenal syndrome' #6 'acute kidney' #7 injury OR failure OR disease OR impairment OR insufficiency #8 'hepato-renal syndrome' #9 #6 AND #7 #10 'acute renal' #11 #7 AND #10 #12 #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #8 OR #9 OR #11 #13 'risk factor' OR 'risk prediction' OR severity OR staging OR stage OR progress* OR prognosis OR diagnos* OR oliguria OR 'serum creatinine' OR 'urine output' OR rifle OR akin OR aware OR awaken #14 #12 AND #13 #15 kdigo OR 'kidney disease improving global outcomes' #16 #14 AND #15
RQ 2 & 3: Trajectories and AKD	PubMed	((("Acute Kidney Injury"[Mesh] OR (("Acute Kidney"[tiab] OR "Acute Renal"[tiab]) AND (Injury[tiab] OR failure[tiab] OR impairment[tiab] OR insufficiency[tiab]))) AND ("Transient acute"[tiab:~2] OR "Transient AKI"[tiab:~2] OR "Persistent acute"[tiab:~2] OR "Persistent AKI"[tiab:~2] OR "Progressive acute"[tiab:~2] OR "Progressive AKI"[tiab:~2] OR "Recurrent acute"[tiab:~2] OR "Recurrent AKI"[tiab:~2] OR "Recovery acute"[tiab:~2] OR "Recovery AKI"[tiab:~2] OR "reversal acute"[tiab:~2] OR "reversal AKI"[tiab:~2] OR " resolving acute"[tiab:~2] OR " resolving AKI"[tiab:~2] OR " reversible acute"[tiab:~2] OR " reversible AKI"[tiab:~2] OR " prognostic value acute"[tiab:~2] OR " prognostic value AKI"[tiab:~2])) OR "acute kidney disease" OR AKD) NOT ("addresses" OR "autobiography"[pt] OR "bibliography"[pt] OR "biography"[pt] OR "case reports" OR "comment"[pt] OR "congresses" OR "dictionary"[pt] OR "directory"[pt] OR "festschrift"[pt] OR "government publications" OR "historical article"[pt] OR "interview"[pt] OR "lectures" OR "legal cases" OR "legislation"[pt] OR "news"[pt] OR "newspaper article"[pt] OR "patient education handout"[pt] OR

		“periodical index”[pt] OR "comment on" OR (“review”[pt] NOT “systematic review”[pt]) OR ("Animals"[Mesh] NOT "Humans"[Mesh]) OR rats[tw] OR cow[tw] OR cows[tw] OR chicken*[tw] OR horse[tw] OR horses[tw] OR mice[tw] OR mouse[tw] OR bovine[tw] OR sheep OR ovine OR murinae OR “animal model”)
	Embase	#1 'acute kidney failure'/exp OR 'acute kidney failure' #2 ('acute kidney' OR 'acute renal') AND (injury OR failure OR impairment OR insufficiency) #3 'transient' AND next AND 'acute' OR ('transient' AND next AND 'aki') OR ('persistent' AND next AND 'acute') OR ('persistent' AND next AND 'aki') OR ('progressive' AND next AND 'acute') OR ('progressive' AND next AND 'aki') OR ('recurrent' AND next AND 'acute') OR ('recurrent' AND next AND 'aki') OR ('reversal' AND next AND 'acute') OR ('reversal' AND next AND 'aki') OR ('resolving' AND next AND 'acute') OR ('resolving' AND next AND 'aki') OR ('reversible' AND next AND 'acute') OR ('reversible' AND next AND 'aki') OR ('prognostic value' AND next AND 'acute') OR ('prognostic value' AND next AND 'aki') #4 #1 OR #2 #5 #3 AND #4 #6 'acute kidney disease' #7 #5 OR #6
RQs 4 & 5: Diagnostic tests	PubMed	Focused: ((((((("Decision Support Systems, Clinical"[Mesh] OR "clinical decision support system"[tiab] OR "NINJA"[tiab] OR "sniffer*"[tiab] OR "furosemide stress test"[tiab] OR "furosemide respons*"[tiab]) OR (("real time"[tiab] OR "real-time"[tiab] OR "realtime"[tiab] OR "kinetic"[tiab]) AND ("eGFR"[tiab] OR "GFR"[tiab] OR "glomerular filtration rate"[tiab]))) OR

		(("Kidney Tubular Necrosis, Acute"[Mesh] OR "tubular necrosis"[tiab] OR ("urine"[tiab] AND ("microscop*"[tiab] OR "marker"[tiab] OR "cast"[tiab] OR "sediment"[tiab] OR "granul*"[tiab]))) AND ("score"[tiab] OR "scoring"[tiab] OR "scale"[tiab] OR "biops*"[tiab] OR "histolog*"[tiab] OR "interstitial"[tiab]))) OR (("Ultrasonography"[Mesh] OR "ultrasound"[tiab] OR "ultrasonograph*"[tiab]) AND ("hydronephro*"[tiab] OR "perfusion"[tiab] OR "resistive index"[tiab]))) OR (("Magnetic Resonance Imaging"[Mesh] OR "magnetic resonance imag*"[tiab]) AND ("fibrosis"[tiab] OR "fibrotic"[tiab]))) AND ("Acute Kidney Injury"[Mesh] OR (("Acute Kidney"[tiab] OR "Acute Renal"[tiab]) AND (Injury[tiab] OR failure[tiab] OR disease[tiab] OR impairment[tiab] OR insufficiency[tiab])))) NOT (("address"[pt] OR "autobiography"[pt] OR "bibliography"[pt] OR "biography"[pt] OR "case reports"[pt] OR "comment"[pt] OR "congress"[pt] OR "dictionary"[pt] OR "directory"[pt] OR "festschrift"[pt] OR "government publication"[pt] OR "historical article"[pt] OR "interview"[pt] OR "lecture"[pt] OR "legal case"[pt] OR "legislation"[pt] OR "news"[pt] OR "newspaper article"[pt] OR "patient education handout"[pt] OR "periodical index"[pt] OR "comment"[ti] OR "Editorial" [Publication Type] OR "ephemera"[pt] OR "in vitro techniques"[mh] OR "introductory journal article"[pt] OR ("Animals"[Mesh] NOT "Humans"[Mesh]) OR rats[tw] OR rat[tw] OR cow[tw] OR cows[tw] OR chicken*[tw] OR horse[tw] OR horses[tw] OR mice[tw] OR mouse[tw] OR bovine[tw] OR sheep[tw] OR ovine[tw] OR murinae[tw] OR cats[tw] OR cat[tw] OR dog[tw] OR dogs[tw] OR rodent[tw] OR "Case Reports" [Publication Type] OR ("Review" [Publication Type] NOT "Systematic Review" [Publication Type]))) <u>Specified scores and models</u>: ((((((((((("Renal angina index"[tiab]) OR (("Nephrotoxic Injury Negated" AND "just-in-Time Action") OR "NINJA"[tiab])) OR ("Fluid Overload Kidney Injury Score" OR "Kidney Injury Score" OR "FOKIS"[tiab])) OR

		("Simple Postoperative AKI Risk Score")) OR ("Simple Postoperative AKI Risk Index")) OR (SPARK[tiab] AND score[tiab])) OR ("Cleveland Clinic Score")) OR (("Mehta Score") OR (Mehta[tiab] AND score[tiab]))) OR (AKICS[tiab] OR (("Acute kidney injury following cardiac surgery" OR "AKI following cardiac surgery") AND "score"[tiab]))) OR ("Mehran Score" OR (Mehran[tiab] AND score[tiab]))) AND (("Acute Kidney Injury"[Mesh] OR (("Acute Kidney"[tiab] OR "Acute Renal"[tiab]) AND (Injury[tiab] OR failure[tiab] OR disease[tiab] OR impairment[tiab] OR insufficiency[tiab]))))) NOT ((("address"[pt] OR "autobiography"[pt] OR "bibliography"[pt] OR "biography"[pt] OR "case reports"[pt] OR "comment"[pt] OR "congress"[pt] OR "dictionary"[pt] OR "directory"[pt] OR "festschrift"[pt] OR "government publication"[pt] OR "historical article"[pt] OR "interview"[pt] OR "lecture"[pt] OR "legal case"[pt] OR "legislation"[pt] OR "news"[pt] OR "newspaper article"[pt] OR "patient education handout"[pt] OR "periodical index"[pt] OR "comment"[ti] OR "Editorial" [Publication Type] OR "ephemera"[pt] OR "in vitro techniques"[mh] OR "introductory journal article"[pt] OR ("Animals"[Mesh] NOT "Humans"[Mesh]) OR rats[tw] OR rat[tw] OR cow[tw] OR cows[tw] OR chicken*[tw] OR horse[tw] OR horses[tw] OR mice[tw] OR mouse[tw] OR bovine[tw] OR sheep[tw] OR ovine[tw] OR murinae[tw] OR cats[tw] OR cat[tw] OR dog[tw] OR dogs[tw] OR rodent[tw] OR "Case Reports" [Publication Type] OR ("Review" [Publication Type] NOT "Systematic Review" [Publication Type]))))
	Embase	Focused: #6 ('decision support system' OR 'ninja' OR 'sniffer') AND ('acute kidney failure'/exp OR 'acute kidney failure') #10 'furosemide stress test' AND ('acute kidney failure'/exp OR 'acute kidney failure') #11 ('real time' OR 'real-time' OR 'realtime' OR 'kinetic') AND 'glomerulus filtration rate' AND ('acute kidney failure'/exp OR 'acute kidney failure')

		#12 ('acute kidney tubule necrosis' OR 'acute tubular injury') AND ('urine' OR 'microscopy') AND ('score' OR 'scoring system' OR 'scale' OR 'biopsy' OR 'histology') AND ('acute kidney failure'/exp OR 'acute kidney failure') #13 'echography' AND ('hydronephrosis' OR 'perfusion' OR 'resistive index') AND ('acute kidney failure'/exp OR 'acute kidney failure') #14 'nuclear magnetic resonance imaging' AND 'fibrosis' AND ('acute kidney failure'/exp OR 'acute kidney failure') #15 #6 OR #10 OR #11 OR #12 OR #13 OR #14 #16 #15 AND 'human'/de #17 #16 AND ('clinical trial'/de OR 'cohort analysis'/de OR 'comparative study'/de OR 'controlled clinical trial'/de OR 'controlled study'/de OR 'diagnostic test accuracy study'/de OR 'feasibility study'/de OR 'logistic regression analysis'/de OR 'longitudinal study'/de OR 'major clinical study'/de OR 'meta analysis'/de OR 'meta analysis topic'/de OR 'model'/de OR 'multicenter study'/de OR 'multivariate logistic regression analysis'/de OR 'observational study'/de OR 'predictive model'/de OR 'prospective study'/de OR 'randomized controlled trial'/de OR 'retrospective study'/de OR 'risk model'/de OR 'statistical model'/de OR 'systematic review'/de OR 'validation study'/de) #18 #17 AND ('Article'/it OR 'Article in Press'/it OR 'Preprint'/it) <u>Specified scores and models</u>: #1 'renal angina index'/exp OR 'renal angina index' #2 'nephrotoxic injury negated by just-in-time action' #3 'fluid overload kidney injury score' OR fokis #4 'simple' AND 'postoperative' AND 'aki' AND 'risk score' #5 'cleveland clinic score' #6 'mehta score' #7 akics #8 ('aki following cardiac surgery' OR 'acute kidney injury following cardiac surgery') AND 'score' #9 'mehran score' #11 #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 #12 #11 AND 'human'/de

		#13	#12 AND ('clinical study'/de OR 'clinical trial'/de OR 'cohort analysis'/de OR 'comparative effectiveness'/de OR 'comparative study'/de OR 'controlled clinical trial'/de OR 'controlled study'/de OR 'diagnostic test accuracy study'/de OR 'double blind procedure'/de OR 'intervention study'/de OR 'logistic regression analysis'/de OR 'longitudinal study'/de OR 'major clinical study'/de OR 'meta analysis'/de OR 'meta analysis topic'/de OR 'multicenter study'/de OR 'multivariate logistic regression analysis'/de OR 'observational study'/de OR 'pilot study'/de OR 'practice guideline'/de OR 'predictive model'/de OR 'prospective study'/de OR 'randomized controlled trial'/de OR 'randomized controlled trial topic'/de OR 'retrospective study'/de OR 'risk model'/de OR 'systematic review'/de OR 'validation process'/de OR 'validation study'/de)
		#14	#13 AND ('Article'/it OR 'Article in Press'/it)
RQs 5, 6, 12, & 13: Biomarkers, fluids, bicarbonate, rhabdomyolysis etc.	PubMed	1	"Cystatin C"[Mesh] OR Cystatin-C[tiab] OR "kidney injury molecule-1"[tiab] OR "KIM-1"[tiab] OR NGAL[tiab] OR "neutrophil gelatinase-associated lipocalin"[tiab] OR TIMP2[tiab] OR IGFBP7[tiab] OR "tissue inhibitor of metalloproteinases-2"[tiab] OR "insulin-like growth factor-binding protein 7"[tiab] OR "fatty acid binding protein"[tiab] OR "FABP"[tiab] OR "Tissue Inhibitor of Metalloproteinase-2"[Mesh] OR "Lipocalin-2"[Mesh] OR "Lipocalin-2"[tiab] OR "Fatty Acid-Binding Proteins"[Mesh]
		2	"Acute Kidney Injury"[Mesh] OR (("Acute Kidney"[tiab] OR "Acute Renal"[tiab]) AND (Injury[tiab] OR failure[tiab] OR disease[tiab] OR impairment[tiab] OR insufficiency[tiab]))
		3	AND/1, 2
		4	Hydroxyethyl starch*[tiab] OR "Hydroxyethyl Starch Derivatives"[Mesh] OR "HES"[tiab] OR Pentafraction[tiab] OR Hespan[tiab] OR Plasmasteril[tiab] OR HAES-steril[tiab] OR Hemohes[tiab] OR Pentaspan[tiab] OR Pentastarch[tiab] OR Elohes[tiab] OR Hetastarch[tiab] OR Hespan[tiab] OR Hextend[tiab] OR Volulyte[tiab] OR Voluven[tiab]
		5	("Fluid Therapy"[Mesh] OR "Rehydration Solutions"[Mesh] OR "World Health Organization oral rehydration solution"[Supplementary Concept] OR rehydrat*[tiab] OR

			hydrat*[tiab] OR colloid*[tiab] OR hydrocolloid*[tiab] OR crystalloid*[tiab] OR albumin*[tiab] OR albumen*[tiab] OR starch*[tiab] OR dextran*[tiab] OR isotonic[tiab] OR ringer*[tiab] OR gelatin*[tiab] OR pentastarch*[tiab] OR saline[tiab] OR (isotonic[tiab] AND saline[tiab] AND solution*[tiab]) OR (blood[tiab] AND expan*[tiab]) OR (plasma[tiab] AND volume[tiab] AND expan*[tiab]) OR (volume[tiab] AND expan*[tiab])) AND (oral[tiab] OR "per os"[tiab]) AND (intravenous[tiab] OR IV[tiab])
		6	("Acidosis"[Mesh] OR acidos*[tiab] OR acidemi*[tiab] OR acidaemi*[tiab]) AND ("Bicarbonates"[Mesh] OR bicarbon*[tiab] OR carbonic[tiab])
		7	"Tumor Lysis Syndrome"[Mesh] OR ((tumor[tiab] OR tumour[tiab]) AND lysis[tiab]) OR "Rhabdomyolysis"[Mesh] OR rhabdomyolys*[tiab] OR ((pigment[tiab] OR haem*[tiab] OR heme*[tiab] OR hemoglobin[tiab] OR myoglobin[tiab]) AND nephropath*[tiab])
		8	OR/3-7
		9	(((("randomized controlled trial"[pt] OR "controlled clinical trial"[pt] OR randomized[tiab] OR placebo[tiab] OR randomly[tiab] OR trial[tiab] OR groups [tiab]) NOT "review"[Filter]) OR systematic[sb]) NOT (animals[mh] NOT humans[mh])) AND (
		10	random*[tiab] OR trial[tiab] OR systematic*[tiab] OR Cochrane[tiab] OR "meta"[tiab] OR "literature review"[tiab] OR PRISMA[tiab] OR AMSTAR[tiab]
		11	AND/8, 9, 10
	Embase	#1	'cystatin c'/exp OR 'cystatin c' OR 'kidney injury molecule 1 protein'/exp OR 'kidney injury molecule 1 protein' OR 'kim 1 protein'/exp OR 'kim 1 protein' OR 'neutrophil gelatinase associated lipocalin'/exp OR 'neutrophil gelatinase associated lipocalin' OR (('tissue inhibitor of metalloproteinase 2'/exp OR 'tissue inhibitor of metalloproteinase 2') AND ('insulin like growth factor binding protein 7'/exp OR 'insulin like growth factor binding protein 7')) OR 'fatty acid binding protein'/exp OR 'fatty acid binding protein'
		#2	'acute kidney failure' OR 'kidney injury' OR 'acute kidney disease'
		#3	#1 AND #2

		#4 'hydroxyethyl starch' #5 'fluid therapy' OR 'rehydration' OR 'oral rehydration solution' OR 'plasma substitute' OR 'crystalloid' OR 'isotonic solution' OR 'ringer lactate solution' OR 'volume expansion' #6 'oral drug administration' AND 'intravenous drug administration' #7 #5 AND #6 #8 ('acidosis' OR 'acidemia') AND 'bicarbonate' #9 'tumor lysis syndrome' OR 'rhabdomyolysis' OR 'pigment nephropathy' #10 'fluid therapy' OR 'rehydration' OR 'oral rehydration solution' OR 'plasma substitute' OR 'crystalloid' OR 'isotonic solution' OR 'ringer lactate solution' OR 'volume expansion' OR 'allopurinol' OR 'rasburicase' OR 'febuxostat' OR 'renal replacement therapy' OR 'ultrafiltration' #11 #9 AND #10 #12 #3 OR #4 OR #7 OR #8 OR #11 #13 #12 AND 'human'/de #14 #13 AND ('clinical trial'/de OR 'cohort analysis'/de OR 'comparative effectiveness'/de OR 'comparative study'/de OR 'controlled clinical trial'/de OR 'controlled study'/de OR 'double blind procedure'/de OR 'major clinical study'/de OR 'meta analysis'/de OR 'phase 2 clinical trial'/de OR 'phase 3 clinical trial'/de OR 'pilot study'/de OR 'randomized controlled trial'/de OR 'systematic review'/de)
RQ 7: Amino acids	PubMed	1 "Cardiac Surgical Procedures"[Mesh] OR "Cardiovascular Surgical Procedures"[Mesh] OR "Coronary Artery Bypass"[Mesh] OR "Myocardial Revascularization"[Mesh] OR "Cardiac Valve Annuloplasty"[Mesh] OR ((cardiac OR cardiovascular OR cardio* OR coronary OR valve) AND (surger* OR surgical)) 2 ((intravenous OR infus*) AND amino acid*) OR (("Infusions, Intravenous"[Mesh] OR "Injections, Intravenous"[Mesh] OR "Parenteral Nutrition"[Mesh]) AND "Amino Acids"[Mesh]) 3 AND 1, 2 4 ("randomized controlled trial"[pt] OR "controlled clinical trial"[pt] OR randomized[tiab] OR placebo[tiab] OR randomly[tiab] OR trial[tiab] OR groups [tiab] OR random*[tiab] OR trial[tiab] OR systematic*[tiab] OR systematic[sb] OR Cochrane[tiab] OR

		"meta"[tiab] OR "literature review"[tiab] OR PRISMA[tiab] OR AMSTAR[tiab]) NOT (animals[mh] NOT humans[mh])
	Embase	#1. 'cardiovascular surgery'/exp OR 'cardiovascular surgery' OR 'heart valve replacement'/exp OR 'heart valve replacement' OR 'coronary artery bypass graft'/exp OR 'coronary artery bypass graft' #2. 'intravenous drug administration' OR 'infusion' OR 'parenteral drug administration' #3. 'amino acid' #4. #1 AND #2 AND #3 #5. #4 AND ('clinical trial'/de OR 'cohort analysis'/de OR 'comparative effectiveness'/de OR 'comparative study'/de OR 'controlled clinical trial'/de OR 'controlled study'/de OR 'double blind procedure'/de OR 'major clinical study'/de OR 'meta analysis'/de OR 'phase 2 clinical trial'/de OR 'phase 3 clinical trial'/de OR 'pilot study'/de OR 'randomized controlled trial'/de OR 'systematic review'/de) #6. #5 AND ('Article'/it OR 'Article in Press'/it)
RQ 8: Alerts	PubMed	(((("Acute Kidney Injury"[Mesh] OR "Acute Kidney"[tiab] OR "Acute Renal"[tiab]) AND (Injury[tiab] OR failure[tiab] OR disease[tiab] OR impairment[tiab] OR insufficiency[tiab])) AND ("Automation"[Mesh] OR "Automation, Laboratory"[Mesh] OR alarm* OR alert* OR information system* OR messag* OR notif* OR remind* OR reporting system* OR warn* OR "Clinical Alarms"[Mesh] OR "Decision Support Systems, Clinical"[Mesh] OR "Reminder Systems"[Mesh])) NOT (animals[mh] NOT humans[mh])) AND ("randomized controlled trial"[pt] OR "controlled clinical trial"[pt] OR (systematic[sb] NOT "review"[Filter]) OR randomized[tiab] OR placebo[tiab] OR randomly[tiab] OR trial[tiab] OR groups [tiab] OR random*[tiab] OR trial[tiab] OR systematic*[tiab] OR Cochrane[tiab] OR "meta"[tiab] OR "literature review"[tiab] OR PRISMA[tiab] OR AMSTAR[tiab])
	Embase	#1 'acute kidney failure'

		#2 'computer assisted diagnosis' OR 'automation' OR 'alarm monitoring' OR 'information system' OR 'disease notification' OR 'reminder system' OR 'alarm monitor' OR 'decision support system' #3 #1 AND #2 #4 #3 AND 'human'/de #5 #4 AND ('clinical trial'/de OR 'controlled study'/de OR 'major clinical study'/de OR 'meta analysis'/de OR 'randomized controlled trial'/de OR 'randomized controlled trial topic'/de OR 'systematic review'/de) #6 #4 AND ('clinical trial'/de OR 'controlled study'/de OR 'major clinical study'/de OR 'meta analysis'/de OR 'randomized controlled trial'/de OR 'randomized controlled trial topic'/de OR 'systematic review'/de)
RQ 9: Fluid removal	PubMed	("Acute Kidney Injury"[Mesh] OR (("Acute Kidney"[tiab] OR "Acute Renal"[tiab]) AND (Injury[tiab] OR failure[tiab] OR disease[tiab] OR impairment[tiab] OR insufficiency[tiab]))) AND (“Renal Replacement Therapy”[Mesh] OR (“renal replacement”[tiab] OR “kidney replacement”) OR (Hemofiltration[Mesh] OR Hemofiltration[tiab] OR Hemodiafiltration[tiab] OR Ultrafiltration[tiab] OR hemodialysis[tiab] or dialysis[tiab])) OR ("Diuretics"[Mesh] OR diuretic*[tiab] OR "Furosemide"[Mesh] or furosemide[tiab]) AND (("randomized controlled trial"[pt] OR "controlled clinical trial"[pt] OR trial[tiab] OR random*[tiab] OR systematic*[tiab] OR systematic[sb] OR Cochrane[tiab] OR "meta"[tiab] OR "literature review"[tiab] OR PRISMA[tiab] OR AMSTAR[tiab])) NOT ("addresses" OR "autobiography"[pt] OR "bibliography"[pt] OR "biography"[pt] OR "case reports" OR "comment"[pt] OR "congresses" OR "dictionary"[pt] OR "directory"[pt] OR "festschrift"[pt] OR "government publications" OR "historical article"[pt] OR "interview"[pt] OR "lectures" OR "legal cases" OR "legislation"[pt] OR "news"[pt] OR "newspaper article"[pt] OR "patient education handout"[pt] OR "periodical index"[pt] OR "comment on" OR ("review"[pt] NOT "systematic review"[pt]) OR ("Animals"[Mesh] NOT "Humans"[Mesh]) OR rats[tw]

		OR cow[tw] OR cows[tw] OR chicken*[tw] OR horse[tw] OR horses[tw] OR mice[tw] OR mouse[tw] OR bovine[tw] OR sheep OR ovine OR murinae OR "animal model" OR "Case-Control Studies"[Mesh] OR "Cohort Studies"[Mesh] OR "Regression Analysis"[Mesh] OR "Retrospective Studies"[Mesh] OR "Observational Studies as Topic"[Mesh] OR "Observational Study" [Publication Type] OR "Case Reports" [Publication Type]0
	Embase	#1 'acute kidney failure'/exp OR 'acute kidney failure' #2 'renal replacement therapy'/exp OR 'renal replacement therapy' OR 'hemodialysis'/exp OR 'hemodialysis' OR 'peritoneal dialysis'/exp OR 'peritoneal dialysis' #3 'diuretic agent' OR 'furosemide' #4 #2 OR #3 #5 #1 AND #4 #7 #5 AND 'human'/de #8 #7 AND ('meta analysis'/de OR 'randomized controlled trial'/de OR 'randomized controlled trial topic'/de OR 'systematic review'/de) #9 #8 AND ('article'/it OR 'article in press'/it OR 'preprint'/it) #10 random*:ti OR random*:ab OR random*:kw OR systematic*:ti OR systematic*:ab OR systematic*:kw OR 'meta analys*':ti OR 'meta analys*':ab OR 'meta analys*':kw OR metaanalys*:ti OR metaanalys*:ab OR metaanalys*:kw #11 #9 AND #10
RQ 10: Hepatorenal syndrome	PubMed	("Hepatorenal Syndrome"[Mesh] OR "hepatorenal"[tiab] OR "hepato-renal"[tiab]) AND ((("Terlipressin "[Mesh] OR "terlipressin"[tiab])) OR (("Midodrine"[Mesh] OR "Midodrine"[tiab]) AND ("Octreotide"[Mesh] OR "octreotide"[tiab])) OR (("Epinephrine"[Mesh] OR "Norepinephrine"[Mesh] OR "Epinephrine"[tiab] OR "Norepinephrine"[tiab])) OR ("Fenoldopam"[Mesh] OR "Fenoldopam"[tiab]))
	Embase	#21 'hepatorenal syndrome' #22 'terlipressin' #23 'midodrine' AND 'octreotide' #24 ('epinephrine' OR 'noradrenalin')

		#25 'fenoldopam' #26 #22 OR #23 OR #24 #27 #26 AND ('clinical trial'/de OR 'controlled clinical trial'/de OR 'controlled study'/de OR 'double blind procedure'/de OR 'major clinical study'/de OR 'meta analysis'/de OR 'meta analysis topic'/de OR 'randomized controlled trial'/de OR 'randomized controlled trial topic'/de OR 'single blind procedure'/de OR 'systematic review'/de) #28 #25 AND ('clinical trial topic'/de OR 'cohort analysis'/de OR 'comparative effectiveness'/de OR 'comparative study'/de OR 'controlled clinical trial'/de OR 'controlled clinical trial topic'/de OR 'controlled study'/de OR 'double blind procedure'/de OR 'experimental design'/de OR 'intention to treat analysis'/de OR 'major clinical study'/de OR 'meta analysis'/de OR 'meta analysis topic'/de OR 'multicenter study'/de OR 'multicenter study topic'/de OR 'network meta analysis'/de OR 'observational study'/de OR 'phase 2 clinical trial topic'/de OR 'pilot study'/de OR 'prospective study'/de OR 'randomized controlled trial'/de OR 'randomized controlled trial topic'/de OR 'retrospective study'/de OR 'systematic review'/de OR 'systematic review topic'/de) #29 #27 OR #28 #30 #21 AND #29 #31 #30 AND 'human'/de
RQs 14, 15, & 16: Pediatric theophylline, nitric oxide, and malaria	PubMed	((((Theophylline[tiab] OR "Theophylline"[Mesh] OR "Aminophylline"[Mesh] OR Aminophylline[tiab]) AND (neonat*[tiab] OR infant*[tiab] OR preterm[tiab] OR premature[tiab] OR newborn*[tiab] OR "new-born*"[tiab] OR "Infant, Newborn"[Mesh] OR baby[tiab] OR babies[tiab])) OR (("Nitric Oxide"[Mesh] OR “nitric oxide”[tiab] OR “Nitrogen Monoxide”[tiab:~1] OR “Mononitrogen Monoxide”[tiab:~1] OR “Nitrate Vasodilator”[tiab:~1]) AND ("Acute Kidney Injury"[Mesh] OR (("Acute Kidney"[tiab] OR "Acute Renal"[tiab]) AND (Injury[tiab] OR failure[tiab] OR disease[tiab] OR impairment[tiab] OR insufficiency[tiab])) OR “fluid balance"[tiab]))

		OR (("Malaria"[Mesh] OR malaria[tiab] OR Plasmodium[tiab] OR "marsh fever"[tiab]) AND ("Acetaminophen"[Mesh] OR Acetaminophen*[tiab] OR Paracetamol[tiab] OR Tylenol[tiab] OR Anacin[tiab] OR Panadol[tiab] OR Hydroxyacetanilide[tiab])))) NOT (animals[mh] NOT humans[mh])
	Embase	#1 'theophylline derivative'/exp OR 'theophylline derivative' OR 'theophylline'/exp OR 'theophylline' OR 'aminophylline'/exp OR 'aminophylline' #2 'newborn' OR 'infant' OR 'prematurity' OR 'baby' #3 #1 AND #2 #4 'nitric oxide' NOT 'nitric oxide synthase inhibitor' #5 'acute kidney failure' OR 'acute kidney disease' #6 #4 AND #5 #7 'malaria' #8 'paracetamol' #9 #7 AND #8 #10 #3 OR #6 OR #9 #11 #10 AND ('clinical trial'/de OR 'cohort analysis'/de OR 'comparative effectiveness'/de OR 'comparative study'/de OR 'controlled clinical trial'/de OR 'controlled study'/de OR 'major clinical study'/de OR 'meta analysis'/de OR 'randomized controlled trial'/de OR 'systematic review'/de)
RQ 17: Stewardship, pharmacist	PubMed	("Acute Kidney Injury"[Mesh] OR (("Acute Kidney"[tiab] OR "Acute Renal"[tiab]) AND (Injury[tiab] OR failure[tiab] OR disease[tiab] OR impairment[tiab] OR insufficiency[tiab]))) AND (((multimodal[tiab] OR antimicrobial[tiab] OR antibiotic*[tiab]) AND steward*[tiab]) OR "Antimicrobial Stewardship"[Mesh] OR "Drug Monitoring"[Mesh] OR (drug[tiab] AND monitor*[tiab]) OR "Drug Interactions"[Mesh] OR (drug[tiab] AND interact*[tiab]) OR "Aminoglycosides"[Mesh] OR aminoglycoside*[tiab] OR gentamicin[tiab] OR amikacin[tiab] OR streptomycin[tiab] OR "Vancomycin"[Mesh] OR vancomycin[tiab] OR "beta-Lactams"[Mesh] OR "beta Lactam

		Antibiotics"[Mesh] OR (beta[tiab] AND lactam[tiab]) OR cepha*[tiab] or cefa*[tiab] OR aztreonam[tiab])) AND ("randomized controlled trial"[pt] OR "comparative study"[pt] OR "multicenter study"[pt] OR "observational study"[pt] OR "clinical trial"[pt] OR "controlled clinical trial"[pt] OR "Prospective Studies"[Mesh] OR "Follow-Up Studies"[Mesh] OR "Cohort Studies"[Mesh] OR "Longitudinal Studies"[Mesh] OR "registries"[Mesh] OR random* OR crossover OR parallel \l OR "clinical trial" OR "cohort stud*"[tiab] OR systematic*[tiab] OR systematic[sb] OR Cochrane[tiab] OR "meta"[tiab] OR "literature review"[tiab] OR PRISMA[tiab] OR AMSTAR[tiab]0 NOT ("addresses" OR "autobiography"[pt] OR "bibliography"[pt] OR "biography"[pt] OR "case reports" OR "comment"[pt] OR "congresses" OR "dictionary"[pt] OR "directory"[pt] OR "festschrift"[pt] OR "government publications" OR "historical article"[pt] OR "interview"[pt] OR "lectures" OR "legal cases" OR "legislation"[pt] OR "news"[pt] OR "newspaper article"[pt] OR "patient education handout"[pt] OR "periodical index"[pt] OR "comment on" OR ("review"[pt] NOT "systematic review"[pt]) OR ("Animals"[Mesh] NOT "Humans"[Mesh]) OR rats[tw] OR cow[tw] OR cows[tw] OR chicken*[tw] OR horse[tw] OR horses[tw] OR mice[tw] OR mouse[tw] OR bovine[tw] OR sheep OR ovine OR murinae OR "animal model")
	Embase	#1 'acute kidney failure'/exp OR 'acute kidney failure' #2 (multimodal OR 'antiinfective agent') AND 'stewardship' OR 'drug monitoring' OR 'drug interaction' OR 'drug surveillance program' OR (('aminoglycoside antibiotic agent' OR 'vancomycin' OR 'beta lactam antibiotic') AND ('drug dose comparison' OR 'dosage schedule comparison')) #3 #1 AND #2 #4 #3 AND 'human'/de #5 #4 AND ('clinical trial'/de OR 'clinical trial topic'/de OR 'comparative effectiveness'/de OR 'comparative study'/de OR 'controlled study'/de OR 'double blind procedure'/de OR 'meta analysis'/de OR 'multicenter study topic'/de OR 'phase 3 clinical trial'/de OR 'phase 3 clinical trial topic'/de OR 'randomized

		controlled trial'/de OR 'randomized controlled trial topic'/de OR 'systematic review'/de) #6 #5 AND ('article'/it OR 'article in press'/it OR 'preprint'/it)
RQ 18: Drug holds	PubMed	("Withholding Treatment"[Mesh] OR Discontinue* OR omission OR omit* OR cease OR cessation OR stop* OR withhold* OR suspen*) AND (("Angiotensin-Converting Enzyme Inhibitors"[Mesh] OR "Angiotensin Receptor Antagonists"[Mesh] OR "Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR ("Anti-Inflammatory Agents, Non-Steroidal"[Mesh] AND "Surgical Procedures, Operative"[Mesh]) OR "Glucagon-Like Peptide-1 Receptor Agonists"[Mesh])) OR (("nonsteroidal"[tiab] OR "non-steroidal"[tiab]) AND ("surgery"[tiab] OR "surgical"[tiab] OR "operat*"[tiab])) NOT (animals[mh] NOT humans[mh]))
	Embase	#1 'dipeptidyl carboxypeptidase inhibitor' OR 'angiotensin receptor antagonist' OR 'sodium glucose cotransporter 2 inhibitor' OR 'glucagon like peptide receptor agonist' #2 'nonsteroid antiinflammatory agent' AND 'surgery' #3 #1 OR #2 #4 'treatment withdrawal' #5 #3 AND #4 #6 #5 AND 'human'/de #7 #6 AND ('clinical trial'/de OR 'clinical trial topic'/de OR 'major clinical study'/de OR 'meta analysis'/de OR 'meta analysis topic'/de OR 'phase 3 clinical trial topic'/de OR 'randomized controlled trial'/de OR 'single blind procedure'/de OR 'systematic review'/de)
RQs 19, & 20: Contrast-induced nephropathy	PubMed	(((("Contrast Media"[Mesh] OR ((intravascular OR intra-vascular OR intravenous OR intra-venous OR intraarter* OR intra-arter*) AND (contrast OR iodin*))) AND ((contrast AND nephropathy) OR "Acute Kidney Injury"[Mesh] OR (("Acute Kidney"[tiab] OR "Acute Renal"[tiab]) AND (Injury[tiab] OR failure[tiab] OR disease[tiab] OR impairment[tiab] OR insufficiency[tiab])))) NOT ((animals[mh] NOT humans[mh]))) AND

		(("Models, Statistical"[Mesh] OR "Regression Analysis"[Mesh] OR "Logistic Models"[Mesh] OR "Linear Models"[Mesh] OR "Risk Factors"[Mesh] OR "model*"[tiab] OR "regression"[tiab] OR "logistic"[tiab] OR "linear"[tiab] OR "associate*"[tiab] OR "risk score]"[tiab] "risk factor*"[tiab]) OR ('clinical trial'/de OR 'cohort analysis'/de OR 'comparative effectiveness'/de OR 'comparative study'/de OR 'controlled clinical trial'/de OR 'controlled study'/de OR 'major clinical study'/de OR 'meta analysis'/de OR 'randomized controlled trial'/de OR 'systematic review'/de OR "Meta-Analysis"[pt]))
	Embase	#1 'contrast radiography' OR 'contrast medium' #2 'acute kidney failure' OR 'acute kidney disease' #4 #1 AND #2 #5 #4 AND 'human'/de #6 #5 AND ('clinical trial'/de OR 'clinical trial topic'/de OR 'cohort analysis'/de OR 'comparative effectiveness'/de OR 'comparative study'/de OR 'controlled clinical trial'/de OR 'controlled study'/de OR 'diagnostic test accuracy study'/de OR 'double blind procedure'/de OR 'intention to treat analysis'/de OR 'logistic regression analysis'/de OR 'major clinical study'/de OR 'meta analysis'/de OR 'meta analysis topic'/de OR 'multivariate logistic regression analysis'/de OR 'observational study'/de OR 'practice guideline'/de OR 'proportional hazards model'/de OR 'prospective study'/de OR 'randomized controlled trial'/de OR 'randomized controlled trial topic'/de OR 'retrospective study'/de OR 'single blind procedure'/de OR 'systematic review'/de OR 'validation study'/de) #7 #6 AND 'article'/it
RQs 22, 23, and 24: RRT	PubMed	("Acute Kidney Injury"[Mesh] OR (("Acute Kidney"[tiab] OR "Acute Renal"[tiab]) AND (Injury[tiab] OR failure[tiab] OR disease[tiab] OR impairment[tiab] OR insufficiency[tiab]))) AND (“Renal Replacement Therapy”[Mesh] OR (“renal replacement”[tiab] OR “kidney replacement”) OR

		(Hemofiltration[Mesh] OR Hemofiltration[tiab] OR Hemodiafiltration[tiab] OR Ultrafiltration[tiab] OR hemodialysis[tiab] OR dialysis[tiab])) AND (((early[tiab] OR earlier[tiab] OR timing[tiab] OR start[tiab] OR initial[tiab] OR initiat*[tiab] OR begin[tiab] OR commence[tiab] OR prophylactic[tiab] OR accelerat*[tiab] OR late[tiab] OR delay*[tiab] OR standard[tiab])) OR ((continuous[tiab] OR continual[tiab] OR intermittent[tiab] OR technique*[tiab] OR technic*[tiab] OR sustained[tiab] OR peritoneal[tiab]) OR ((dose[tiab] OR dosing[tiab] OR prescription[tiab] OR intens*[tiab] OR “urea reduction ratio”[tiab] OR “kt/V”[tiab] OR volume[tiab]))) AND (("randomized controlled trial"[pt] OR "controlled clinical trial"[pt] OR randomized[tiab] OR placebo[tiab] OR randomly[tiab] OR trial[tiab] OR groups [tiab] OR random*[tiab] OR trial[tiab] OR systematic*[tiab] OR systematic[sb] OR Cochrane[tiab] OR "meta"[tiab] OR "literature review"[tiab] OR PRISMA[tiab] OR AMSTAR[tiab]) NOT (animals[mh] NOT humans[mh]))
	Embase	#1 'renal replacement therapy'/exp OR 'renal replacement therapy' OR 'hemodialysis'/exp OR 'hemodialysis' OR 'peritoneal dialysis'/exp OR 'peritoneal dialysis' #2 'acute kidney failure' #4 'clinical trial'/de OR 'controlled study'/de OR 'major clinical study'/de OR 'meta analysis'/de OR 'randomized controlled trial'/de OR 'randomized controlled trial topic'/de OR 'systematic review'/de #5 #3 AND #4 #8 early OR earlier OR timing OR start OR initial OR initiate OR begin OR commence OR prophylaxis OR prophylactic OR accelerated OR late OR delayed OR standard #11 continuous OR continual OR intermittent OR technique OR technic OR sustained OR peritoneal

		#14 dose OR dosing OR prescription OR intense OR intensity OR 'urea reduction ratio' OR 'kt/v' OR volume #15 #8 OR #11 OR #14 #16 #5 AND #15 #17 #16 AND 'human'/de #18 'randomized controlled trial' OR 'systematic review' OR 'meta analysis' #19 #17 AND #18 #20 #19 AND ('Article'/it OR 'Article in Press'/it OR 'Preprint'/it)
RQ 25: RRT anticoagulation	PubMed	("Acute Kidney Injury"[Mesh] OR (("Acute Kidney"[tiab] OR "Acute Renal"[tiab]) AND (Injury[tiab] OR failure[tiab] OR disease[tiab] OR impairment[tiab] OR insufficiency[tiab]))) AND (“Renal Replacement Therapy”[Mesh] OR (“renal replacement”[tiab] OR “kidney replacement”) OR (Hemofiltration[Mesh] OR Hemofiltration[tiab] OR Hemodiafiltration[tiab] OR Ultrafiltration[tiab] OR hemodialysis[tiab] or dialysis[tiab])) AND ("Anticoagulants"[Mesh] OR "Factor Xa Inhibitors"[Mesh] OR "Heparinoids"[Mesh] OR "Sodium Citrate"[Mesh] OR "Heparin, Low-Molecular-Weight"[Mesh] OR "Antithrombins"[Mesh] OR "Epoprostenol"[Mesh] OR "argatroban" [Supplementary Concept] OR "nafamostat" [Supplementary Concept] OR “anticoagula*”[tiab] OR “anti-coagula*”[tiab] OR “Factor Xa”[tiab] OR “heparin”[tiab] OR “LMWH”[tiab] OR “Regional citrate anticoagulant*”[tiab] OR “antithrombin*”[tiab] OR “anti-thrombin*”[tiab] OR “Direct thrombin inhibitor*”[tiab] OR “Epoprostenol”[tiab] OR “Argatroban”[tiab] OR “Nafamostat”[tiab] OR "Protamines"[Mesh] OR “protamine*”[tiab] OR “prostacyclin”[tiab]) NOT ("addresses" OR "autobiography"[pt] OR "bibliography"[pt] OR "biography"[pt] OR "case reports" OR "comment"[pt] OR "congresses" OR "dictionary"[pt] OR "directory"[pt] OR "festschrift"[pt] OR "government publications" OR "historical article"[pt] OR "interview"[pt] OR "lectures" OR "legal cases" OR "legislation"[pt] OR "news"[pt] OR

		"newspaper article"[pt] OR "patient education handout"[pt] OR "periodical index"[pt] OR "comment on" OR ("review"[pt] NOT "systematic review"[pt]) OR ("Animals"[Mesh] NOT "Humans"[Mesh]) OR rats[tw] OR cow[tw] OR cows[tw] OR chicken*[tw] OR horse[tw] OR horses[tw] OR mice[tw] OR mouse[tw] OR bovine[tw] OR sheep OR ovine OR murinae OR "animal model")
	Embase	#1 'acute kidney failure'/exp OR 'acute kidney failure' #2 'renal replacement therapy'/exp OR 'renal replacement therapy' OR 'hemodialysis'/exp OR 'hemodialysis' OR 'peritoneal dialysis'/exp OR 'peritoneal dialysis' #3 'anticoagulant agent' OR 'regional citrate anticoagulation' OR 'heparin' OR 'low molecular weight heparin' OR 'protamine' OR 'thrombin inhibitor' OR 'blood clotting factor 10a inhibitor' OR 'prostacyclin' OR 'argatroban' OR 'nafamstat' #5 #1 AND #2 AND #3 #6 #5 AND 'human'/de #7 #6 AND ('clinical trial'/de OR 'clinical trial topic'/de OR 'comparative effectiveness'/de OR 'comparative study'/de OR 'controlled clinical trial'/de OR 'controlled study'/de OR 'double blind procedure'/de OR 'major clinical study'/de OR 'meta analysis'/de OR 'meta analysis topic'/de OR 'randomized controlled trial'/de OR 'randomized controlled trial topic'/de OR 'systematic review'/de) #8 #7 AND ('Article'/it OR 'Article in Press'/it OR 'Preprint'/it)
RQ 26: Post-AKI care	PubMed	("Acute Kidney Injury"[Mesh] OR (("Acute Kidney"[tiab] OR "Acute Renal"[tiab]) AND (Injury[tiab] OR failure[tiab] OR disease[tiab] OR impairment[tiab] OR insufficiency[tiab]))) AND ("Aftercare"[Mesh] OR "Continuity of Patient Care"[Mesh] OR “after-care”[tiab] OR “aftercare”[tiab] OR (care[tiab] AND (survivor*[tiab] OR "Survivors"[Mesh])) OR “post-discharge”[tiab] OR “postdischarge”[tiab] OR “post-hospital*”[tiab] OR “posthospital*”[tiab] OR "Patient Discharge"[Mesh] OR “post-AKI”[tiab] OR “post-AKD”[tiab] OR “post-ARF”[tiab] OR "Patient Care Team"[Mesh] OR “Patient Care Team"[tiab] OR “multidisciplinary care team”[tiab] OR “care model*”[tiab] OR "Case

		Management"[Mesh] OR “case manage*”[tiab] OR "Office Visits"[Mesh] OR “office visit*”[tiab]) AND ("randomized controlled trial"[pt] OR "comparative study"[pt] OR "multicenter study"[pt] OR "observational study"[pt] OR "clinical trial"[pt] OR "controlled clinical trial"[pt] OR "Prospective Studies"[Mesh] OR "Follow-Up Studies"[Mesh] OR "Cohort Studies"[Mesh] OR "Longitudinal Studies"[Mesh] OR "registries"[Mesh] OR random* OR crossover OR parallel \l OR "clinical trial" OR "cohort stud*"[tiab] OR systematic*[tiab] OR systematic[sb] OR Cochrane[tiab] OR "meta"[tiab] OR "literature review"[tiab] OR PRISMA[tiab] OR AMSTAR[tiab]) NOT ("addresses" OR "autobiography"[pt] OR "bibliography"[pt] OR "biography"[pt] OR "case reports" OR "comment"[pt] OR "congresses" OR "dictionary"[pt] OR "directory"[pt] OR "festschrift"[pt] OR "government publications" OR "historical article"[pt] OR "interview"[pt] OR "lectures" OR "legal cases" OR "legislation"[pt] OR "news"[pt] OR "newspaper article"[pt] OR "patient education handout"[pt] OR "periodical index"[pt] OR "comment on" OR ("review"[pt] NOT "systematic review"[pt]) OR ("Animals"[Mesh] NOT "Humans"[Mesh]) OR rats[tw] OR cow[tw] OR cows[tw] OR chicken*[tw] OR horse[tw] OR horses[tw] OR mice[tw] OR mouse[tw] OR bovine[tw] OR sheep OR ovine OR murinae OR "animal model")
	Embase	#1 'acute kidney failure'/exp OR 'acute kidney failure' #2 'aftercare' OR 'patient care team' OR 'multidisciplinary team' OR 'case management' OR 'ambulatory care' #4 #1 AND #2 #5 #4 AND 'human'/de #6 #5 AND ('Article'/it OR 'Article in Press'/it OR 'Preprint'/it) #7 #6 AND ('clinical article'/de OR 'clinical practice guideline'/de OR 'clinical trial'/de OR 'cohort analysis'/de OR 'comparative effectiveness'/de OR 'comparative study'/de OR 'controlled clinical trial'/de OR 'controlled study'/de OR 'double blind procedure'/de OR 'feasibility study'/de OR 'intervention study'/de OR 'longitudinal study'/de OR 'major clinical study'/de OR 'medical record review'/de OR 'meta analysis'/de OR 'multicenter study'/de OR 'multicenter

		study topic'/de OR 'multivariate logistic regression analysis'/de OR 'observational study'/de OR 'pilot study'/de OR 'practice guideline'/de OR 'prospective study'/de OR 'quasi experimental study'/de OR 'randomized controlled trial'/de OR 'randomized controlled trial topic'/de OR 'retrospective study'/de OR 'systematic review'/de OR 'total quality management'/de)
RQ 27: Post-AKI medications	PubMed	("Acute Kidney Injury"[Mesh] OR (("Acute Kidney"[tiab] OR "Acute Renal"[tiab]) AND (Injury[tiab] OR failure[tiab] OR disease[tiab] OR impairment[tiab] OR insufficiency[tiab]))) AND ("Angiotensin-Converting Enzyme Inhibitors"[Mesh] OR "Angiotensin Receptor Antagonists"[Mesh] OR "Anti-Inflammatory Agents, Non-Steroidal"[Mesh] OR "finerenone" [Supplementary Concept] OR "finerenone"[tiab] OR "Glucagon-Like Peptide-1 Receptor Agonists"[Mesh])) AND ("randomized controlled trial"[pt] OR "comparative study"[pt] OR "multicenter study"[pt] OR "observational study"[pt] OR "clinical trial"[pt] OR "controlled clinical trial"[pt] OR "Prospective Studies"[Mesh] OR "Follow-Up Studies"[Mesh] OR "Cohort Studies"[Mesh] OR "Longitudinal Studies"[Mesh] OR "registries"[Mesh] OR random* OR crossover OR parallel \l OR "clinical trial" OR "cohort stud*"[tiab] OR systematic*[tiab] OR systematic[sb] OR Cochrane[tiab] OR "meta"[tiab] OR "literature review"[tiab] OR PRISMA[tiab] OR AMSTAR[tiab]) NOT ("addresses" OR "autobiography"[pt] OR "bibliography"[pt] OR "biography"[pt] OR "case reports" OR "comment"[pt] OR "congresses" OR "dictionary"[pt] OR "directory"[pt] OR "festschrift"[pt] OR "government publications" OR "historical article"[pt] OR "interview"[pt] OR "lectures" OR "legal cases" OR "legislation"[pt] OR "news"[pt] OR "newspaper article"[pt] OR "patient education handout"[pt] OR "periodical index"[pt] OR "comment on" OR ("review"[pt] NOT "systematic review"[pt]) OR ("Animals"[Mesh] NOT "Humans"[Mesh]) OR rats[tw] OR cow[tw] OR cows[tw] OR chicken*[tw] OR horse[tw] OR horses[tw] OR mice[tw] OR mouse[tw] OR bovine[tw] OR sheep OR ovine OR murinae OR "animal model")

	Embase	#1 'acute kidney failure'/exp OR 'acute kidney failure' #2 'dipeptidyl carboxypeptidase inhibitor'/mj OR 'angiotensin receptor antagonist'/mj OR 'nonsteroid antiinflammatory agent'/mj OR 'finerenone'/mj OR 'glucagon like peptide receptor agonist'/mj #4 #1 AND #2 #5 #4 AND 'human'/de #6 #5 AND ('article'/it OR 'article in press'/it OR 'preprint'/it) #7 #6 AND ('clinical article'/de OR 'clinical practice guideline'/de OR 'clinical trial'/de OR 'cohort analysis'/de OR 'comparative effectiveness'/de OR 'comparative study'/de OR 'controlled clinical trial'/de OR 'controlled study'/de OR 'double blind procedure'/de OR 'feasibility study'/de OR 'intervention study'/de OR 'longitudinal study'/de OR 'major clinical study'/de OR 'medical record review'/de OR 'meta analysis'/de OR 'multicenter study'/de OR 'multicenter study topic'/de OR 'multivariate logistic regression analysis'/de OR 'observational study'/de OR 'pilot study'/de OR 'practice guideline'/de OR 'prospective study'/de OR 'quasi experimental study'/de OR 'randomized controlled trial'/de OR 'randomized controlled trial topic'/de OR 'retrospective study'/de OR 'systematic review'/de OR 'total quality management'/de)
RQ 28: Remote ischemic preconditioning		[Included studies found from two recent existing systematic reviews: Hariri G, Collet L, Duarte L, et al. Prevention of cardiac surgery-associated acute kidney injury: a systematic review and meta-analysis of non-pharmacological interventions. Crit Care. 2023 Sep 12;27(1):354. doi: 10.1186/s13054-023-04640-1. Martín-Fernández M, Casanova AG, Jorge-Monjas P, et al. A wide scope, pan-comparative, systematic meta-analysis of the efficacy of prophylactic strategies for cardiac surgery-associated acute kidney injury. Biomed Pharmacother. 2024 Sep;178:117152. doi: 10.1016/j.biopha.2024.117152.

Appendix B. Concurrence with Institute of Medicine (IOM) standards for guideline development and Appraisal of Guidelines, Research and Evaluation (AGREE) reporting checklist

Table S2. Guideline development checklist - IOM standards for development of trustworthy clinical practice guidelines (1)

IOM Standard	Description	Addressed in 2020 KDIGO BP in CKD guideline
Establishing transparency	Clear description on the process of guideline development.	See *Methods for Guideline Development*
Management of conflicts of interests	Disclosure of a comprehensive conflict of interests of the Work Group against a set-criteria and a clear strategy to manage conflicts of interest	See *Work Group Financial Disclosures*
Guideline group composition and guideline development	Appropriate clinical and methodological expertise in the Work Group The processes of guideline development are transparent and allow for involvement of all Work Group Members	For guideline group composition – see *Work Group Membership* For guideline development process see *Methods for Guideline Development*
Establishing evidence foundations for rating strength of recommendations	Rationale is provided for the rating the strength of the recommendation and the transparency for the rating the quality of the evidence.	See *Methods for Guideline Development*
Articulation of recommendations	Clear and standardized wording of recommendations	All recommendations were written to standards of GRADE and were actionable statements. Please see *Methods for Guideline Development*
External review	An external review of relevant experts and stakeholders was conducted. All comments received from external review are considered for finalization of the guideline.	An external public review was undertaken [April 2026].
Updating	An update for the guidelines is planned, with a provisional timeframe provided.	The KDIGO clinical practice guideline will be updated. However, no set timeframe has been provided.

Table S3. Adapted systematic review reporting standards checklist - IOM standards for systematic reviews (2)

Appropriate IOM systematic review standards*	Addressed in 2020 KDIGO diabetes in CKD guideline
Methods	
Include a research protocol with appropriate eligibility criteria (PICO format)	See *Table 16 clinical question and systematic review topics in PICO format*
Include a search strategy	See *Appendix A*
Include a study selection and data extraction process	See guideline development process see *Methods for Guideline Development – Literature searching and article selection, data extraction*
Methods on critical appraisal	See *Methods for Guideline Development – Critical appraisal of studies*
Methods of synthesize of the evidence	See *Methods for Guideline Development – Evidence synthesis and meta-analysis*
Results	
Study selection processes	See *Methods for Guideline Development – Figure 57 – Search yield and study flow diagram*
Appraisal of individual studies quality	The summary of findings tables in Appendix C & D provide an assessment of risk of bias for all studies in a comparison between intervention and comparator.
Meta-analysis results	See *Appendix C & D* for summary of findings tables for meta-analysis results for all critical and important outcomes
Table and figures	See *Appendix C & D* for summary of findings tables

References

1. Institute of Medicine Committee on Standards for Developing Trustworthy Clinical Practice Guidelines. Clinical practice guidelines we can trust. Graham R, Mancher M, editors. National Academies Press Washington, DC; 2011.
2. Institute of Medicine Committee on Standards for Systematic Reviews of Comparative Effectiveness R. In: Eden J, Levit L, Berg A, Morton S, editors. Finding What Works in Health Care: Standards for Systematic Reviews. Washington (DC): National Academies Press (US) Copyright 2011 by the National Academy of Sciences. All rights reserved; 2011.

Table S4. AGREE checklist (3)

CHECKLIST ITEM AND DESCRIPTION	REPORTING CRITERIA	Page #
DOMAIN 1: SCOPE and PURPOSE		
1. OBJECTIVES Report the overall objective(s) of the guideline. The expected health benefits from the guideline are to be specific to the clinical problem or health topic.	☐Health intent(s) (i.e., prevention, screening, diagnosis, treatment, etc.) ☐ Expected benefit(s) or outcome(s) ☐ Target(s) (e.g., patient population, society)	See *Methods for Guideline Development – Aim*
2. QUESTIONS Report the health question(s) covered by the guideline, particularly for the key recommendations	☐ Target population ☐ Intervention(s) or exposure(s) ☐ Comparisons (if appropriate) ☐ Outcome(s) ☐ Health care setting or context	See *Methods for Guideline Development – Table 16*
3. POPULATION Describe the population (i.e., patients, public, etc.) to whom the guideline is meant to apply	☐ Target population, sex, and age ☐ Clinical condition (if relevant) ☐ Severity/stage of disease (if relevant) ☐ Comorbidities (if relevant) ☐ Excluded populations (if relevant)	See *Methods for Guideline Development – Table 16*
DOMAIN 2: STAKEHOLDER INVOLVEMENT		
4. GROUP MEMBERSHIP Report all individuals who were involved in the development process. This may include members of the steering group, the research team involved in selecting and reviewing/rating the evidence, and individuals involved in formulating the final recommendations.	☐ Name of participant ☐ Discipline/content expertise (e.g., neurosurgeon, methodologist) ☐ Institution (e.g., St. Peter's hospital) ☐ Geographical location (e.g., Seattle, WA) ☐ A description of the member's role in the guideline development group	See *Work Group Membership*
5. TARGET POPULATION PREFERENCES AND VIEWS Report how the views and preferences of the target population were sought/considered and what the resulting outcomes were.	☐ Statement of type of strategy used to capture patients'/publics' views and preferences (e.g., participation in the guideline development group, literature review of values and preferences) ☐ Methods by which preferences and views were sought (e.g., evidence from literature, surveys, focus groups) ☐ Outcomes/information gathered on patient/public information ☐ How the information gathered was used to inform the guideline development process and/or formation of the recommendations	See *Methods for Guideline Development – Patient preferences and values*
6. TARGET USERS Report the target (or intended) users of the guideline.	☐ The intended guideline audience (e.g., specialists, family physicians, patients, clinical or institutional leaders/administrators) ☐ How the guideline may be used by its target audience (e.g., to inform clinical decisions, to inform policy, to inform standards of care)	See *Methods for Guideline Development – Aim*

CHECKLIST ITEM AND DESCRIPTION	REPORTING CRITERIA	Page #
DOMAIN 3: RIGOUR OF DEVELOPMENT		
7. SEARCH METHODS Report details of the strategy used to search for evidence	☐ Named electronic database(s) or evidence source(s) where the search was performed (e.g., MEDLINE, EMBASE, PsychINFO, CINAHL) ☐ Time periods searched (e.g., January 1, 2004, to March 31, 2008) ☐ Search terms used (e.g., text words, indexing terms, subheadings) ☐ Full search strategy included (e.g., possibly located in appendix)	See *Methods for Guideline Development – Literature searching and article selection* See *Appendix A*
8. EVIDENCE SELECTION CRITERIA Report the criteria used to select (i.e., include and exclude) the evidence. Provide rationale where appropriate.	☐ Target population (patient, public, etc.) ☐ Study design ☐ Comparisons (if relevant) ☐ Outcomes ☐ Language (if relevant) ☐ Context (if relevant)	*Methods for Guideline Development – Literature searching and article selection; Table 16*
9. STRENGTHS & LIMITATIONS OF THE EVIDENCE Describe the strengths and limitations of the evidence. Consider from the perspective of the individual studies and the body of evidence aggregated across all the studies. Tools exist that can facilitate the reporting of this concept	☐ Study design(s) included in body of evidence ☐ Study methodology limitations (sampling, blinding, allocation concealment, analytical methods) ☐ Appropriateness/relevance of primary and secondary outcomes considered ☐ Consistency of results across studies ☐ Direction of results across studies ☐ Magnitude of benefit versus magnitude of harm ☐ Applicability to practice context	See *Methods for Guideline Development – Critical appraisal of studies; See Table 16 and Appendixes C and D*
10. FORMULATION OF RECOMMENDATIONS Describe the methods used to formulate the recommendations and how final decisions were reached. Specify any areas of disagreement and the methods used to resolve them.	☐ Recommendation development process (e.g., steps used in modified Delphi technique, voting procedures that were considered) ☐ Outcomes of the recommendation development process (e.g., extent to which consensus was reached using modified Delphi technique, outcome of voting procedures) ☐ How the process influenced the recommendations (e.g., results of Delphi technique influence final recommendation, alignment with recommendations, and the final vote)	See *Methods for Guideline Development – Developing the recommendations*

CHECKLIST ITEM AND DESCRIPTION	REPORTING CRITERIA	Page #
11. COMDINERATIONS OF BENEFITS AND HARMS Report the health benefits, side effects, and risks that were considered when formulating the recommendations.	☐ Supporting data and report of benefits ☐ Supporting data and report of harms/side effects/risks ☐ Reporting of the balance/trade-off between benefits and harms/side effects/risks ☐ Recommendations reflect considerations of both benefits and harms/side effects/risks	See *Methods for Guideline Development – Balance of benefits and harms*
12. LINK BETWEEN RECOMMENDATIONS AND EVIDENCE Describe the explicit link between the recommendations and the evidence on which they are based.	☐ How the guideline development group linked and used the evidence to inform recommendations ☐ Link between each recommendation and key evidence (text description and/or reference list) ☐ Link between recommendations and evidence summaries and/or evidence tables in the results section of the guideline	See *Methods for Guideline Development – Developing the recommendations; Grading the strength of the recommendations; The overall quality of evidence*
13. EXTERNAL REVIEW Report the methodology used to conduct the external review.	☐ Purpose and intent of the external review (e.g., to improve quality, gather feedback on draft recommendations, assess applicability and feasibility, disseminate evidence) ☐ Methods taken to undertake the external review (e.g., rating scale, open-ended questions) ☐ Description of the external reviewers (e.g., number, type of reviewers, affiliations) ☐ Outcomes/information gathered from the external review (e.g., summary of key findings) ☐ How the information gathered was used to inform the guideline development process and/or formation of the recommendations (e.g., guideline panel considered results of review in forming final recommendations)	An external public review was undertaken [April 2026].
14. UPDATING PROCEDURE Describe the procedure for updating the guideline.	☐ A statement that the guideline will be updated ☐ Explicit time interval or explicit criteria to guide decisions about when an update will occur ☐ Methodology for the updating procedure	The KDIGO clinical practice guideline will be updated. However, no set timeframe has been determined.
DOMAIN 4: CLARITY OF PRESENTATION		

CHECKLIST ITEM AND DESCRIPTION	REPORTING CRITERIA	Page #
15. SPECIFIC AND UNAMBIGUOUS RECOMMENDATIONS Describe which options are appropriate in which situations and in which population groups, as informed by the body of evidence.	☐ A statement of the recommended action ☐ Intent or purpose of the recommended action (e.g., to improve quality of life, to decrease side effects) ☐ Relevant population (e.g., patients, public) ☐ Caveats or qualifying statements, if relevant (e.g., patients or conditions for whom the recommendations would not apply) ☐ If there is uncertainty about the best care option(s), the uncertainty should be stated in the guideline	See *Guidelines*
16. MANAGEMENT OF OPTIONS Describe the different options for managing condition or health issue	☐ Description of management options ☐ Population or clinical situation most appropriate to each option	See *Guidelines*
17. IDENTIFIABLE KEY RECOMMENDATIONS Present the key recommendations so that they are easy to identify.	☐ Recommendations in a summarized box, typed in bold, underlined, or presented as flow charts or algorithms ☐ Specific recommendations grouped together in one section	See *Guidelines*
DOMAIN 5: APPLICABILITY		
18. FACILITATORS AND BARRIERS TO APPLICATION Describe the facilitators and barriers to the guideline's application.	☐ Types of facilitators and barriers that were considered ☐ Methods by which information regarding the facilitators and barriers to implementing recommendations were sought (e.g., feedback from key stakeholders, pilot testing of guidelines before widespread implementation) ☐ Information/description of the types of facilitators and barriers that emerged from the inquiry (e.g., practitioners have the skills to deliver the recommended care, sufficient equipment is not available to ensure all eligible members of the population receive mammography) ☐ How the information influenced the guideline development process and/or formation of the recommendations	See *Guidelines*

CHECKLIST ITEM AND DESCRIPTION	REPORTING CRITERIA	Page #
19. IMPLEMENTATION ADVICE/TOOLS Provide advice and/or tools on how the recommendations can be applied in practice.	☐ Additional materials to support the implementation of the guideline in practice. For example: ☐ Guideline summary documents ☐ Links to check lists, algorithms ☐ Links to how-to manuals ☐ Solutions linked to barrier analysis (see Item 18) ☐ Tools to capitalize on guideline facilitators (see Item 18) ☐ Outcome of pilot test and lessons learned	See *Guidelines*
20. RESOURCE IMPLICATIONS Describe any potential resource implications of applying the recommendations.	☐ Types of cost information that were considered (e.g., economic evaluations, drug acquisition costs) ☐ Methods by which the cost information was sought (e.g., health economist part of guideline panel, use of health technology assessments for specific drugs, etc.) ☐ Information/description of the cost information that emerged from the inquiry (e.g., specific drug acquisition costs per treatment course) ☐ How the information gathered was used to inform the guideline development process and/or formation of the recommendations	See *Guidelines*
21. MONITORING/ AUDITING CRITERIA Provide monitoring and/or auditing criteria to measure the application of guideline recommendations.	☐ Criteria to assess guideline implementation or adherence to recommendations ☐ Criteria for assessing impact of implementing the recommendations ☐ Advice on the frequency and interval of measurement ☐ Operational definitions of how the criteria should be measured	See *Guidelines*
DOMAIN 6: EDITORIAL INDEPENDENCE		
22. FUNDING BODY Report the funding body's influence on the content of the guideline.	☐ The name of the funding body or source of funding (or explicit statement of no funding) ☐ A statement that the funding body did not influence the content of the guideline	See *Work Group Financial Disclosures*
23. COMPETING INTERESTS Provide an explicit statement that all group members have declared whether they have any competing interests.	☐ Types of competing interests considered ☐ Methods by which potential competing interests were sought ☐ A description of the competing interests ☐ How the competing interests influenced the guideline process and development of recommendations	See *Work Group Financial Disclosures*

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Appendix C. Evidence profiles cited in the guideline text

Table S5. Serum cystatin C in adults

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI	4 (338) [1]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	AUC 0.82 (0.78-0.85)	Critical
Severe AKI (stage 2-3)	0								none	Critical
RRT	0								none	Critical
Death	0								none	Critical
MAKE90	0								none	Critical
Duration of AKI	0								none	Important
CKD G3-G5	0								none	Important
CKD G5D (dialysis)	0								none	Important
Balance of Potential Benefits and Harms: In adults, sCystatin C has moderate* accuracy to predict post-major surgery AKI incidence.								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = acute kidney injury; AUC = area under the curve; CKD = chronic kidney disease; MAKE = major adverse kidney injury; RRT = renal replacement therapy; sCystatin C = serum cystatin C

* *A priori* defined as AUC 0.76-0.85

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Table S6. Serum cystatin C in children

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI	16 (2213) [1]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	AUC 0.80 (0.76-0.85) Sn 0.76 (0.66-0.84) Sp 0.77 (0.69-0.84)	Critical
Severe AKI (stage 2-3)	4 (692) [1]	Some limitations	Consistent	Direct	Imprecise*	Undetected	None	Low	AUC 0.83 (0.75-0.92)	Critical
RRT	0								none	Critical
Death	0								none	Critical
MAKE90	0								none	Critical
Duration of AKI	0								none	Important
CKD G3-G5	0								none	Important
CKD G5D (dialysis)	0								none	Important
Balance of Potential Benefits and Harms: In pediatrics, sCystatin C has moderate † sensitivity and specificity with moderate ‡ accuracy to predict AKI incidence and stages.								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = acute kidney injury; AUC = area under the curve; CKD = chronic kidney disease; MAKE = major adverse kidney injury; RRT = renal replacement therapy; sCystatin C = serum cystatin C; Sn = sensitivity; Sp = specificity

* Range of AUC 95% CI >0.2 (larger than the ranges for low, moderate, and high accuracy).
† *A priori* defined as 76-89% sensitivity or specificity
‡ *A priori* defined as AUC 0.76-0.85

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Table S7. Comparison of different methods to estimate baseline serum creatinine on prioritized outcomes in adults

Study / PMID / Country	Specific Outcome	AKI Definition	Baseline SCr Estimate Method	Events/N (%)	Effect Size (CI)	Analysis Quality
Lee 2023[32] 36935538 S Korea Severe trauma N=3228 57 yr	Mortality, in-hospital	KDIGO SCr	MDRD eGFR=75	AKI 276/678 (40.7%) No AKI 183/2550 (7.2%)	**adjOR 5.4 (4.2, 6.8)** AUC 0.75 (0.72, 0.77) Sn 25.3% Sp 98.1%	Good Also reported data by AKI Stage
			Trauma MDRD ?=121	AKI 383/1488 (25.7%) No AKI 76/1740 (4.4%)	**adjOR 4.6 (3.5, 6.1)** AUC 0.80 (0.77, 0.82) Sn 41.6% Sp 95.2%	
			Admission SCr	AKI 248/482 (51.4%) No AKI 211/2746 (7.7%)	**adjOR 9.0 (7.1, 11.5)** AUC 0.74 (0.71, 0.76) Sn 22.9% Sp 98.5%	
			First-day nadir SCr	AKI 307/968 (31.7%) No AKI 152/2260 (6.7%)	**adjOR 4.3 (3.4, 5.4)** AUC 0.76 (0.73, 0.78) Sn 27.9% Sp 98.3%	
	RRT	KDIGO SCr	MDRD eGFR=75	AKI 90/1488 (6.0%) No AKI 20/1740 (1.1%)	**adjOR 14.5 (8.5, 24.7)** AUC 0.86 (0.82, 0.91) Sn 64.5% Sp 96.9%	
			Trauma MDRD ?=121	AKI 104/1488 (7.0%) No AKI 6/1740 (0.3%)	**adjOR 13.8 (5.9, 32.2)** AUC 0.89 (0.86, 0.92) Sn 75.2% Sp 92.3%	
			Admission SCr	AKI 78/482 (16.2%) No AKI 32/2746 (1.2%)	**adjOR 13.1 (8.3, 20.6)** AUC 0.82 (0.77, 0.86) Sn 57.3% Sp 97.3%	
			First-day nadir SCr	AKI 93/968 (9.6%) No AKI 17/2260 (0.8%)	**adjOR 9.5 (5.5, 16.3)** AUC 0.86 (0.82, 0.91) Sn 65.5% Sp 96.7%	
Thongprayoon 2015[33] 26032233 US ICU admission N=4020 65 yr	Mortality, 60-day	KDIGO SCr	**Minimum SCr during admission:** Stage 3	77/185 (41.6%)	**OR 11.4 (8.17, 15.9)**	Poor (unadjusted analysis) Poor reporting, incomplete analyses
			Stage 2	63/230 (27.4%)	**OR 6.02 (4.33, 8.37)**	
			Stage 1	138/789 (17.5%)	**OR 3.38 (2.66, 4.31)**	
			No AKI	166/2816 (5.9%)	(reference)	
			Pre-admission: 180-7 days: Stage 3	57/147 (38.8%)	**OR 8.82 (6.15, 12.7)**	
			Stage 2	53/160 (33.1%)	**OR 6.90 (4.82, 9.87)**	
			Stage 1	130/638 (20.4%)	**OR 3.56 (2.81, 4.53)**	
			No AKI	206/3075 (6.7%)	(reference)	

Study / PMID / Country	Specific Outcome	AKI Definition	Baseline SCr Estimate Method	Events/N (%)	Effect Size (CI)	Analysis Quality
Warnock 2021[34] 34419950 US Hospital admission N=~27,000 (43,433 admissions) 58 yr	Mortality, in-patient	KDIGO SCr	Pre-admission ("ambulatory") SCr	AKI: NR/NR (NR%) No AKI: NR/NR (NR%) [reported data very unclear]	**adjOR 1.64 (1.43, 1.88)** [NS per study, which implicitly used a P<0.001 threshold] AUC 0.74	Poor Did not account for multiple admissions. Incomplete and unclear reporting. Overall study included 62,380 patients (100,570 admissions). They did not report the number of patients among the 43,433 admissions analyzed. These included only those with pre-admission SCr analyses. Numbers related to AKI categories very unclear. Unclear if the RRT numbers make sense (eg, adjOR 1.27 for AKI vs. no AKI) NRI: net reclassification improvement, an index that attempts to quantify how well a new model reclassifies subjects - either appropriately or inappropriately - as compared to an old model (range from -1 to +1)
			Minimum SCr in rolling 48-hour window	AKI: NR/NR (NR%) No AKI: NR/NR (NR%)	**adjOR 6.57 (6.59, 7.58)** AUC 0.82 **NRI 0.31 (0.26, 0.35) vs. pre-admission**	
	RRT	KDIGO SCr	Pre-admission ("ambulatory") SCr	AKI: NR/NR (NR%) No AKI: NR/NR (NR%)	**adjOR 1.27 (1.10, 1.46)** [NS per study] AUC 0.86	
			Minimum SCr in rolling 48-hour window	AKI: NR/NR (NR%) No AKI: NR/NR (NR%)	**adjOR 7.16 (5.95, 8.61))** AUC 0.90 **NRI 0.32 (0.27, 0.36) vs. pre-admission**	
Graversen 2022[35] 35373126 Denmark General population (inpatient and outpatient SCr values) N=2,095,850 73 yr	Mortality, 30-day	KDIGO SCr	Most recent outpatient SCr 8-365 days prior	AKI: NR/61,579 (18%)		Fair (groups comparable, no major concerns, some lack of clarity and precision, mostly qualitative comparisons) Unclear how the eGFR=75 method group overlaps with other groups; the numbers don't clearly parse out
			… or eGFR=75 if no prior SCr available	AKI: NR/81,686 (14%)		
			AKI with this method only	*AKI: NR/2240 (7%)*		
			Mean outpatient SCr between 8-365 days prior	AKI: NR/61,189 (18%)		
			… or eGFR=75 if no prior SCr available	AKI: NR/81,137 (14%)		
			Median outpatient SCr between 8-365 days prior	AKI: NR/62,597 (17%)		
			… or eGFR=75 if no prior SCr available	AKI: NR/82,525 (14%)		
			Median outpatient SCr between 8-90 days prior, if available, or between 8-365 days prior	AKI: NR/61,399 (18%)		
			… or eGFR=75 if no prior SCr available	AKI: NR/81,331 (14%)		
			eGFR=75, no prior SCr available	AKI: NR/37,659 (23%)		

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Table S8. Comparison of different methods to estimate baseline serum creatinine on prioritized outcomes in children

Study / PMID / Country	Specific Outcome	AKI Definition	Baseline SCr Estimate Method	Events/N (%)	Effect Size (CI)	Analysis Quality
Batte 2020[36] 32993548 Uganda Malaria, children N=1078 2.8 y [0.5-12]	Mortality, duration NR	KDIGO SCr [1.5x baseline]	Schwartz eGFR=120	AKI 51/~338 (15.1%) No AKI 25/~735 (3.4%)	**RR 4.41 (2.78, 7.00)** AUC 0.69 (0.64, 0.75) Sn 67.1% Sp 71.3%	Poor Unadjusted
			Pottel eGFR=120	AKI 59/~434 (13.6%) No AKI 17/~630 (2.7%)	**RR 5.13 (3.03, 8.68)** AUC 0.70 (0.65, 0.75) Sn 77.6% Sp 62.4%	
			Schwartz eGFR=137 (population mean)	AKI 59/~468 (12.6%) No AKI 17/~607 (2.8%)	**RR 4.50 (2.66, 7.62)** AUC 0.68 (0.63, 0.73) Sn 77.6% Sp 59.2%	
			Published UL by age (ht-independent)	AKI 33/~168 (19.6%) No AKI 43/~915 (4.7%)	**RR 4.17 (2.74, 6.36)** AUC 0.65 (0.59, 0.71) Sn 43.4% Sp 86.5%	

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Table S9. AKI definitions and serum creatinine and urinary output criteria: Marker criteria in adults

Study PMID	AKI Stage	AKI Definition	SCr Criteria	SCr Timing	UO Criteria	Timing	Estimate of Baseline SCr	Comments
Amathieu 2017[1] 28586126	3 stages	KDIGO	≥0.3 mg/dl increase, or ≥50% increase (Stages not explicitly stated)	48 hr NR	I: <0.5 ml/kg/hr II: ≤0.5 ml/kg/hr III: <0.3 ml/kg/hr anuria	≥6 hr ≥12 hr ≥24 hr ≥12 hr	Prehospital <1 yr or first hospital or first ICU value	
Baggot 2023[2] 37786499	Any AKI	KDIGO	≥26.5 µmol/l (0.3 mg/dl) increase, or >50% increase	48 hr 7 d	<0.5 ml/kg/hr	12 hr	Pre-admission	
Bianchi 2021[3] 34735011	3 stages	KDIGO	I: ≥0.3 mg/dl increase, or 50-90% increase II: 100-190% increase III: ≥4 mg/dl, or ≥200% increase, or eGFR <35 ml/min/1.73 m^2 (Schwartz), or RRT	48 hr 7 d	I: <0.5 ml/kg/hr II: ≤0.5 ml/kg/hr III: <0.3 ml/kg/hr anuria	6-12 hr ≥12 hr ≥24 hr ≥12 hr	Lowest w/in 12 mo, excluding w/in 24 hr of unscheduled admission, or lowest pre-ICU admission	
Hessler 2024[4] 37851903	3 stages	KDIGO	I: ≥26.5 µmol/l (0.3 mg/dl) increase, or 50-90% increase II: 100-190% increase III: ≥354 µmol/l (4 mg/dl), or ≥200% increase, or RRT	48 hr 7 d	I: <0.5 ml/kg/hr II: ≤0.5 ml/kg/hr III: <0.3 ml/kg/hr anuria [kg = real weight]	6-12 hr ≥12 hr ≥24 hr ≥12 hr	Pre-operative	
		KDIGO, predicted body weight (PBW) adjusted	Same	Same	[kg = PBW] Male: PBW (kg) = 50 + 0.91 × (height in cm – 152.4) Female: PBW (kg) = 45. 5 + 0.91 × (height in cm – 152.4)			
Hong 2023[5] 36728812	Any AKI	KDIGO	≥0.3 mg/dl increase, or ≥50% increase	48 hr 7 d	(none)		Pre-admission (7-365 days prior)	
		cROCK	≥25% increase	7 d	(none)			
		APKD	≥0.2 mg/dl increase, or ≥10% increase	48 hr 7 d				
Kaddourah 2017[6] 27959707 AWARE study	3 stages	KDIGO	I: ≥0.3 mg/dl increase, or 50-90% increase II: 100-190% increase III: ≥4 mg/dl, or ≥200% increase, or eGFR <35 ml/min/1.73 m^2 (Schwartz), or RRT	48 hr 7 d	I: <0.5 ml/kg/hr II: ≤0.5 ml/kg/hr III: <0.3 ml/kg/hr anuria	6-12 hr ≥12 hr ≥24 hr ≥12 hr	Pre-admission (within 3 mo) or eGFR 120 ml/min/1.73 m^2 (Schwartz)	

Study PMID	AKI Stage	AKI Definition	SCr Criteria	SCr Timing	UO Criteria	Timing	Estimate of Baseline SCr	Comments
Kaddourah 2019[7] 30676490	Stage 3	KDIGO	≥4 mg/dl, or ≥200% increase, or eGFR <35 ml/min/1.73 m^2, or RRT	7 d	<0.3 ml/kg/hr Anuria	≥24 hr ≥12 hr	"Baseline" or eGFR 120 ml/min/1.73 m^2 (method NR)	
Kellum 2015[8] 25568178	Maximum AKI severity	KDIGO	NR	NR	NR	NR	Unclear	
Koeze 2017[9] 28219327	Stage 3 (Failure)	KDIGO	SCr ≥354.6 μmol/l (4 mg/dl), or >300% increase, or RRT	7 d	<0.3 ml/kg/hr Anuria	24 hr ≥12 hr	MDRD eGFR 75 ml/min/1.73 m^2	
		AKIN	SCr ≥354 μmol/l (4 mg/dl) and ≥44 μmol/l (0.5 mg/dl) increase, or >300% increase, or RRT	7 d	<0.3 ml/kg/hr Anuria	24 hr ≥12 hr		
		RIFLE	SCr ≥350 μmol/l (4 mg/dl), or ≥44 μmol/l (0.5 mg/dl) increase, or >300% increase, or GFR >75% decline	7 d	<0.3 ml/kg/hr Anuria	24 hr ≥12 hr		
Lex 2014[10] 24206964	3 stages	pRIFLE CrCl	Risk: ≥25% decrease Injury: ≥ 50% decrease Failure:≥75% decrease or <35 mL/min/1.73 m^2	NR	(none)	(NA)	24 hr pre-op	eCrCl per Schwartz
		AKIN CrCl	I: ≥0.3 mg/dL increase, or 50-100% increase II: 100-200% increase III: >200% increase or >4.0 mg/dL or RRT	48 hr	(none)	(NA)		
		KDIGO CrCl	"Combining AKIN and pRIFLE" (otherwise undefined)	NR	(none)	(NA)		This definition does not align with the numbers of patients diagnosed with AKI (fewer per KDIGO than per pRIFLE)
Lu 2024[11] 38971227	Stage 1	KDIGO (SCr only)	I: ≥26.5 μmol/l (0.3 mg/dl) increase, or 50-90% increase	NR	(none)		Pre-operative (implied)	
		"Small increase"	5-26.4 μmol/L (0.06-0.29 mg/dl) increase	Post-op	(none)		Pre-operative	
Machado 2024[12] 39135063	3 stages	KDIGO	I: ≥0.3 mg/dl increase, or 50-90% increase II: 100-190% increase III: ≥4 mg/dl, or ≥200% increase, or eGFR <35 ml/min/1.73 m^2 (Schwartz), or RRT	48 hr 7 d	I: <0.5 ml/kg/hr II: ≤0.5 ml/kg/hr III: <0.3 ml/kg/hr anuria	6-12 hr ≥12 hr ≥24 hr ≥12 hr	Pre-ICU admission (<6 mo) or lowest in ICU	
		UO-AKI	Same as KDIGO		I: 0.2-0.3 mL/kg/hr II: 0.1-0.2 mL/kg/hr III: <0.1 mL/kg/hr	6 hr		Defined based on study data

Study PMID	AKI Stage	AKI Definition	SCr Criteria	SCr Timing	UO Criteria	Timing	Estimate of Baseline SCr	Comments
Nateghi Haredasht 2022[13] 35441981	4 stages	KDIGO (SCr only)	1a: ≥0.3 mg/dl increase 1b: ≥50% increase 2: ≥100% increase 3: ≥200% increase	48 hr 48 hr 48 hr 48 hr	(none)		Minimum "within a rolling 48-h window" inpatient	
	4 stages	AKIN (SCr only)	1a: ≥0.3 mg/dl increase 1b: ≥50% increase 2: ≥100% increase 3: ≥200% increase	48 hr 7 d 7 d 7 d	(none)			
Priyanka 2021[14] 32033818	3 stages	"KDIGO" (SCr % increase only)	I: 50-90% increase II: 100-190% increase III: >200% increase	72 hr max	I: <0.5 ml/kg/hr II: ≤0.5 ml/kg/hr III: <0.3 ml/kg/hr anuria	6-12 hr >12 hr ≥24 hr ≥12 hr	Undefined	
Qin 2016[15] 27569230	Any AKI	KDIGO	≥26.5 µmol/l (0.3 mg/dl) increase, or >50% increase	48 hr 7 d	NR [<0.5 ml/kg/hr]	NR [12 hr]	NR	
Quan 2016[16] 27941063	3 stages	KDIGO	Not explicitly reported	NR	Not explicitly reported	NR	Pre-surgery	
Ruth 2022[17] 35145645	Any AKI	KDIGO	Not explicitly stated		Not explicitly stated		Lowest ≤3 mo pre-ICU admission or eGFR 120 ml/min/1.73 m^2 (method NR)	
Sutherland 2015[18] 25649155	3 stages	pRIFLE eGFR	I (Risk): ≥25% decrease II (Injury): ≥ 50% decrease III (Failure):≥75% decrease or <35 mL/min/1.73 m^2	NR	(none)		Most recent SCr w/in 3 mo, including admission	
		AKIN SCr, modified	I: ≥0.3 mg/dL increase, or ≥50% increase II: ≥100% increase III: ≥200% increase	48 hr	(none)			
		KDIGO SCr, modified	I: ≥0.3 mg/dl increase, or 50-90% increase II: ≥100% increase III: ≥200% increase, or eGFR <35 ml/min/1.73 m^2 (Schwartz), if age <18 yo	48 hr 7 d	(none)			
Tai 2022[19] 34076880	Any AKI	KDIGO	≥26.5 µmol/l (0.3 mg/dl) increase, or >50% increase	NR	<0.5 ml/kg/hr	6-12 hr	Unclear ("not sourced")	
		pROCK	≥20 µmol/l (0.23 mg/dl), or ≥30% increase	NR	(none, implicitly)			
		pRIFLE	CrCl ≥25% decrease	NR	<0.5 ml/kg/hr	8 hr		
Vanmassenhove 2021[20] 34045582	Stage 3 (study states Stage 2)	"KDIGO" SCr	≥4 mg/dl, or ≥200% increase	ICU stay	N/A		In order: Lowest, pre-hospitalization Lowest, pre-ICU eGFR 75 ml/min/1.73 m^2 (MDRD)	
		UO-1	N/A		≤6 ml/kg	12 hr		
		UO-2	N/A		≤0.5 ml/kg per every hour	12 hr		

Study PMID	AKI Stage	AKI Definition	SCr Criteria	SCr Timing	UO Criteria	Timing	Estimate of Baseline SCr	Comments
Wainstein 2022[21] 35442972	Any AKI 3 stages	KDIGO	I: ≥26.5 μmol/l (0.3 mg/dl) increase, or 50-90% increase II: 100-190% increase III: ≥353.6 μmol/l (4 mg/dl), or ≥200% increase, or RRT	48 hr 7 d	(none)		Minimum within 7 day moving window, in hospital	
		eKDIGO (extended)	I: ≥26.5 μmol/l (0.3 mg/dl) ***increase***, or ≥26.5 μmol/l (0.3 mg/dl) ***decrease***, or ***increase*** of 50-90% baseline, or ***decrease*** to 50-90% baseline II: ***increase*** of 100-190% baseline, or ***decrease*** to 100-190% baseline III: ***increase*** to ≥353.6 μmol/l (4 mg/dl), or ***decrease*** to <353.6 μmol/l, or ***increase*** of ≥200% baseline, or ***decrease*** to <200% baseline RRT	48 hr 7 d	(none)			Includes evidence of recovering AKI at the time of hospitalization
		deKDIGO	I: ≥26.5 μmol/l (0.3 mg/dl) ***decrease***, or ***decrease*** to 50-90% baseline II: ***decrease*** to 100-190% baseline III: ***decrease*** to <353.6 μmol/l, or ***decrease*** to <200% baseline	48 hr 7 d	(none)			Focus just on those diagnosed by decrease in SCr
Wang H 2022[22] 35854309	Any AKI	KDIGO (% increase only)	≥50% increase (any increase in terms of μmol/l)	Index date	(none)		Lowest of median 8-365 d prior or actual 1-7 d prior	Baseline SCr's differ in the two groups. Unclear whether this may bias analysis.
		KDIGO (μmol/l recent increase only)	≥26.5 μmol/l (0.3 mg/dl) increase (recent), <u>*but <1.5x*</u>	Index date			48 hr prior	
Wang HE 2013[23] 23355628	3 stages	KDIGO SCr	I: ≥0.3 mg/dl increase, or 50-90% increase II: 100-190% increase III: ≥4 mg/dl, or ≥200% increase, or eGFR <35 ml/min/1.73 m^2, or RRT	NR	(none)		Lowest of first 3 hospital SCr [prehospital SCr not available]	
		Delta-Creatinine	I: 0.3-0.69 mg/dl increase II: 0.7-1.19 mg/dl increase III: ≥1.2 mg/dl increase, or RRT	NR				
Wei 2021[24] 32791515	3 stages	KDIGO SCr	I: ≥26.5 μmol/l (0.3 mg/dl) increase, or 50-90% increase II: 100-190% increase III: ≥354 μmol/l (4 mg/dl), or ≥200% increase, or RRT	48 hr 7 d	(none)		Mean across 90 days pre-ICU admission or lowest during hospitalization	

Study PMID	AKI Stage	AKI Definition	SCr Criteria	SCr Timing	UO Criteria	Timing	Estimate of Baseline SCr	Comments
		pROCK	I: 20-39 μmol/l increase, and 30-59% increase II: 40-79 μmol/l increase, and 60-119% increase III: ≥80 μmol/l increase, and ≥120% increase	7 d	(none)			
White 2023[25] 37432520	Any AKI 3 stages	KDIGO	Not explicitly stated		Not explicitly stated		eGFR 75 ml/min/1.73 m^2 (CKD-EPI)	

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Table S10. AKI definitions and serum creatinine and urinary output criteria: Marker criteria in children

Study PMID	AKI Stage	AKI Definition	SCr Criteria	SCr Timing	UO Criteria	Timing	Estimate of Baseline SCr	Comments
Batte 2020[26] 32993548	3 stages	KDIGO (% increase only)	I: 50-90% increase II: 100-190% increase III: ≥200% increase	NR	(none)		Back-calculated from Schwartz eGFR=120 Back-calculated from Pottel eGFR=120 Back-calculated from Schwartz eGFR=137 (population mean) Published UL by age (ht-independent)	Omitted analyses of study population specific methods (eg, average of local children)
Graversen 2022[27] 35373126	Any AKI	KDIGO	≥0.3 mg/dl increase, or ≥50% increase	48 hr 7 d	(none)		Most recent outpatient SCr 8-365 days prior Mean outpatient SCr between 8-365 days prior Median outpatient SCr between 8-365 days prior Median outpatient SCr between 8-90 days prior, if available, or between 8-365 days prior eGFR=75 if no prior SCr data	
Lee 2023[28] 36935538	3 stages	KDIGO	I: ≥0.3 mg/dl increase, or 50-90% increase II: 100-190% increase III: ≥4 mg/dl, or ≥200% increase, or eGFR <35 ml/min/1.73 m^2 if <18 yr, or RRT	48 hr 7 d	(none)		Back-calculated from MDRD eGFR=75 Back-calculated from trauma MDRD eGFR=121 (The description is not completely clear) Admission SCr First day (24 hr) nadir SCr	
Thongprayoon 2015[29] 26032233	3 stages	KDIGO	I: ≥0.3 mg/dl increase, or 50-90% increase II: 100-190% increase III: ≥4 mg/dl, or ≥200% increase, or RRT	NR	(none)		Minimum SCr during admission Pre-admission SCr: 180-7 days Pre-admission SCr: 365-7 days [Not analyzed] Pre-admission SCr: 180-0 days [Not analyzed] Pre-admission SCr: 180-0 days [Not analyzed]	
Warnock 2021[30] 34419950	Any AKI	KDIGO	≥0.3 mg/dl increase, or ≥50% increase	48 hr 7 d	(none)		Pre-admission ("ambulatory") SCr, 365-7 days (implied) Minimum SCr in rolling 48-hour window	Poorly defined

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Table S11. Comparison of different AKI definitions (criteria) on prioritized outcomes in adults

Study PMID Country AKI Cause* Total N Age	Specific Outcome	AKI Definition	Events/N (%)	Effect Size (CI)	Analysis Quality
Koeze 2017[1] 28219327 Netherlands ICU admission (CKD 4.2%, 1.7% on cRRT) N=1376 60 y	Mortality, in-hospital	**KDIGO:** UO criteria only	AKI: 19/236 (8.1%) No AKI: NR		Poor (inadequate analysis; inconsistent numbers, poor reporting) Omitted ICU mortality
		SCr and UO criteria	AKI: 17/72 (23.6%) No AKI: NR		
		AKIN: UO criteria only	AKI: 17/230 (7.4%) No AKI: NR		
		SCr and UO criteria	AKI: 19/80 (23.8%) No AKI: NR		
		RIFLE: UO criteria only	AKI: 24/265 (9.1%) No AKI: NR		
		SCr and UO criteria	AKI: 13/61 (21.3%) No AKI: NR		
		AKIN, KDIGO, and RIFLE	51/212 (24.1%)		
		AKIN and KDIGO, not RIFLE	11/70 (15.7%)		
		AKIN only, not KDIGO or RIFLE	3/14 (21.4%)		
		No AKI (per AKIN, KDIGO, or RIFLE	64/1078 (5.9%)		
Baggot 2023[2] 37786499 UK Surgical wards (68% *no* surgery) N=2149 71 y (IQR 55-82)	Mortality, in-hospital	KDIGO UO criteria alone	AKI 58/1568 (3.7%) No AKI 12/581 (2.1%)	OR 1.82 (0.97, 3.42)	Poor (unadjusted)
		KDIGO SCr criteria alone	AKI 33/218 (15.1%) No AKI 37/1931 (1.9%)	**OR 9.13 (5.58, 14.95)**	
		KDIGO SCr and UO criteria	AKI 29/172 (16.8%) No AKI 41/1977 (2.1%)	**OR 8.13 (5.19, 12.74)**	
		KDIGO SCr or UO (calculated)	AKI 62/1614 (3.8%) No AKI 8/535 (1.5%)	**OR 2.63 (1.25, 5.53)**	

Study PMID Country AKI Cause* Total N Age	Specific Outcome	AKI Definition	Events/N (%)	Effect Size (CI)	Analysis Quality
Qin 2016[3] 27569230 China ICU admission N=1058 62 y	Mortality, in-hospital	**Stage 1:**			Good (adjusted, no major concerns)
		KDIGO UO criteria alone	18/60 (30.0%)	**OR 2.64 (1.47, 4.74)**	
		KDIGO SCr criteria alone	56/257 (21.8%)	**OR 2.06 (1.38, 3.08)**	
		KDIGO SCr and UO criteria	43/238 (18.1%)	**OR 2.15 (1.36, 3.38)**	
		Stage 2:			
		KDIGO UO criteria alone	18/44 (40.9%)	**OR 4.27 (2.27, 8.02)**	
		KDIGO SCr criteria alone	44/148 (29.7%)	**OR 3.13 (2.01, 4.88)**	
		KDIGO SCr and UO criteria	42/154 (27.3%)	**OR 3.65 (2.28, 5.84)**	
		Stage 3:			
		KDIGO UO criteria alone	68/109 (62.4%)	**OR 10.2 (6.62, 15.8)**	
		KDIGO SCr criteria alone	62/149 (41.6%)	**OR 5.27 (3.46, 8.05)**	
		KDIGO SCr and UO criteria	93/194 (47.9%)	**OR 8.96 (5.89, 13.6)**	
		No AKI:			
		KDIGO UO criteria alone	118/845 (14.0%)	(reference)	
		KDIGO SCr criteria alone	60/504 (11.9%)	(reference)	
		KDIGO SCr and UO criteria	44/472 (9.3%)	(reference)	
		AKI diagnosis (Stages 1-3):			
		KDIGO UO criteria alone	NR	**adjOR 2.89 (1.96, 4.25) [vs. no AKI]**	
		KDIGO SCr criteria alone	NR	adjOR 1.32 (0.90, 1.94) [vs. no AKI]	
		AKI diagnosis (Stages 1-3):			
		KDIGO SCr stage more severe than UO stage	NR	**adjOR 1.59 (1.05, 2.42) [vs. no AKI]**	
		KDIGO SCr stage same as UO stage	NR	**adjOR 3.92 (2.20, 6.97) [vs. no AKI]**	
		KDIGO SCr stage less severe than UO stage	NR	**adjOR 7.58 (4.14, 13.9) [vs. no AKI]**	
	RRT	**AKI diagnosis (Stages 1-3):**			
		KDIGO SCr stage more severe than UO stage	43/416 (10.3%)	**OR 0.08 (0.05, 0.14)**	
		KDIGO SCr stage same as UO stage	54/93 (58.1%)	(reference)	
		KDIGO SCr stage less severe than UO stage	20/77 (26.0%)	**OR 0.25 (0.13, 0.49)**	
Hessler 2024[4] 37851903 Germany Cardiac surgery w/24 hr ICU admission N=3279 69 y	Mortality, in-hospital	**Stage 1:**			No major concerns
		KDIGO (real body weight)	16/1137 (1.4%)	P 0.73	
		KDIGO (predicted body weight)	17/1036 (1.6%)		
		Stage 2:			
		KDIGO (real body weight)	17/275 (6.2%)	P 0.083	
		KDIGO (predicted body weight)	16/140 (11.4%)		
		Stage 3:			
		KDIGO (real body weight)	71/236 (30.1%)	P 1.00	
		KDIGO (predicted body weight)	71/239 (29.7%)		
		AKI diagnosis (Stages 1-3):			
		KDIGO (real body weight)	104/1648 (6.3%)	P 0.28 (real vs pred) AUC 0.883 (mortality)	
		KDIGO (predicted body weight)	104/1415 (7.3%)	AUC 0.895 (mortality)	

Study PMID Country AKI Cause* Total N Age	Specific Outcome	AKI Definition	Events/N (%)	Effect Size (CI)	Analysis Quality
Huang 2021[5] 34458286 China ICU admission N=2984 67 y	Stage 2-3 AKI (UO criteria)	UO [6-12 hr] ≥1.1 ml/kg/hr (inflection point after multivariate analysis of UO [6-12 hr] vs. progression to stage 2-3 AKI)	NR	**AdjOR 0.37 (0.24, 0.48)**	No major concerns, but unclear reporting
		UO [6-12 hr] 0.5-1.1 ml/kg/hr	NR (reference)		
	Stage 3 AKI (UO criteria)	UO [6-12 hr] ≥1.1 ml/kg/hr	NR	**AdjOR 0.31 (0.23, 0.41)**	
		UO [6-12 hr] 0.5-1.1 ml/kg/hr	NR (reference)		
	RRT	UO [6-12 hr] ≥1.1 ml/kg/hr	NR (1.14%)	**RR 0.41 (0.21, 0.79)** **[CI based on reported P]**	
		UO [6-12 hr] 0.5-1.1 ml/kg/hr	NR (2.77%)	**P 0.008 (reported)**	
	Mortality, 30 day	UO [6-12 hr] ≥1.1 ml/kg/hr	NR (11.05%)	**RR 0.60 (0.44, 0.81)** **[CI based on reported P]**	
		UO [6-12 hr] 0.5-1.1 ml/kg/hr	NR (18.42%)	**P <0.001 (reported)**	
Amathieu 2017[6] 28586126 US Chronic liver disease and ICU admission N=3458 ~51 y	Mortality, in-hospital	**KDIGO UO criteria only:**			Poor (unadjusted, indirect comparisons) Note: article has table with further break down of AKI categories SCr crossed with UO criteria (for hospital mortality)
		AKI (any)	572/2553 (22.4%)	**OR 3.93 (2.98, 5.16)**	
		Stage 1	67/450 (14.9%)	**OR 2.38 (1.65, 3.43)**	
		Stage 2	224/1419 (15.8%)	**OR 2.55 (1.90, 3.42)**	
		Stage 3	281/684 (41.1%)	**OR 9.48 (7.03, 12.79)**	
		No AKI	62/905 (6.9%)	(reference)	
		KDIGO SCr criteria only:			
		AKI (any)	531/2002 (26.5%)	**OR 4.74 (3.79, 5.93)**	
		Stage 1	164/1007 (16.3%)	**OR 2.56 (1.97, 3.32)**	
		Stage 2	139/492 (28.3%)	**OR 5.17 (3.91, 6.85)**	
		Stage 3	228/503 (45.3%)	**OR 10.9 (8.34, 14.2)**	
		No AKI	103/1456 (7.1%)	(reference)	
		KDIGO SCr and UO criteria:			
		AKI (any)	604/2854 (21.2%)	**OR 5.14 (3.52, 7.49)**	
		Stage 1	50/578 (8.7%)	**OR 1.81 (1.13, 2.89)**	
		Stage 2	190/1389 (13.7%)	**OR 3.03 (2.04, 4.51)**	
		Stage 3	364/887 (41.0%)	**OR 13.3 (9.01, 19.7)**	
		No AKI	30/604 (5.0%)	(reference)	

Study PMID Country AKI Cause* Total N Age	Specific Outcome	AKI Definition	Events/N (%)	Effect Size (CI)	Analysis Quality
Priyanka 2021[7] 32033818 US Cardiac surgery N=6637 66 y	Mortality, 180 day	**KDIGO UO criteria only:**			Poor (unadjusted) Data also reported for 30-, 90-, and 180-day outcomes.
		AKI (any)	353/4983 (7.1%)	**OR 2.35 (1.75, 3.16)**	
		Stage 1	38/1013 (3.8%)	OR 1.20 (0.78, 1.84)	
		Stage 2	206/3219 (6.4%)	**OR 2.11 (1.54, 2.87)**	
		Stage 3	109/751 (14.5%)	**OR 5.23 (3.71, 7.37)**	
		No AKI	52/1654 (3.1%)	(reference)	
		KDIGO SCr criteria only:			
		AKI (any)	286/2562 (11.2%)	**OR 4.18 (3.35, 5.21)**	
		Stage 1	190/2095 (9.1%)	**OR 3.32 (2.62, 4.20)**	
		Stage 2	55/344 (16.0%)	**OR 6.33 (4.50, 8.90)**	
		Stage 3	41/123 (33.3%)	**OR 16.6 (11.0, 25.2)**	
		No AKI	119/4075 (2.9%)	(reference)	
		KDIGO SCr or **UO criteria:**			
		AKI (any)	378/5389 (7.0%)	**OR 3.41 (2.30, 5.07)**	
		No AKI	27/1248 (2.2%)	(reference)	
		KDIGO SCr and UO criteria:			
		AKI (any)	261/2156 (12.1%)	**OR 4.15 (3.36, 5.12)**	
		No AKI	144/4481 (3.2%)	(reference)	
	RRT, 30 day	**KDIGO UO criteria only:**			90- and 180-d RRT are not cumulative (omits those who stopped RRT)
		AKI (any)	168/4983 (3.4%)	**OR 7.18 (3.52, 14.6)**	
		Stage 1	6/1013 (0.59%)	OR 1.23 (0.42, 3.54)	
		Stage 2	57/3219 (1.8%)	**OR 3.71 (1.77, 7.79)**	
		Stage 3	105/751 (14.0%)	**OR 33.4 (16.2, 69.0)**	
		No AKI	8/1654 (0.48%)	(reference)	
		KDIGO SCr criteria only:			
		AKI (any)	166/2562 (6.5%)	**OR 28.2 (14.8, 53.4)**	
		Stage 1	74/2095 (3.5%)	**OR 14.9 (7.7, 28.9)**	
		Stage 2	40/344 (11.6%)	**OR 53.5 (26.5, 108.0)**	
		Stage 3	52/123 (42.3%)	**OR 298 (145, 609)**	
		No AKI	10/4075 (0.25%)	(reference)	
		KDIGO SCr or **UO criteria:**			
		AKI (any)	175/5389 (3.2%)	**OR 41.9 (5.86, 299)**	
		No AKI	1/1248 (0.08%)	(reference)	
		KDIGO SCr and UO criteria:			
		AKI (any)	159/2156 (7.4%)	**OR 20.9 (12.6, 34.6)**	
		No AKI	17/4481 (0.38%)	(reference)	

Study PMID Country AKI Cause* Total N Age	Specific Outcome	AKI Definition	Events/N (%)	Effect Size (CI)	Analysis Quality
Priyanka 2021 32033818 (continued)	MAKE, 180 day	**KDIGO UO criteria only:**			
		AKI (any)	584/4983 (11.7%)	**OR 1.85 (1.49, 2.28)**	
		Stage 1	96/1013 (9.5%)	**OR 1.46 (1.09, 1.94)**	
		Stage 2	283/3219 (8.8%)	**OR 1.34 (1.07, 1.68)**	
		Stage 3	205/751 (27.3%)	**OR 5.22 (4.06, 6.71)**	
		No AKI	111/1654 (6.7%)	(reference)	
		KDIGO SCr criteria only:			
		AKI (any)	795/2562 (31.0%)	**OR 6.32 (5.44, 7.33)**	
		Stage 1	365/2095 (17.4%)	**OR 2.96 (2.51, 3.50)**	
		Stage 2	92/344 (26.7%)	**OR 5.12 (3.92, 6.70)**	
		Stage 3	67/123 (54.5%)	**OR 16.8 (11.5, 24.5)**	
		No AKI	271/4075 (6.7%)	(reference)	
		KDIGO SCr or **UO criteria:**			
		AKI (any)	1010/5389 (18.7%)	**OR 4.91 (3.72, 6.47)**	
		No AKI	56/1248 (4.5%)	(reference)	
		KDIGO SCr and UO criteria:			
		AKI (any)	740/2156 (34.3%)	**OR 6.66 (5.77, 7.69)**	
		No AKI	326/4481 (7.3%)	(reference)	
Quan 2016[8] 27941063 Canada Major noncardiac surgery N=4229 60 yr	Mortality, 30 day	**KDIGO UO criteria only:**			Good (adjusted, no serious limitations) Note: NS between UO only St 2 & 3
		Stage 1	NR	**adjOR 5.27 (3.05, 9.10)**	
		Stage 2	NR	**adjOR 8.62 (3.14, 23.7)**	
		Stage 3	NR	**adjOR 5.00 (1.65 ,15.2)**	
		No AKI	NR	(reference)	
		KDIGO SCr criteria only:			
		Stage 1	25/263 (9.5%)	adjOR 1.55 (0.33 ,7.39)	
		Stage 2	6/39 (15.4%)	adjOR 1.75 (0.83 ,3.68)	
		Stage 3	5/39 (12.8%)	**adjOR 2.84 (1.41 ,5.70)**	
		No AKI		(reference)	
		KDIGO SCr or UO criteria:			
		Stage 1	6/187 (3.2%)	**adjOR 9.90 (2.66 ,36.80)**	
		Stage 2	25/1319 (1.9%)	**adjOR 4.89 (1.67 ,14.3)**	
		Stage 3	49/1201 (4.1%)	**adjOR 7.85 (2.76 ,22.3)**	
		No AKI	4/1522 (0.3%)	(reference)	
Nateghi Haredasht 2022[9] 35441981 Greece ≥7 day hospital stay (and ≥5 SCr measures) N=7242 77 yr	Mortality, in-hospital	**KDIGO-4 SCr criteria only:**			No major concerns (although very high death rates)
		Stage 1a	185/486 (38.1%)	**AdjOR 4.69 (2.98, 7.16)**	
		Stage 1b	448/1138 (39.4%)	**AdjOR 11.9 (9.28, 15.2)**	
		Stage 1 (standard KDIGO)	633/1624 (39.0%)	**AdjOR 9.73 (7.70, 12.3)**	
		Stage 2	327/584 (56.0%)	**AdjOR 39.8 (30.6, 52.0)**	
		Stage 3	139/215 (64.7%)	**AdjOR 62.9 (45.2, 88.2)**	
		No AKI	145/5713 (2.5%)	(reference)	
		AKIN-4 SCr criteria only:			
		Stage 1a	350/797 (43.9%)	**AdjOR 10.4 (8.25, 13.1)**	
		Stage 1b	222/422 (52.6%)	**AdjOR 22.8 (17.8, 29.2)**	
		Stage 1 (standard AKIN)	572/1219 (46.9%)	**AdjOR 14.7 (12.1, 17.8)**	
		Stage 2	68/127 (53.5%)	**AdjOR 24.3 (16.5, 35.8)**	
		Stage 3	15/24 (62.5%)	**AdjOR 37.5 (16.1, 94.0)**	
		No AKI	246/6172 (4.0%)	(reference)	

Study PMID Country AKI Cause* Total N Age	Specific Outcome	AKI Definition	Events/N (%)	Effect Size (CI)	Analysis Quality
Lu 2024[10] 38971227 Sweden Surgical aortic valve replacement N=12,591 70 yr	CKD, up to 13 y f/up (mean 5.5 y, max 14 y)	KDIGO Stage 1 (SCr only)	NR/2753 (standardized cumulative incidence 16.2% [13.9-18.8])	Diff 9.0% (6.7-11.3)	Poor Numbers inconsistent across text and supplemental tables. Naïve analyses relying on visual inspection of figures. Seem to present every basic analysis in R (except estimation of effect sizes). Also graphical data of SCr change vs. 30-day mortality, without cutpoints (not extracted) [Also very long term mortality data]
		Small increase SCr (5-26.4 μmol/L)	NR/5764 (10% [8.5-11.7%])	Diff 2.8% (1.0-4.5)	
		No change SCr (<small increase)	NR/4074 (7.2% [5.8-8.9])	(reference)	
Vanmassenhove 2021[11] 34045582 Belgium ICU admission N=13,403 61 yr	Mortality, ICU	KDIGO Stage 3 (SCr only)	NR/~1770 (31.3 per 1000 d w/AKI)	**adjHR 1.81 (1.56, 2.09)**	Good
		UO-1: UO ≤6 ml/kg (12 hr) only	NR/~4597 (27.2 per 1000 d w/AKI)	**adjHR 2.59 (2.18, 3.09)**	
		UO-2: ≤0.5 ml/kg/hr (12 hr) only	NR/~1903 (36.6 per 1000 d w/AKI)	**adjHR 2.83 (2.44,3.28)**	
		SCr (St 3) **and** UO-1	NR/~5187 (26.5 per 1000 d w/AKI)	**adjHR 2.54 (2.14, 3.02)**	
		SCr (St 3) **and** UO-2	NR/~2801 (32.8 per 1000 d w/AKI)	**adjHR 2.62 (2.27, 3.04)**	
Bianchi 2021[12] 34735011 Switzerland ICU admission (>6 hr) N=15,551 65 yr	Mortality, 90 day	**KDIGO UO criteria only:**			Poor (unadjusted)
		Stage 1	410/2635 (15.6%)	**OR 1.59 (1.38-1.83)**	
		Stage 2	1120/6365 (17.6%)	**OR 1.84 (1.64-2.06)**	
		Stage 3	867/2107 (41.1%)	**OR 6.03 (5.29-6.86)**	
		No AKI	462/4444 (10.4%)	(reference)	
		KDIGO SCr criteria only:			
		Stage 1	747/3406 (21.9%)	**OR 2.23 (2.01-2.48)**	
		Stage 2	334/1233 (27.1%)	**OR 2.95 (2.57-3.40)**	
		Stage 3	766/1842 (41.6%)	**OR 5.67 (5.06-6.34)**	
		No AKI	1012/9070 (11.2%)	(reference)	
		KDIGO SCr or **UO criteria:**			
		Stage 1	452/3013 (15.0%)	**OR 1.94 (1.66-2.27)**	
		Stage 2	1074/6304 (17.0%)	**OR 2.26 (1.97-2.59)**	
		Stage 3	1045/2772 (37.7%)	**OR 6.67 (5.78-7.69)**	
		No AKI	288/3462 (8.3%)	(reference)	
	RRT in ICU	**Any AKI**			
		UO Criteria only	0/5630 (0%)		
		SCr Criteria only	30/989 (3.0%)		
		KDIGO SCr **and** UO criteria	1130/5524 (20.5%)		
Wang HE 2013[13] 23355628 US Hospitalized N=15,096 (19,249 hospitalizations) 55 yr	Mortality, in-hospital	**KDIGO SCr criteria only:**			Poor Average of 1.25 hospitalizations per patient, not accounted for in analyses Unadjusted, incomplete results reporting. Supplementary material is incorrect
		Any AKI	NR/4491	unadjHR 1.76 (NR)	
		Stage 1	NR/2987	unadjHR 1.34 (NR)	
		Stage 2	NR/504	unadjHR 1.85 (NR)	
		Stage 3	NR/1000	unadjHR 2.37 (NR)	
		No AKI	NR/14552	(reference)	
		Delta-SCr only:			
		Any AKI	NR/4091	unadjHR 1.77 (NR)	
		Stage 1	NR/2551	unadjHR 1.58 (NR)	
		Stage 2	NR/618	unadjHR 1.85 (NR)	
		Stage 3	NR/922	unadjHR 1.89 (NR)	
		No AKI	NR/14952	(reference)	

Study PMID Country AKI Cause* Total N Age	Specific Outcome	AKI Definition	Events/N (%)	Effect Size (CI)	Analysis Quality
Kellum 2015[14] 25568178 US ICU admission N=32,045 60 yr	Mortality, in-hospital	Maximum AKI severity detected by: UO criteria alone	1761/14,177 (12.4%)	**OR 2.90 (2.60, 3.24)**	Poor. Unadjusted, unclear definitions
		SCr criteria alone	788/4694 (16.8%)	**OR 3.92 (3.48, 4.43)**	
		UO **and** SCr criteria	1597/4995 (32.0%)	**OR 10.5 (9.30, 11.9)**	
		No AKI	350/8179 (4.3%)	(reference)	
	Mortality, 30 day	Maximum AKI severity detected by: UO criteria alone	1822/14,177 (12.9%)	**OR 2.69 (2.41, 3.00)**	
		SCr criteria alone	808/4694 (17.2%)	**OR 3.79 (3.35, 4.29)**	
		UO **and** SCr criteria	1375/4995 (27.5%)	**OR 6.93 (6.17, 7.78)**	
		No AKI	425/8179 (5.2%)	(reference)	
	RRT, in-hospital	Maximum AKI severity detected by: UO criteria alone	304/14,177 (2.1%)	**OR 0.066 (0.058, 0.075)**	
		SCr criteria alone	232/4694 (4.9%)	**OR 0.16 (0.13, 0.18)**	
		UO **and** SCr criteria	1251/4995 (25.0%)	*(reference)*	
		No AKI	4/8179 (0.05%)	.	
Hong 2023[15] 36728812 US ICU admission N=32,794 68 yr (30% with CKD G3)	Mortality, in-hospital	KDIGO SCr criteria alone AKI	6260/25,293 (24.7%,)	**OR 2.30 (1.93, 2.74)**	Suspect Number of participants in Table 2 sum to 41,153, 25% more than reported included participants. Unadjusted Data presented here are meta-analyzed data from across 3 databases (MIMIC IV, MIMIC III, eICU). MIMIC IV & eICU also analyzed by Machado 2024 39135063 Unclear analysis omitted: HR for [AKI per APKD but not (KDIGO or cROCK)] vs. [No AKI per APKD (ignoring KDIGO or cROCK)]
		No AKI	2016/15,860 (12.7%)	(reference)	
		cROCK SCr criteria alone AKI	4828/18,230 (26.5%)	**OR 2.04 (1.66, 2.52)**	
		No AKI	3448/22,923 (15.0%)	(reference)	
		APKD SCr criteria alone AKI	7241/32,293 (22.4%)	**OR 2.24 (1.87, 2.67)**	
		No AKI	1035/8860 (11.7%)	(reference)	

Study PMID Country AKI Cause* Total N Age	Specific Outcome	AKI Definition	Events/N (%)	Effect Size (CI)	Analysis Quality
Machado 2024[16] 39135063 US ICU admission N= 35,845 (MIMIC IV), 100,917 (eICU validation set) 65 yr (MIMIC IV), 63 yr (eICU)	Mortality, in-hospital	**KDIGO SCr** or **UO criteria:**			Good (crude n/N often missing) Red are analyses run in validation set (eICU) Databases also analyzed (poorly) by Hong 2023 36728812 Also reported "net reclassification improvement (NRI)", the percent who were reclassified up or down, correctly and incorrectly.
		Stage 1	NR	adjOR 1.0 (0.8, 1.2) **adjOR 1.69 (1.42-2.01)**	
		Stage 2	NR	**adjOR 1.7 (1.3, 2.1)** **adjOR 1.37 (1.17-1.60)**	
		Stage 3	NR	**adjOR 3.3 (2.7, 3.9)** **adjOR 2.67 (2.36-3.02)**	
		No AKI	NR	(reference) AUC ROC (overall): 0.69 (0.68, 0.70)	
		KDIGO UO criteria only:			
		Stage 1	376/8377 (4.5%)	adjOR 0.9 (0.7, 1.1)	
		Stage 2	1317/14,155 (9.3%)	**adjOR 1.5 (1.3, 1.8)**	
		Stage 3	1423/7004 (20.3%)	**adjOR 2.9 (2.5, 3.5)**	
		No AKI	205/6309 (3.2%)	(reference) AUC ROC (overall): 0.68 (0.67, 0.69)	
		UO-AKI or SCr (as per KDIGO):			
		Stage 1 (0.2–0.3 mL/kg/hr)	NR	**adjOR 1.4 (1.3, 1.6)** **adjOR 1.76 (1.61-1.94)**	
		Stage 2 (0.1–0.2 mL/kg/hr)	NR	**adjOR 2.3 (2.0, 2.6)** **adjOR 2.05 (1.87-2.26)**	
		Stage 3 (<0.1 mL/kg/hr)	NR	**adjOR 5.0 (4.4, 5.7)** **adjOR 2.94 (2.71-3.19)**	
		No AKI	NR	(reference) AUC ROC (overall): 0.75 (0.74, 0.76)	
		UO-AKI criteria only:			
		Stage 1 (0.2–0.3 mL/kg/hr)	489/7140 (6.8%)	**adjOR 1.3 (1.2, 1.5)** Sn 80% / Sp 54%	
		Stage 2 (0.1–0.2 mL/kg/hr)	762/6042 (12.6%)	**adjOR 2.1 (1.9, 2.4)** Sn 65% / Sp 74%	
		Stage 3 (<0.1 mL/kg/hr)	1397/4394 (31.8%)	**adjOR 4.8 (4.3, 5.4)** Sn 42% / Sp 91%	
		No AKI	673/18,268 (3.7%)	(reference) AUC ROC (overall): 0.74 (0.73, 0.75)	
Machado 2024 39135063 (continued)	Mortality, 90 day	**KDIGO SCr or UO criteria:**			
		Stage 1	NR	adjHR 0.97 (0.87, 1.09)	
		Stage 2	NR	**adjHR 1.48 (1.33, 1.64)**	
		Stage 3	NR	**adjHR 2.43 (2.18, 2.71)**	
		No AKI	NR	(reference)	
		UO-AKI or SCr (as per KDIGO):			
		Stage 1 (0.2–0.3 mL/kg/hr)	NR	**adjHR 1.23 (1.14, 1.33)**	
		Stage 2 (0.1–0.2 mL/kg/hr)	NR	**adjHR 1.76 (1.63, 1.90)**	
		Stage 3 (<0.1 mL/kg/hr)	NR	**adjHR 3.08 (2.86, 3.32)**	
		No AKI	NR	(reference)	
	RRT	**KDIGO SCr or UO criteria**	NR	AUC ROC 0.81 (0.79–0.82)	
		UO-AKI or SCr (as per KDIGO)	NR	AUC ROC 0.85 (0.83–0.86)	
	Mortality, in-hospital	KDIGO SCr only	6198/12,309 (50%)		Fair

Study PMID Country AKI Cause* Total N Age	Specific Outcome	AKI Definition	Events/N (%)	Effect Size (CI)	Analysis Quality
Wainstein 2022[17] 35442972 54 countries COVID-19, hospital N=75,670 67 yr		deKDIGO SCr (focus just on decrease in SCr)	2692/10,909 (25%)		Adjusted main analysis. Incomplete data/analysis for KDIGO and deKDIGO data
		eKDIGO SCr (includes decrease in SCr)	8890/23,226 (38%)	**adjOR 1.78 (1.70, 1.85)**	
		No AKI (per eKDIGO)	9794/50,866 (19%)	(reference)	
White 2023[18] 37432520 Australia Sepsis, ICU admission N=23,555 61 yr	Mortality, in-hospital	**Any AKI**			Poor Mostly unadjusted analyses (crude data only reported). Adjusted analysis has unclear comparator. Also crude results reported for Stages 1, 2, and 3 (Supplement tables S16-S18), not extracted here. Also data for partial and no renal recovery
		KDIGO UO criteria alone	738/5952 (12.4%)	**OR 1.79 (1.60, 1.99)**	
		KDIGO SCr criteria alone	702/4642 (15.1%)	**OR 2.25 (2.01, 2.51)**	
		KDIGO SCr **and** UO criteria	914/2857 (32.0%)	**OR 5.94 (5.33, 6.62)**	
		No AKI	742/10,104 (7.3%)	(reference)	
	RRT	**Any AKI**			
		KDIGO UO criteria alone	211/5952 (3.5%)	**OR 1.60 (1.32, 1.93)**	
		KDIGO SCr criteria alone	857/4642 (18.5%)	**OR 9.85 (8.47, 11.5)**	
		KDIGO SCr **and** UO criteria	1460/2857 (51.1%)	**OR 45.5 (39.1, 52.9)**	
		No AKI	227/10,104 (2.2%)	(reference)	
	Not Full renal recovery	**Any AKI** KDIGO UO criteria alone	822/5952 (13.8%)	**OR 2.69 (2.40, 3.00)**	
		KDIGO SCr criteria alone	2857/4642 (61.5%)	**OR 26.8 (24.2, 29.7)**	
		KDIGO SCr **and** UO criteria	2231/2857 (78.1%)	**OR 59.7 (52.8, 67.5)**	
		No AKI	569/10,104 (5.6%)	(reference)	
	MAKE-30	**Any AKI** KDIGO UO criteria alone	820/5952 (13.8%)	**OR 2.05 (1.85, 2.28)**	
				adjOR 0.34 (0.32, 0.36) vs. SCr alone	
		KDIGO SCr criteria alone	1265/4642 (27.3%)	**OR 4.82 (4.36, 5.32)**	
		KDIGO SCr **and** UO criteria	1780/2857 (62.3%)	**OR 21.3 (19.2, 23.6)**	
		No AKI	729/10,104 (7.2%)	(reference)	
Parsh 2016[19] 27179735 US Undergoing PCI (contrast) N=119,554 65 yr	Mortality, in-patient	Increase SCr ≥0.2 mg/dL & ≥70%	AKI: 2056/119,554 (1.7%)	adjOR 15.5 (NR) Sn 33.7% Sp 98.7%	Good Although no estimates of variance Note that in pre-analyses (likelihood ratio testing), **KDIGO definition** and **SCr ≥0.5 mg/dl increase** (ignoring % change) were note selected for further analysis.
		Increase SCr ≥0.3 mg/dL & ≥50%	**AKI: 3371/119,554 (2.8%)**	**adjOR 12.6 (NR)** **Sn 42.6% Sp 97.7%** **Optimal based on cost of FN 10x cost of FP**	
		Increase SCr ≥0.2 mg/dL & ≥50%	AKI: 3443/119,554 (2.9%)	adjOR 12.4 (NR) Sn 42.6% Sp 97.7%	
		Increase SCr ≥0.3 mg/dL & ≥40%	*AKI: 4519/119,554 (3.8%)*	*adjOR 10.8 (NR)* *Sn 47.0% Sp 96.8%* *Optimal based on cost of FN 20x cost of FP*	
		Increase SCr ≥0.2 mg/dL & ≥40%	AKI: 4519/119,554 (3.8%)	adjOR 10.8 (NR) Sn 47.0% Sp 96.7%	
		Increase SCr ≥0.3 mg/dL & ≥30%	AKI: 6312/119,554 (5.3%)	adjOR 9.31 (NR) Sn 52.6% Sp 95.3%	
		Increase SCr ≥0.2 mg/dL & ≥30%	AKI: 7520/119,554 (6.3%)	adjOR 8.65 (NR) Sn 53.5% Sp 94.3%	
		Increase SCr ≥0.2 mg/dL & ≥20%	AKI: 11,286/119,554 (9.4%)	adjOR 7.71 (NR) Sn 59.9% Sp 91.2%	
		Increase SCr ≥0.2 mg/dL & ≥10%	AKI: 12,206/119,554 (10.1%)	adjOR 7.79 (NR) Sn 63.3% Sp 90.4%	

Study PMID Country AKI Cause* Total N Age	Specific Outcome	AKI Definition	Events/N (%)	Effect Size (CI)	Analysis Quality
		Increase SCr ≥0.8 mg/dL or ≥20%	AKI: 13,223/119,554 (11.1%)	adjOR 7.20 (NR) Sn 60.9% Sp 89.6%	
		Increase SCr ≥0.7 mg/dL or ≥20%	AKI: 13,235/119,554 (11.1%)	adjOR 7.29 (NR) Sn 61.2% Sp 89.6%	
		Increase SCr ≥0.6 mg/dL or ≥20%	AKI: 13,247/119,554 (11.1%)	adjOR 7.28 (NR) Sn 61.2% Sp 89.5%	
		Increase SCr ≥0.5 mg/dL or ≥20%	AKI: 13,282/119,554 (11.1%)	adjOR 7.26 (NR) Sn 61.5% Sp 89.5%	
		Increase SCr ≥0.4 mg/dL or ≥20%	AKI: 13,306/119,554 (11.1%)	adjOR 7.21 (NR) Sn 61.5% Sp 89.5%	
		Increase SCr ≥0.3 mg/dL or ≥20%	AKI: 13,665/119,554 (11.4%)	adjOR 7.36 (NR) Sn 63.2% Sp 89.2%	
		Increase SCr ≥0.2 mg/dL or ≥20%	AKI: 14,239/119,554 (11.9%)	adjOR 7.39 (NR) Sn 64.9% Sp 88.7%	
Swan 2021[20] 34362089 US Hospital admission (9.4% with CKD) N=83,938 (126,367 admissions) 63 years	Mortality, in-hospital	KDIGO SCr Stage 1 (a or b) within 24 hr	900/9357 admissions (9.6%)	**adjOR 16.9 (P <0.01)**	Poor Did not account for multiple admissions. Some lack of clarity about distinction between the 2 48 hr – 7 day definitions Note that the 4 AKI definitions are *non-overlapping* Also data in Supplement Figures S6 and S7 for Stage 1a and Stage 1b criteria specifically with narrower time frames
		KDIGO SCr Stage 1 (a or b) during 24-48 hr	234/6936 admissions (3.4%)	**adjOR 5.5 (P <0.01)**	
		KDIGO SCr Stage 1b during 48 hr – 7 d	61/2509 admissions (2.4%)	**adjOR 4.2 (P <0.01)**	
		KDIGO SCr Stage 1a during 48 hr – 7 d	32/3004 admissions (1.1%)	**adjOR 1.6 (P <0.01)**	
		No AKI	574/104, 534 admissions (0.6%)	(reference)	
Xu 2020[21] 32349004 China Hospitalized, CKD (eGFR <60 ml/min) N=21,661 68 yr	Mortality, in-hospital	KDIGO SCr (not cROCK AKI)	4/221 (1.8%)	adjHR 2.12 (0.65, 6.88)	Good Supplement Table S3 also has subgroup analyses for in-hospital mortality based on ICU admission, with similar findings
		cROCK SCr (not KDIGO AKI)	48/1512 (3.2%)	**adjHR 5.53 (3.75, 8.16)**	
		KDIGO SCr and cROCK AKI	405/3145 (12.9%)	**adjHR 11.6 (8.48, 15.9)**	
		No AKI (either KDIGO or cROCK))	84/16,783 (0.5%)	(reference)	
	Mortality, post-discharge (2.8 yr, median)	KDIGO SCr (not cROCK AKI)	14/113 (12.4%)	adjHR 1.07 (0.63, 1.82)	
		cROCK SCr (not KDIGO AKI)	117/702 (16.7%)	**adjHR 1.51 (1.25, 1.83)**	
		KDIGO SCr and cROCK AKI	274/1438 (19.1%)	**adjHR 1.96 (1.72, 2.25)**	
		No AKI (either KDIGO or cROCK))	1010/8611 (11.7%)	(reference)	
	CKD progression (both GFR reduction ≥25% and increase in CKD stage), 90 days	KDIGO SCr (not cROCK AKI)	0/17	NA	
		cROCK SCr (not KDIGO AKI)	18/167 (10.8%)	**adjOR 5.65 (3.05, 10.5)**	
		KDIGO SCr and cROCK AKI	31/226 (13.7%)	**adjOR 6.88 (4.03, 11.7)**	
		No AKI (either KDIGO or cROCK))	52/1973 (2.6%)	(reference)	
Yang 2015[22] 26466051 China Hospitalized, AKI N=7604 61 yr	Mortality, in-hospital	**KDIGO SCr:**			Poor Unclear definitions, analyses, and eligibility criteria
		Stage 3	NR/1301	**adjOR 1.95 (1.38, 2.75)**	
		Stage 2	NR/959	**adjOR 1.89 (1.46, 2.44)**	
		Stage 1	NR/1427	(reference)	
		[Any stage]	704/3687 (19.1%)	[reference vs. Expanded]	
		Expanded SCr criteria (not KDIGO criteria):			
		Stage 3	NR/870	**adjOR 2.05 (1.54, 2.74)**	
		Stage 2	NR/991	**adjOR 1.89 (1.53, 2.32)**	
		Stage 1	NR/2056	(reference)	
		[Any stage]	223/3917 (5.7%)	**OR 0.26 (0.22, 0.30)** (vs. KDIGO SCr criteria)	
	RRT	KDIGO SCr (any stage)	638/3687 (17.3%)	(reference)	

Study PMID Country AKI Cause* Total N Age	Specific Outcome	AKI Definition	Events/N (%)	Effect Size (CI)	Analysis Quality
		Expanded SCr, not KDIGO (any stage)	258/3917 (6.6%)	**OR 0.34 (0.29, 0.39)**	
Wang H 2022[23] 35854309 Scotland	AKD, 90 d	SCr ≥26.5, but <1.5x (baseline w/in 48 hr)	NR/6342	**adjOR 0.47 (0.44, 0.49)**	Fair Incomplete data reporting Unclear why 13,154 (23%) missing
		SCr >1.5x (baseline w/in 1 yr)	NR/37,410	(reference)	
General population (inpatient and outpatient SCr values) N=43,752 75 yr (median)	CKD, 2 yr (among those who entered AKD, but excluding initiating chronic KRT during AKD)	SCr ≥26.5, but <1.5x (baseline w/in 48 hr)	NR/1701	**adjOR 0.71 (0.63, 0.79)**	Extracted more complete data set, including those who had non-recovery starting during AKI or AKD (and those who died) Unclear if this is a biased comparison of those with and without SCr w/in 48 hr
		SCr >1.5x (baseline w/in 1 yr)	NR/14,880	(reference)	

SCr (or UO) criteria alone = only the SCr (or UO) data were considered; the alternate lab measure, UO (or SCr), was either ignored or not available.
SCr (or UO) criteria only = SCr (or UO) was abnormal but the alternative lab measure, UO (or SCr) was normal; there was isolated creatinemia (or isolated oliguria).

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Table S12. Comparison of different AKI definitions (criteria) on prioritized outcomes in children

Study PMID Country AKI Cause* Total N Age	Specific Outcome	AKI Definition	Events/N (%)	Effect Size (CI)	Analysis Quality
Lex 2014[24] 24206964 Hungary Pediatric cardiac surgery N=1489 54% <1 y	RRT	pRIFLE CrCl	AKI 88/481 (18.3%) No AKI 0/1005 (0%)	Sn 91% Sp 72% AUC (adj) 0.87 (0.83, 0.91)	Poor (mostly unadjusted, inadequate analysis, some data and reporting unclear) Also data by age group (neonate, infant, >1), not extracted Unclear how "No AKI" defined. We assumed this is per pRIFLE, but 3 patients missing.
		AKIN CrCl	AKI 102/285 (35.8%)	Sn 60% Sp 87% AUC (adj) 0.75 (0.69, 0.81)	
		KDIGO CrCl	AKI 102/409 (24.9%)	Sn 24% Sp 77% AUC (adj) 0.50 (0.44, 0.55)	
	Mortality, in-hospital	pRIFLE CrCl	AKI 43/481 (8.9%)	**OR 12.2 (5.70, 26.2)** Sn 78% Sp 70% AUC (adj) 0.78 (0.71, 0.85)	
			No AKI 8/1005 (0.8%)	(reference)	
			AKI Stage 3	**OR 13.6 (7.0, 26.3)**	
		AKIN CrCl	AKI 43/285 (15.1%)	Sn 71% Sp 84% AUC (adj) 0.81 (0.74, 0.88)	
			AKI Stage 3	**OR 38.3 (20.6, 70.9)**	
		KDIGO CrCl	AKI 50/409 (12.2%)	Sn 82% Sp 75% AUC (adj) 0.81 (0.75, 0.87)	
			AKI Stage 3	**OR 18.8 (9.6, 36.6)**	
Wei 2021[25] 32791515 China Pediatric ICU admission N=3338 1.0 y (IQR 0.3, 2.9) 1 mo – 1 y 49% 1-5 y 37% 5-12 y 12% 12-18 1.3%	Mortality, in-hospital	KDIGO AKI (any) SCr	NR	**adjHR 9..05 (5.38, 15.2)**	No major concerns, but incomplete reporting
		pROCK AKI (any) SCr	32/257 (12.5%)	**adjHR 10.5 (6.24, 17.7)**	
		No AKI (per pROCK) SCr	32/3081 (1.0%)	(reference)	
		KDIGO Stage 1 SCr	NR	**adjHR 5.32 (2.53, 11.2)**	
		pROCK Stage 1 SCr	NR	**adjHR 8.28 (4.17, 16.4)**	
		KDIGO Stage 2 SCr	NR	**adjHR 12.1 (5.66, 26.0)**	
		pROCK Stage 2 SCr	NR	**adjHR 10.1 (4.60, 22.2)**	
		KDIGO Stage 3 SCr	NR	**adjHR 13.6 (6.91, 26.7)**	
		pROCK Stage 3 SCr	NR	**adjHR 17.1 (8.05, 36.5)**	
Tai 2022[26] 34076880 Australia Pediatric ICU N=7505 Median 2.0 (IQR 0.4, 7.3) All ≤16 yr	Mortality, ICU	KDIGO	87/690 (12.6%)	**OR 5.57 (4.25, 7.31)**	Poor, unadjusted
		No AKI	172/6815 (2.5%)	(reference)	
		pROCK	60/356 (16.9%)	**OR 7.08 (5.19, 9.66)**	
		No AKI	199/7149 (2.8%)	(reference)	
		pRIFLE	41/154 (26.6%) (all also met KDIGO criteria)	**OR 11.9 (8.10, 17.4)**	
		No AKI	218/7351 (3.0%)	(reference)	
		KDIGO + pROCK criteria	53/334 (15.9%)	**OR 6.38 (4.61, 8.82)**	
		No AKI	206/7171 (2.9%)	(reference)	
		KDIGO + pROCK + pRIFLE criteria	21/60 (35.0%)	**OR 16.3 (9.45, 28.1)**	
		No AKI	238/7445 (3.2%)	(reference)	
	Mortality, 90-day	**KDIGO SCr:**			Good Also subgroup analyses for 41. ICU and no ICU, with similar results as overall Also HRs for combinations of various definitions and stages
		Any stage	387/9544 (4.0%)	**adjHR 2.57 (2.28, 2.90)**	
		Stage 1	157/5438 (2.9%)	**OR 1.70 (1.43, 2.00)**	
		Stage 2 or 3	230/4106 (5.6%)	**OR 3.38 (2.93, 3.90)**	
		No AKI	1468/85,171 (1.7%)	(reference)	
		pROCK SCr:			
		Any stage	311/4891 (6.4%)	**adjHR 3.57 (3.15, 4.04)**	

Study PMID Country AKI Cause* Total N Age	Specific Outcome	AKI Definition	Events/N (%)	Effect Size (CI)	Analysis Quality
Xu 2018[27] 30054338 China Pediatric in-hospital N=94,715 1 mo – 18 yr (33% ≤1 yr, 49% 1-10 yr, 18% 10-18)		Stage 1	NR		
		Stage 2 or 3	NR		
		No AKI	1544/89,824 (1.7%)		
		pRIFLE SCr:			
		Any stage	291/14,468 (2.0%)	**adjHR 2.22 (1.99, 2.48)**	
		Stage 1	259/10,362 (2.5%)	**OR 1.48 (1.30, 1.69)**	
		Stage 2 or 3	230/4106 (5.6%)	**OR 3.43 (2.97, 3.96)**	
		No AKI	1366/80,310 (1.7%)	(reference)	
Kaddourah 2017[28] 27959707 AWARE study Multinational (US 78%, Canada 8%, Australia 4%, China 3%, Italy 2%, Indonesia 2%, UK 2%, Serbia 2%, S Korea 1%) Pediatric ICU N=5237 5.5 yr (IQR 18 mo – 12 yr)	Mortality, 28 day	**Stage 3:**			Poor (unadjusted, incomplete reporting and analysis) [RRs and 95% CI's based on simply dividing the risks in each arm and the P value
		KDIGO UO only	NR (32.6%)	**RR~1.8 (1.1, 2.9)** **P 0.028**	
		KDIGO SCr only	NR (18.5%)		
		Stage 2:			
		KDIGO UO only	NR (4.8%)	RR~0.7 (0.3, 1.9) P 0.51	
		KDIGO SCr only	NR (6.7%)		
		Stage 1:			
		KDIGO UO only	NR (2.5%)	**RR~0.5 (0.3, 0.97)** **P 0.040**	
		KDIGO SCr only	NR (5.0%)		
		No AKI:			
		KDIGO UO only	NR (3.0%)	RR~1.1 (0.8, 1.5) P 0.45	
		KDIGO SCr only	NR (2.7%)		
Kaddourah 2019[29] 30676490 AWARE study 9 countries (4 continents) Pediatric ICU admission N=3318 ~5.6 y (3 mo – 25 y)	Mortality, 28 day	Stage 3: KDIGO SCr criteria only	23/343 (6.7%)	**OR 2.40 (1.49, 3.87)** **vs. No AKI** **OR 0.12 (0.06, 0.23)** **vs. SCr and UO**	Poor, unadjusted with substantive differences between groups, including median age
		Stage 3: KDIGO UO criteria only	7/90 (7.8%)	**OR 2.82 (1.26, 6.28)** **vs. No AKI** **OR 0.14 (0.05, 0.35)** **vs. SCr and UO**	
		Stage 3: KDIGO SCr and UO criteria	24/63 (38.1%)	**OR 20.6 (11.8, 35.8)** **vs. No AKI**	
		No AKI	82/2822 (2.9%)	(reference)	
	RRT	Stage 3: KDIGO SCr criteria only	11/343 (3.2%)	**OR 9.32 (3.93, 22.1)** **vs. No AKI** **OR 0.02 (0.01, 0.05))** **vs. SCr and UO**	
		Stage 3: KDIGO UO criteria only	9/90 (10.0%)	**OR 31.2 (12.4, 79.0)** **vs. No AKI** **OR 0.08 (0.03, 0.18)** **vs. SCr and UO**	
		Stage 3: KDIGO SCr and UO criteria	37/63 (58.7%)	**OR 400 (180, 889)** **vs. No AKI**	
		No AKI	10/2822 (0.4%)	(reference)	
Ruth 2022[30] 35145645 Multinational Pediatric ICU N=1262	Mortality, ICU	**Any AKI**			Poor, unadjusted
		KDIGO UO criteria alone	9/443 (2.0%)		
		KDIGO SCr criteria alone	39/645 (6.0%)		
		KDIGO SCr **and** UO criteria	33/174 (19.0%)		
	RRT, in ICU	**Any AKI**			
		KDIGO UO criteria alone	4/443 (0.9%)		

Study PMID Country AKI Cause* Total N Age	Specific Outcome	AKI Definition	Events/N (%)	Effect Size (CI)	Analysis Quality
Median 6 (IQR 1.3, 13.6)		KDIGO SCr criteria alone	13/645 (2.0%)		
		KDIGO SCr **and** UO criteria	46/174 (26.4)		
Sutherland 2015[31] 25649155 US Pediatric hospitalization N=14,795 Median 6.7 yr (IQR 2-14)	Mortality, in-hospital	**ICU population:** pRIFLE eGFR Stage 1	NR (5.5% [3.3, 7.5))	LR 3.08 (1.64 to 5.79), NS after Bonferroni correction	Poor Unadjusted, some missing results
		pRIFLE eGFR Stage 2	NR (10.6% [6.8, 14.5])	LR 6.06 (3.26, 11.3)	
		pRIFLE eGFR Stage 3	NR (35.9% [29.2, 42.6])	LR 20.49 (12.0, 35.0)	
		pRIFLE eGFR No AKI	NR (1.8% [0.9, 2.6])	(reference)	
		AKIN SCr Stage 1	NR (8.7% [5.5, 11.9])	LR 3.83 (NR)	
		AKIN SCr Stage 2	NR (13.2% [8.7, 17.8])	LR 5.82 (NR)	
		AKIN SCr Stage 3	NR (30.4% [23.9, 36.9])	LR 13.4 (NR	
		AKIN SCr No AKI	NR (2.3% [1.4, 3.2])	(reference)	
		KDIGO SCr Stage 1	NR (6.4% [3.7, 9.2])	**LR 2.85 (1.59, 5.13), P<0.05 after Bonferroni correction**	
		KDIGO SCr Stage 2	NR (10.0% [5.8, 14.1])	**LR 4.42 (2.47, 7.89)**	
		KDIGO SCr Stage 3	NR (32.3% [26.2, 38.4])	**LR 14.34 (9.18, 22.4)**	
		KDIGO SCr No AKI	NR (2.3% [1.3, 3.2])	(reference)	
		Non-ICU population: pRIFLE eGFR Stage 1	NR (0.7% [0.4, 0.9])	LR 0.91 (0.55, 1.50)	
		pRIFLE eGFR Stage 2	NR (0.7% [0.3, 1.1])	LR 0.96 (0.51, 1.81)	
		pRIFLE eGFR Stage 3	NR (1.4% [0.8, 2.0])	LR 1.96 (1.17, 3.31)	
		pRIFLE eGFR No AKI	NR (0.8% [0.5, 0.9])	(reference)	
		AKIN SCr Stage 1	NR (0.8% [0.5, 1.1])	LR 1.00 (NR)	
		AKIN SCr Stage 2	NR (0.6% [0.2, 1.0])	LR 0.77 (NR)	
		AKIN SCr Stage 3	NR (0.8% [0.2, 1.4])	LR 1.02 (NR)	
		AKIN SCr No AKI	NR (0.8% [0.6, 1.0])	(reference)	
		KDIGO SCr Stage 1	NR (0.6% [0.3, 0.9])	LR 0.73 (0.41, 1.30)	
		KDIGO SCr Stage 2	NR (0.7% [0.2, 1.1])	LR 0.82 (0.41, 1.64)	
		KDIGO SCr Stage 3	NR (1.0% [0.5, 1.5])	LR 1.22 (0.69, 2.13)	
		KDIGO SCr No AKI	NR (0.8% [0.6, 1.0])	(reference)	

Table S13. AKI trajectories: Summary of trajectories addressed by eligible studies

Trajectory	Definition of Trajectory	Studies
Rapid/Transient	< or ≤48 hr	Jensen 2014 Ozrazgat-Baslanti 2021 Heung 2016 Meersch 2024
Rapid/Transient	≤3 d	Truche 2018
Persistent	≥48 to 7 d	Jensen 2014 Meersch 2024
Persistent	3-10 d	Heung 2016
Persistent	> or ≥3 d	Koyner-2022 / 2023 Truche 2018
Persistent (AKD)	>7 d	Wang H 2022
Persistent, no recovery	>48 hr, no recovery pre-discharge	Ozrazgat-Baslanti 2021
Persistent, recovery	>48 hr, recovery pre-discharge	Ozrazgat-Baslanti 2021
Persistent severe	≥3 d & RRT (or death [Gómez]), AKI stage 2 or 3	Gómez 2025 Koyner 2022 / 2023
Complete recovery	SCr return to <50% above baseline	Korenkevych 2016 Peerapornratana 2023
Complete recovery	SCr return to <25% above baseline	Pannu 2013
Complete recovery	SCr return to <20% above baseline	Quiroga 2022
Complete recovery	SCr return to <5-55% above baseline, multiple analyses	Pannu 2013
Complete recovery	To no AKI per KDIGO	Truche 2018
Partial recovery	SCr return to >50% above baseline but no RRT	Korenkevych 2016
Worsening AKI (upward trajectory)	Max SCr after min SCr in-hospital	Warnock 2021 Warnock 2015
Resolving AKI (downward trajectory)	Max SCr before min SCr in-hospital	Warnock 2021 Warnock 2015
Resolving AKI (downward trajectory)	Decrease by at least 1 AKI stage compared to previous day	Truche 2018
Recurrent AKI	post-discharge, with rehospitalization	Liu 2019
Recurrent AKI	In-hospital	Warnock 2015 Rutledge 2024 (**NEONATAL**) Vincent 2024 (**NEONATAL**) Adegboyega 2022 (**NEONATAL**)
Recurrent AKI	After ≥48 hr post-AKI resolution	Ruth 2022 (**PEDIATRIC**)
Recurrent AKI	not defined/other	Ponce 2020 (not defined) Walker 2021 (w/in 6 yr) Altawalbeh 2023 (w/in 5 yr)
"Complete" recurrent	AKI after ≥75% return to SCr baseline	Adegboyega 2022 (**NEONATAL**)
"Incomplete" recurrent	AKI after 25-74% return to SCr baseline	Adegboyega 2022 (**NEONATAL**)
Late/early and Transient/Persistent combos		Ruth 2022 (**PEDIATRIC**)

Table S14. AKI trajectories: Comparison of different AKI definitions on prioritized outcomes

Study / PMID / Country Population / N / Age / Years	AKI Definition	Outcome	Population	Events/N (%)	Effect Size (CI)	Analysis Quality
Jensen 2014[1] 38765592 Denmark General (in- and outpatient) N=59,133 (AKI) Median 72 yr 1990-2018 36% with CKD	AKI: KDIGO (SCr) Rapid reversal: <48 hr, back to no AKI per KDIGO Persistent: 2-7 d (Omitting AKD, see RQ 3)	CKD, incident, 20 yr risk [No previous CKD]	Persistent AKI	2667/12,106 (22.0%)	**adjHR 1.13 (1.08, 1.19)**	Good Subgroup analyses with generally similar patterns (age, sex, AKI stage, location, AKI cause, baseline eGFR)
			Rapid reversal AKI	3990/22,467 (17.8%)	[REFERENCE]	
		ESKD, incident, 20 yr risk [36% w/prior CKD]	Persistent AKI	899/22,500 (4.0%)	**adjHR 1.19 (1.08, 1.30)**	
			Rapid reversal AKI	915/36,398 (2.5%)	[REFERENCE]	
Koyner 2022 / 2023[2,3] 36450235 / 36996299 US In-patient (LOS ≥3 d), AKI stage 2 or 3 N= 126,528 (AKI) 66 yr 2017-19	AKI: KDIGO (SCr) Persistent-severe: Stage 3 ≥3 d or death or w/in 3 d), or Stage 2 or 3 w/dialysis w/in 3 d Alternative persistent-severe: Stage 3 ≥3 d (only), omitting death or dialysis criteria Not persistent-severe: Stage 2 or 3, not persistent-severe	Mortality, in-hospital	Persistent-severe AKI	9711/30,916 (31.4%)	**adjOR 4.58 (4.41, 4.75)**	Good Although, *Alternative definition analysis problematic:* Note that "Alternative persistent-severe AKI" analyses exclude those who died w/in 3 days and those with stage 2 w/dialysis w/in 3 d
			Alternate persistent-severe AKI	2578/17,285 (14.9%)	**adjOR 1.79 (1.70, 1.89)**	
			Not persistent-severe AKI	8071/95,612 (8.4%)	[REFERENCE for all]	
			ICU subset			
			Persistent-severe AKI	7910/18,858 (41.9%)	**adjOR 4.42 (4.23, 4.62)**	
			Alternate persistent-severe AKI	2062/8322 (24.8%)	**adjOR 1.87 (1.76, 1.99)**	
			Not persistent-severe AKI	5854/39,089 (15.0%)	[REFERENCE for all]	
			Non-ICU subset			
			Persistent-severe AKI	1801/12,058 (14.9%)	**adjOR 4.89 (4.54, 5.25)**	
			Alternate persistent-severe AKI	516/8963 (5.8%)	**adjOR 1.58 (1.42, 1.75)**	
			Not persistent-severe AKI	2217/56,523 (3.9%)	[REFERENCE for all]	
		ICU use	Persistent-severe AKI	18,856/30,916 (61.0%)	**adjOR 2.15 (2.09, 2.21)**	
			Alternate persistent-severe AKI	8322/17,285 (48.1%)	**adjOR 1.28 (1.24, 1.32)**	
			Not persistent-severe AKI	39,089/95,612 (40.9%)	[REFERENCE for all]	
		Dialysis, in-hospital	Persistent-severe AKI	1580/30,916 (7.5%)	**adjOR 15.66 (13.87, 17.67)**	
			Not persistent-severe AKI	345/95,612 (0.4%)	[REFERENCE]	
		Hospital LOS	Persistent-severe AKI	15.4 d (N=30,916)	**AdjMD 3.33 (3.33, 3.34) d** **adjRatio 1.32 (1.31, 1.33)**	
			Alternate persistent-severe AKI	NR	**adjRatio 1.38 (1.36, 1.39)**	
			Not persistent-severe AKI	10.0 d (N=95,612)	[REFERENCE for all]	
		ICU LOS	Persistent-severe AKI	9.0 d (N=18,858)	**AdjMD 2.56 (2.56, 2.57) d** **adjRatio 1.45 (1.43, 1.47)**	
			Alternate persistent-severe AKI	NR	**adjRatio 1.42 (1.39, 1.45)**	
			Not persistent-severe AKI	5.5 d (N=39,089)	[REFERENCE for all]	
		Inpatient readmission, 30 d	Persistent-severe AKI	4030/30,916 (19.0%)	**adjOR 1.07 (1.02, 1.11)**	
			Not persistent-severe AKI	15,264/95612 (17.4%)	[REFERENCE]	
			ICU subset			
			Persistent-severe AKI	1935/10,948 (17.7%)	adjOR 1.00 (0.94, 1.06)	
			Not persistent-severe AKI	5584/33,235 (16.8%)	[REFERENCE]	
			Non-ICU subset			
			Persistent-severe AKI	2095/10,257 (20.4%)	**adjOR 1.12 (1.06, 1.18)**	
			Not persistent-severe AKI	9680/54,306 (17.8%)	[REFERENCE]	

Study / PMID / Country Population / N / Age / Years	AKI Definition	Outcome	Population	Events/N (%)	Effect Size (CI)	Analysis Quality
Warnock 2021[4] 34419950 US In-patient N=37,827 (AKI) 57 yr 2016-18	AKI: KDIGO (SCr) Resolving AKI: Max SCr before min SCr in-hospital (downward trajectory) "In-hospital" AKI: Max SCr after min SCr in-hospital (upward trajectory)	Mortality, in-patient	In-hospital AKI (upward trajectory)	1664/19,376 admissions (8.6%) [n=11,138] 3.07 per pt-year	**adjIRR 1.82 (1.62, 2.04)** **P<0.05 (more) vs. resolving AKI**	Fair Did not account for multiple admissions (62,380 patients with 100,570 admissions; 1.61/pt)
			Resolving AKI (downward trajectory)	1428/42,458 admissions (3.4%) [n=36,689] 1.40 per pt-year	adjIRR=0.93 (0.83, 1.05)	
			No AKI	417/38,736 admissions (1.1%) [n=24,553] 0.80 per pt-year	[REFERENCE for all]	
		RRT, in-patient	In-hospital AKI (upward trajectory)	638/19,376 admissions (3.3%) [n=11,138] 0.97 per pt-year	**adjIRR 6.83 (2.70, 6.02)** **P<0.05 (less) vs. resolving AKI**	
			Resolving AKI (downward trajectory)	1252/42,458 admissions (3.0%) [n=36,689] 1.03 per pt-year	**adjIRR 8.61 (3.82, 8.42)**	
			No AKI	417/38,736 admissions (1.1%) [n=24,553] 0.80 per pt-year	[REFERENCE for all]	
Ozrazgat-Baslanti 2021[5] 34876451 US In-patient N=54,212 (AKI) 60 yr 2012-19	AKI: KDIGO (SCr) Persistent AKI w/o recovery: AKI >48 hr, no recovery pre-discharge Persistent AKI w/recovery: AKI >48 hr, with recovery pre-discharge Rapid reversal: Resolved ≤48 hr to no AKI per KDIGO	Mortality, in-hospital	Persistent AKI w/o recovery	3918/14,122 (28%)	**adjOR 26.75 (25.15, 28.46)**	Fair Unclear how they accounted for multiple admissions (if there were any) Definition of recovery vague (only in-hospital data, apparently dependent on LOS) 1-year analyses omit in-hospital events (eg, death) Analyses of mild and severe AKI trajectories not extracted
			Persistent AKI w/recovery	376/8590 (4.4%)	**adjOR 3.33 (2.97, 3.73)**	
			Rapid reversal	680/31,500 (2.2%)	**adjOR 1.98 (1.82, 2.16)**	
			No AKI	2825/301,466 (0.9%)	[REFERENCE for all]	
			ICU admission: Persistent AKI w/o recovery	3488/8573 (41%)	**adjOR 12.50 (11.59, 13.47)**	
			Persistent AKI w/recovery	346/5860 (5.9%)	**adjOR 1.17 (1.03, 1.32)**	
			Rapid reversal	585/13,278 (4.4%)	adjOR 1.01 (0.91, 1.11)	
			No AKI	2070/51,058 (4.1%)	[REFERENCE for all]	
			No ICU admission: Persistent AKI w/o recovery	430/5549 (7.7%)	**adjOR 19.19 (16.78, 21.95)**	
			Persistent AKI w/recovery	30/2730 (1.1%)	**adjOR 3.40 (2.53, 4.59)**	
			Rapid reversal	95/18,222 (0.5%)	**adjOR 1.80 (1.50, 2.17)**	
			No AKI	755/250,408 (0.3%)	[REFERENCE for all]	
		Mortality, 1-year (omitting in-hospital death)	Persistent AKI w/o recovery	1907/10,204 (19%)	**adjHR 5.63 (5.40, 5.86)**	
			Persistent AKI w/recovery	1720/8214 (21%)	**adjHR 1.90 (1.80, 2.03)**	
			Rapid reversal	4072/30,820 (13%)	**adjHR 1.60 (1.50, 1.67)**	
			No AKI	19,411/298,641 (6.5%)	[REFERENCE for all]	
			ICU admission: Persistent AKI w/o recovery	1062/5085 (21%)	**adjHR 5.10 (4.86, 5.35)**	
			Persistent AKI w/recovery	1135/5514 (21%)	**adjHR 1.50 (1.39, 1.63)**	
			Rapid reversal	1888/12,693 (15%)	**adjHR 1.30 (1.22, 1.38)**	
			No AKI	5022/48988 (10%)	[REFERENCE for all]	
			No ICU admission: Persistent AKI w/o recovery	845/5119 (17%)	**adjHR 4.70 (4.35, 4.98)**	
			Persistent AKI w/recovery	585/2700 (22%)	**adjHR 3.60 (3.23, 4.04)**	
			Rapid reversal	2184/18,127 (12%)	**adjHR 1.90 (1.77, 2.02)**	
			No AKI	14,389/249,653 (5.8%)	[REFERENCE for all]	

Study / PMID / Country Population / N / Age / Years	AKI Definition	Outcome	Population	Events/N (%)	Effect Size (CI)	Analysis Quality
Liu 2019[6] 30482577 US In-hospital (survived to hospital discharge) N=38,659 (AKI) 69 yr 2006-14	AKI: KDIGO (SCr) Recurrent AKI: AKI episode w/rehospitalization after discharge from hospital (not explicitly stated that patients had recovered from index AKI) Non-recurrent AKI: No recurrence	Mortality, duration not stated	Recurrent AKI	NR/11,048 (NR%)	**adjHR 1.66 (1.56, 1.77)**	Poor Unclear how they accounted for multiple admissions (if there were any) Unclear definition of recurrent AKI Incomplete reporting of analysis of interest
			Non-recurrent AKI	NR/27,611 (NR%)	[REFERENCE]	
Warnock 2015[7] 26571483 US In-patient N= 20,591 (AKI) 57 yr 2009-13	AKI: KDIGO (SCr) "Transient" AKI: Max SCr before min SCr in-hospital (downward trajectory), no time component reported "Hospital-associated" AKI: Max SCr after min SCr in-hospital (upward trajectory), no time component reported Recurrent ("2HA") AKI: 2nd AKI after recovery, in-hospital	Mortality, in-patient (28 d max)	Recurrent AKI	267/2243 (12%)	adjHR 1.02 (0.84, 1.22)	Poor Unclear definitions of categories
			"Hospital-associated" (upward trajectory)	1121/6247 (18%)	**adjHR 2.12 (1.88, 2.39)**	
			"Transient" (downward trajectory)	606/12,101 (5.1%)	adjHR 1.09 (0.95, 1.24)	
			No AKI	506/29,989 (1.7%)	[REFERENCE for all]	
Korenkevych 2016[8] 26181482 US In-hospital, post-surgical (any, ≥1 d) N=17,990 Median ~56 yr 2000-10	AKI: KDIGO (SCr) Complete renal recovery: SCr return to <50% above baseline (no timeframe reported) Partial renal recovery: SCr remains >50% above baseline but no RRT No renal recovery: "Implies" need for RRT at hospital discharge	Mortality, 90 d	Complete recovery:			Good Although, definition of trajectories unclear without time frame
			Stage 1 AKI	481/9441 (5.1%)	**adjOR 1.48 (1.30, 1.68)**	
			Stage 2 AKI	214/2613 (8.2%)	**adjOR 2.16 (1.81, 2.56)**	
			Stage 3 AKI	199/1193 (17%)	**adjOR 5.73 (4.74, 6.93)**	
			Partial recovery:			
			Stage 1 AKI	238/2521 (9.4%)	**adjOR 2.73 (2.31, 3.23)**	
			Stage 2 AKI	180/717 (25%)	**adjOR 8.81 (7.17, 10.8)**	
			Stage 3 AKI	283/792 (36%)	**adjOR 16.4 (13.6, 19.7)**	
			No recovery	235/713 (33%)	**adjOR 14.7 (12.1, 17.9)**	
			No AKI	615/28,309 (2.2%)	[REFERENCE for all]	

Study / PMID / Country Population / N / Age / Years	AKI Definition	Outcome	Population	Events/N (%)	Effect Size (CI)	Analysis Quality
Peerapornratana 2023[9] 36848723 US ICU admission N=17,605 (AKI) 60 yr 2000-08	AKI: KDIGO (SCr) Recovery from AKI: SCr return to ≤150% of baseline, no RRT Non-recovery AKI: SCr remain >150% of baseline or RRT w/in 48 hr of discharge	Mortality, 15 yr	AKI non-recovery	1949/3372 (58%)	**adjHR 1.84 (1.73, 1.94)**	Fair Unclear timeframe in definitions (based on LOS)
			AKI recovery	6435/14,233 (45%)	**adjHR 1.37 (1.31, 1.43)**	
			No AKI	3676/12,121 (30%)	[REFERENCE for all]	
Heung 2016[10] 26690912 US In-hospital N=16,018 (AKI, excluding unknown recovery) 63 yr 2011	AKI: KDIGO (SCr) Fast recovery: Return to <0.3 mg/dl above baseline at ≤2 d Intermediate recovery: 3-10 d Slow or no recovery: Elevated SCr on day 10	CKD (≥G3), 1 yr	Slow or no AKI recovery	997/1874 (53%) overall	NR	Fair Incomplete data for analyzed AKI stage subgroups Category of unknown (no f/up SCr measure available) omitted here Adjusted HR's are death-censored
			Stage 1 AKI	NR	**adjRR 2.65 (2.51, 2.80)** **adjHR 3.25 (3.02, 3.50)**	
			Stage 2 AKI	NR	**adjRR 3.31 (2.85, 3.84)** **adjHR 3.81 (3.09, 4.68)**	
			Stage 3 AKI	NR	**adjRR 3.59 (3.27, 3.94)** **adjHR 2.97 (2.55, 3.46)**	
			Persistent ("Intermediate recovery")	823/2072 (40%) overall	NR	
			Stage 1 AKI	NR	**adjRR 2.00 (1.88, 2.12)** **adjHR 2.32 (2.16, 2.49)**	
			Stage 2 AKI	NR	**adjRR 1.91 (1.49, 2.45)** **adjHR 1.66 (1.24, 2.23)**	
			Stage 3 AKI	NR	**adjRR 2.20 (1.91, 2.53)** **adjHR 1.61 (1.31, 1.97)**	
			Fast recovery	3341/12,072 (28%) overall		
			Stage 1 AKI	NR	**adjRR 1.43 (1.39, 1.48)** **adjHR 1.49 (1.43, 1.54)**	
			Stage 2 AKI	NR	**adjRR 1.80 (1.46, 2.23)** **adjHR 1.32 (1.01, 1.72)**	
			Stage 3 AKI	NR	**adjRR 1.96 (1.64, 2.34)** **adjHR 1.52 (1.23, 1.89)**	
			No AKI	13,625/87,715 (16%) overall	[REFERENCE for all]	
Ponce 2020[11] 32495022 Brazil ICU admission N=5428 (AKI) 63 yr 2011-18	AKI: KDIGO (SCr) Recurrent AKI: Not defined	Mortality, in-hospital (possibly 30 d)	Recurrent AKI	NR/481 (NR%)	**adjHR 1.54 (1.12, 2.93)**	Poor Terms not defined. Primary outcome defined as in-hospital, but Table reports 30-d mortality. Incomplete data for analysis of interest
			Non-recurrent AKI (undefined comparator)	NR/4947 (NR%) (34% for all AKI)	[REFERENCE]	

Study / PMID / Country Population / N / Age / Years	AKI Definition	Outcome	Population	Events/N (%)	Effect Size (CI)	Analysis Quality
Truche 2018[12] 30560526 France ICU admission N=5242 (AKI) 70 yr 1996-2015	AKI: KDIGO (SCr) AKI recovery: Decrease by at least 1 AKI stage compared to previous day AKI recovery (alternative): Full recovery of AKI Transient AKI: Recovery (1 stage) w/in 3 d) Persistent AKI: >3 d	Mortality, 28-d ICU	AKI recovery (stage decrease)	431/3085 (14%)	**adjHR 0.54 (0.46, 0.63)**	Fair Incomplete reporting of data for analyses of interest
			No AKI recovery	876/2157 (41%)	[REFERENCE]	
			AKI recovery (full)	NR/2184 (NR%)	**adjHR 0.55 (0.47, 0.66)**	
			No AKI recovery	NR/3058 (NR%)	[REFERENCE]	
			Transient AKI (≤3 d)	NR/NR (NR%)	**adjHR 0.80 (0.67, 0.95)**	
			Persistent AKI (>3 d)	NR/NR (NR%) N=3584 total	[REFERENCE]	
Walker 2021[13] 33623694 UK General (in- and outpatient) N=3524 (AKI) 69 yr 2009	AKI: KDIGO (SCr) Recurrent AKI: AKI w/in previous 6 yr	Mortality, median 3.9 yr	Recurrent AKI	725/880 (82%)	**adjHR 1.25 (1.14, 1.37)**	Good Note: People with recurrent AKI had a mean of 2.9 AKI episodes. Very high death rate
			No recurrent AKI	1776/2644 (67%)	[REFERENCE]	
Pannu 2013[14] 23124779 Canada In-hospital (non-AKI: survive >90 post-admission) N=3231 (AKI) 62 yr 2002-2007	AKI: KDIGO *Stage 2* (SCr) AKI recovery: SCr to w/in 25% of baseline (and no RRT), 30-150 d post-admission (unclear) No recovery: Not AKI recovery	Mortality, >90 d	No AKI recovery	347/984 (35%)	**adjHR 1.26 (1.10, 1.43)**	Moderate Timeframe for AKI recovery unclear (ERT omitted 880 with missing data, not classified as recovered or not)
			AKI recovery	720/2247 (32%)	[REFERENCE]	
			Alternative AKI recovery definitions: AKI recovery to <5% from baseline	519/1554 (33%)	[REFERENCE]	
			5-15% from baseline	118/404 (29%)	adjHR 0.98 (0.80, 1.21)	
			15-25% from baseline	83/289 (29%)	adjHR 0.85 (0.67, 1.07)	
			25-35% from baseline	63/203 (31%)	adjHR 0.95 (0.73, 1.24)	
			35-45% from baseline	60/163 (37%)	adjHR 1.18 (0.90, 1.55)	
			45-55% from baseline	33/102 (32%)	adjHR 1.12 (0.79, 1.60)	
			>55% from baseline	176/473 (37%)	**adjHR 1.40 (1.17, 1.67)**	
		ESKD/SCr doubling, >90 d	No AKI recovery	267/984 (27%)	**adjHR 4.13 (3.38, 5.04)**	
			AKI recovery	168/2247 (7.5%)	[REFERENCE]	
			Alternative AKI recovery definitions: AKI recovery to <5% from baseline	110/1554 (7.1%)	[REFERENCE]	
			5-15% from baseline	33/404 (8.2%)	adjHR 1.25 (0.85, 1.86)	
			15-25% from baseline	25/289 (8.7%)	adjHR 1.26 (0.81, 1.94)	
			25-35% from baseline	31/203 (15%)	**adjHR 2.22 (1.48, 3.32)**	
			35-45% from baseline	27/163 (17%)	**adjHR 2.54 (1.66, 3.90)**	
			45-55% from baseline	17/102 (17%)	**adjHR 2.43 (1.44, 4.08)**	
			>55% from baseline	181/473 (38%)	**adjHR 7.70 (6.01, 9.86)**	

Study / PMID / Country Population / N / Age / Years	AKI Definition	Outcome	Population	Events/N (%)	Effect Size (CI)	Analysis Quality
Meersch 2024[15] 38285051 Multinational (30 countries) Major surgery w/ICU stay (survived 90 days) N=1492 (AKI) ~61 yr 2020-21	AKI: KDIGO (SCr or UO) Transient AKI: <48 hr Persistent AKI: >48 hr, to 7 d (implied)	AKD (eGFR <60 ml/min/1.73 m2)	**Transient AKI**	NR/1100 (NR%)	NR	Poor Excludes those who died w/in 90 days Incomplete reporting for analyses of interest (ERT omitted early AKI analyses as the comparator was unclear)
			AKI Stage 1	NR/848 (NR%)	**adjOR 2.09 (1.69, 2.56)**	
			AKI Stage 2	NR/239 (NR%)	**adjOR 2.62 (1.84, 3.65)**	
			AKI Stage 3	NR/13 (NR%)	adjOR 2.17 (0.34, 8.12)	
			Persistent AKI	NR/392 (38.3%)	NR	
			AKI Stage 1	NR/168 (NR%)	**adjOR 8.13 (5.89, 11.2)**	
			AKI Stage 2	NR/129 (NR%)	**adjOR 6.85 (4.71, 9.85)**	
			AKI Stage 3	NR/95 (NR%)	**adjOR 6.97 (4.52, 10.6)**	
			No AKI	NR/8018 (NR%)	[REFERENCE for all]	
Ruth 2022[16] 35145645 Multinational Pediatric ICU N=1262 (AKI) 6 yr (3 mo – 25 yr) [IQR 1.3-13.6] Dates not reported	AKI: KDIGO (SCr or UO) Early AKI: AKI w/in 48 hr in ICU Late AKI: AKI starting >48 hr in ICU Transient AKI: Resolved w/in 48 hr Persistent AKI: Lasted >48 hr or died Recurrent AKI: 2 distinct AKI events (recovered [<Stage 1] with 2nd ≥48 hr post-recovery)	MAKE, 28 d	Recurrent AKI	29/65 (45%)	**adjOR 3.82 (2.07, 7.06)**	Good
			Late & Persistent AKI	11/34 (32%)	**adjOR 2.60 (1.16, 5.83)**	
			Late & Transient AKI	23/134 (17%)	adjOR 0.78 (0.46, 1.34)	
			Early & Persistent AKI	211/325 (65%)	**adjOR 6.40 (4.79, 8.54)**	
			Early & Transient AKI	158/704 (22%)	[REFERENCE for all]	
Quiroga 2022[17] 33342021 Spain Hospitalized (AKI diagnosis), survived hospitalization N=1190 (AKI) 77 yr 2013-14	AKI: KDIGO (SCr) AKI recovery: SCr <1.2x baseline by 7 d Persistent AKI (no recovery): SCr ≥1.2x baseline at 7 d	Mortality, 30 d	Persistent AKI	21/321 (6.5%)	**adjOR 2.6 (1.2, 5.4)**	Poor Exclude in-hospital deaths
			AKI recovery	25/869 (2.8%)	[REFERENCE]	
Altawalbeh 2023[18] 36731422 Jordan Hospitalized with AKI diagnosis N=1105 (AKI) 70 yr 2010-19	AKI: KDIGO (SCr) [Sample selected based on ICD codes] Recurrent AKI: Within 5 years of index AKI (not explicitly defined)	Mortality, 5 yr	Recurrent AKI	72/220 (33%)	**adjHR 1.71 (1.24, 2.35)**	Moderate Incomplete definition of recurrent AKI
			No recurrent AKI	170/885 (19%)	[REFERENCE]	

Study / PMID / Country Population / N / Age / Years	AKI Definition	Outcome	Population	Events/N (%)	Effect Size (CI)	Analysis Quality
Rutledge 2024[19] 38329754 US, Canada, Australia, India Neonatal ICU, receiving IV fluids N=605 (AKI) <14 days of age 2014	AKI: KDIGO (SCr or UO) Recurrent AKI: AKI after return of SCr to baseline value (no UO criteria applied). Minimum time between episodes not reported but median of 18 days of life (vs. 4 days of life for 1[st] event)	Mortality, in-hospital	Recurrent AKI	16/133 (12%)	adjHR 1.4 (0.6, 3.0)	Good
			No recurrent AKI	43/472 (9.1%)	[REFERENCE]	
		Hospital discharge (LOS)	Recurrent AKI	Median (IQR) 60 d (25-109)	**adjHR 0.7 (0.6, 0.9)** i.e., longer LOS	
			No recurrent AKI	18 d (9-45)	[REFERENCE]	
Vincent 202420 37932405 US Neonatal ICU (admit ≤48 hr of life) N=164 (AKI) <48 hr of life 2020-21	AKI: KDIGO (SCr) Recurrent AKI: AKI after return of SCr to baseline value. Minimum time between episodes not reported	Mortality, in-hospital	Recurrent AKI	7/64 (11%)	**adjOR 3.85 (1.1, 13.3)**	Good
			No recurrent AKI	14/100 (14%)	[REFERENCE]	
		Hospital LOS	Recurrent AKI	Median (IQR) 98 d (55, 127)	**adj Median Difference (IQR) −25.9 d [−26.3, −32.6]**	Authors suggest that increased risk of death associated with recurrent AKI may be due to survivor bias (only survivors of first AKI can have recurrent AKI)
			No recurrent AKI	44 d (17, 82)	[REFERENCE]	
Adegboyega 2022[21] 34593979 US Gestational age ≤28 wk, Neonatal ICU, survived >48 hr N=116 (AKI) Newborn 2014-18	AKI: KDIGO (SCr), baseline is trough after typical SCr decrease after premature birth Recurrent AKI: AKI after SCr decreases ≥0.3 mg/dl Minimum time between episodes not reported Complete recurrent AKI: AKI after ≥75% return to SCr baseline Incomplete recurrent AKI: AKI after 25-74% return to SCr baseline	Mortality, in-hospital	Recurrent AKI	9/49 (19%)	**adjOR 4.55 (1.12, 18.5)** *crude OR 1.04 (0.39, 2.71)*	Moderate Although small with multiple comparisons and incomplete reporting of complete and incomplete recurrent AKI data No explanation for large change in OR between crude and adjusted analyses
			Complete recurrent AKI	NR/28	adjOR 4.71 (0.99, 22.2) *crude OR 1.20 (0.41, 3.68)*	
			Incomplete recurrent AKI	NR/21	adjOR 4.28 (0.70, 26.1) *crude OR 0.79 (0.20, 3.15)*	
			No recurrent AKI	12/67 (18%)	[REFERENCE for all]	
		Hospital LOS	Recurrent AKI	Median (IQR) 117 d (81–139)	**adjMD 36 d (24.9, 47.1)**	
			Complete recurrent AKI	NR	**adjMD 31 d (18.22, 43.8)**	
			Incomplete recurrent AKI	NR	**adjMD 46 d (31.6, 60.4)**	
			No recurrent AKI	85 d (58–106)	[REFERENCE for all]	
Wang H 2022[22] 35854309 Scotland General population (inpatient and outpatient SCr values) N=29,330 (mortality), 14,486 (CKD) 75 yr (median) 2010-18	AKI: KDIGO (SCr), baseline is lowest SCr in prior 7 days or median of days 8-365 prior "Early recovery": AKI, recovered w/in 7 d. SCr <1.2x baseline "Delayed recovery: AKD within 8-90 d	Mortality, 1 year	AKD (delayed)	NR/NR	**adjHR 1.20 (1.13, 1.26)** [higher death rate]	Fair Incomplete data reporting Unclear why ~1/2 or more of total sample are not included
			AKI (early)	NR/NR	[REFERENCE]	
		CKD, de novo, 1 year	AKD (delayed)	NR/NR	**adjHR 2.21 (1.91, 2.57)** [higher CKD rate]	
			AKI (early)	NR/NR	[REFERENCE]	

Study / PMID / Country Population / N / Age / Years	AKI Definition	Outcome	Population	Events/N (%)	Effect Size (CI)	Analysis Quality
Gómez 2025[23] 39966171 US Critically ill (ICU) N=65,119 (Stage 2/3 AKI) 68 yr (median) 2010-18	AKI: KDIGO (SCr and UO), hierarchical selection of baseline, per KDIGO Persistent severe AKI: AKI Stage 2 or 3, and Stage 3 ≥72 hr, or dialysis, or death after Stage 3 (If Stage 2, must progress to Stage 3 w/in 48 hr prior to above criteria) No persistent severe AKI: AKI Stage 2 or 3 who did not meet criteria	Mortality, 90 d	Persistent severe AKI	4294/8059 (53%)	**adjHR 1.49 (1.42, 1.56)** [higher death rate]	Good Numerous other comparisons reported in appendix
			Severe AKI, not persistent	14,258/57,060 (25%)	[REFERENCE]	
		Mortality, in hospital	Persistent severe AKI	3409/8059 (42%)	**adjHR 2.21 (2.06, 2.38)** [higher death rate]	
			Severe AKI, not persistent	8070/57,060 (14%)	[REFERENCE]	
		Renal recovery (SCr to ≤150% baseline w/o RRT)	Persistent severe AKI	NR	**adjHR 0.14 (0.13, 0.15)** [lower recovery rate]	
			Severe AKI, not persistent	NR	[REFERENCE]	

adjIRR = adjusted incident rate ratio

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Table S15. Outcomes in children and adults with AKD

Study / PMID / Country Population / N / Age / Years	AKD Definition	Outcome	Population	Events/N (%)	Effect Size (CI)	Analysis Quality
Su 2023[24] 36531983 **Systematic Review** N=67,173 (k=21 studies) N<1000 (k=15) N<100 (k=5) ≥18 y Known previous AKI (k=17) No previous known AKI (k=3) Unspecified (k=1)	KDIGO (15 studies) Other (6 studies)	Mortality, all-cause 1 mo – 6 yr k=21	AKD (KDIGO definition)	NR	**Summary OR 4.02 (3.18, 5.08)** Range 1.11 – 14.4 (I^2 91%)	Adequate SR However, meta-analyses combined disparate comparisons Certainty of evidence = Moderate
			No AKD (variably w/ or w/o prior AKI)	NR		
			AKD (any definition)	NR/NR (26.5%)	**Summary OR 3.62 (2.64, 4.95)** Range 1.11 – 14.4 (I^2 99%) **Summary HR 1.39 (1.25, 1.55)** Range 1.12 – 14.4 (I^2 99%)	
			No AKD (variably w/ or w/o prior AKI)	NR/NR (7.8%)		
			Previous CKD<50% (k=14)	NR	Summary OR 3.61 (2.51, 5.19)	
			Previous CKD>50% (k=4)	NR	Summary OR 4.23 (2.06, 8.70) P 0.70 btw (ERT estimate)	
			Surgery (k=4)	NR	*Summary OR 6.87 (3.23, 14.59)*	
			No surgery (k=17)	NR	*Summary OR 3.16 (2.24, 4.61)* *P 0.069 btw (ERT estimate)*	
			ICU admission (k=8)	NR	Summary OR 4.75 (2.80, 8.07)	
			No ICU admission (k=13)	NR	Summary OR 3.11 (2.10, 4.61) P 0.21 btw (ERT estimate)	
			Known previous AKI (k=8)	NR	**Summary OR 3.51 (2.25, 5.46)**	
			No known previous AKI (k=18)	NR	**Summary OR 3.64 (2.69, 4.93)** P 0.89 btw (ERT estimate)	
		ESKD 7.7 yr, mean k=12	AKD	NR/NR (1.3%)	**Summary OR 6.58 (3.75, 11.56)** Range 1.70 – 22.3 (I^2 95%)	
			No AKD (variably w/ or w/o prior AKI)	NR/NR (0.14%)		
			Previous CKD<50% (k=8)	NR	Summary OR 4.87 (2.51, 9.43)	
			Previous CKD>50% (k=3)	NR	Summary OR 9.31 (2.83, 30.64) P 0.35 btw (ERT estimate)	
			Surgery (k=2)	NR	**Summary OR 11.68 (7.53, 18.12)**	
			No surgery (k=10)	NR	**Summary OR 4.51 (4.09, 4.97)** **P <0.001 btw (ERT estimate)**	
		Incident or progressive CKD 7.8 yr, mean	AKD	NR/NR (37.2%)	**Summary OR 4.22 (2.79, 6.39)** Range 1.70 – 22.3 (I^2 97%)	
			No AKD (variably w/ or w/o prior AKI)	NR/NR (7.5%)		
Sawhney 2022[25] 35398477 Canada (Alberta), Denmark, UK General (post-hospitalization) N~101,529 (AKD) ≥18 yr 2010-14 [Sawhney 2024 similar, not extracted]	KDIGO AKD = SCr rise 8-90 days post prior SCr AKI-7 = SCr rise <7 days post prior SCr	Mortality, 1 yr	AKD	~38,100/~101,529	MA 39.1% (34.5, 43.8) (calculated by ERT)	Moderate Article makes qualitative conclusions based on very difficult to parse granular data that is not summarized.
			AKI-7	~22,875/~63,628	MA 37.0% (32.6, 41.6) (calculated by ERT)	

Study / PMID / Country Population / N / Age / Years	AKD Definition	Outcome	Population	Events/N (%)	Effect Size (CI)	Analysis Quality
James 2019[26] 30951162 Canada (Alberta) General (in- and outpatient) N=48,794 (AKD) Mean 52 yr 2008-2014 Included in Su 2023 SR (although some results differ)	AKD without AKI: eGFR <60 mL/min/1.73 m2 and previously ≥60 mL/min/1.73 m2 or ND (albuminuria absent or ND) AKI: KDIGO (SCr)	Mortality, all-cause, 6 yr (median f/up)	AKD without known AKI	10,940/42,487 (25.8%)	**adjHR 1.42 (1.39, 1.45)**	Good
			AKI	7964/15,777 (50.5%)	**adjHR 3.23 (3.16, 3.31)**	
			No kidney disease	67,655/921,116 (7.3%)	[REFERENCE for all]	
			CKD with AKD w/o AKI	2977/6307 (47.2%)	**adjHR 1.92 (1.85, 2.00)**	
			CKD with AKI	3585/5015 (71.5%)	**adjHR 3.30 (3.19, 3.41)**	
			CKD without AKI/AKD	34,816/118,397 (29.4%)	**adjHR 1.37 (1.35, 1.39)**	
		CKD, incident, 6 yr (median f/up)	AKD without known AKI	15,884/42,487 (37.4%)	**adjHR 3.17 (3.1, 3.23)**	
			AKI	3066/15,777 (19.4%)	**adjHR 1.32 (1.26, 1.38)**	
			No kidney disease	68,355/921,116 (7.4%)	[REFERENCE for all]	
		CKD progression, 6 yr (median f/up)	CKD with AKD w/o AKI	3123/6307 (49.5%)	**adjHR 1.38 (1.33, 1.44)**	
			CKD with AKI	2064/5015 (41.2%)	**adjHR 1.38 (1.33, 1.44)**	
			CKD without AKI/AKD	34,816/118,397 (29.4%)	[REFERENCE for all]	
		ESKD, 6 yr (median f/up)	AKD without known AKI	236/42,487 (0.6%)	**adjHR 8.56 (7.32, 10.01)**	
			AKI	135/15,777 (0.9%)	**adjHR 9.89 (8.12, 12.03)**	
			No kidney disease	653/921,116 (0.1%)	[REFERENCE for all]	
			CKD with AKD w/o AKI	256/6307 (4.1%)	**adjHR 65.95 (55.91, 77.80)**	
			CKD with AKI	176/5015 (3.5%)	**adjHR 52.20 (43.20, 63.08)**	
			CKD without AKI/AKD	1590/118,397 (1.3%)	**adjHR 21.54 (19.44, 23.87)**	
Jensen 2014 38765592 Denmark General (in- and outpatient) N=110,449 (AKD) Median 72 yr 1990-2018 36% with CKD	AKD: KDIGO (SCr) ≥7 d AKI, rapid reversal: KDIGO (SCr) <48 hr AKI, persistent: KDIGO (SCr) 2-7 d	CKD, incident, 20 yr risk [No previous CKD]	AKD	13,417/65,905 (20.4%)	**adjHR 1.36 (1.30, 1.41)**	Good Subgroup analyses with generally similar patterns (age, sex, AKI stage, location, AKI cause, baseline eGFR)
			AKI, persistent	2667/12,106 (22.0%)	**adjHR 1.13 (1.08, 1.19)**	
			AKI, rapid reversal	3990/22,467 (17.8%)	[REFERENCE for all]	
		ESKD, incident, 20 yr risk [36% w/prior CKD]	AKD	4183/109,133 (3.8%)	**adjHR 1.56 (1.44, 1.69)**	
			AKI, persistent	899/22,500 (4.0%)	**adjHR 1.19 (1.08, 1.30)**	
			AKI, rapid reversal	915/36,398 (2.5%)	[REFERENCE for all]	

Study / PMID / Country Population / N / Age / Years	AKD Definition	Outcome	Population	Events/N (%)	Effect Size (CI)	Analysis Quality
Xu L 2023[27] 37915910 China In-patient AKI N=10,203 (AKD) 61 yr 2012-18	AKD: KDIGO (SCr, UO) AKD w/AKI: Stage ≥1 for >7 days AKD w/o AKD: "SCr increased slowly but lasted >7 days (subacute AKD w/o meeting AKI criteria)" AKD stage not defined	Mortality, 14 mo median	AKD w/AKI	NR/5623 (NR%)	**adjHR 4.51 (4.32, 4.71)**	Good, but some poor reporting and incomplete analyses Unclear whether the mortality HRs are for hospital or long-term mortality; survival curves are for long-term
			AKD w/o AKI	NR/4580 (NR%)	**adjHR 2.25 (2.13, 2.39)**	
			AKI recovery	NR/5895 (NR%)	**adjHR 1.18 (1.09, 1.26)**	
			No kidney disease	NR/54,943 (NR%)	[REFERENCE for all]	
			Stage 3 AKD	NR/1440 (NR%)	**adjHR 16.6 (15.8, 17.5)**	
			Stage 2 AKD	NR/2506 (NR%)	**adjHR 6.54 (6.20, 6.91)**	
			Stage 1 AKD	NR/6538 (NR%)	**adjHR 2.77 (2.64, 2.91)**	
			No kidney disease	NR/54,943 (NR%)	[REFERENCE for all]	
		CKD, incident, 14 mo median	AKD w/AKI	589/5623 (10.5%)	**adjHR 2.49 (2.37, 2.62)**	
			AKD w/o AKI	410/4580 (9.0%)	**adjHR 2.17 (2.05, 2.30)**	
			AKI recovery	386/5895 (6.6%)	**adjHR 1.64 (1.54, 1.73)**	
			No kidney disease	1788/54,943 (3.3%)	[REFERENCE for all]	
			Stage 3 AKD	NR/1440 (NR%)	**adjHR 8.60 (7.93, 9.34)**	
			Stage 2 AKD	NR/2506 (NR%)	**adjHR 4.73 (4.44, 5.05)**	
			Stage 1 AKD	NR/6538 (NR%)	**adjHR 2.33 (2.21, 2.45)**	
			No kidney disease	1788/54,943 (3.3%)	[REFERENCE for all]	
		ESKD, incident, 14 mo median	AKD w/AKI	246/5623 (4.4%)	**RR (calc) 2.46 (1.96, 3.08)**	
			AKD w/o AKI	116/4580 (2.5%)	**RR (calc) 1.42 (1.10, 1.85)**	
			AKI recovery	105/5895 (1.8%)	[REFERENCE for all]	
			No kidney disease	444/54,943 (0.8%)	.	
		Length of stay, hospital	AKD w/AKI	20±11 days	**MD (calc) 6 (5.6, 6.4) days**	
			AKD w/o AKI	18±10 days	**MD (calc) 4 (3.6, 4.4) days**	
			AKI recovery	14±9 days	[REFERENCE for all]	
			No kidney disease	13±7 days	.	
See 2021[28] 33906191 Australia In-patient (eGFR≥60 at baseline and alive 30 days post-discharge) N=1260 (AKD) 55 yr 2012-19 Included in Su 2023 SR	AKD: KDIGO (SCr)	Mortality, median ~2 yr	AKD w/AKI	249/783 (31.8%)	**adjHR 1.52 (1.20, 1.92)**	Fair Exclude those who died pre-30 days post-discharge Some unclear reporting
			AKD w/o AKI	182/477 (38.2%)	**adjHR 1.57 (1.19, 2.07)**	
			AKI alone	418/2596 (16.1%)	**adjHR 1.57 (1.33, 1.84)**	
			No kidney disease	3457/32,262 (10.7%)	[REFERENCE for all, implied]	
		MAKE, median ~2 yr	AKD w/AKI	384/783 (49.0%)	**adjHR 2.51 (2.16, 2.91)**	
			AKD w/o AKI	288/477 (60.4%)	**adjHR 2.26 (1.89, 2.70)**	
			AKI alone	627/2596 (24.2%)	**adjHR 1.52 (1.35, 1.72)**	
			No kidney disease	5250/32,262 (16.3%)	[REFERENCE for all, implied]	
		CKD, incident, median ~2 yr	AKD w/AKI	212/783 (27.1%)	**adjHR 3.18 (2.60, 3.89)**	
			AKD w/o AKI	146/477 (30.6%)	**adjHR 2.69 (2.11, 3.43)**	
			AKI alone	267/2596 (10.3%)	**adjHR 1.51 (1.27, 1.79)**	
			No kidney disease	2296/32,262 (7.1%)	[REFERENCE for all, implied]	
		ESKD, median ~2 yr	AKD w/AKI	4/783 (0.51%)	**adjHR 24.8 (5.93, 104)**	
			AKD w/o AKI	4/477 (0.84%)	**adjHR 12.6 (1.48, 108)**	
			AKI alone	9/2596 (0.35%)	**adjHR 8.22 (1.96, 34.5)**	
			No kidney disease	25/32,262 (0.08%)	[REFERENCE for all, implied]	
Pan 202[29] 37138740 Taiwan ICU admission (≥2 days) N=8975 (AKD) 64 yr 2001-2018	AKD: ADQI 16 (2017) AKI: KDIGO Stage 0: SCr <1.5x Stage 1: SCr 1.5-1.99x Stage 2: SCr 2-2.99x Stage 3: SCr ≥3x	Mortality, 180 d	AKD post-d/c (no AKI in ICU)	227/5178 (4.4%)	**adjOR 2.25 ((1.71, 2.97)**	Fair Some reporting unclarity
			AKD post-ICU AKI	88/3797 (2.3%)	**adjOR 1.34 (1.00, 1.78)**	
			AKI (in ICU), recovered	115/7248 (1.6%)	[REFERENCE for all]	
			Stage 3 AKD	NR (NR%)	**adjOR 1.96 (1.25, 3.06)**	
			Stage 2 AKD	NR (NR%)	adjOR 1.41 (0.92, 2.16)	
			Stage 1 AKD	NR (NR%)	adjOR 0.95 (0.64, 1.42)	
			Stage 0 AKD	NR (NR%)	[REFERENCE for all]	

Study / PMID / Country Population / N / Age / Years	AKD Definition	Outcome	Population	Events/N (%)	Effect Size (CI)	Analysis Quality
Diaz 2024[30] 39049453 Spain General (in- and outpatient) N=42,326 62 yr 2012-16	AKD: Unclear definition AKI: RIFLE	Mortality, 3 mo (after AKI episode)	AKD w/prior CKD	79/12,079 (5.6%)		Poor Unadjusted No statistical analyses AKI patients were much older (74 vs 62) with more comorbidities
			AKD, no prior CKD	994/30,252 (3.3%)		
			AKI (no AKD)	1699/5646 (30.1%)		
		Mortality, mean 4.8 yr	AKD w/prior CKD	3125/12,079 (25.9%)		
			AKD, no prior CKD	3380/30,252 (11.2%)		
			AKI (no AKD)	3423/5646 (60.6%)		
		CKD, incident, mean 4.8 yr	AKD, no prior CKD	14,714/30,252 (43.8%)		
			AKI (no AKD)	430/5646 (12.7%)		
		eGFR decrease ≥50% (w/CKD)	AKD w/prior CKD	5713/12,074 (63.1%)		
Chen 2022[31] 35444219 Taiwan In-patient N=4741 (AKD) 69 yr 2014-2019	AKD: ADQI 16 (2017) Stage 0: SCr <1.5x Stage 1: SCr 1.5-1.99x Stage 2: SCr 2-2.99x Stage 3: SCr ≥3x	MAKE, 1 yr	**Baseline eGFR <60 mL/min/1.73 m2**			Good overall Prolonged RRT for subgroup with higher baseline RRT omitted (mostly 0 events per stage (6/448 [1.3%] for stage 3 AKD)
			Stage 1-3 AKD	NR/1093 (NR%)	**adjOR 3.79 (3.40, 4.58)**	
			Stage 3 AKD	NR/544 (NR%)	**adjOR 7.60 (5.9, 9.8)**	
			Stage 2 AKD	NR/236 (NR%)	**adjOR 2.73 (2.0, 3.7)**	
			Stage 1 AKD	NR/313 (NR%)	**adjOR 1.87 (1.4, 2.5)**	
			Stage 0 AKD	NR/957 (NR%)	[REFERENCE for all]	
			Baseline eGFR ≥60 mL/min/1.73 m2			
			Stage 1-3 AKD	NR/1361 (NR%)	**adjOR 4.63 (3.88, 5.52)**	
			Stage 3 AKD	NR/448 (NR%)	**adjOR 13.8 (10.5, 18.0)**	
			Stage 2 AKD	NR/414 (NR%)	**adjOR 4.43 (3.5, 5.7)**	
			Stage 1 AKD	NR/499 (NR%)	**adjOR 2.01 (1.6, 2.6)**	
			Stage 0 AKD	NR/1330 (NR%)	[REFERENCE for all]	
		RRT, any, 1 yr	**Baseline eGFR <60 mL/min/1.73 m2**			
			Stage 1-3 AKD	401/1093 (36.7%)	**adjOR 3.29 (2.63, 4.11)**	
			Stage 3 AKD	293/544 (53.9%)	**adjOR 6.27 (4.85 ,8.10)**	
			Stage 2 AKD	53/236 (22.5%)	**adjOR 1.87 (1.29, 2.70)**	
			Stage 1 AKD	55/313 (17.6%)	adjOR 1.21 (0.85, 1.71)	
			Stage 0 AKD	160/957 (16.7%)	[REFERENCE for all]	
			Baseline eGFR ≥60 mL/min/1.73 m2			
			Stage 1-3 AKD	128/1361 (9.4%)	**adjOR 10.5 (5.97, 18.4)**	
			Stage 3 AKD	107/448 (23.9%)	**adjOR 31.4 (17.6, 56.1)**	
			Stage 2 AKD	19/414 (4.6%)	**adjOR 5.05 (2.49, 10.30**	
			Stage 1 AKD	2/499 (0.4%)	adjOR 0.40 (0.09, 1.78)	
			Stage 0 AKD	14/1330 (1.1%)	[REFERENCE for all]	
		Prolonged RRT, 1 yr	**Baseline eGFR <60 mL/min/1.73 m2**			
			Stage 1-3 AKD	86/1093 (7.9%)	**adjOR 2.16 (1.45, 3.23)**	
			Stage 3 AKD	72/544 (13.2%)	**adjOR 3.48 (2.29, 5.31)**	
			Stage 2 AKD	6/236 (2.5%)	adjOR 0.82 (0.34, 2.00)	
			Stage 1 AKD	8/313 (2.6%)	adjOR 0.69 (0.35, 1.51)	
			Stage 0 AKD	38/957 (4.0%)	[REFERENCE for all]	

Study / PMID / Country Population / N / Age / Years	AKD Definition	Outcome	Population	Events/N (%)	Effect Size (CI)	Analysis Quality
Xiao 2020[32] 32973230 China In-patient N=1359 (AKD) 55 yr 2015 Included in Su 2023 SR	AKD: KDIGO (SCr) ≥7 d AKI: KDIGO (SCr)	Mortality, 90 d (from hospital admission)	**All patients** AKD	260/1359 (19.1%)	**adjHR 1.98 (1.43, 2.75)**	Poor Incomplete reporting Deaths <7 days post-admission are graphed for patients with AKD (death <7 days also an exclusion criterion)
			AKI	NR/1197 (NR%)	[REFERENCE]	
			Baseline eGFR <60 mL/min/1.73 m2 AKD	NR/150 (~23%), per survival curve	adjHR NR (survival curve implies larger HR than ≥60 subgroup)	
			AKI	NR/158 (~4%)	[REFERENCE]	
			Baseline eGFR ≥60 mL/min/1.73 m2 AKD	NR/1209 (~18%)	**adjHR 1.94 (1.44, 2.62)**	
			AKI	NR/1039 (~6%), per survival curve	[REFERENCE]	
Deng 2022[33] 35676902 China Pediatric In-patient, AKI N=419 (AKD) Median ~8 (1 mo – 18 yr) 2015-20	AKD: KDIGO (SCr) ≥6 d AKD stages based on AKI stage on day 7 (or nearest) AKI: KDIGO (SCr)	MAKE, 30 d	AKD Stage 1	23/186 (12%)	adjOR 1.06 (0.61, 1.86)	Good AKD likely 7-90 d, despite text “between 6 and 90 days”
			AKD Stage 2-3	158/233 (68%) [107 stage 2, 126 stage 3]	**adjOR 12.2 (7.38, 20.1)**	
			AKI (Stage 0 AKD)	68/571 (12%)	[REFERENCE for all]	
		MAKE at 90 d	AKD Stage 1	8/186 (4.3%)	adjOR 0.53 (0.22, 1.26)	
			AKD Stage 2-3	47/233 (20%)	**adjOR 2.49 (1.26, 4.91)**	
			AKI (Stage 0 AKD)	40/571 (7.0%)	[REFERENCE for all]	
Chisavu 2024[34] 38892856 Romania Pediatric In-patient, AKI N=125 (AKD) 9 [IQR 4–14] (2-18 yr) 2014-22	AKD: KDIGO (SCr) ≥7 d AKI: KDIGO (SCr)	Mortality, median 22.6 mo	AKD	NR/125	**adjOR 2.65 (1.38, 5.10)**	Moderate Incomplete reporting
			AKI	NR/611	[REFERENCE]	
		Mortality, in-hospital (excluding death during AKI [w/in 7 d])	AKD	NR/125	**adjOR 2.62 (1.14, 5.99)**	
			AKI	NR/611	[REFERENCE]	
		Mortality, post-discharge	AKD	NR/125	adjOR 2.47 (0.90, 6.81)	
			AKI	NR/611	[REFERENCE]	
		CKD, median 22.6 mo	AKD	NR/125	**adjHR 3.01 (1.18, 7.71)**	
			AKI	NR/611	[REFERENCE]	
Sawhney 2017[35] 27555107 Scotland Hospitalized (all abnormal eGFR and 20% sample of normal) N=923 AKD ~76 (w/AKI, any) 2003-13	AKD: KDIGO (SCr) ≥7 d	Mortality, 1-10 yr	**Baseline eGFR ≥60 mL/min/1.73 m2** AKD	NR/683	**adjHR 1.42 (1.26, 1.59)**	Moderate No direct or indirect comparison with AKI (recovery w/in 7 d) Incomplete reporting on AKD
			No AKI	NR/7741	[REFERENCE]	
			Baseline eGFR 45-59 mL/min/1.73 m2 AKD	NR/133	**adjHR 1.33 (1.09, 1.63**	
			No AKI	NR/3242	[REFERENCE]	
			Baseline eGFR 30-44 mL/min/1.73 m2 AKD	NR/72	adjHR 1.24 (0.96, 1.60)	
			No AKI	NR1337	[REFERENCE]	
			Baseline eGFR <30 mL/min/1.73 m2 AKD	NR/35	adjHR 1.26 (0.84, 1.87)	
			No AKI	NR/425	[REFERENCE]	

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Table S16. Prediction models in adults (all models analyzed together)

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	**Description of Findings**	**Importance of Outcome**
Incident AKI	30 (NR) [1]	Serious limitations	Inconsistent†	Indirect‡	Precise	Undetected	None	Low	Hospitalized patients Summary AUC 0.75 (0.72-0.79)	Critical
	20 (NR) [1]	Serious limitations	Inconsistent†	Indirect‡	Precise	Undetected	None	Low	ICU patients Summary AUC 0.82 (0.78-0.85)	
	37 (NR) [1]	Serious limitations	Inconsistent†	Indirect‡	Precise	Undetected	None	Low	Postoperative patients Summary AUC 0.78 (0.76-0.81)	
	12 (NR) [1]	Serious limitations	Inconsistent†	Indirect‡	Precise	Undetected	None	Low	Post-contrast patients Summary AUC 0.78 (0.72-0.83)	
AKI Stage 3	16 (NR) [1]	Serious limitations	Inconsistent†	Indirect‡	Precise	Undetected	None	Low	All populations Summary AUC 0.84 (0.81-0.87)	Critical
Mortality	11 (NR) [1]	Serious limitations	Inconsistent†	Indirect‡	Precise	Undetected	Sparse	Low	All populations Summary AUC 0.80 (0.76-0.83)	Critical
RRT	26 (NR) [1]	Serious limitations	Inconsistent†	Indirect‡	Precise	Undetected	None	Low	All populations Summary AUC 0.82 (0.79-0.85)	Critical
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Balance of Potential Benefits and Harms: Prediction models, overall, have moderate accuracy § to predict AKI outcomes, including mortality								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = acute kidney injury; AUC = area under curve; CKD = chronic kidney disease; ICU = intensive care unit, NR = not reported, RRT = renal replacement therapy

* All prediction models analyzed together. Analyses from external validation studies included here. Internal validation studies yielded similar, though typically slightly more accurate, estimates.
† I^2 ≥96%. In addition, substantial heterogeneity in scores, populations, and settings.
‡ Analyses do not apply to any specific score.
§ *A priori* defined as AUC 0.76-0.85

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Table S17. Mehta score in adults undergoing cardiac surgery

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI	3 (33110) [1-2,4]	Some limitations	Inconsistent*	Direct	Precise	Undetected	None	Low	Summary AUC 0.72 (0.70–0.74)	Critical
AKI Stage 3	2 (21370) [1-2]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	Summary AUC 0.79 (0.78–0.80)	Critical
Mortality	1 (343) [6]	Serious limitations	NA	Direct	Precise	Undetected	Sparse	Very low	AUC 0.65 (0.59–0.71)	Critical
RRT	3 (2083) [2-3,5]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	Summary AUC 0.78 (0.69–0.87)	Critical
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Balance of Potential Benefits and Harms: The Mehta score has low † to moderate ‡ accuracy to predict AKI outcomes and low accuracy to predict mortality								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = acute kidney injury; RRT = renal replacement therapy; CKD = chronic kidney disease; AUC = area under curve

* I^2 = 73%
† *A priori* defined as AUC 0.51-0.75
‡ *A priori* defined as AUC 0.76-0.85

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Table S18. SPARK score in adults undergoing noncardiac surgery

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI	3 (45,872) [1-3]	No serious	Consistent	Direct	Precise	Undetected	None	High	AUC 0.70 (0.63, 0.76)	Critical
AKI Stage 3	3 (45,872) [1-3]	No serious	Consistent	Direct	Imprecise*	Undetected	None	Moderate	AUC 0.70 (0.48, 0.86)	Critical
Mortality	0								No evidence	Critical
RRT	0								No evidence	Critical
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Balance of Potential Benefits and Harms: The Mehta score has low‡ accuracy to predict AKI outcomes								**Certainty of Overall Evidence:** High		

Abbreviations: AKI = acute kidney injury; RRT = renal replacement therapy; CKD = chronic kidney disease; AUC = area under curve

* Range of AUC 95% CI >0.2 (larger than the ranges for low, moderate, and high accuracy).
‡ *A priori* defined as AUC 0.51-0.75

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Table S19. Prediction models in children (all models analyzed together)

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Incident AKI	4 (NR) [1]	Serious limitations	Inconsistent†	Indirect‡	Precise	Undetected	None	Low	Various populations Summary AUC 0.84 (0.77-0.91)	Critical
AKI Stage 3	0								No evidence	Important
Mortality	0								No evidence	Important
RRT	0								No evidence	Important
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Balance of Potential Benefits and Harms: Prediction models, overall, have moderate accuracy § to predict AKI, without evidence for other outcomes.								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = acute kidney injury; AUC = area under curve; CKD = chronic kidney disease; ICU = intensive care unit, NR = not reported, RRT = renal replacement therapy

* All prediction models analyzed together. Analyses from external validation studies included here. Internal validation studies yielded a similar estimate.
† I^2 = 95%. In addition, substantial heterogeneity in scores, populations, and settings.
‡ Analyses do not apply to any specific score.
§ *A priori* defined as AUC 0.76-0.85

References

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Table S20. Alerts versus no alerts in adults at risk of AKI

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Mortality	13 [1-6] (29979) [RCT]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	RR 0.96 (0.91-1.02) 6 fewer per 1000 (from 14 fewer to 1 more)	Critical
Incident AKI	3 (11758) [2-3,5] [RCT]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	RR 0.90 (0.78–1.04) 21 fewer per 1000 (from 34 fewer to 7 more)	Critical
AKI Stage 3	2 (3938) [2,5] [RCT]	Some limitations	Inconsistent	Direct	Precise	Undetected	None	Very Low	RR 1.04 (0.79–1.36) 4 more per 1000 (from 9 fewer to 16 more)	Important
MAKE	1 (7820) [3] [RCT]	Some limitations	N/A	Direct	Precise	Undetected	Sparse	Low	adjOR 0.83 (0.67–1.03) 16 fewer per 1000 (from 31 fewer to 3 more)	Critical
RRT	13 [1,3-6] (28786) [RCT]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	RR 1.02 (0.92–1.13) 0 more per 1000 (from 6 fewer to 5 more)	Critical
Progression of AKI [NB: not *a priori* outcome]	5 (16574) [1,4] [RCT]	No limitations	Consistent	Direct	Precise	Undetected	None	High	RR 0.95 (0.89–1.02) 9 fewer per 1000 (from 20 fewer to 3 more)	Important
Recovery of kidney function [not *a priori* outcome]	4 (5270) [1,4] [RCT]	Serious limitations	Consistent	Direct	Precise	Undetected	None	Moderate	RR 1.01 (0.97–1.05) 7 more per 1000 (from 19 fewer to 33 more)	Important
Duration of AKI (AKI at discharge)	4 (3335) [1] [RCT]	Serious limitations	Consistent	Direct	Precise	Undetected	None	Very Low	RR 1.39 (0.94–2.05) 112 more per 1000 (from 17 fewer to 301 more)	Important
ICU admission, unplanned	0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	No data	Important
Balance of Potential Benefits and Harms: There are no differences in critical outcomes with use of electronic alerts via clinical decision support systems. No suggestion of worse outcomes.								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = acute kidney injury; ICU = intensive care unit; MAKE = major adverse kidney event; N/A = not applicable; OR = odds ration; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy.

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Table S21. Furosemide stress test in adults who are critically ill or undergoing surgery

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Mortality	3 (290) [1,7]	Not serious	Inconsistent	Direct	Sn: Imprecise* Sp: Imprecise*	Undetected	None	Low	Sn 0.38 (0.18-0.63) Sp 0.64 (0.41-0.82)	Critical
Incident AKI	3 (4760) [1-3]	Not serious	Sn: Inconsistent Sp: Consistent	Direct	Sn: Imprecise* Sp: Precise	Undetected	None	Low	Sn 0.68 (0.38-0.88) Sp 0.82 (0.63-0.92)	Critical
AKI Stage 3	12 (5335) [1,2-4,7]	Not serious	Sn: Consistent Sp: Some inconsistency	Direct	Sn: Precise Sp: Precise	Detected †	None	Low	Sn 0.74 (0.61-0.83) Sp 0.80 (0.62-0.91)	Critical
RRT	8 (1303) [1,5-6]	Not serious	Sn: Consistent Sp: Some inconsistency	Direct	Sn: Precise Sp: Precise	Undetected	None	Moderate	Sn 0.75 (0.51-0.90) Sp 0.72 (0.57-0.83)	Critical
Duration of AKI	0								none	Important
CKD G3-G5	0								none	Important
CKD G5D	0								none	Important
Balance of Potential Benefits and Harms: FST has low ‡ sensitivity and moderate § specificity for incident AKI and Stage 3 AKI, with low sensitivity and specificity for RRT. FST is inaccurate to predict death.								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = acute kidney injury; CKD = chronic kidney disease; G = grade; RRT = renal replacement therapy; Sn = sensitivity; Sp = specificity

* Lower bound of 95% confidence interval <50%
† Per Deeks asymmetry test, P <0.043, such that larger studies tended to have smaller diagnostic odds ratios.
‡ A priori defined as 51-75% sensitivity or specificity
§ A priori defined as 76-89% sensitivity or specificity

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Table S22. Furosemide stress test in children in the ICU or undergoing surgery

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Mortality	2 (92) [4-5]	Not serious	Sn: Consistent Sp: Some inconsistency	Direct	Sn: Imprecise* Sp: Imprecise*	Undetected	None	Low	Sn 0.62 (0.38-0.82) Sp 0.54 (0.43-0.65)	Critical
Incident AKI	3 (1134) [1-3]	Not serious	Consistent	Direct	Sn: Precise Sp: Precise	Undetected	None	High	Sn 0.66 (0.54-0.77) Sp 0.67 (0.54-0.78)	Critical
AKI Stage 3	5 (1088) [4-8]	Not serious	Consistent	Direct	Sn: Precise Sp: Precise	Undetected	None	High	Sn 0.70 (0.62-0.77) Sp 0.80 (0.51-0.94)	Critical
RRT	1 (41) [5]	Not serious	N/A	Direct	Sn: Imprecise* Sp: Precise	Undetected	None	Very low	Sn 0.83 (0.36-0.996) Sp 1.00 (0.90-1.00)	Critical
Duration of AKI	0								none	Important
CKD G3-G5	0								none	Important
CKD G5D	0								none	Important
Balance of Potential Benefits and Harms: FST has low† test accuracy for incident AKI, Stage 3 AKI, and mortality, with possible high test accuracy to predict RRT								**Certainty of Overall Evidence:** High		

Abbreviations: AKI = acute kidney injury; CKD = chronic kidney disease; FST = furosemide stress test; G = grade; ICU = intensive care unit; N/A = not applicable; RRT = renal replacement therapy; Sn = sensitivity; Sp = specificity

* Lower bound of 95% confidence interval <50%
† A priori defined as 51-75% sensitivity or specificity

References

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Table S23. Kinetic eGFR in adult medical or surgical patients

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI	3 (4273) [1-3]	No serious	Consistent	Direct	Precise	Undetected	None	High	AUC 0.81 (0.73, 0.86)	**Critical**
AKI Stage 3	0								No evidence	**Critical**
Mortality	2 (4126) [2,3]	No serious	Consistent	Direct	Highly imprecise	Undetected	None	Very low	AUC 0.70 (0.06, 0.99)	**Critical**
RRT	1 (107) [2]	No serious	N/A	Direct	Precise	Undetected	Sparse	Low	AUC 0.902 (0.82, 0.97) Sn* 0.87 Sp* 0.85	**Critical**
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Balance of Potential Benefits and Harms: Kinetic eGFR has moderate accuracy to predict incident AKI and may have high accuracy to predict RRT †								**Certainty of Overall Evidence:** HIgh		

Abbreviations: AKI = acute kidney injury; AUC = area under the curve; CKD = chronic kidney disease; eGFR = estimated glomerular filtration rate; N/A = not applicable; RRT = renal replacement therapy; Sn = sensitivity; Sp = specificity

* Cutoff = 30 mL/min
† *A priori* , accuracy arbitrarily categorized as High—AUC >0.85, Moderate—AUC 0.76-0.85, Low—AUC 0.51-0.75.

References

1. Dumnicka P, Mazur-Laskowska M, Ceranowicz P, et al. Acute Changes in Serum Creatinine and Kinetic Glomerular Filtration Rate Estimation in Early Phase of Acute Pancreatitis. J Clin Med. 2022;11(20):6159. Published 2022 Oct 19. doi:10.3390/jcm11206159
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Table S24. Kinetic eGFR with KDIGO urine output criteria in adults who are critically ill in the ICU with sepsis

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI	1 (2492) [1]	No serious	N/A	Direct	Unclear	Undetected	Sparse, but large	Low	Sn* 93.2% Sp* 73.0% Accuracy 85.7%	**Critical**
AKI Stage 3	0								No evidence	**Critical**
Mortality	1 (2492) [1]	No serious	N/A	Direct	Unclear	Undetected	Sparse, but large	Low	AUC 0.604	**Critical**
RRT	1 (2492) [1]	No serious	N/A	Direct	Unclear	Undetected	Sparse, but large	Low	AUC 0.802	**Critical**
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Balance of Potential Benefits and Harms: Kinetic eGFR with KDIGO UO criteria has high sensitivity with low specificity to predict incident AKI,† moderate accuracy to predict RRT,‡ and low accuracy to predict mortality‡								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = acute kidney injury; AUC = area under the curve; CKD = chronic kidney disease; eGFR = estimated glomerular filtration rate; G = grade; ICU = intensive care unit; N/A = not applicable; RRT = renal replacement therapy; Sn = sensitivity; Sp = specificity, UO = urine output.

*Cutoff not specified

† *A priori*, sensitivity and specificity ≥90% defined as high, 51% to 75% as low.

‡ *A priori*, test accuracy defined as moderate for AUC 0.76-0.85 and low for AUC 0.51-0.75.

References

1. Lijović L, Pelajić S, Hawchar F, et al. Diagnosing acute kidney injury ahead of time in critically ill septic patients using kinetic estimated glomerular filtration rate. J Crit Care. 2023;75:154276. doi:10.1016/j.jcrc.2023.154276

Table S25. Kinetic eGFR with different calculations based on serum creatinine timing in children

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Incident AKI	1 (175) [1]	No serious	N/A	Direct	Unclear	Undetected	Sparse	Very Low	Orthotopic heart transplant Preop and 1st postop AUC 0.65	**Critical**
	1 (175) [1]	No serious	N/A	Direct	Unclear	Undetected	Sparse	Very Low	Orthotopic heart transplant 1st and 2nd postop AUC 0.68	
	1 (60) [2]	Serious	N/A	Direct	Sn: Precise Sp: Imprecise*	Undetected	Sparse	Very Low	PICU Sn † 93% (80.0, 98.2) Sp † 76% (49.8, 92.2) PPV 90.9% / NPV 81.3%	
AKI Stage 3	0								No evidence	**Critical**
AKD	1 (803) [3]	No serious	N/A	Direct	Precise	Undetected	Sparse, but large	Low	Within the first 24 h of AKI KeGFR1 ≤39.18 AUC 0.78 (0.75, 0.81) Sn 76.2% / Sp 68.6% AUC (adjusted[1]) 0.80 In the 24 h prior to AKD KeGFR2 ≤40.41 AUC 0.84 (0.80, 0.88) Sn 66.5% / Sp 91.1% AUC (adjusted) 0.83	**Critical**
MAKE30	1 (803) [3]	No serious	N/A	Direct	Precise	Undetected	Sparse, but large	Low	Within the first 24 h of AKI KeGFR1 ≤21.28 AUC 0.70 (0.67–0.73) Sn 55.9% / Sp 81.7% AUC (adjusted) 0.76 In the 24 h prior to AKD KeGFR2 ≤25.69 AUC 0.82 (0.77–0.86) Sn 66.7% / Sp 87.7% AUC (adjusted) 0.84	**Critical**
Mortality	0								No evidence	**Critical**
RRT	0								No evidence	**Critical**
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Balance of Potential Benefits and Harms: Kinetic eGFR has variable accuracy in different populations of children								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = acute kidney injury; AKD = acute kidney disease; AUC = area under curve; CKD = chronic kidney disease; eGFR= estimated glomerular filtration rate; G = grade; keGFR = kinetic estimated glomerular filtration rate; MAKE = major adverse kidney event; N/A = not applicable; NPV = negative predictive value; PICU = pediatric intensive care unit; PPV = positive predictive value; RRT = renal replacement therapy, SCr = serum creatinine; Sn = sensitivity; Sp = specificity

* Lower bound of 95% confidence interval <50%

† Threshold not specified.

References

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Table S26. Urinary TIMP-2*IGFBP7 guided bundles vs. control in adults undergoing surgery

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	4 (775) [1] [RCT]	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	OR 0.90 (0.49 – 1.65) 6 fewer per 1000 (Uri`e)	Critical
AKI, any stage	4 (775) [1] [RCT]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	OR 0.70 (0.43 – 1.15) 163 fewer per 1000 (from 309 fewer to 81 more)	Critical
AKI, stage 2/3	4 (775) [1] [RCT]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	**OR 0.55 (0.39 – 0.79)** 147 fewer per 1000 (from 199 fewer to 68 fewer)	Critical
RRT	4 (775) [1] [RCT]	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	OR 0.82 (0.37 – 1.85) 12 fewer per 1000 (from 41 fewer to 55 more)	Critical
MAKE (at ≥90 days)	0								No evidence	Critical
Duration of AKI	0									Important
CKD G3-G5 (at ≥90 days)	0									Important
CKD G5D (dialysis) (at ≥90 days)	0								No evidence	Important
Balance of Potential Benefits and Harms: Not difference in most critical outcomes, except lower likelihood of stage 2/3 AKI with biomarker guided bundle. No evidence of harms								Certainty of Overall Evidence: Moderate		

Abbreviations: AKI = Acute Kidney Injury; CKD = Chronic Kidney Injury; MAKE: Major adverse kidney events; OR = odds ratio; RCT = randomized controlled trial; RRT = renal replacement therapy; TIMP-2*IGFBP7 = tissue inhibitor of metalloproteinases 2* insulin-like growth factor-binding protein 7

References

1. Li Z, Tie H, Shi R, Rossaint J, Zarbock A. Urinary [TIMP-2]·[IGFBP7]-guided implementation of the KDIGO bundle to prevent acute kidney injury: a meta-analysis. Br J Anaesth. 2022 Jan;128(1):e24-e26. doi: 10.1016/j.bja.2021.10.015. Epub 2021 Nov 15. PMID: 34794767.

Table S27. Urinary TIMP-2*IGFBP7 in adults

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Incident AKI	19 (3393) [1]	Some limitations	Consistent	Direct	Sn: Precise Sp: Imprecise*	Undetected	None	Low	*uTIMP-2*IGFBP-7: custom threshold* Sn 0.86 (0.75–0.93) Sp 0.58 (0.43–0.71)	Critical
					Sn: Imprecise* Sp: Precise			Low	*uTIMP-2*IGFBP-7: threshold* ≥0.3 Sn 0.68 (0.58-0.76) Sp 0.74 (0.64-0.81)	
					Sn: Imprecise* Sp: Precise			Low	*uTIMP-2*IGFBP-7: threshold* ≥2.0 Sn 0.19 (0.12-0.27) Sp 0.97 (0.96-0.98)	
Severe AKI (stage 2-3)	19 (3393) [1]	Some limitations	Consistent	Direct	Sn: Imprecise* Sp: Precise	Undetected	None	Low	*uTIMP-2*IGFBP-7: custom threshold* Sn 0.68 (0.38–0.87) Sp 0.90 (0.81–0.96)	Critical
					Sn: Precise Sp: Imprecise*			Low	*uTIMP-2*IGFBP-7: threshold* ≥0.3 Sn 0.90 (0.76-0.96) Sp 0.53 (0.35-0.70)	
					Sn: Imprecise* Sp: Precise			Low	*uTIMP-2*IGFBP-7: threshold* ≥2.0 Sn 0.33 (0.16-0.57) Sp 0.95 (0.89-0.97)	
RRT	0								none	Critical
Death	0								none	Critical
MAKE90	0								none	Critical
Duration of AKI	0								none	Important
CKD G3-G5	0								none	Important
CKD G5D (dialysis)	0								none	Important

Balance of Potential Benefits and Harms:
In adults,
Urinary TIMP-2*IGFBP-7 custom threshold has high † sensitivity but poor ‡ specificity to predict AKI incidence.
Urinary TIMP-2*IGFBP-7 custom threshold has poor ‡ sensitivity but high † specificity to predict AKI stage.

Urinary TIMP-2*IGFBP-7 >0.3 has poor ‡ sensitivity and specificity to predict AKI incidence.
Urinary TIMP-2*IGFBP-7 >0.3 has high † sensitivity but poor ‡ specificity to predict AKI stage.

Urinary TIMP-2*IGFBP-7 >2 has poor ‡ sensitivity but high † specificity to predict AKI incidence.
Urinary TIMP-2*IGFBP-7 >2 has high † sensitivity and specificity to predict AKI stage.

Certainty of Overall Evidence:
Low

Abbreviations: AKI = acute kidney injury; CKD = chronic kidney disease; MAKE = major adverse kidney injury; RRT = renal replacement therapy; Sn = sensitivity; Sp = specificity; TIMP-2*IGFBP-7 = tissue inhibitor of metalloproteinases 2*insulin-like growth factor-binding protein 7

* Lower bound of 95% confidence interval <50%.

† *A priori*, high sensitivity or specificity defined as ≥90%.
‡ *A priori*, low sensitivity or specificity defined as 51-75%.

References

1. Pan HC, Yang SY, Chiou TT, et al. Comparative accuracy of biomarkers for the prediction of hospital-acquired acute kidney injury: a systematic review and meta-analysis. *Crit Care*. 2022;26(1):349. Published 2022 Nov 12. doi:10.1186/s13054-022-04223-6

Table S28. Urinary TIMP-2*IGFBP7 in children

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Incident AKI	7 (880) [1]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	AUC 0.79 (0.72-0.85)	Critical
Severe AKI (stage 2-3)	5 (949) [1]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	AUC 0.79 (0.75-0.83)	Critical
RRT	1 (NR) [1]	Some limitations	Consistent	Direct	Imprecise*	Undetected	None	Low	AUC 0.67 (0.50-0.84)	Critical
Death	1 (NR) [1]	Some limitations	Consistent	Direct	Imprecise*	Undetected	None	Low	AUC 0.79 (0.61-0.97)	Critical
MAKE90	0								none	Critical
Duration of AKI	0								none	Important
CKD G3-G5	0								none	Important
CKD G5D (dialysis)	0								none	Important
Balance of Potential Benefits and Harms: In pediatrics, uTIMP2*IGFBP7 has low † to moderate ‡ accuracy to predict AKI outcomes and mortality.								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = acute kidney injury; AUC = area under the curve; CKD = chronic kidney disease; MAKE = major adverse kidney injury; RRT = renal replacement therapy; uTIMP2*IGFBP7 = urine tissue inhibitor of metalloproteinases 2* insulin-like growth factor-binding protein 7

* Range of AUC 95% CI >0.2 (larger than the ranges for low, moderate, and high accuracy).
† *A priori* defined as AUC 0.51-0.75
‡ Arbitrarily defined as AUC 0.76-0.85

References

1. Meena J, Thomas CC, Kumar J, Mathew G, Bagga A. Biomarkers for prediction of acute kidney injury in pediatric patients: a systematic review and meta-analysis of diagnostic test accuracy studies. *Pediatr Nephrol*. 2023;38(10):3241-3251. doi:10.1007/s00467-023-05891-4

Table S29. Urinary NGAL in adults

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Incident AKI	35 (8374) [1]	Some limitations	Consistent	Direct	Sn: Precise Sp: Precise	Undetected	None	Moderate	Sn 0.77 (0.72-0.81) Sp 0.81 (0.77-0.84)	Critical
Severe AKI (stage 2-3)	6 (NR) [1]	Some limitations	Consistent	Direct	Sn: Imprecise* Sp: Precise	Undetected	None	Low	Sn 0.68 (0.45-0.84) Sp 0.83 (0.72-0.91)	Critical
RRT	4 (NR) [1]	Some limitations	Consistent	Direct	Sn: Imprecise* Sp: Precise	Undetected	None	Low	Sn 0.65 (0.47-0.79) Sp 0.89 (0.78-0.95)	Critical
Death	0								none	Critical
MAKE90	0								none	Critical
Duration of AKI	0								none	Important
CKD G3-G5	0								none	Important
CKD G5D (dialysis)	0								none	Important
Balance of Potential Benefits and Harms: In adults, uNGAL has low to moderate† sensitivity and moderate† specificity to predict AKI outcomes.								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = acute kidney injury; CKD = chronic kidney disease; MAKE = major adverse kidney injury; RRT = renal replacement therapy; Sn = sensitivity; Sp = specificity; uNGAL = urine neutrophil gelatinase-associated lipocalin

* Lower bound of 95% confidence interval <50%.
† *A priori*, low sensitivity or specificity defined as 51-75% and moderate sensitivity or specificity defined as 76-89%.

References

1. Pan HC, Yang SY, Chiou TT, et al. Comparative accuracy of biomarkers for the prediction of hospital-acquired acute kidney injury: a systematic review and meta-analysis. *Crit Care*. 2022;26(1):349. Published 2022 Nov 12. doi:10.1186/s13054-022-04223-6

Table S30. Urinary NGAL in children

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI	28 (3459) [1]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	AUC 0.82 (0.77-0.88) Sn 0.75 (0.64-0.83) Sp 0.85 (0.80-0.89)	Critical
Severe AKI (stage 2-3)	9 (2088) [1]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	AUC 0.80 (0.72-0.87)	Critical
RRT	3 (935) [1]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	AUC 0.78 (0.70-0.87)	Critical
Death	5 (646) [1]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	AUC 0.87 (0.84-0.90)	Critical
MAKE90	0								none	Critical
Duration of AKI	0								none	Important
CKD G3-G5	0								none	Important
CKD G5D (dialysis)	0								none	Important
Balance of Potential Benefits and Harms: In pediatrics, uNGAL has moderate* accuracy to predict AKI incidence. uNGAL has moderate* accuracy to predict AKI outcomes and mortality.								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = acute kidney injury; AUC = area under the curve; CKD = chronic kidney disease; G = grade; MAKE = major adverse kidney injury; RRT = renal replacement therapy; Sn = sensitivity; Sp = specificity; uNGAL = urine neutrophil gelatinase-associated lipocalin

* *A priori* defined as 76-89% sensitivity or specificity or AUC 0.76-0.85

References

1. Meena J, Thomas CC, Kumar J, Mathew G, Bagga A. Biomarkers for prediction of acute kidney injury in pediatric patients: a systematic review and meta-analysis of diagnostic test accuracy studies. *Pediatr Nephrol*. 2023;38(10):3241-3251. doi:10.1007/s00467-023-05891-4

Table S31. Urinary KIM-1 in adults

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI	14 (2993) [1]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	Sn 0.76 (0.70-0.81) Sp 0.79 (0.75-0.83)	Critical
Severe AKI (stage 2-3)	0								none	Critical
RRT	0								none	Critical
Death	0								none	Critical
MAKE90	0								none	Critical
Duration of AKI	0								none	Important
CKD G3-G5	0								none	Important
CKD G5D (dialysis)	0								none	Important
Balance of Potential Benefits and Harms: In adults, uKIM-1 has moderate* sensitivity and specificity to predict AKI incidence.								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = acute kidney injury; CKD = chronic kidney disease; MAKE = major adverse kidney injury; RRT = renal replacement therapy; Sn = sensitivity; Sp = specificity; uKIM-1 = urine kidney injury molecule-1

* *A priori* defined as 76-89% sensitivity or specificity.

References

1. Pan HC, Yang SY, Chiou TT, et al. Comparative accuracy of biomarkers for the prediction of hospital-acquired acute kidney injury: a systematic review and meta-analysis. *Crit Care*. 2022;26(1):349. Published 2022 Nov 12. doi:10.1186/s13054-022-04223-6

Table S32. Urinary KIM-1 in children

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Incident AKI	12 (1593) [1]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	AUC 0.70 (0.63-0.75)	Critical
Severe AKI (stage 2-3)	6 (944) [1]	Some limitations	Consistent	Direct	Imprecise*	Undetected	None	Low	AUC 0.73 (0.62-0.83)	Critical
RRT	1 (NR) [1]	Some limitations	Consistent	Direct	Imprecise*	Undetected	None	Low	AUC 0.76 (0.60-0.91)	Critical
Death	1 (NR) [1]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	AUC 0.78 (0.70-0.87)	Critical
MAKE90	0								none	Critical
Duration of AKI	0								none	Important
CKD G3-G5	0								none	Important
CKD G5D (dialysis)	0								none	Important
Balance of Potential Benefits and Harms: In pediatrics, uKIM-1 has low † to moderate ‡ accuracy to predict AKI outcomes and mortality.								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = acute kidney injury; AUC = area under the curve; CKD = chronic kidney disease; MAKE = major adverse kidney injury; NR = not reported; RRT = renal replacement therapy; uKIM-1 = urine kidney injury molecule-1

* Range of AUC 95% CI >0.2 (larger than the ranges for low, moderate, and high accuracy).
† *A priori* defined as AUC 0.51-0.75
‡ *A priori* defined as AUC 0.76-0.85

References

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Table S33. Urinary L-FABP in adults

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI	10 (1340) [1]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	Sn 0.70 (0.62-0.77) Sp 0.81 (0.77-0.84)	Critical
Severe AKI (stage 2-3)	0								none	Critical
AKI requiring RRT	0								none	Critical
Death	0								none	Critical
MAKE90	0								none	Critical
Duration of AKI	0								none	Important
CKD G3-G5	0								none	Important
CKD G5D (dialysis)	0								none	Important
Balance of Potential Benefits and Harms: In adults, uL-FABP has low* sensitivity and moderate † specificity to predict AKI incidence.								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = Acute Kidney Injury; CKD = chronic kidney disease; MAKE = major adverse kidney injury; RRT = renal replacement therapy; Sn = sensitivity; Sp = specificity; uL-FABP = urine liver fatty acid–binding protein

* *A priori* defined as 51-75% sensitivity or specificity
† *A priori* defined as 76-89% sensitivity or specificity

References

1. Pan HC, Yang SY, Chiou TT, et al. Comparative accuracy of biomarkers for the prediction of hospital-acquired acute kidney injury: a systematic review and meta-analysis. *Crit Care*. 2022;26(1):349. Published 2022 Nov 12. doi:10.1186/s13054-022-04223-6

Table S34. Urinary L-FABP in children

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI	5 (703) [1]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	AUC 0.80 (0.73-0.88)	Critical
Severe AKI (stage 2-3)	3 (453) [1]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	AUC 0.85 (0.80-0.90)	Critical
AKI requiring RRT	0								none	Critical
Death	0								none	Critical
MAKE90	0								none	Critical
Duration of AKI	0								none	Important
CKD G3-G5	0								none	Important
CKD G5D (dialysis)	0								none	Important
Balance of Potential Benefits and Harms: In pediatrics, uL-FABP has moderate* accuracy to predict AKI incidence and stages.								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = Acute Kidney Injury; AUC = area under the curve; CKD = chronic kidney disease; G = grade; MAKE = major adverse kidney injury; RRT = renal replacement therapy; uL-FABP = urine liver fatty acid–binding protein

* *A priori* defined as AUC 0.76-0.85

References

1. Meena J, Thomas CC, Kumar J, Mathew G, Bagga A. Biomarkers for prediction of acute kidney injury in pediatric patients: a systematic review and meta-analysis of diagnostic test accuracy studies. *Pediatr Nephrol*. 2023;38(10):3241-3251. doi:10.1007/s00467-023-05891-4

Table S35. Urinary microscopy score in adults admitted to the surgical ICU after noncardiac surgery

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
AKI, incident	0								No evidence	**Critical**
AKI, Stage 2-3	1 (661) [1]	No serious	N/A	Direct	Sn: Precise Sp: Precise	Undetected	Sparse, but large	Low	**UMS ≥1** Sn 15.6% (9.7, 21.5) Sp 93.8% (91.7, 95.9) PPV 41.8% (28.8, 54.8) NPV 79.5% (76.3, 82.7) **UMS ≥3** Sn 5.4% (1.7, 9.1) Sp 98.4% (97.3, 99.5) PPV 50.0% (25.5, 74.5) NPV 78.4% (75.2, 81.6)	**Critical**
Mortality	0								No evidence	**Critical**
RRT	0								No evidence	**Critical**
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Balance of Potential Benefits and Harms: The urinary microscopy score does not indicate an increased risk of severe AKI (sensitivity <50%), but has high specificity*								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = acute kidney injury; CKD = chronic kidney disease; G = grade; N/A = not applicable; NPV = negative predictive value; PPV = positive predictive value; RRT = renal replacement therapy; SICU = surgical intensive care unit; Sn = sensitivity; Sp = specificity, UMS = urinary microscopy score.

* *A priori* defined as ≥90% sensitivity or specificity

References

1. Li N, Zhou WJ, Chi DX, et al. Association between urine microscopy and severe acute kidney injury in critically ill patients following non-cardiac surgery: a prospective cohort study. Ann Palliat Med. 2022;11(7):2327-2337. doi:10.21037/apm-21-3085

Table S36. Muddy brown granular casts in adults with Stage ≥2 AKI

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI (acute tubular injury)	1 (270) [1]	No serious	N/A	Direct	Unclear	Undetected	Sparse	Very Low	Sn 78% Sp 100% PPV 100% NPV 62%	**Critical**
AKI Stage 3	0								No evidence	**Critical**
Mortality	0								No evidence	**Critical**
RRT	0								No evidence	**Critical**
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Balance of Potential Benefits and Harms: Muddy brown granular casts have moderate* sensitivity and high † specificity to predict acute tubular injury								**Certainty of Overall Evidence:** Very Low		

Abbreviations: AKI = acute kidney injury; CKD = chronic kidney disease; G = grade; N/A = not applicable; NPV = negative predictive value; PPV = positive predictive value; RRT = renal replacement therapy; Sn = sensitivity; Sp = specificity

* *A priori* defined as 76-89% sensitivity or specificity
† *A priori* defined as ≥90% sensitivity or specificity

References

1. Varghese V, Rivera MS, Alalwan A, et al. Concomitant Identification of Muddy Brown Granular Casts and Low Fractional Excretion of Urinary Sodium in AKI. Kidney360. 2022;3(4):627-635. Published 2022 Jan 19. doi:10.34067/KID.0005692021

Table S37. Urine eosinophils in adults with both a urine eosinophils test and biopsy of native kidney performed as part of their work-up for AKI

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	**Description of Findings**	**Importance of Outcome**
Incident AKI (acute interstitial nephritis)	1 (566) [1]	No serious	N/A	Direct	Sn: Precise Sp: Precise	Undetected	Sparse, but large	Low	**UEs >1%** Sn 30.8% (22.2, 40.9) Sp 68.2% (63.7, 72.2) PPV 15.6% (11.1, 21.7) NPV 83.7% (79.7, 87.1) LR+ 0.97 LR− 1.01 **UEs >5%** Sn 19.8% (12.9, 29.1) Sp 91.2% (88.3, 93.4) PPV 30.0% (19.9, 42.5) NPV 85.6% (82.2, 88.4) LR+ 2.3 LR− 0.9	**Critical**
AKI Stage 3	0								No evidence	**Critical**
Mortality	0								No evidence	**Critical**
RRT	0								No evidence	**Critical**
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Balance of Potential Benefits and Harms: Urinary eosinophils do not indicate an increased risk of acute interstitial nephritis (sensitivity <50%), but with a >5% threshold have high specificity*								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = acute kidney injury; CKD = chronic kidney disease; G = grade; LR = likelihood ratio; N/A = not applicable; NPV = negative predictive value; PPV = positive predictive value; RRT = renal replacement therapy; Sn = sensitivity; Sp = specificity; UEs = urine eosinophils

* *A priori* defined as ≥90% sensitivity or specificity

References

1. Muriithi AK, Nasr SH, Leung N. Utility of urine eosinophils in the diagnosis of acute interstitial nephritis. Clin J Am Soc Nephrol. 2013;8(11):1857-1862. doi:10.2215/CJN.01330213

Table S38. Renal angina index in children in the ICU with sepsis

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Incident AKI	18 [1-9] (13098)	Not serious	Consistent	Direct	Sn: Precise Sp: Precise	Undetected	None	High	Sn: 0.76 (0.67–0.83) Sp: 0.78 (0.70-0.85)	Critical
Severe AKI (stage 2-3)*	10 (4271)* [3-5,7,10-15]	Serious	Consistent	Direct	Sn: Precise Sp: Precise	Undetected	None	Moderate	Sn: 0.85 (0.77–0.90) Sp: 0.74 (0.56-0.87)	Critical
AKI Stage 3	4 (1484) [3-4,7,16]	Not serious	Consistent	Direct	Sn: Precise Sp: Precise	Undetected	None	High	Sn: 0.94 (0.87–0.97) Sp: 0.76 (0.59-0.87)	Critical
Mortality	7 (3385) [3-5,7,9,11,14]	Not serious	Consistent	Direct	Sn: Precise Sp: Imprecise†	Undetected	None	Moderate	Sn: 0.74 (0.55–0.86) Sp: 0.68 (0.45-0.85)	Critical
RRT	6 (1810) [3-5,7,9,14]	Not serious	Consistent	Direct	Sn: Precise Sp: Imprecise†	Undetected	None	Moderate	Sn: 0.84 (0.74–0.91) Sp: 0.61 (0.38-0.80)	Critical
Duration of AKI	0								none	Important
CKD G3-G5 (at ≥90 days)	0								none	Important
CKD G5D (dialysis) (at ≥90 days)	0								none	Important
Balance of Potential Benefits and Harms: RAI has moderate ‡ to high § sensitivity to predict AKI outcomes and death. RAI has moderate specificity to predict incident AKI and stages, but poor specificity to predict RRT or death								**Certainty of Overall Evidence:** High		

Abbreviations: AKI = acute kidney injury; CKD = chronic kidney disease; G = grade; ICU = intensive care unit; RRT = renal replacement therapy; Sn = sensitivity; Sp = specificity;

* 2 studies that reported AKI stage 2-3 only were conducted in adults. No significant difference in findings between pediatric and adult studies.
† Lower bound of 95% confidence interval <50%
‡ *A priori* defined as 76-89% sensitivity or specificity
§ *A priori* defined as ≥90% sensitivity or specificity

References

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4. Ribeiro-Mourão F, Vaz AC, Azevedo A, et al. Assessment of the renal angina index for the prediction of acute kidney injury in patients admitted to a European pediatric intensive care unit. Pediatr Nephrol. 2021;36(12):3993-4001. doi:10.1007/s00467-021-05116-6
5. Stanski NL, Krallman KA, Chima RS, Goldstein SL. A risk-stratified assessment of biomarker-based acute kidney injury phenotypes in children. Pediatr Res. 2023;93(5):1354-1360. doi:10.1038/s41390-022-02233-2
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Table S39. Balanced crystalloid vs. high molecular weight hydroxyethyl starch in critically ill adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	3* (605) [1] [RCT]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	**OR 0.76 (0.58, 0.99)*** 102 fewer per 1000 (from 179 fewer to 4 fewer)	**Critical**
AKI	1† (533) [1, 2] [RCT]	Some limitations	Consistent	Direct	Precise	Undetected	Sparse direct evidence†	Low	**OR 0.55 (0.34, 0.88) †** 167 fewer per 1000 (from 228 fewer to 58 fewer)	**Critical**
AKI, Stage 3	0								No evidence	**Critical**
RRT	2‡ (564) [1] [RCT]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	**OR 0.51 (0.27, 0.93) ‡** 132 fewer per 1000 (from 197 fewer to 19 fewer)	**Critical**
MAKE90	0								No evidence	**Critical**
CKD G5D	0								No evidence	Important
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
Serious AEs	0								No evidence	**Critical**

Balance of Potential Benefits and Harms:	Certainty of Overall Evidence:
In critically ill adults, high molecular weight hydroxyethyl starch increases the risk of death, incidence of AKI, and risk of RRT compared with crystalloids (balanced or normal saline)	Moderate

Abbreviations: AEs = adverse events; AKI = acute kidney injury; BC = balanced crystalloid; CKD = chronic kidney disease; H-HES = high molecular weight hydroxyethyl starch; L-HES = low molecular weight hydroxyethyl starch; MAKE = major adverse kidney events; NS = normal saline; OR = odds ratio; RCT = randomized controlled trial; RRT = renal replacement therapy

* Based on network meta-analysis with 49 trials with 40,910 participants.
† Based on network meta-analysis with 13 trials with 27,190 participants. Single study consistent with network meta-analysis (OR 0.55; 95% CI 0.38, 0.81)
‡ Based on network meta-analysis with 14 trials with 29,789 participants.

References

1. Liu C, Mao Z, Hu P, et al. Fluid resuscitation in critically ill patients: a systematic review and network meta-analysis. *Ther Clin Risk Manag*. 2018;14:1701-1709. doi:10.2147/TCRM.S175080
2. Brunkhorst FM, Engel C, Bloos F, et al. Intensive insulin therapy and pentastarch resuscitation in severe sepsis. N Engl J Med. 2008; 358(2):125–139.

Table S40. Balanced crystalloid vs. low molecular weight hydroxyethyl starch in critically ill adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	5* (1701) [1] [RCT]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	**OR 0.88 (0.78, 0.99)*** 18 fewer per 1000 (from 36 fewer to 1 fewer)	**Critical**
AKI	1† (798) [1, 2] [RCT]	Some limitations	Consistent†	Direct	Imprecise†	Undetected	Sparse direct evidence	Low	OR 0.91 (0.73, 1.21) † 25 fewer per 1000 (from 76 fewer to 59 more)	**Critical**
AKI, Stage 3	0								No evidence	**Critical**
RRT	2‡ (843) [1] [RCT]	Some limitations	Consistent	Direct	Imprecise	Undetected	None	Low	OR 0.74 (0.52, 1.15) ‡ 29 fewer per 1000 (from 53 fewer to 17 more)	**Critical**
MAKE90	0								No evidence	**Critical**
CKD G5D	0								No evidence	Important
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
Serious AEs	0								No evidence	**Critical**
Balance of Potential Benefits and Harms: In critically ill adults, low molecular weight hydroxyethyl starch increases the risk of death, but without evidence of increased risk of AKI or RRT compared with crystalloids (balanced or normal saline)								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AEs = adverse events; AKI = acute kidney injury; BC = balanced crystalloid; CKD = chronic kidney disease; H-HES = high molecular weight hydroxyethyl starch; L-HES = low molecular weight hydroxyethyl starch; MAKE = major adverse kidney events; NS = normal saline; OR = odds ratio; RCT = randomized controlled trial; RRT = renal replacement therapy

* Based on network meta-analysis with 49 trials with 40,910 participants.
† Based on network meta-analysis with 13 trials with 27,190 participants. Single study consistent with network meta-analysis (OR 0.79; 95% CI 0.58, 1.05)
‡ Based on network meta-analysis with 14 trials with 29,789 participants.

References

1. Liu C, Mao Z, Hu P, et al. Fluid resuscitation in critically ill patients: a systematic review and network meta-analysis. *Ther Clin Risk Manag*. 2018;14:1701-1709. Published 2018 Sep 12. doi:10.2147/TCRM.S175080
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Table S41. Crystalloids vs. albumin in critically ill adults

Outcome	# Studies* (Patients)	Methodological Study Limitations	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Other Consider-ations	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	Surgery 3 (1744) [1]	No serious	Consistent	Direct	Very imprecise	Undetected	None	Very low	Iso- or hyper-oncotic: OR 0.84 (0.19, 3.70)†	Critical
	Sepsis 0 Iso [2] 1 Hyper (96)[2]	 N/A Serious	 N/A Consistent ‡	 Indirect Direct	 Imprecise Precise	 N/A Undetected	 No direct Sparse	 Very low Low	 Iso: OR 1.00 (0.77, 1.30) # Hyper: OR 0.90 (0.74, 1.09) #	
	Trauma 1 Iso (17)[2] 1 Hyper (141)[2]	 No serious No serious	 Consistent ‡ Consistent ‡	 Direct Direct	 Imprecise Very imprecise	 Undetected Undetected	 Sparse Sparse	 Low Very low	 Iso: OR 0.89 (0.47, 1.71) # Hyper: OR 0.67 (0.13, 3.43) #	
	TBI 0 [2]	N/A	N/A	Indirect	Very imprecise	N/A	No direct	Very low	Iso: OR 0.71 (0.17, 2.98) #	
AKI	Surgery 4 (1749) [1]	No serious	Consistent ‡	Direct	Very imprecise	Undetected	None	Very low	Iso- or hyper-oncotic: OR 1.01 (0.41, 2.42) §	Critical
	Sepsis 0 Iso [2]	N/A	N/A	Indirect	Imprecise	N/A	No direct	Very low	Iso: OR 0.95 (0.68, 1.33)	
	Trauma Hyper 0 [2]	N/A	N/A	Indirect	Imprecise	N/A	No direct	Very low	Hyper: OR 0.83 (0.13, 1.69) #	
AKI, Stage 3	0								No evidence	Critical
RRT	0								No evidence	Critical
MAKE90	0								No evidence	Critical
AKI duration	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Serious AEs	0								No evidence	Important
Balance of Potential Benefits and Harms: No evidence of a difference in AKI or death in various groups of critically ill patients between crystalloids and albumin (iso-oncontic or hyper-oncotic)								Certainty of Overall Evidence: Very low		

Abbreviations: AEs = adverse events; AKI: acute kidney injury; CKD: chronic kidney disease; Hyper = balanced crystalloids vs. hyper-oncotic albumin; Iso = balanced crystalloids vs. iso-oncotic albumin; MAKE90 = major adverse kidney events in 90 days; MD = mean difference; N/A = not applicable; OR = odds ratio; RCT = randomized controlled trial; RRT = renal replacement therapy.

* All randomized controlled trials. Numbers of studies and patients shown are those with direct comparisons between fluids.
† Network meta-analysis[2] found similarly imprecise OR estimates: Iso-oncotic 0.99 (0.48, 2.03); Hyper-oncotic 2.53 (0.09, 74.5).
‡ Direct (within-studies) and indirect (in the network meta-analysis) evidence are consistent with each other (the network was coherent).
# Network meta-analysis estimate.
§ Network meta-analysis[2] found similarly very imprecise OR estimate: Iso-oncotic 1.19 (0.27, 5.15)

References

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Table S42. Gelatin vs. albumin/crystalloids in children with hypovolemia

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	3 (924) [1]	No limitations	Consistent	Direct	Imprecise	Undetected	None	Moderate	RR 1.21 (0.94, 1.57) 24 more per 1000 (from 12 fewer to 113 more)	Critical
AKI	1 (60) [1]	Serious limitations	N/A	Direct	Imprecise	Undetected	Sparse	Very low	RR 0.36 (0.04, 3.23) 62 fewer per 1000 (from 186 fewer to 61 more)	Critical
AKI, Stage 3	0								No evidence	Critical
RRT	0								No evidence	Critical
MAKE90	0								No evidence	Critical
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Serious AEs	0								No evidence	Important
Balance of Potential Benefits and Harms: There is no evidence that the choice of gelatin vs. albumin or crystalloids affects the risk of death in children, with an unclear effect on incident AKI and no trial evidence for other outcomes								Certainty of Overall Evidence: Moderate		

Abbreviations: AEs = adverse events; AKI = acute kidney injury; CKD: Chronic Kidney Injury; MAKE90: Major adverse kidney events in 90 days; N/A = not applicable; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

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Table S43. Buffered crystalloids vs. normal saline in critically ill adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Death	21 (36387) [1,4-6] [RCT]	Serious	Consistent	Direct	Precise	No serious	None	Moderate	**RR 0.96 (0.92, 1.00)** 8 fewer per 1000 (from 15 fewer to 0)	**Critical**
AKI	12 (29241) [1,4,7-9] [RCT]	Serious	Consistent	Direct	Precise	No serious	None	Moderate	**RR 0.93 (0.87, 0.99)** 8 fewer per 1000 (from 14 fewer to 1 fewer)	**Critical**
AKI, Stage 3	0								No evidence	**Critical**
RRT	8 (32536) [1,10] [RCT]	Serious	Inconsistent	Direct	Imprecise*	Undetected	None	Low	RR 0.92 (0.84, 1.01) 4 fewer per 1000 (from 9 fewer to 1 fewer)	**Critical**
MAKE30	2 (16776) [11,12] [RCT]	Serious	Consistent	Direct	Imprecise*	Undetected	None	Moderate	OR 0.93 (0.85, 1.01) 11 fewer per 1000 (From 24 fewer to 2 more)	**Critical**
CKD G5D	0								No evidence	Important
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
Serious AE	0								No evidence	**Critical**
Balance of Potential Benefits and Harms: In critically ill adults, balanced fluids reduce the risk of death, and incidence of AKI, also possibly need for RRT and MAKE30, with no evidence regarding harms, compared with normal saline								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AE = adverse event; AKI = acute kidney injury; CKD = chronic kidney disease; MAKE = major adverse kidney injury; OR = odds ratio; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

* Imprecise from the perspective of there being a clinical effect (of similar magnitude of other outcomes). RRT P = 0.076; MAKE30 P = 0.099.

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Table S44. Buffered crystalloids vs. normal saline in critically ill children in the ICU with severe dehydration, sepsis, or septic shock

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Death	10 (1316) [1-10] [RCT]	Serious	Inconsistent	Direct	Imprecise	No serious	None	Very Low	OR 0.92 (0.69, 1.23) 17 fewer per 1000 (from 66 fewer to 49 more)	**Critical**
AKI	8 (1568) [5-8,11-14] [RCT]	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	OR 0.89 (0.57, 1.41) 20 fewer per 1000 (from 68 fewer to 46 more)	**Critical**
AKI, Stage 3	0								No evidence	**Critical**
RRT	4 (896) [5-6,9,12] [RCT]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	**OR 0.53 (0.34, 0.84)** 56 fewer per 1000 (from 81 fewer to 18 fewer)	**Critical**
MAKE30	2 (72) [5,7] [RCT]	Serious	Consistent	Direct	Very imprecise	Undetected	None	Very Low	OR 0.47 (0.08, 2.81) 57 fewer per 1000 (From 99 fewer to 195 more)	**Critical**
CKD G5D	0								No evidence	Important
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
Serious AE	0								No evidence	**Critical**
Balance of Potential Benefits and Harms: In critically ill children, balanced fluids reduces the need for RRT, but with no difference in mortality, MAKE30 and AKI outcomes, with no evidence regarding harms, compared with normal saline								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AE = adverse event; AKI = acute kidney injury; CKD = chronic kidney disease; ICU = intensive Care Unit; MAKE30 = major adverse kidney injury in 30 days; OR = odd ratio; RCT = randomized controlled trial; RRT = renal replacement therapy

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Table S45. Intravenous bicarbonate vs. usual care in critically ill adults* with acidosis

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Death, 28 d †	4 (1140)[1-3,6] [RCT]	Some limitations	Inconsistent	Direct	Precise	Undetected	None	Moderate	RR 0.88 (0.72,1.07) 46 fewer per 1000 (107 fewer to 27 more)	Critical
AKI Stage 3	2 (46)[5,6] [RCT]	No limitations	Consistent	Direct	Very imprecise	Undetected	None	Low	RR 0.89 (0.43, 1.82) 46 fewer per 1000 (from 342 fewer to 238 more)	Critical
RRT	4 (1071)[2,3,5,6] [RCT]	Some limitations	Consistent	Direct	Precise	Undetected	None	High	**RR 0.69 (0.60, 0.80)** **131 fewer per 1000** **(168 fewer to 84 fewer)**	Critical
MAKE90	1 (627)[3]	No limitations	N/A	Direct	Precise	Undetected	Sparse, but large	Low	RR 0.99 (0.91, 1.07) 8 fewer per 1000 (76 fewer to 59 more)	Critical
ICU LOS	4 (1071)[3,5-7] [RCT]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	MD 0.41 days (-0.68, 1.49)	Important
Hospital LOS	4 (730)[2,3,5,6] [RCT]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	MD 0.05 days (-2.49, 2.60)	Important
Serious AE	3 (1041)[2,3,5] [RCT]	Some limitations	Consistent	Direct	Imprecise	Undetected	Sparse	Very low	Severe electrolyte abnormality (1 study): RR 1.85 (0.41, 8.32) 142 more per 1000 (from 1220 fewer to 98 more) Infection (2 studies): RR 1.02 (0.80, 1.29) 6 more per 1000 (60 fewer to 87 more)	Critical
Discontinuation due to AE	0								No evidence	Important
Balance of Potential Benefits and Harms: Intravenous bicarbonate for acidosis reduces the risk of RRT with no evidence of effect on mortality, other kidney outcomes or harms.								**Certainty of Overall Evidence:** High		

Abbreviations: AE = adverse event; AKI = acute kidney injury; MAKE = major adverse kidney injury; MD = mean difference; N/A = not applicable; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy.

* One study randomized 50 children.[7] Combined with other studies.
† Analyses at other time points were similar in effect size and were not statistically significant.

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Table S46. Restricted vs. liberal fluid therapy in abdominal surgery in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Publication Bias	Precision of the Estimate	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Mortality 30 days*	7 (3803) [1-7] [RCT]	No limitations	Consistent	Direct	Undetected	Very imprecise	None	Very low	RR 0.91 (95% CI 0.39, 2.15) 1 fewer per 1000 (from 9 fewer to 16 more)	Critical
Incident AKI	5 (3396) [2, 4-7] [RCT]	No limitations	Consistent	Direct	Undetected	Precise	Sparse, but large†	Moderate	**RR 1.67 (95% CI 1.27, 2.20)** 29 more per 1000 (from 12 more to 52 more)	Critical
RRT	5 (3344) [1-3, 6,8] [RCT]	No limitations	Consistent	Direct	Undetected	Imprecise	None	Moderate	RR 1.96 (95% CI 0.73, 5.25) 7 more per 1000 (from 0 more to 38 more)	Critical
MAKE	0								No evidence	Critical
AKI Stage 3	1 (2983) [6] [RCT]	No limitations	N/A	Direct	Undetected	Precise	Sparse, but large	Moderate	**RR 1.79 (1.03, 3.13)** 10 more per 1000 (from 1 more to 20 more)	Important
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Serious AE, any	2 (3124) [1, 6] [RCT]	No limitations	Unclear†	Direct	Undetected	Imprecise	Sparse, but large†	Low	RR 0.99 (95% CI 0.72, 1.36) 0.5 fewer per 1000 (from 15 fewer to 14 more)	Critical
Serious AE: Pulmonary edema or CHF	7 (3727) [1-4,6-8] [RCT]	Serious limitations	Consistent	Direct	Undetected	Imprecise	Very imprecise	Very low	RR 0.92 (95% CI 0.26, 3.31) 2 fewer per 1000 (from 17 fewer to 53 more)	Critical
Balance of Potential Benefits and Harms: Restrictive fluid therapy increases the risk of incident AKI, stage 3 AKI, and possibly RRT compared with liberal fluid therapy, without indication of a difference in fluid-related adverse events.								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AE = adverse events; AKI = acute kidney injury; CI = confidence interval; CHF = congestive heart failure (or cardiac failure); CKD = chronic kidney disease; HR = hazard ratio; MAKE = major adverse kidney event; N/A = not applicable; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

* Except Myles et al.[6], 90 days.
† Meta-analysis recapitulates Myles et al.[6], which provided >95% of meta-analytic weight.

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Table S47. High vs low MAP targets in critically ill adults with chronic hypertension*

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Mortality	2 (1184) [1-2] [RCT]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	RR **1.19 (1.01, 1.40)** 78 more per 1000 (from 7 more to 164 more)	Critical
Incident AKI	1 (340) [3] [RCT]	Serious	N/A	Direct	Precise	Undetected	Sparse (1 study)	Low	RR **0.75 (0.59, 0.95)** 130 fewer per 1000 (from 213 fewer to 26 fewer)	Critical
RRT	2 (1466) [2-3] [RCT]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	RR **0.83 (0.71, 0.97)** 54 fewer per 1000 (from 92 fewer to 9 fewer)	Critical
MAKE	0								No evidence	Important
AKI Stage 3	0								No evidence	Important
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Serious AE	0								No evidence	Important
Balance of Potential Benefits and Harms: Treating patients with high MAP goals † is associated with lower risk of AKI and need for RRT, but higher risk of mortality								**Certainty of Overall Evidence:** Low		

Abbreviations: AE = adverse events; AKI = acute kidney injury; CKD = chronic kidney disease; MAKE = major adverse kidney event; MAP = mean arterial pressure; N/A = not applicable; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy.

* Included studies with ≥80% of patients with chronic hypertension
† Lamontagne et al. 2016: Target MAP: Usual care vs. 60-65 mmHg (with mean achieved MAPs of 73 vs. 68 mmHg)
Lamontagne et al. 2020: Target MAP: 75-80 vs. 60-65 mmHg (with mean achieved MAPs of 79 vs. 70 mmHg)
Asfar et al. 2014: Target MAP: 80-85 vs. 65-70 mmHg (with mean achieved MAPs of 85 vs. 75 mmHg)

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Table S48. High vs low MAP targets in adults with chronic hypertension* undergoing cardiac surgery

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Mortality	1 (194) [1] [RCT]	No limitations	N/A	Direct	Very imprecise	Undetected	Sparse (1 study)	Very low	RR 9.09 (0.50, 166.7) 40 more per 1000 (from 3 fewer to 829 more)	Critical
Incident AKI	1 (194) [1] [RCT]	No limitations	N/A	Direct	Precise	Undetected	Sparse (1 study)	Low	**RR 4.59 (1.02, 20.71)**[b] 73 more per 1000 (from 0 more to 402 more)	Critical
RRT	1 (194) [1] [RCT]	No limitations	N/A	Direct	Very imprecise	Undetected	Sparse (1 study)	Very low	RR 1.02 (0.15, 7.10) 0 more per 1000 (from 17 fewer to 124 more)	Critical
MAKE	0								No evidence	Important
AKI Stage 3	0								No evidence	Important
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Serious AE	0								No evidence	Important

Balance of Potential Benefits and Harms: Treating patients with high MAP goals (target MAP ≥70 mm Hg) results in *higher* risk of AKI,† with no evidence of an effect on mortality or RRT	**Certainty of Overall Evidence:** Low

Abbreviations: AE = adverse events; AKI = acute kidney injury; CKD = chronic kidney disease; MAKE = major adverse kidney event; MAP = mean arterial pressure; N/A = not applicable; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy.

* Included studies/analyses with ≥80% of patients with chronic hypertension.
† The study authors cautiously consider that this finding may have been due to "extensive use of adrenergic vasoconstrictors".

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Table S49. High vs low MAP targets in adults with chronic hypertension* undergoing noncardiac surgery

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Mortality	1 (646) [1] [RCT]	Serious	N/A	Direct	Highly imprecise	Undetected	Sparse (1 study)	Very low	RR 1.11 (0.45, 2.70) 3 more per 1000 (from 17 fewer to 52 more)	Critical
Incident AKI	2 (7652) [1-2] [RCT]	Serious	Inconsistent	Direct	Imprecise	Undetected	None	Very low	RR 0.92 (0.64, 1.32) 11 fewer per 1000 (from 52 fewer to 46 more)	Critical
RRT	1 (7006) [2] [RCT]	Serious	N/A	Direct	Highly imprecise	Undetected	Sparse (1 large study)	Very low	RR 1.00 (0.51, 1.96) 0 more per 1000 (from 2 fewer to 4 more)	Critical
MAKE	0								No evidence	Important
AKI Stage 3	2 (7652) [1-2] [RCT]	Serious	N/A[b]	Direct	Precise	Undetected	Sparse (1 large study) †	Low	RR 0.72 (0.46, 1.12) 4 fewer per 1000 (from 7 fewer to 2 more)	Important
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Serious AEs	0								No evidence	Important
Balance of Potential Benefits and Harms: No evidence of difference in mortality or AKI outcomes between patients treated with high (target MAP ≥80 mm Hg) or low MAP goals								**Certainty of Overall Evidence:** Very low		

Abbreviations: AEs = adverse events; AKI = acute kidney injury; CKD = chronic kidney disease; MAKE = major adverse kidney event; MAP = mean arterial pressure; N/A = not applicable; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy.

* Included studies with ≥80% of patients with chronic hypertension
† One small study had 0 events

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Table S50. Individualized vs. standard MAP targets in adults with chronic hypertension undergoing noncardiac surgery

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Mortality	3 (4673) [1-3] [RCT]	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	RR 0.77 (0.52, 1.15) 6 fewer per 1000 (from 11 fewer to 4 more)	Critical
Incident AKI	3 (4673) [1-3] [RCT]	Serious	Inconsistent	Direct	Imprecise	Undetected	None	Very Low	RR 0.90 (0.64, 1.27) 8 fewer per 1000 (from 29 fewer to 22 fewer)	Critical
RRT	2 (292) [1 3] [RCT]	Serious	Consistent	Direct	Highly imprecise	Undetected	None	Very low	RR 1.08 (0.46, 2.56) 1 more per 1000 (from 8 fewer to 22 more)	Critical
MAKE	0								No evidence	Important
AKI Stage 3	1 (292)[1] [RCT]	Some limitations	N//A	Direct	Highly imprecise	Undetected	Sparse (but large study)	Very low	AdjRR 0.97 (0.40, 2.34) 2 fewer per 1000 (from 37 fewer to 83 more)	Important
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Serious AEs	3 (4673) [1-3] [RCT]	Serious	Consistent	Direct	N/A*	Undetected	None	Low	No significant difference*	Important
Balance of Potential Benefits and Harms: No evidence of difference in mortality, AKI outcomes, or serious adverse events between patients treated with individualized or standard MAP goals								**Certainty of Overall Evidence:** Low		

Abbreviations: AEs = adverse events; AKI = acute kidney injury; CKD = chronic kidney disease; MAKE = major adverse kidney event; MAP = mean arterial pressure; N/A = not applicable; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy.

* Adverse event data reported only for individual adverse events. Study reported overall no significant difference in adverse events.

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Table S51. Multimodal kidney protection strategies vs. usual care in adults who are critically ill or undergoing cardiovascular surgery

Outcome	Time	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
									Certainty of Evidence	Description of Findings	Importance of Outcome
Death, all-cause	In hospital	3 (331) [1-2, 4] [RCT]	Some limitations	Consistent	Direct	Imprecise	Undetected	None	Low	RR 0.62 (0.27, 1.39)	Critical
	Day 30	2 (1452) [3,6] [RCT]	Some limitations	Consistent	Direct	Imprecise	Undetected	Sparse	Low	RR 1.12 (0.71, 1.78)	Critical
	Day 60	2 (376) [3-4] [RCT]	Some limitations	Consistent	Direct	Imprecise	Undetected	Sparse	Low	RR 0.92 (0.42, 1.97)	Critical
	Day 90	3 (1725) [3, 5-6] [RCT]	Some limitations	Consistent	Direct	Imprecise	Undetected	None	Low	RR 1.04 (0.72, 1.49)	Critical
AKI, incident	72 h	6 (2057) [1-6] [RCT]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	RR 0.87 (0.75, 1.00)	Critical
AKI, stage 2-3	72 h	5 (1946) [1, 3-6] [RCT]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	**RR 0.67 (0.56, 0.79)**	Critical
AKI, stage 3	72 h	3 (829) [3-4, 6] [RCT]	Some limitations	Consistent	Direct	Imprecise	Undetected	None	Low	RR 0.91 (0.65, 1.27)	Critical
RRT	In hospital	4 (785) [1-3, 5] [RCT]	Some limitations	Consistent	Direct	Imprecise	Undetected	None	Low	RR 0.96 (0.52, 1.76)	Critical
	72 h	2 (376) [3-4] [RCT]	Some limitations	Inconsistent	Direct	Imprecise	Undetected	Sparse	Low	RR 0.63 (0.06, 6.11)	Critical
	Day 30	2 (1439) [3, 6] [RCT]	Some limitations	Consistent	Direct	Imprecise	Undetected	Sparse	Low	RR 0.92 (0.58, 1.45)	Critical
	Day 60	2 (362) [3-4] [RCT]	Some limitations	Consistent	Direct	Imprecise	Undetected	Sparse	Low	RR 0.82 (0.08, 8.48)	Critical
	Day 90	3 (1705) [3, 5-6] [RCT]	Some limitations	Consistent	Direct	Imprecise	Undetected	None	Low	RR 0.88 (0.55, 1.39)	Critical
MAKE	Day 30	2 (1400) [3, 6] [RCT]	Some limitations	Consistent	Direct	Imprecise	Undetected	Sparse	Low	RR 1.16 (0.79, 1.70)	Critical
	Day 60	1 (298) [3] [RCT]	Some limitations	Not applicable	Direct	Imprecise	Undetected	None	Low	RR 1.53 (0.77, 3.02)	Critical
	Day 90	5 (1885) [2-6] [RCT]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	**RR 0.80 (0.67, 0.94)**	Critical
Serious AE		0								No evidence	Critical
Duration of AKI		0								No evidence	Important
CKD G3-G5		0								No evidence	Important
CKD G5D		0								No evidence	Important
AE: Fluid overload		0								No evidence	Important
AE: Heart failure		0								No evidence	Important
Balance of Potential Benefits and Harms: Multimodal kidney protection strategies for critically ill patients with cardiovascular diseases reduces the risk of moderate–severe AKI and MAKE90 with no evidence of an effect on all-cause mortality or need for RRT									**Certainty of Overall Evidence:** Moderate		

Abbreviations: AE = adverse event; AKI = acute kidney injury; CKD = chronic kidney disease; MAKE = major adverse kidney injury; OR = odds ratio; RCT = randomized controlled trial; RRT = renal replacement therapy

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Table S52. Furosemide vs. RRT in adults who are critically ill or undergoing cardiovascular surgery

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Mortality	9 (993) [1] [RCT]	Serious limitations	Consistent	Direct	Imprecise	Undetected	None	Low	Summary RR 1.07 (0.79, 1.45) 10 more per 1000 (from 31 fewer to 67 more)	Critical
MAKE90	0								No evidence	Critical
Duration of AKI	0								No evidence	Critical
CKD G5D	0								No evidence	Critical
CKD G3-G5	0								No evidence	Important
Serious AEs	0								No evidence	Critical
Balance of Potential Benefits and Harms: No difference in mortality between interventions								**Certainty of Overall Evidence:** Low		

Abbreviations: AEs = adverse events; AKI = acute kidney disease; CKD = chronic kidney disease; MAKE = major adverse kidney events; RRT = renal replacement therapy; RCT = randomized controlled trial; RR = risk ratio

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Table S53. Prophylactic intravenous amino acids during cardiac surgery in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Sparseness (or Other Considerations)	Summary of Findings		
							Certainty of Evidence	Description of Findings	Importance of Outcome
AKI, any stage	6 (4587) [1] [RCT]	None	Consistent	Direct	Precise	None	High	**RR 0.80 (0.68, 0.95)** 59 fewer per 1000 (from 95 fewer to 15 fewer)	Critical
RRT	4 (4460) [1] [RCT]	None	Consistent	Direct	Imprecise	None	Moderate	OR 0.70 (0.44, 1.19) 4 fewer per 1000 (from 8 fewer to 3 more)	Critical
MAKE90	0*							No evidence	Critical
AKI, stage 3	4 (3734) [2-5] [RCT]	None	Consistent	Direct	Precise	Other*	High	**RR 0.52 (0.35, 0.78)** 8 fewer per 1000 (from 10 fewer to 4 fewer)	Important
Duration of AKI	0							No evidence	Important
Adverse drug reaction	5 (4937) [2,4-7] [RCT]	None	Consistent	Direct	Precise	No events	High	No events observed 0 more per 1000 (from 1 fewer to 1 more)	Critical
Balance of Potential Benefits and Harms: Perioperative i.v. amino acids reduces the risk of AKI (any stage) and stage 3 AKI, but without evidence of a difference in RRT. Adverse drug reactions are very rare.							Certainty of Overall Evidence: High		

Abbreviations: AKI = acute kidney injury; i.v. = intravenous; MAKE: major adverse kidney events; RR = risk ratio; RCT = randomized controlled trial; RRT = renal replacement therapy

* One trial reported MAKE90,[8] but the reported n/N data and RR (0.87; 95% CI 0.73–1.02) were highly discordant and could not be reconciled with plausible alternative data.

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Table S54. Remote ischemic preconditioning with no propofol anesthesia in adults undergoing cardiac surgery

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Mortality, all-cause, 30 or 90 d	7 (1029) [1-7] [RCT]	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	RR 1.01 (0.65, 1.55)* 0 more per 1000 (from 20 fewer to 32 more)	Critical
AKI, any stage	8 (1003) [1-4,6,8-10] [RCT]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	**RR 0.71 (0.59, 0.85)** 103 fewer per 1000 (from 145 fewer to 52 fewer)	Critical
RRT	6 (806) [1-3,5,8,10] [RCT]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	**RR 0.50 (0.28, 0.89)** 42 fewer per 1000 (from 61 fewer to 9 fewer)	Critical
MAKE90	2 (420) [3,7] [RCT]	Very serious	Inconsistent	Direct	Imprecise	Undetected	None	Very low	RR 0.75 (0.45, 1.23) 68 fewer per 1000 (from 147 fewer to 63 more)	Critical
AKI, stages 2-3	3 (664) [2-3,8] [RCT]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	**RR 0.59 (0.40, 0.85)** 79 fewer per 1000 (from 114 fewer to 29 fewer)	Important
AKI, stage 3	3 (664) [2-3,8] [RCT]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	**RR 0.46 (0.25, 0.85)** 52 fewer per 1000 (from 72 fewer to 15 fewer)	Important
Adverse events, treatment-related†	1 (180) [3] [RCT]	Serious	N/A	Direct	Imprecise	Undetected	Sparse, incomplete reporting	Very low	0 events	Critical
Balance of Potential Benefits and Harms: RIPC (without propofol anesthesia) reduces the risk of AKI-related outcomes, without reported treatment-related adverse events								**Certainty of Overall Evidence:** Moderate		

Abbreviations: RIPC = remote ischemic preconditioning; AKI = acute kidney injury; d = day; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy; MAKE: major adverse kidney events, N/A = not applicable

* Findings at 30 and 90 days
† Residual adverse effects on corresponding limb, petechiae.

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Table S55. Remote ischemic preconditioning with no propofol anesthesia in children undergoing cardiac surgery

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI	1 (84) [RCT]	Very serious	N/A	Direct	Imprecise	Undetected	Sparse	Very low	RR 0.66 (0.40, 1.08) P = 0.10 183 fewer per 1000 (from 393 fewer to 27 more)	**Critical**
AKI Stage 3	0								No evidence	**Critical**
Mortality	1 (84) [RCT]	Very serious	N/A	Direct	Imprecise	Undetected	Sparse	Very low	RD 4.4 (-1.6, 10.5)	**Critical**
RRT	0								No evidence	**Critical**
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
AE	1 (84) [RCT]	Very serious	N/A	Direct	Imprecise	Undetected	Sparse	Very low	No AEs reported	Important
Balance of Potential Benefits and Harms: Remote ischemic preconditioning (without propofol anesthesia) may reduce incident AKI in children undergoing cardiac surgery. Effects on mortality and adverse events are unclear								**Certainty of Overall Evidence:** Very low		

Abbreviations: AE = adverse event; AKI = acute kidney injury; CKD = chronic kidney disease; N/A = not applicable; RCT = randomized clinical trial; RD = absolute risk difference; RR = risk ratio; RRT = renal replacement therapy; RIPC = remote ischemic preconditioning

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Table S56. Terlipressin vs. placebo for hepatorenal syndrome in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	11 (1559) [1-2] [RCT]	Serious	Inconsistent	Direct	Precise	Serious*	None	Very low	RR 0.92 (0.82, 1.04) 42 fewer per 1000 (from 94 fewer to 21 more)	**Critical**
Reversal of HRS (AKI recovery)	12 (1718) [1-4] [RCT]	Serious	Consistent	Direct	Precise	Serious†	Large effect	Moderate	**RR 2.09 (1.79, 2.45)** 216 more per 1000 (from 156 more to 287 more)	**Critical**
RRT	2 (649) [2-3] [RCT]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	**RR 0.77 (0.63, 0.95)** 93 fewer per 1000 (from 149 fewer to 20 fewer)	**Critical**
MAKE 90	0								No evidence	**Critical**
Serious adverse events [incl cardiac, GI, resp events]	7 (809) [3,5] [RCT]	Serious	Inconsistent	Direct	Precise	Undetected	Large effect	Moderate	**OR 2.28 (1.58, 3.29)** 805 more per 1000 (from 365 more to 1000)	**Critical**
SAEs: Ischemic	5 (669) [6] [RCT]	Serious	Consistent	Direct	Precise	Undetected	Large effect	High	**RR 2.05 (1.30, 3.25)** 77 more per 1000 (from 22 more to 166 more)	
SAEs: Cardiovascular	4 (288) [5] [RCT]	Serious	Consistent	Direct	Precise	Undetected	Large effect	High	**OR 6.2 (1.5, 25.0)** [CR data missing]	
SAEs: Gastrointestinal	4 (577) [5] [RCT]	Serious	Consistent	Direct	Precise	Undetected	Large effect	High	**OR 4.4 (2.0, 9.8)** [CR data missing]	
SAEs: Respiratory	5 (690) [5] [RCT]	Serious	Consistent	Direct	Precise	Undetected	Large effect	High	**OR 2.7 (1.3, 5.7)** [CR data missing]	
SAEs: SB ischemia	3 (531) [5] [RCT]	Serious	Consistent	Direct	Very imprecise	Undetected	None	Low	OR 4.8 (0.8, 27.5) [CR data missing]	
SAEs: Skin ischemia	2 (243) [5] [RCT]	Serious	Consistent	Direct	Very imprecise	Undetected	None	Low	OR 5.8 (0.7, 51.0) [CR data missing]	
Discontinuation due to AE	1 (90) [3] [RCT	Serious	N/A	Direct	Precise	Undetected	Sparse, but large effect	Low	**RD 18.2 (8.0, 28.4)** 182 more per 1000 (from 80 more to 284 more)	**Critical**
Duration of AKI	0								No evidence	Important
Quality of life	0								No evidence	Important
Balance of Potential Benefits and Harms: Terlipressin increases HRS reversal and decreases risk of RRT, but with a large increase in serious adverse events								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AE = adverse events; AKI = acute kidney injury; HRS = hepatorenal syndrome, RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy; SB = small bowel.

* Per systematic review by Pitre et al. 2022 (35777925): Egger P = 0.0013. The funnel plot suggests that trial with RR >1 (higher death with terlipressin) are missing.
† Based on funnel plot performed by the Evidence Review Team: Egger P = 0.006. The funnel plot suggests that trial with RR <1 (failure to reverse HRS with terlipressin) are missing.

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Table S57. Norepinephrine vs. placebo for hepatorenal syndrome in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	23 (1604)* [1] [RCT]	Serious	Consistent	Indirect*	Precise	Undetected	None	Low	RR 0.95 (0.77, 1.18) 28 fewer per 1000 (from 146 fewer to 114 more)	**Critical**
Reversal of HRS (AKI recovery)	24 (1632)* [1] [RCT]	Serious	Consistent	Indirect*	Precise	Undetected	None	Low	**RR 1.86 (1.40, 2.47)** 113 more per 1000 (from 53 more to 192 more)	**Critical**
RRT	0								No evidence	**Critical**
MAKE (at ≥90 d)	0								No evidence	**Critical**
Serious adverse events	16 (1278) [1]	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	RR 0.76 (0.32, 1.83) 40 fewer per 1000 (from 116 fewer to 141 more)	**Critical**
Discontinuation due to AE	0								No evidence	**Critical**
Duration of AKI	0								No evidence	Important
Quality of life	0								No evidence	Important
Balance of Potential Benefits and Harms: Norepinephrine increases HRS reversal, without evidence of effect on mortality or adverse events								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = acute kidney injury; HRS = hepatorenal syndrome, RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

* N of the whole population in the NMA. No RCTs directly compared norepinephrine vs. placebo.

References

1. Pitre T, Kiflen M, Helmeczi W, et al. The Comparative Effectiveness of Vasoactive Treatments for Hepatorenal Syndrome: A Systematic Review and Network Meta-Analysis. *Crit Care Med.* 2022;50(10):1419-1429. doi:10.1097/CCM.0000000000005595

Table S58. Midodrine + octreotide vs. placebo for hepatorenal syndrome in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	23 (1604)* [1] [RCT]	Serious	Consistent	Indirect	Imprecise	Undetected	None	Very low	RR 1.16 (0.78, 1.72) 99 more per 1000 (from 136 fewer to 447 more)	**Critical**
Reversal of HRS	24 (1632)* [2] [RCT]	Serious	Consistent	Indirect	Imprecise†	Undetected	None	Very low	RR 1.52 (0.98, 2.35) 68 more per 1000 (from 3 more to 177 more)	**Critical**
RRT	0								No evidence	**Critical**
MAKE (at ≥90 d)	0								No evidence	**Critical**
Serious adverse events (Stage ≥3)	16 (1278) [1] [RCT]	Serious	Consistent	Direct	Very imprecise	Undetected	None	Very Low	RR 1.14 (0.42, 3.11) 24 more per 1000 (from 99 fewer to 359 more)	**Critical**
Discontinuation due to AE	0								No evidence	**Critical**
Duration of AKI	0								No evidence	Important
Quality of life	0								No evidence	Important
Balance of Potential Benefits and Harms: Midodrine + octreotide may increase the likelihood of reversing HRS, with no evidence of an effect on death or adverse events								**Certainty of Overall Evidence:** Very Low		

Abbreviations: AKI = acute kidney injury; HRS = hepatorenal syndrome, RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

* N of the whole population in the NMA. No RCTs directly compared midodrine + octreotide vs. placebo.
† Imprecise if consider the effect size to be a consistent with a clinical effect (P = 0.061).

References

1. Pitre T, Kiflen M, Helmeczi W, et al. The Comparative Effectiveness of Vasoactive Treatments for Hepatorenal Syndrome: A Systematic Review and Network Meta-Analysis. *Crit Care Med.* 2022;50(10):1419-1429. doi:10.1097/CCM.0000000000005595

Table S59. Norepinephrine vs. terlipressin hepatorenal syndrome in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	9 (486) [1] [RCT]	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	HR 1.33 (0.87, 2.00) 170 more per 1000 (from 68 fewer to 483 more)	**Critical**
Reversal of HRS (AKI recovery)	10 (518) [2] [RCT]	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	RR 0.89 (0.74, 1.06) 30 fewer per 1000 (from 83 fewer to 15 more)	**Critical**
RRT	0								No evidence	**Critical**
MAKE (at ≥90 d)	0								No evidence	**Critical**
Serious adverse events	6 (NR) [2] [RCT]	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	RR 0.68 (0.29, 1.60) [CR data missing]	**Critical**
Ischemic adverse events	5 (293) [3] [RCT]	Serious	Consistent	Direct	Precise	Undetected	Large effect size	High	**RR 3.33 (1.47, 7.53)** [Terlipressin vs. norepinephrine] 95 more per 1000 (w/terlipressin) (from 19 more to 267 more)	
Discontinuation due to AE	2 (NR) [4] [RCT]	Serious	Inconsistent	Direct	Very imprecise	Undetected	None	Very Low	OR 0.53 (0.11, 2.63) [CR data missing]	**Critical**
Duration of AKI	0								No evidence	Important
Quality of life	0								No evidence	Important
Balance of Potential Benefits and Harms: There is no evidence of a difference between treatment of HRS with terlipressin or norepinephrine on AKI recovery or death, but increased risk of ischemic adverse events with terlipressin								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = acute kidney injury; AE = adverse events, HRS = hepatorenal syndrome, RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

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Table S60. Midodrine + octreotide vs. terlipressin for hepatorenal syndrome in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Death	1 (48) [1] [RCT]	Serious	Consistent [in NMA)	Direct	Imprecise*	Undetected	Sparse	Low	RR 1.37 (0.95, 1.97) 73 more per 1000 (from 12 fewer to 198 more)	**Critical**
Reversal of HRS (AKI recovery)	1 (48) [1] [RCT]	Serious	Inconsistent [w/indirect in NMA]	Direct	Precise	Undetected	Sparse	Very low	**RR 0.41 (0.19, 0.86)** 415 fewer per 1000 (from 563 fewer to 99 fewer) †	**Critical**
RRT	0								No evidence	**Critical**
MAKE (at ≥90 d)	0								No evidence	**Critical**
Serious adverse events	1 (48) [1] [RCT]	Serious	Consistent [in NMA)	Direct	Very imprecise	Undetected	Sparse	Very Low	RR 1.01 (0.38, 2.73) 3 more per 1000 (from 161 fewer to 449 more)	**Critical**
Discontinuation due to AE	0								No evidence	**Critical**
Duration of AKI	0								No evidence	Important
Quality of life	0								No evidence	Important
Balance of Potential Benefits and Harms: Terlipressin may be more effective to reverse HRS than midodrine + octreotide, with no evidence of difference in death rate or serious adverse events								**Certainty of Overall Evidence:** Very Low		

* Imprecise if consider the effect size to be a consistent with a clinical effect
† Network RR nonsignificant: RR 0.72 (0.50, 1.06)

Abbreviations: AKI = acute kidney injury; HRS = hepatorenal syndrome, RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

References

1. Pitre T, Kiflen M, Helmeczi W, et al. The Comparative Effectiveness of Vasoactive Treatments for Hepatorenal Syndrome: A Systematic Review and Network Meta-Analysis. *Crit Care Med.* 2022;50(10):1419-1429. doi:10.1097/CCM.0000000000005595

Table S61. Midodrine + octreotide vs. norepinephrine for hepatorenal syndrome in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	3 (133) [1-2] [RCT]	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	RR 1.06 (0.78, 1.44) 24 more per 1000 (from 88 fewer to 176 more)	**Critical**
Reversal of HRS	3 (133) [1-2] [RCT]	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	RR 0.78 (0.56, 1.10) 62 fewer per 1000 (from 123 fewer to 28 more)	**Critical**
RRT	0								No evidence	**Critical**
MAKE (at ≥90 d)	0								No evidence	**Critical**
Serious adverse events (Stage ≥3)	2 (83) [1] [RCT]	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	NMA RR 1.49 (0.89, 2.50) 64 more per 1000 (from 14 fewer to 194 more)	**Critical**
Discontinuation due to AE	0								No evidence	**Critical**
Duration of AKI	0								No evidence	Important
Quality of life	0								No evidence	Important
Balance of Potential Benefits and Harms: No evidence of differences in AKI or other critical outcomes, including serious adverse events.								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = acute kidney injury; HRS = hepatorenal syndrome, RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

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Table S62. Rasburicase (urate oxidase) vs. allopurinol in tumor lysis syndrome in children

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	4 (458)[1] [1 RCT, 3 NRCS]	Very serious	Consistent	Direct	Imprecise	Undetected	None	Very Low	RR 0.19 (0.02, 1.50) 18 fewer per 1000 (from 22 fewer to 11 more)	Critical
Death due to TLS	4 (448)[1] [1 RCT, 3 NRCS]	Very serious	Consistent	Direct	Precise	Undetected	Very strong effect	Moderate	**RR 0.05 (0.00, 0.90)** 44 fewer per 1000 (from 46 fewer to 5 fewer)	Critical
Resolution of TLS	0								No evidence	Critical
Incident AKI	0								No evidence	Critical
RRT	6 (1040)[1] [1 RCT, 5 NRCS]	Very serious	Consistent	Direct	Precise	Undetected	None	Low[1]	**RR 0.27 (0.10, 0.76)** 82 fewer per 1000 (from 102 fewer to 27 fewer)	Critical
MAKE90	0								No evidence	Critical
CKD G5D	0								No evidence	Critical
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
Serious AE	4 (397)[1-5] [1 RCT, 3 NRCS]	Very serious	Consistent	Direct	Imprecise	Undetected	None	Very low	6/174 (3.4%) vs. 2/184 (1.1%)* RR 3.17 (0.65, 15.51) 24 more per 1000 (from 9 fewer to 56 more)	Critical
Discontinuation due to AE	0								No evidence	Important
Balance of Potential Benefits and Harms: Rasburicase greatly reduces death due to TLS and risk of RRT compared with allopurinol, and a potentially large, though imprecise, effect on overall death, However, rasburicase has a potentially large, though imprecise higher risk of serious adverse events compared with allopurinol.								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AE = adverse effect; AKI = acute kidney injury; CCT = clinical controlled trial, RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy; TLS = tumor lysis syndrome; MAKE = major adverse kidney event; CKD = chronic kidney disease

* Serious AEs on rasburicase included 1 disseminated intravascular coagulopathy and 5 grade 3 (severe) allergies (in one study). In addition, one child had methemoglobinemia, which resolved without treatment. The child was subsequently found to have glucose-6-phosphate dehydrogenase deficiency. Serious AEs on allopurinol included 1 sepsis and 1 intracerebral hemorrhage (both in a single study, which did not clearly identify these as adverse events, per se.

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Table S63. Theophylline vs. control in neonates with perinatal asphyxia

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Publication Bias	Precision of the Estimate	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	5 (356) [1] [RCT]	No limitations	Consistent	Direct	Undetected	Imprecise	None	Low	OR 0.86 (0.46 – 1.62) 17 fewer per 1000 (from 71 fewer to 69 more)	Critical
AKI, Stage 3	0								No evidence	Critical
AKI, incident	7 (458) [1] [RCT]	No limitations	Consistent	Direct	Undetected	Precise	None	High	**OR 0.24 (0.16 – 0.36)** 314 fewer per 1000 (from 372 fewer to 239 fewer)	(Not prioritized)
RRT	0								No evidence	Critical
MAKE (at ≥90 d)	0								No evidence	Critical
Hospital duration	0								No evidence	Important
Serious AEs (Stage ≥3)	0								No evidence	Critical
Discontinuation due to AEs	0								No evidence	Critical
Balance of Potential Benefits and Harms: Theophylline greatly reduces risk of AKI, but not death, in neonates with asphyxia								Certainty of Overall Evidence: High		

Abbreviations: AEs = adverse events; AKI = acute kidney injury; MAKE: Major adverse kidney events; RCT = randomized controlled trial; OR = odds ratio; RRT = renal replacement therapy

References

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Table S64. Nitric oxide vs. placebo in children undergoing cardiopulmonary bypass[a]

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death, all-cause, ≤42 days	5 (1686) [2-3, 5-7] [RCT]	No limitations	Consistent	Direct	Imprecise	Undetected	Sparse	Low	RR 1.11 (0.53, 2.34)	Critical
AKI, incident, 48 hr	1 (1364) [7] [RCT]	No limitations	Single study	Direct	Precise	Undetected	None	Low	RR 1.16 (0.88, 1.53)	Critical
AKI, stage 3	0								No evidence	Critical
RRT[b], post-op	1 (1364) [7] [RCT]	No limitations	Single study	Direct	Precise	Undetected	Sparse	Low	RR 0.94 (0.71, 1.25)[b]	Critical
MAKE90	0								No evidence	Critical
LOS, days, ICU	4 (1675) [1,3-4,7] [RCT]	No limitations	Consistent	Direct	Precise	Undetected	None	Moderate	MedDiff -0.55 (-1.21, 0.11]	Important
LOS, days, hospital	6 (1739) [1-4, 6-7] [RCT]	No limitations	Consistent	Direct	Precise	Undetected	None	Moderate	MedDiff -0.31 [-1.23, 0.61]	Important
Serious AE, post-op	1 (40) [RCT][6]	No limitations	Single study	Direct	Imprecise	Undetected	None	Very low	RR 3.67 (0.42, 32.30)	Critical
Discontinuation due to AE	0								No evidence	Important
Balance of Potential Benefits and Harms: Nitric oxide for children with cardiopulmonary bypass surgery does not affect the risk of AKI, RRT, death, or length of stay, with imprecise estimates of serious adverse events.								**Certainty of Overall Evidence:** Low		

Abbreviations: AE = adverse events; AKI = acute kidney injury; ICU = intensive care unit; MAKE90 = major adverse kidney events by 90 days; LOS = length of stay; MedDiff = med difference; post-op = post-operative, RCT = randomized controlled trial; RR = relative risk; RRT = renal replacement therapy.

[a] All children underwent cardiopulmonary bypass (CPB). Two studies included all children undergoing CPB. Other studies specified surgeries, including tetralogy of Fallot; corrective heart surgery with high pulmonary flow or pressure, or congenital heart lesions; hypoplastic left heart syndrome or variant; Fontan surgery; and open congenital heart disease surgery.
[b] An additional study (James 2016) reported peritoneal dialysis, which was commonly performed for reasons other than AKI (RR 0.92, 95% CI 0.56, 1.52).

References

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Table S65. Vancomycin infusion: continuous vs. intermittent infusion in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Publication Bias	Precision of the Estimate	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Mortality	9 (1973) [1-2] [1 RCT 8 NRCS]	Serious	Consistent	Direct	Undetected	Precise	None	Moderate	OR: 1.07 (95% CI 0.83–1.39**)** 7 more per 1000 (from 17 fewer to 38 more)	**Critical**
Mortality, drug related	0								No evidence	**Critical**
Incident AKI	13 (2292) [1-3] [2 RCT 11 NRCS]	Serious	Consistent	Direct	Undetected	Precise	None	Moderate	**OR 0.47 (95% CI 0.34–0.64)** 52 fewer per 1000 (from 65 fewer to 35 fewer)	**Critical**
AKI requiring RRT	0								No evidence	**Critical**
MAKE	0								No evidence	**Critical**
Progression of AKI	0								No evidence	Important
Recovery of kidney function	0								No evidence	Important
Duration of AKI	0								No evidence	Important
Balance of Potential Benefits and Harms: Continuous infusion prevents AKI compared with intermittent infusion, but does not affect mortality								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = acute kidney injury; ICU = intensive care unit; MAKE = major adverse kidney event; N/A = not applicable; NRCS = nonrandomized comparative study; OR = odds ratio; RCT = randomized controlled trial; RRT = renal replacement therapy

References

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2. Maluangnon C, Tongyoo S, Permpikul C. Continuous Vancomycin Infusion versus Intermittent Infusion in Critically Ill Patients. Infect Drug Resist. 2022 Dec 28;15:7751-7760. doi: 10.2147/IDR.S395385. PMID: 36597455; PMCID: PMC9805718.
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Table S66. Vancomycin dosing: AUC vs. trough dosing in adults*

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Publication Bias	Precision of the Estimate	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Mortality	1 (517) [1] [NRCS]	Very serious	N/A	Direct	Undetected	Very imprecise	Sparse	Very low	RR 1.06 (95% CI 0.55, 2.00) 36 more per 1000 (from 324 fewer to 720 more)	**Critical**
Mortality, drug related	0								No evidence	**Critical**
Incident AKI	12 (7548) [1-3] [12 NRCS]	Very serious	Consistent	Direct	Serious*	Precise	None	Low	**RR 0.68 (95% CI 0.59; 0.77)** 73 fewer per 1000 (from 94 fewer to 49 fewer)	**Critical**
AKI requiring RRT	0								No evidence	**Critical**
MAKE	0								No evidence	**Critical**
Progression of AKI	0								No evidence	Important
Recovery of kidney function	0								No evidence	Important
Duration of AKI	0								No evidence	Important
Balance of Potential Benefits and Harms: AUC-based dosing prevents AKI compared with trough-based dosing, with unclear effects on other outcomes								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = Acute Kidney Injury; AUC = area under the curve; ICU = intensive care unit; MAKE = major adverse kidney event; N/A = not applicable; NRCS = nonrandomized comparative study; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy; controlled trial

* Egger P =0.022. With trim and fill, imputed RR 0.71 (95% CI 0.62, 0.81). Certainty of evidence was not further downgraded since the imputed effect estimate was similar.

References

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2. Abdelmessih E, Patel N, Vekaria J, Crovetto B, SanFilippo S, Adams C, Brunetti L. Vancomycin area under the curve versus trough only guided dosing and the risk of acute kidney injury: Systematic review and meta-analysis. Pharmacotherapy. 2022 Sep;42(9):741-753. doi: 10.1002/phar.2722. Epub 2022 Aug 5. PMID: 35869689; PMCID: PMC9481691.
3. Knight JM, Iso T, Perez KK, et al. Risk of Acute Kidney Injury Based on Vancomycin Target Trough Attainment Strategy: Area-Under-the-Curve-Guided Bayesian Software, Nomogram, or Trough-Guided Dosing. Ann Pharmacother. 2024;58(2):110-117. doi:10.1177/10600280231171373

Table S67. Clinical decision support systems/therapeutic drug monitoring for vancomycin in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Publication Bias	Precision of the Estimate	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Mortality	8 (17959) [1] [RCT]	No serious	Consistent	Direct	Undetected	Precise	None	High	RR: 1.00 (95% CI 0.93–1.07) 0 fewer per 1000 (from 10 fewer to 10 more)	Critical
Mortality, drug related	0								No evidence	Critical
Incident AKI	1 (259) [2] [NRCS]	No serious	N/A	Direct	Undetected	Precise	Sparse, but large	Low	**adjOR 0.63 (95% CI 0.42–0.95)** 265 fewer per 1000 (from 415 fewer to 36 fewer)	Critical
RRT	9 (18355) [1] [RCT]	No serious	Consistent	Direct	Undetected	Imprecise*	None	Moderate	RR: 1.11 (95% CI 0.99–1.24) 7 more per 1000 (from 1 fewer to 14 more)	Critical
MAKE	0								No evidence	Critical
Progression of AKI	4 (14439) [1] [RCT]	No serious	Consistent	Direct	Undetected	Precise	None	High	RR 0.98 (95% CI 0.92–1.05) 4 fewer per 1000 (from 15 fewer to 9 more)	Important
Recovery of kidney function	3 (3135) [1] [RCT]	Serious	Consistent	Direct	Undetected	Precise	None	Moderate	RR 0.99 (95% CI 0.93– 1.06) 5 fewer per 1000 (from 36 fewer to 31 more)	Important
AKI at discharge (Duration of AKI)	4 (3335) [1] [RCT]	Serious	Consistent	Direct	Undetected	Precise	None	Very Low	RR 1.39 (95% CI 0.94–2.05) 112 more per 1000 (from 17 fewer to 301 more)	Important
Balance of Potential Benefits and Harms: Clinical decision support systems (therapeutic drug monitoring) for vancomycin may reduce incidence of AKI but may increase the risk of RRT, with no effect on mortality or other sequelae of AKI								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = acute kidney injury; CI = confidence interval; MAKE = major adverse kidney event; N/A = not applicable; OR = odds ratio; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

* Imprecise assuming a true effect (P = 0.069)

References

1. Altobaishat O, Abouzid M, Amin AM, Bani-Salameh A, Tanashat M, Abdullah Bataineh O, Turkmani M, Abuelazm M, Mohamed MMB. The effect of clinical decision support systems on clinical outcomes in acute kidney injury: a systematic review and meta-analysis of randomized controlled trials. Ren Fail. 2024 Dec;46(2):2400552. doi: 10.1080/0886022X.2024.2400552. Epub 2024 Sep 9. PMID: 39252153; PMCID: PMC11389631.
2. Hambrick HR, Greco KF, Weller E, Ganapathi L, Lehmann LE, Sandora TJ. Impact of decreasing vancomycin exposure on acute kidney injury in stem cell transplant recipients. Infect Control Hosp Epidemiol. 2022;43(10):1375-1381. doi:10.1017/ice.2021.454

Table S68. Pharmacist intervention for vancomycin monitoring in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Publication Bias	Precision of the Estimate	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Mortality	10 (1321) [*,1] [1 RCT, 9 NRCS]	Very serious	Consistent	Direct	Undetected‡	Precise	None	Low	**OR: 0.47 (95% CI 0.31, 0.72)** 71 fewer per 1000 (from 92 fewer to 38 fewer)	Critical
Mortality, drug related	0								No evidence	Critical
Incident AKI	35 (19,434) [† 1,2] [1 RCT, 15 NRCS, 19 pre-post]	Very serious	Consistent	Direct	Undetected‡	Precise	None	Low	**RR 0.59 (95% CI 0.48, 0.72)** 39 fewer per 1000 (from 50 fewer to 15 fewer)	Critical
	2 pediatric (1487) [1] [pre-post]	Very serious	Inconsistent	Direct	Undetected	Very imprecise	None	Very low	OR 0.30 (95% CI 0.04, 2.34) 33 fewer per 1000 (from 45 fewer to 63 more)	
RRT	0								No evidence	Critical
MAKE	0								No evidence	Critical
Progression of AKI	0								No evidence	Important
Recovery of kidney function	0								No evidence	Important
Duration of AKI	0								No evidence	Important
Balance of Potential Benefits and Harms: Pharmacist intervention for vancomycin monitoring reduces the risk of death and AKI in adults. There is insufficient evidence in children.								**Certainty of Overall Evidence:** Low		

Abbreviation: AKI = acute kidney injury; MAKE = major adverse kidney injury; NRCS = nonrandomized comparative study; OR = odds ratio; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

* No pediatric studies.
† 2 of 35 studies (N=1487, combined), reported as a conference abstract, was conducted in children.
‡ Egger's test used to assess publication bias in Kunming 2023 35285970 (mortality Egger P = 0.30, AKI Egger P = 0.21).
\# With unclear denominators.3

References

1. Kunming P, Xiaotian J, Qing X, Chenqi X, Xiaoqiang D, Qian Zhou L. Impact of pharmacist intervention in reducing vancomycin-associated acute kidney injury: A systematic review and meta-analysis. Br J Clin Pharmacol. 2023 Feb;89(2):526-535. doi: 10.1111/bcp.15301. Epub 2022 Apr 2. PMID: 35285970.
2. Truong J, Smith SR, Veillette JJ, Forland SC. Individualized Pharmacokinetic Dosing of Vancomycin Reduces Time to Therapeutic Trough Concentrations in Critically Ill Patients. J Clin Pharmacol. 2018 Sep;58(9):1123-1130. doi: 10.1002/jcph.1273. Epub 2018 Jun 29. PMID: 29957824.
3. Lee P, Mason C, Ahmad-Hossain H, et al. Evaluation of a Pharmacy Directed Vancomycin and Monitoring Pilot Program at an Academic Pediatric Hospital. Open Forum Infect Dis. 2016;3(suppl_1):1923.
4. Olson J, Stockmann C, Hersh AL, et al. Optimizing vancomycin prescribing through a pharmacist driven monitoring intervention at a children's hospital. Open Forum Infect Dis. 2016;3(suppl_1):1924.

Table S69. Antibiotic monitoring, not specific to vancomycin, in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Publication Bias	Precision of the Estimate	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death, in hospital	2 (455) [1,2] [1 RCT, 1 NRCS]	Some limitations	Inconsistent	Direct	Undetected	Mixed	1 study per intervention	Low	Autokinetics RR 0.87 (0.63, 1.21) β-lactam TDM unadjusted OR 0.65 (0.31, 1.36) β-lactam TDM propensity adjusted **OR 0.022 (0.005, 0.094)**	Critical
Death, drug-related	0								No evidence	Critical
AKI	3 (2845) [1-3] [1 RCT, 2 NRCS]	Some limitations	Consistent	Direct	Undetected	Imprecise	1 study per intervention	Low	No significant differences with or without antibiotic monitoring	Critical
RRT	0								No evidence	Critical
MAKE90	0								No evidence	Critical
AKI duration	0								No evidence	Important
Balance of Potential Benefits and Harms: Three approaches to antibiotic monitoring did not affect AKI-related outcomes, with no evidence of harms								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = acute kidney injury; MAKE90 = major adverse kidney injury at 90 days; N/A = not applicable; NRCS = nonrandomized comparative study; OR = odds ratio; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy; TDM = therapeutic drug monitoring

References

1. Roggeveen LF, Guo T, Fleuren LM, Driessen R, Thoral P, van Hest RM, Mathot RAA, Swart EL, de Grooth HJ, van den Bogaard B, Girbes ARJ, Bosman RJ, Elbers PWG. Right dose, right now: bedside, real-time, data-driven, and personalised antibiotic dosing in critically ill patients with sepsis or septic shock-a two-centre randomised clinical trial. Crit Care. 2022 Sep 5;26(1):265. doi: 10.1186/s13054-022-04098-7. PMID: 36064438; PMCID: PMC9443636.
2. Kunz Coyne AJ, Alshaer M, Casapao AM, Venugopalan V, Isache C, Ferreira J, Jankowski CA. Effectiveness and Safety of Beta-Lactam Antibiotics with and without Therapeutic Drug Monitoring in Patients with Pseudomonas aeruginosa Pneumonia or Bloodstream Infection. Antimicrob Agents Chemother. 2022 Oct 18;66(10):e0064622. doi: 10.1128/aac.00646-22. Epub 2022 Sep 8. PMID: 36073943; PMCID: PMC9578394.
3. Cotner SE, Rutter WC, Burgess DR, Wallace KL, Martin CA, Burgess DS. Influence of β-Lactam Infusion Strategy on Acute Kidney Injury. Antimicrob Agents Chemother. 2017 Sep 22;61(10):e00871-17. doi: 10.1128/AAC.00871-17. PMID: 28760891; PMCID: PMC5610533.

Table S70. Conventional vs. prolonged infusion of meropenem in neonates

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Publication Bias	Precision of the Estimate	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	1 (102) [1] [RCT]	Some limitations	N/A	Direct	Undetected	Precise	Sparse	Very low	**RR 0.44 (0.20, 0.97)** 176 fewer (from 335 fewer to 18 fewer)	Critical
Death, drug-related	0								No evidence	Critical
AKI	1 (102) [1] [RCT]	Some limitations	N/A	Direct	Undetected	Precise	Sparse	Very low	**RR 0.25 (0.07, 0.83)** 176 fewer (from 311 fewer to 42 fewer)	Critical
RRT	0								No evidence	Critical
MAKE90	0								No evidence	Critical
AKI duration	0								No evidence	Important
Balance of Potential Benefits and Harms: Slow infusion of meropenem reduces the risk of neonatal death and AKI compared with conventional infusion								**Certainty of Overall Evidence:** Very low		

Abbreviations: AKI = Acute Kidney Injury; MAKE90 = major adverse kidney injury at 90 days; N/A = not applicable; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy.

References

1. Shabaan AE, Nour I, Elsayed Eldegla H, Nasef N, Shouman B, Abdel-Hady H. Conventional Versus Prolonged Infusion of Meropenem in Neonates With Gram-negative Late-onset Sepsis: A Randomized Controlled Trial. Pediatr Infect Dis J. 2017 Apr;36(4):358-363. doi: 10.1097/INF.0000000000001445. PMID: 27918382.

Table S71. Pharmacist intervention for all drugs in adults at risk for AKI

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Publication Bias	Precision of the Estimate	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Mortality	0								No evidence	Critical
Mortality, drug related	0								No evidence	Critical
Incident AKI	1 (150) [1] [RCT]	Some limitations	N/A	Direct	Undetected	Precise	Sparse, but adequately powered	Low	**RR 0.39 (0.19, 0.79)** 187 fewer (from 248 fewer to 64 fewer)	Critical
RRT	0								No evidence	Critical
MAKE	0								No evidence	Critical
Progression of AKI	0								No evidence	Important
Recovery of kidney function	0								No evidence	Important
Duration of AKI	0								No evidence	Important
Balance of Potential Benefits and Harms: Pharmacist intervention for drug monitoring may reduce the risk of AKI in patients in the ICU								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = acute kidney injury; ICU = intensive care unit; MAKE = major adverse kidney event; N/A = not applicable; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

References

1. Polat EC, Koc A, Demirkan K. The role of the clinical pharmacist in the prevention of drug-induced acute kidney injury in the intensive care unit. J Clin Pharm Ther. 2022 Dec;47(12):2287-2294. doi: 10.1111/jcpt.13811. Epub 2022 Nov 16. PMID: 36394173.

Table S72. Prophylactic hydration vs no hydration for contrast-induced nephropathy in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Incident AKI	7 (3258)[1-3] [RCT]	No serious	Consistent	Direct	Precise	Undetected	None	High	**RR 0.64 (0.54–0.76)** 58 fewer per 1000 (from 64 fewer to 43 fewer)	Critical
Mortality	6 (2475)[2]	No serious	Consistent	Direct	Precise	Undetected	None	High	**RR 0.57 (0.33–0.98)** 11 fewer per 1000 (from 17 fewer to 0.5 fewer)	
RRT	3* (1074)[2]	No serious	Consistent	Direct	Imprecise	Undetected	None	Moderate	**RR 0.39 (0.12–1.23)** 5 fewer per 1000 (from 8 fewer to 2 more)	
Duration of AKI	0								No evidence	Important
Balance of Potential Benefits and Harms: Hydration prior to contrast imaging lowers the risk of incident AKI, mortality, and possibly RRT, in people at increased risk of AKI								**Certainty of Overall Evidence:** High		

Abbreviations: AKI = acute kidney injury; SR = systematic review; RR = relative risk; RRT = renal replacement therapy

* 4 additional studies (including Arslan 2021[3]) had no participants who required RRT and are thus excluded from meta-analysis.

References

1. Cai Q, Jing R, Zhang W, Tang Y, Li X, Liu T. Hydration Strategies for Preventing Contrast-Induced Acute Kidney Injury: A Systematic Review and Bayesian Network Meta-Analysis. J Interv Cardiol. 2020 Feb 11;2020:7292675. doi: 10.1155/2020/7292675. PMID: 32116474; PMCID: PMC7036123.
2. Jiang Y, Chen M, Zhang Y, Zhang N, Yang H, Yao J, Zhou Y. Meta-analysis of prophylactic hydration versus no hydration on contrast-induced acute kidney injury. Coron Artery Dis. 2017 Dec;28(8):649-657. doi: 10.1097/MCA.0000000000000514. PMID: 28692484.
3. Arslan S, Yildiz A, Dalgic Y, Batit S, Kilicarslan O, Ser OS, Dalgic SN, Kocas C, Abaci O. Avoiding the emergence of contrast-induced acute kidney injury in acute coronary syndrome: routine hydration treatment. Coron Artery Dis. 2021 Aug 1;32(5):397-402. doi: 10.1097/MCA.0000000000000966. PMID: 33060531.

Table S73. Oral hydration vs no hydration for contrast-induced nephropathy in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Incident AKI	0 (Direct) [60 (21,293) in NMA])[1] [RCT]	No serious	N/A	Indirect	Very imprecise	Undetected	No studies	Very low	OR 0.66 (0.22–2.15)* 36 fewer per 1000 (from 82 fewer to 121 more)	Critical
Mortality	Not analyzed								No evidence	Critical
RRT	Not analyzed								No evidence	Critical
Duration of AKI	Not analyzed								No evidence	Important
Balance of Potential Benefits and Harms: No evidence regarding the effect of oral hydration to prevent contrast-induced nephropathy and sequelae, but point estimate suggests possible reduction in incident AKI.								**Certainty of Overall Evidence:** Very low		

Abbreviations: AKI = acute kidney injury; N/A = not applicable; NMA = network meta-analysis; RR = relative risk; RRT = renal replacement therapy

* Based on network meta-analysis of 60 trials that evaluate prophylactic fluid for intravenous contrast.

References

1. Cai Q, Jing R, Zhang W, Tang Y, Li X, Liu T. Hydration Strategies for Preventing Contrast-Induced Acute Kidney Injury: A Systematic Review and Bayesian Network Meta-Analysis. J Interv Cardiol. 2020 Feb 11;2020:7292675. doi: 10.1155/2020/7292675. PMID: 32116474; PMCID: PMC7036123.

Table S74. Intravenous hypotonic (0.45%) saline vs no hydration for contrast-induced nephropathy in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Incident AKI	0 (Direct) [60 (21,293) in NMA])[1] [RCT]	No serious	N/A	Indirect	Imprecise	Undetected	No studies	Very low	OR 1.72 (0.47–5.56)* 75 more per 1000 (from 56 fewer to 478 more)	Critical
Mortality	Not analyzed								No evidence	Critical
RRT	Not analyzed								No evidence	Critical
Duration of AKI	Not analyzed								No evidence	Important
Balance of Potential Benefits and Harms: No evidence regarding the effect of i.v. 0.45% saline to prevent contrast-induced nephropathy and sequelae, but point estimate suggests possible *increase* in incident AKI.								**Certainty of Overall Evidence:** Very low		

Abbreviations: AKI = acute kidney injury; i.v. = intravenous; N/A = not applicable; NMA = network meta-analysis; RR = relative risk; RRT = renal replacement therapy

* Based on network meta-analysis of 60 trials that evaluate prophylactic fluid for intravenous contrast.

References

1. Cai Q, Jing R, Zhang W, Tang Y, Li X, Liu T. Hydration Strategies for Preventing Contrast-Induced Acute Kidney Injury: A Systematic Review and Bayesian Network Meta-Analysis. J Interv Cardiol. 2020 Feb 11;2020:7292675. doi: 10.1155/2020/7292675. PMID: 32116474; PMCID: PMC7036123.

Table S75. Intravenous isotonic (0.9%) saline versus hypotonic (0.45%) saline for contrast-induced nephropathy in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI	2 (1808)[1] [RCT]	Some serious	Consistent	Direct	Imprecise	Undetected	None	Low	OR 0.32 (0.09–1.02)* 71 fewer per 1000 (from 95 fewer to 2 more)	Critical
Mortality	Not analyzed								No evidence	Critical
RRT	Not analyzed								No evidence	Critical
Duration of AKI	Not analyzed								No evidence	Important

Balance of Potential Benefits and Harms:	**Certainty of Overall Evidence:**
Prophylactic intravenous normal saline (0.9%) may reduce the incidence of AKI after intravascular contrast compared with half-normal (0.45%) saline.	Low

Abbreviations: AKI = acute kidney injury; N/A = not applicable; NMA = network meta-analysis; RR = relative risk; RRT = renal replacement therapy

* Based on network meta-analysis of 60 trials that evaluate prophylactic fluid for intravenous contrast.

References

1. Cai Q, Jing R, Zhang W, Tang Y, Li X, Liu T. Hydration Strategies for Preventing Contrast-Induced Acute Kidney Injury: A Systematic Review and Bayesian Network Meta-Analysis. J Interv Cardiol. 2020 Feb 11;2020:7292675. doi: 10.1155/2020/7292675. PMID: 32116474; PMCID: PMC7036123.

Table S76. Intravenous sodium bicarbonate versus hypotonic (0.45%) saline for contrast-induced nephropathy in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI	1 (72)[1] [RCT]	Some serious	N/A	Direct	Precise	Undetected	Sparse	Low	OR 0.24 (0.10–0.51)* 80 fewer per 1000 (from 94 fewer to 51 fewer)	Critical
Mortality	Not analyzed								No evidence	Critical
RRT	Not analyzed								No evidence	Critical
Duration of AKI	Not analyzed								No evidence	Important
Balance of Potential Benefits and Harms: Prophylactic intravenous sodium bicarbonate reduces the incidence of AKI after intravascular contrast compared with half-normal (0.45%) saline.								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = acute kidney injury; N/A = not applicable; NMA = network meta-analysis; RR = relative risk; RRT = renal replacement therapy

* Based on network meta-analysis of 60 trials that evaluate prophylactic fluid for intravenous contrast.

References

1. Cai Q, Jing R, Zhang W, Tang Y, Li X, Liu T. Hydration Strategies for Preventing Contrast-Induced Acute Kidney Injury: A Systematic Review and Bayesian Network Meta-Analysis. J Interv Cardiol. 2020 Feb 11;2020:7292675. doi: 10.1155/2020/7292675. PMID: 32116474; PMCID: PMC7036123.

Table S77. Sodium bicarbonate vs. sodium chloride for prevention of contrast-induced AKI in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI	33 (13135)[1-31] [RCT]	All studies: Some limitations	Inconsistent	Direct	Precise	Undetected	None	Low	**RR 0.81 (0.66, 0.99)** 22 fewer per 1000 (from 38 fewer to 1 fewer)	Critical
	9 (2816)[1,4,5,7,11,12,15,17,31] [RCT]	Subset with no limitations	Consistent	Direct	Imprecise [a]	Undetected	None	Moderate	RR 0.85 (0.67, 1.06) 16 fewer per 1000 (from 36 fewer to 7 more)	
	9 (7264)[3,9,13,20,21,26,27,29] [RCT]	Subset with some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	RR 1.07 (0.94, 1.22) 8 more per 1000 (from 6 fewer to 24 more)	
	18 (10,080)[1,3-5,7,9,11-13,15,17,20,21,26,27,29,31] [RCT] [b]	Subset with no or some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	RR 0.99 (0.88, 1.13) 1 fewer per 1000 (from 13 fewer to 14 more)	
	15 (3055)[2,6,8,10,14,16,18,19,22-25,28,30] [RCT]	Subset with serious limitations	Consistent	Direct	Precise	Undetected	None	Low	**RR 0.55 (0.34, 0.90)** 58 fewer per 1000 (from 85 fewer to 13 fewer)	
Duration of AKI	0								No evidence	Important
RRT	25 (11,043)[1-5,7,9-16,18-20,22,24-29] [RCT] [b]	Some limitations	Consistent	Direct	Imprecise	Undetected	Rare events	Very low	RR 1.14 (0.80, 1.63) 0 more per 1000 (from 2 fewer to 7 more)	Critical
	7 (2124)[1,4,5,7,11,12,15] [RCT] [b]	Subset with no limitations	Consistent	Direct	Highly imprecise	Undetected	Rare events	Very low	RR 1.02 (0.14, 7.17) 0 more per 1000 (From 2 fewer to 12 more)	
	9 (7051)[3,9,13,20,26,27,29] [RCT] [b]	Subset with some limitations	Consistent	Direct	Imprecise	Undetected	Rare events	Very low	RR 1.23 (0.81, 1.85) 3 more per 1000 (from 2 fewer to 10 more)	
	16 (9175)[1,3-5,7,9,11-13,15,20,26,27,29] [RCT] [b]	Subset with no or some limitations	Consistent	Direct	Imprecise	Undetected	Rare events	Very low	RR 1.21 (0.81, 1.82) 2 more per 1000 (from 1 fewer to 8 more)	
	9 (1868)[2,10,14,16,18,19,22,24,25,28] [RCT] [b]	Subset with serious limitations	Consistent	Direct	Imprecise	Undetected	Rare events	Very low	RR 0.91 (0.42, 1.93) 1 fewer per 1000 (from 9 fewer to 15 more)	
Balance of Potential Benefits and Harms: In people at increased risk of AKI who are undergoing intravascular contrast imaging, the risk of incident AKI, and possibly RRT, is similar with prophylactic hydration with either intravenous isotonic sodium bicarbonate or sodium chloride solutions.[c]								**Certainty of Overall Evidence:** Moderate		

[a] Similar finding with, but with less power than, overall analysis. Imprecise for the conclusion of a clinically significant effect.
[b] Any risk of bias, 12 of 25 studies had no events; Low risk of bias: 5/7 studies had no events; Moderate risk of bias: 3/9 studies had no events; High risk of bias: 4/9 studies had no events.
[c] Prioritizing low and moderate risk of bias study findings.

Abbreviations: AKI = acute kidney injury; RR = relative risk; RRT, renal replacement therapy.

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Table S78. RASi discontinuation or hold in adults undergoing cardiac procedures with intravascular contrast

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Publication Bias	Precision of the Estimate	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Death, All cause	2 (428)[1,2] [RCT]	Serious	Consistent	Direct	Undetected	Highly imprecise	Rare events	Very Low	RR N/A* 9 fewer per 1000 (from 25 fewer to 6 more)	Critical
AKI, incident	4 (2042)[1-4] [RCT]	Very serious	Consistent	Direct	Undetected	Imprecise	None	Low	OR 0.76 (0.48, 1.19) 12 fewer per 1000 (from 27 fewer to 10 more)	Critical
AKI, stage 3	0								No evidence	Critical
MAKE90	0								No evidence	Critical
RRT	3 (1728)[1-3] [RCT]	Very serious	Consistent	Direct	Undetected	Highly imprecise	Rare events	Very low	RR N/A* 1 more per 1000 (from 2 fewer to 4 more)	Critical
Duration of AKI	0								No evidence	Important
Hospital admission, unplanned	1 (208)[1] [RCT]	Serious	N/A	Direct	Undetected	Imprecise	Sparse, rare events	Very Low	RR N/A* 29 fewer per 1000 (from 67 fewer to 8 more)	Important
Poor control of underlying conditions, in hospital:										Critical
Hypertension requiring antihypertensive	1 (61)[4] [RCT]	Serious	N/A	Direct	Undetected	Highly imprecise	Sparse, rare events	Very low	RR N/A* 67 more per 1000 (from 38 fewer to 172 more)	
Myocardial infarction	2 (1508)[1,3] [RCT]	Serious	Consistent	Direct	Undetected	Imprecise	Sparse, rare events	Low	RR 0.91 (0.57, 1.47) † 5 fewer per 1000 (from 28 fewer to 19 more)	
Heart Failure	2 (1508)[1,3] [RCT]	Serious	Consistent	Direct	Undetected	Imprecise	Sparse, rare events	Low	RR 0.93 (0.45, 1.92) † 2 fewer per 1000 (from 18 fewer to 15 more)	
Stroke	1 (208)[1] [RCT]	Serious	N/A	Direct	Undetected	Highly imprecise	Sparse, rare events	Very low	RR N/A* 10 fewer per 1000 (from 36 fewer to 17 more)	

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Publication Bias	Precision of the Estimate	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	**Description of Findings**	**Importance of Outcome**
Unstable angina, in-hospital	1 (1300)[3] [RCT]	Some limitations	N/A	Direct	Undetected	Imprecise	Sparse	Very low	RR 1.18 (0.68, 2.06) 6 more per 1000 (from 15 fewer to 27 more)	
MACE, in-hospital (cardiac death, nonfatal MI, UA, HF)	1 (1300)[3] [RCT]	Some limitations	N/A	Direct	Undetected	Imprecise	Sparse	Very low	RR 1.04 (0.73, 1.47) 3 more per 1000 (from 28 fewer to 34 more)	
Balance of Potential Benefits and Harms: No significant differences in potential benefits and harms between stopping or maintaining RASi for people undergoing cardiac procedures with intravascular contrast								Certainty of Overall Evidence: Low		

Abbreviations: AKI = acute kidney injury; HF = heart failure; HR = hazard ratio; MACE = major adverse cardiovascular events; N/A = not applicable; OR = odds ratio; RASi = renin-angiotensin system inhibitors; RCT = randomized controlled trial; RD= risk difference; RR = risk ratio; RRT = renal replacement therapy; UA = unstable angina.

* 0 vs. 1-3 events across studies
† Study with 0 events omitted.

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Table S79. RASi discontinuation for noncardiac surgery in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death, All cause	2 (2469) [1] [RCT]	No serious	Consistent	Direct	**Imprecise**	**Undetected**	None	Moderate	OR 0.85 (0.39, 1.86) 2 fewer per 1000 (from 7 fewer to 10 more)	Critical
AKI, incident	2 (2476) [1] [RCT]	No serious	Consistent	Direct	Imprecise	Undetected	None	Moderate	OR 1.01 (0.78, 1.30) 1 fewer per 1000 (from 24 fewer to 23 more)	Critical
AKI, stage 3	0								No evidence	Critical
MAKE90	0								No evidence	Critical
RRT	0								No evidence	Critical
Duration of AKI	0								No evidence	Important
Hospital admission, unplanned	0								No evidence	Important
Poor control of underlying condition:										Critical
Hypertension	6 (3365) [1] [RCT]	Serious	Inconsistent	Direct	Precise	Undetected	None	Low	OR 1.41 (0.94, 2.10) 34 more per 1000 (from 5 fewer to 92 more)	
Myocardial infarction	2 (2469) [1] [RCT]	No limitations	Inconsistent	Direct	Highly Imprecise	Undetected	None	Very low	OR 1.68 (0.30, 9.30) 2 more per 1000 (from 2 fewer to 27 more)	
MACE	1 (2222) [1] [RCT]	No limitations	N/A	Direct	Imprecise	Undetected	Sparse, but large	Low	OR 0.99 (95%CI 0.67, 1.47) 0 more per 1000 (from 16 fewer to 22 more)	
Myocardial injury	1 (241) [1] [RCT]	No limitations	N/A	Direct	Imprecise	Undetected	Sparse	Very low	OR 1.33 (0.80, 2.21) 136 more per 1000 (from 83 fewer to 500 more)	
Stroke	2 (2469) [1] [RCT]	No limitations	Consistent	Direct	Highly Imprecise	Undetected	None	Very low	OR 1.24 (0.30, 5.19) 1 more per 1000 (from 2 fewer to 10 more)	
Heart Failure	1 (247) [1] [RCT]	No limitations	N/A	Direct	Highly Imprecise	Undetected	Sparse, rare event	Very low	RR N/A* 16 more per thousand (from 11 fewer to 43 more)	
Balance of Potential Benefits and Harms: Evidence does not find significant differences in potential benefits and harms for stopping RASi for people undergoing noncardiac surgery								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = acute kidney injury; MACE = major adverse cardiovascular events; MAKE = major adverse kidney events; N/A = not applicable; OR = odds ratio; RASi = renin angiotensin system inhibitors; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy.

* 2 vs 0 events across studies.

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Table S80. RASi discontinuation for cardiac surgery in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Publication Bias	Precision of the Estimate	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death, All cause	1 (121) [1] [RCT]	Some limitations	N/A	Direct	Undetected	Highly imprecise	Sparse	Very low	RR 0.98 (0.06–15.37) 0 per 1000 (from 15 fewer to 236 more)	Critical
AKI, incident	1 (121) [1] [RCT]	Some limitations	N/A	Direct	Undetected	Highly imprecise	Sparse	Very low	RR 0.98 (0.06–15.37) 0 per 1000 (from 15 fewer to 236 more)	Critical
AKI, stage 3	0								No evidence	Critical
MAKE90	0								No evidence	Critical
RRT	1 (121) [1] [RCT]	Some limitations	N/A	Direct	Undetected	Highly imprecise	Sparse	Very low	Rare event	Critical
Duration of AKI									No evidence	Important
ICU admission, unplanned	1 (121) [1] [RCT]	Some limitations	N/A	Direct	Undetected	Highly imprecise	Sparse	Very low	AdjOR 0.32 (0.02-2.55) 11 fewer per 1000 (from 16 fewer to 34 more)	Important
Poor control of underlying condition:										Critical
Hypertension	1 (121) [1] [RCT]	Some limitations	N/A	Direct	Undetected	Imprecise	Sparse	Very low	AdjOR 1.78 (0.65 - 4.88) 96 more per 1000 (from 64 fewer to 297 more)	
Myocardial infarction	0								No evidence	
MACE	0								No evidence	
Myocardial injury	0								No evidence	
Stroke	1 (121) [1] [RCT]	Some limitations	N/A	Direct	Undetected	Highly imprecise	Sparse	Very low	0 events	
Heart Failure	1 (121) [1] [RCT]	Some limitations	N/A	Direct	Undetected	Highly imprecise	Sparse	Very low	0 events	
Balance of Potential Benefits and Harms: Very low certainty evidence suggests discontinuation of RASis prior to cardiac surgery may result in no difference in mortality, AKI, unplanned hospital admission, and control of underlying condition (HTN).								**Certainty of Overall Evidence:** Very low		

Abbreviations: AKI = acute kidney injury; ICU = intensive care unit; HTN = hypertension; MACE = major adverse cardiovascular event; MAKE = major adverse kidney event; N/A = not applicable; OR = odds ratio; RASi = renin angiotensin system inhibitors; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy.

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Table S81. Discontinuation of multiple drugs for acute illness in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death, All cause	0								No evidence	Critical
AKI, incident:	1 (280) [1] [RCT]	Serious limitations	N/A	Direct	Highly imprecise	Undetected	Sparse	Very low	RR 1.00 (0.26, 3.92) 0 more per 1000 (from 21 fewer to 83 more)	Critical
AKI, stage 3	0								No evidence	Critical
MAKE90	0								No evidence	Critical
RRT	0								No evidence	Critical
Duration of AKI	0								No evidence	Important
Hospital admission, unplanned	1 (280) [1] [RCT]	Serious limitations	N/A	Direct	Imprecise*	Undetected	Sparse	Very low	Prevalence ratio 1.30 (0.96, 1.76) 11 more per 1000 (from 1 fewer to 27 more)	Important
Poor control of underlying condition	0								No evidence	Critical
Balance of Potential Benefits and Harms: Stopping a variety of medications (including RASi, diuretics, NSAIDs, and other drugs) in adults with acute illnesses does not affect risk of AKI, but may increase risk of unplanned hospital admissions								**Certainty of Overall Evidence:** Very Low		

Abbreviations: AKI = acute kidney injury; RASi = renin angiotensin system inhibitors; MAKE = major adverse kidney events; N/A = not applicable; NSAIDs = non-steroidal anti-inflammatory drugs; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy.

* Imprecise assuming that effect size signifies clinically important difference

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Table S82. RASi discontinuation with COVID-19 in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Death, All cause	6 (1136) [1] [RCT]	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	RR 0.81 (0.52, 1.25) 11 fewer per 1000 (from 28 fewer to 14 more)	Critical
AKI, incident	4 (1061) [2] [RCT]	No limitations	Consistent	Direct	Highly Imprecise	Undetected	None	Very low	RR 1.01 (0.47, 2.17) 0 more per 1000 (from 13 fewer to 28 more)	Critical
AKI, stage 3	0								No evidence	Critical
MAKE90	0								No evidence	Critical
RRT	3 (1015) [1] [RCT]	No limitations	Consistent	Direct	Highly Imprecise	Undetected	None	Very low	RR 0.99 (0.45, 2.17) 0 more per 1000 (from 13 fewer to 27 more)	Critical
Duration of AKI	0								No evidence	Important
Hospital admission, unplanned	2 (103) [3-4] [RCT]	Very serious	Consistent	Direct	Highly imprecise	Undetected	None	Very low	RR 2.12 (0.56, 8.02) 134 more per 1000 (from 53 fewer to 842 more)	Important
Poor control of underlying condition:										Critical
Hypertensive crisis	2 (705) [3,5] [RCT]	Very serious	Consistent	Direct	Highly imprecise	Undetected	Sparse	Very low	RR 0.32 (0.03, 3.10) 2 fewer per 1000 (from 3 fewer to 6 more) [study with 0 events omitted]	
Acute Myocardial Infarction	3 (857) [1] [RCT]	No limitations	Inconsistent	Direct	Imprecise	Undetected	None	Low	RR 1.39 (0.77, 2.50) 23 more per 1000 (from 13 fewer to 87 more)	
Heart Failure	3 (857) [1] [RCT]	No limitations	Inconsistent	Direct	Imprecise	Undetected	None	Low	RR 1.03 (0.54, 1.96) 2 more per 1000 (from 23 fewer to 49 more)	
Stroke or TIA	2 (863) [1]	No limitations	Consistent	Direct	Highly Imprecise	Undetected	None	Very low	RR 0.68 (0.17, 2.78) 2 fewer per 1000 (from 6 fewer to 12 more)	
Balance of Potential Benefits and Harms: Evidence does not find significant differences in potential benefits and harms for stopping RASi for people with COVID-19								**Certainty of Overall Evidence:** Very low		

Abbreviations: AKI = acute kidney injury; RR = risk ratio; MAKE = major adverse kidney event; RASi = renin angiotensin system inhibitors; RCT = randomized controlled trial; RRT = renal replacement therapy; TIA = transient ischemic attack.

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Table S83. Early vs. standard RRT in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death, 90 d	6 (4171) [1-2] [RCT]	No serious	Inconsistent	Direct	Imprecise	Undetected	None	Low	RR 0.92 (0.74, 1.13) 36 fewer per 1000 (from 118 fewer to 59 more)	Critical
RRT dependency, 90 d (CKD G5D)	5 (2235) [1-2] [RCT]	No serious	Consistent	Direct	Very imprecise	Undetected	None	Low	RR 1.03 (0.55, 1.90) 2 more per 1000 28 fewer to 56 more	Critical
Time on RRT	1 (94) [3] [RCT]	Very serious	N/A	Direct	Very imprecise	Undetected	Sparse	Very low	Median difference: 0 d	Critical
CKD G5D	1 (212) [2] [RCT]	Very serious	N/A	Direct	Very imprecise	Undetected	Sparse	Very low	OR 0.75 (0.25, 2.25) 47 fewer per 1000 (from 124 fewer to 30 more)	Critical
Serious AE	7 (4546)* [1] [RCT]	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	No significant differences across multiple adverse events	Critical
Length of hospital stay, days	6 (3331) [1-2,4] [RCT]	No serious	Consistent	Direct	Imprecise	Undetected	None	Moderate	MD −0.3 (−2.7, 2.1)	Important
Length of ICU stay, days	6 (3604) [1-4] [RCT]	No serious	Consistent	Direct	Very imprecise	Undetected	None	Low	MD −0.1 (−1.2, 1.1)	Important
CKD G3-G5	0								No evidence	Important
Balance of Potential Benefits and Harms: There is no evidence of differences in 90-day mortality or RRT dependency, other AKI-related outcomes, or serious adverse events								**Certainty of Overall Evidence:** Low		

Abbreviations: AE = adverse event; AKI = acute kidney injury; CKD = chronic kidney disease; d = day; G = grade; ICU = intensive care unit; MD = mean difference; N/A = not applicable; OR = odds ratio; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy; d = days

* Between 5 and 7 studies investigated each adverse event, with between 3191 to 4546 participants

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Table S84. Very late vs. standard RRT in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Death, all-cause	1 (278) [1] [RCT]	Very serious	N/A	Direct	Precise	Undetected	Sparse	Very low	**HR 1.65 (1.09, 2.50)**	Critical
RRT dependency, 60 d	1 (125) [1] [RCT]	Very serious	N/A	Direct	Very imprecise	Undetected	Sparse	Very low	RR 0.50 (0.05, 4.67) 20 fewer per 1000 (from 38 fewer to 147 more)	Critical
Time on RRT	1 (278) [1] [RCT]	Very serious	N/A	Direct	Very imprecise	Undetected	Sparse	Very low	Median difference: 0 d	Critical
CKD G5D	0								No evidence	Critical
Serious AE	0								No evidence	Critical
Length of hospital stay	1 (278) [1] [RCT}	Very serious	N/A	Direct	Very imprecise	Undetected	Sparse	Very low	Median difference: –5 d	Important
Length of ICU stay	1 (278) [1] [RCT}	Very serious	N/A	Direct	Very imprecise	Undetected	Sparse	Very low	Median difference: –2 d	Important
CKD G3-G5	0								No evidence	Important
Balance of Potential Benefits and Harms: Very late RRT increases the likelihood of death, with no evidence of differences in AKI outcomes or length of stay								**Certainty of Overall Evidence:** Very Low		

Abbreviations: AE = adverse event; AKI = acute kidney injury; CKD = chronic kidney disease; d = day; G = grade; HR, hazard ratio; ICU = intensive care unit; N/A = not applicable; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

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1. Gaudry S, Hajage D, Martin-Lefevre L, et al. Comparison of two delayed strategies for renal replacement therapy initiation for severe acute kidney injury (AKIKI 2): a multicentre, open-label, randomised, controlled trial. Lancet. 2021;397(10281):1293-1300. doi:10.1016/S0140-6736(21)00350-0

Table S85. Continuous vs. intermittent renal replacement therapy in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death, all-cause	9* (NR) [1] [RCT]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	RR 1.04 (0.93 to 1.18) 25 more per 1000 (from 44 fewer to 112 more)	Critical
Time on RRT	NR* (NR) [1] [RCT]	No serious	Consistent	Direct	Imprecise	Undetected	None	Moderate	MD -0.88 days (-3.56 to 1.80)	Critical
CKD G5D	0								No evidence	Critical
Quality of life	0								No evidence	Critical
Serious AE	0								No evidence	Critical
Recovery of kidney function	7* (NR) [1] [RCT]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	RR 1.15 (0.91 to 1.45) 45 more per 1000 (from 27 fewer to 134 more)	Critical
Length of Hospital stay	5* (NR) [1] [RCT]	Serious	Consistent	Direct	Very imprecise	Undetected	None	Very low	MD 2.47 days (-2.48 to 7.42)	Important
Length of ICU stay	NR* (NR) [1] [RCT]	Serious	Consistent	Indirect	Very imprecise	Undetected	None	Very low	MD -1.36 days (-5.09 to 2.38)	Important
CKD G3-G5	0								No evidence	Important
Balance of Potential Benefits and Harms: No evidence of differences in AKI or other outcomes of differences between CRRT and intermittent hemodialysis in adults								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AE = adverse event; AKI = acute kidney injury; CKD = chronic kidney disease; CRRT = continuous renal replacement therapy; G = grade; ICU = intensive care unit; MD = mean difference; NR = not reported; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

* Based on network meta-analysis.

References

1. Ye Z, Wang Y, Ge L, et al. Comparing Renal Replacement Therapy Modalities in Critically Ill Patients With Acute Kidney Injury: A Systematic Review and Network Meta-Analysis. Crit Care Explor. 2021;3(5):e0399. Published 2021 May 12. doi:10.1097/CCE.000000000000039

Table S86. Intermittent renal replacement therapy vs peritoneal dialysis in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Death, all-cause	1* (152) [1] [RCT]	Very serious	Consistent [w/network†]	Direct	Imprecise	Undetected	Sparse†	Low	RR 0.97 (0.70, 1.35)† 14 fewer per 1000 (from 171 fewer to 144 more)	Critical
Time on RRT	NR* (NR) [2] [RCT]	Serious	N/A	Indirect [No direct]	Precise	Undetected	Sparse, but large effect	Low	**MD 3.88 days (1.12 to 6.63)**	Critical
CKD G5D	0								No evidence	Critical
Quality of life	0								No evidence	Critical
Serious AE	0								No evidence	Critical
Recovery of kidney function	1* (152) [1] [RCT]	Very serious	Consistent [w/network‡]	Direct	Precise	Undetected	Sparse‡	Low	RR 0.67 (0.41, 1.11)‡ 127 fewer per 1000 (from 282 fewer to 27 more)	Critical
Length of Hospital stay	0								No evidence	Important
Length of ICU stay	NR* (NR) [2] [RCT]	No serious	N/A	Indirect [No direct]	Precise	Undetected	Large effect	Moderate	**MD 11.36 days (3.76 to 18.95)**	Important
CKD G3-G5	0								No evidence	Important
Balance of Potential Benefits and Harms: Indirect evidence that peritoneal dialysis results in much shorter time in dialysis and ICU length of stay. Direct evidence of no effect on death or recovery of renal function.								**Certainty of Overall Evidence:** Low		

Abbreviations: AE = adverse event; AKI = acute kidney injury; CKD = chronic kidney disease; ICU = intensive care unit; MD = mean difference; N/A = not applicable; NR = not reported; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

* Based on network meta-analysis.
† Network meta-analysis found an indirect RR 1.12 (0.85 to 1.46). CoE upgraded one level since study comports with network.
‡ Network meta-analysis found an indirect RR 0.76 (0.49 to 1.18). CoE upgraded one level since study comports with network.

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Table S87. Continuous renal replacement therapy vs. peritoneal dialysis in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Description of Findings	Importance of Outcome
Death, all-cause	3* (179) [1] [RCT]	Serious	Inconsistent	Direct	Precise	Undetected	None	Low	RR 1.16 (0.92 to 1.49) 112 more per 1000 (from 56 fewer to 342 more)	Critical
Time on RRT	NR* (NR) [1] [RCT]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	**RD 3.00 days (2.34 to 3.66)**	Critical
CKD G5D	0								No evidence	Critical
Quality of life	0								No evidence	Critical
Serious AE	0								No evidence	Critical
Recovery of kidney function	2* (NR) [1] [RCT]	Serious	Inconsistent	Direct	Imprecise	Undetected	None	Very low	RR 0.87 (0.60 to 1.27) 78 fewer per 1000 (from 241 fewer to 163 more)	Critical
Length of Hospital stay	0								No evidence	Important
Length of ICU stay	NR* (NR) [1] [RCT]	No serious	Consistent	Direct	Precise	Undetected	None	High	**MD 10.00 days (3.39 to 16.61)**	Important
CKD G3-G5	0								No evidence	Important
Balance of Potential Benefits and Harms: Peritoneal dialysis, compared with CRRT, reduces time on RRT and lengths of stay, with no evidence of an effect on mortality or recovery of renal function								**Certainty of Overall Evidence:** Low		

Abbreviations: AE = adverse event; AKI = acute kidney injury; CKD = chronic kidney disease; CRRT = continuous renal replacement therapy G = grade; ICU = intensive care unit; MD = mean difference; NR = not reported; RCT = randomized controlled trial; RD = risk difference; RR = risk ratio; RRT = renal replacement therapy

* Based on network meta-analysis.

References

1. Ye Z, Wang Y, Ge L, et al. Comparing Renal Replacement Therapy Modalities in Critically Ill Patients With Acute Kidney Injury: A Systematic Review and Network Meta-Analysis. Crit Care Explor. 2021;3(5):e0399. Published 2021 May 12. doi:10.1097/CCE.0000000000000399

Table S88. Intensive high-volume hemofiltration continuous renal replacement therapy (various thresholds) vs. less intensive continuous renal replacement therapy in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Publication bias	Precision of the Estimate	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Death, 30 d	12 (4613) [1-6] [RCT]	Some limitations	Consistent	Direct	Undetected	Precise	None	Moderate	**Any intensity definition:** RR 0.95 (0.85, 1.05) 22 fewer per 1000 (from 67 fewer to 22 more)	Critical
	11 (4413) [1-6] [RCT]	Some limitations	Consistent	Direct	Undetected	Precise	None	Moderate	**≥40 ml/kg/hr:** RR 0.93 (0.83, 1.06) 31 fewer per 1000 (from 76 fewer to 27 more)	
	6 (791) [1-4] [RCT]	Some limitations	Inconsistent	Direct	Undetected	Precise	None	Low	**≥45 ml/kg/hr:** RR 0.98 (0.86, 1.11) 10 fewer per 1000 (from 72 fewer to 57 more)	
Time on RRT	0								No evidence	Critical
CKD G5D	0								No evidence	Critical
Quality of life	0								No evidence	Critical
Complete recovery, 30 d	5 (2402) [1] [RCT]	Some limitations	Consistent	Direct	Undetected	Precise	None	Moderate	**Any intensity definition:** RR 1.12 (0.90, 1.38) 58 more per 1000 (from 48 fewer to 183 more)	Important
	4 (2202) [1] [RCT]	Some limitations	Consistent	Direct	Undetected	Imprecise	None	Low	**≥40 ml/kg/hr:** RR 1.18 (0.94, 1.47) 87 more per 1000 (from 29 fewer to 227 more)	
	1 (106) [1] [RCT]	Some limitations	N/A	Direct	Undetected	Imprecise	Sparse	Very Low	**≥45 ml/kg/hr:** RR 1.03 (0.81, 1.32) 22 more per 1000 (from 136 fewer to 230 more)	
Hospital LOS	5 (2188) [1-2] [RCT]	Some limitations	Inconsistent	Direct	Undetected	Imprecise	None	Very Low	**Any intensity definition:** MD −0.06 (−2.44, 2.32)	Important
	4 (1988) [1-2] [RCT]	Some limitations	Inconsistent	Direct	Undetected	Imprecise	None	Very Low	**≥40 ml/kg/hr:** MD 0.26 (−2.20, 2.71)	
	3 (523) [1-2] [RCT]	Some limitations	Inconsistent	Direct	Undetected	Imprecise	None	Very Low	**≥45 ml/kg/hr:** MD –0.12 (–7.83, 7.59)	
ICU LOS	9 (3140) [1-3,6] [RCT]	Some limitations	Inconsistent	Direct	Undetected	Imprecise	None	Very Low	**Any intensity definition:** MD −0.63 (−3.71, 2.45)	Important
	7 (2663) [1-3,6] [RCT]	Some limitations	Inconsistent	Direct	Undetected	Imprecise	None	Very Low	**≥40 ml/kg/hr:** MD 1.04 (−1.51, 3.59)	

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Publication bias	Precision of the Estimate	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
	5 (754) [1-3] [RCT]	Some limitations	Inconsistent	Direct	Undetected	Imprecise	None	Very Low	**≥45 ml/kg/hr:** MD 2.0 (–2.19, 6.20)	
CKD G3-G5	0								No evidence	Important
Serious AE	1 (444) [6] [RCT]	Some limitations	N/A	Direct	Undetected	Imprecise	Sparse	Very Low	**≥40 ml/kg/hr:** RD 0.01 (0.00, 0.02) 10 more per 1000 (from 4 fewer to 23 more)	Critical
SAE: Arrhythmia requiring treatment	1 (444) [6] [RCT]	Some limitations	N/A	Direct	Undetected	Imprecise	Sparse	Very Low	**≥40 ml/kg/hr:** RR 1.03 (0.79, 1.36) 10 more per 1000 (from 67 fewer to 115 more)	Critical
SAE: Arrhythmia with hemodynamic instability	1 (444) [6] [RCT]	Some limitations	N/A	Direct	Undetected	Imprecise	Sparse	Very Low	**≥40 ml/kg/hr:** RR 1.07 (0.77, 1.48) 17 more per 1000 (from 54 fewer to 113 more)	Critical
Balance of Potential Benefits and Harms: No evidence of a difference in death, complete recovery, with no evidence of increased harms, between HVHF (including specifically ≥40 or ≥45 mL/kg/h) of intensive CRRT and less intensive CRRT dosing.								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AE = adverse events; CKD = chronic kidney disease; CRRT = continuous renal replacement therapy; d = day; G = grade; HVHF = high volume high frequency; ICU = intensive care unit; MD = mean difference; N/A = not applicable; RCT = randomized controlled trial; RD = risk difference; RR = risk ratio; RRT = renal replacement therapy; LOS = length of stay; RD = risk difference; SAE = serious adverse event

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Table S89. Systemic unfractionated heparin vs. regional citrate anticoagulation in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Filter clotting	6 (389) [1] [RCT]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	**RR 1.70 (1.09, 2.65)** 170 more episodes per 1000 patients (from 22 more to 400 more)	**Critical**
AE: Bleeding	8 (1146) [1] [RCT]	Serious	Consistent	Direct	Precise	Undetected	Large effect	High	**RR 3.09 (2.08, 4.60)** 96 more per 1000 (from 50 more to 141 more)	**Critical**
AE: Stroke	0								No evidence	**Critical**
Discontinuation due to AE	3 (412) [RCT]	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	RR 2.13 (0.67, 6.67) 55 more per 1000 (from 16 fewer to 274 more)	**Critical**

Balance of Potential Benefits and Harms:	**Certainty of Overall Evidence:**
Compared with regional citrate anticoagulation, systemic UFH increases the risk of both filter clotting and bleeding events, with no evidence of a difference in discontinuation due to adverse events	Moderate

Abbreviations: AE = adverse events; CRRT = continuous renal replacement therapy; ICU = intensive care unit; MD = mean difference; RCT = randomized controlled trial; RR = risk ratio

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1. Zhou Z, Liu C, Yang Y, Wang F, Zhang L, Fu P. Anticoagulation options for continuous renal replacement therapy in critically ill patients: a systematic review and network meta-analysis of randomized controlled trials. Crit Care. 2023;27(1):222. Published 2023 Jun 7. doi:10.1186/s13054-023-04519-1

Table S90. Regional unfractionated heparin (plus protamine) vs. regional citrate anticoagulation in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Filter clotting	3 (252) [1]	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	RR 1.01 (0.61, 1.67) 6 more per 1000 filters (from 226 fewer to 388 more)	**Critical**
AE: Bleeding	3 (252) [1]	Serious	Consistent	Direct	Very imprecise	Undetected	None	Very Low	RR 1.82 (0.23, 14.43)* 8 more per 1000 (from 13 fewer to 29 more)	**Critical**
AE: Stroke	0								No evidence	**Critical**
Discontinuation due to AE	0								No evidence	**Critical**
Balance of Potential Benefits and Harms: There is no difference in the risk of filter clotting, with an uncertain effect on bleeding, between regional unfractionated heparin and regional citrate								**Certainty of Overall Evidence:** Low		

Abbreviations: AE = adverse events; CRRT = continuous renal replacement therapy; ICU = intensive care unit; MD = mean difference; NA = not applicable; RCT = randomized controlled trial; RR = risk ratio

* Direct evidence found RR 0.34 (0.01, 8.32)

References

1. Zhou Z, Liu C, Yang Y, Wang F, Zhang L, Fu P. Anticoagulation options for continuous renal replacement therapy in critically ill patients: a systematic review and network meta-analysis of randomized controlled trials. Crit Care. 2023;27(1):222. Published 2023 Jun 7. doi:10.1186/s13054-023-04519-1

Table S91. Regional citrate anticoagulation vs. low molecular weight heparin in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Filter clotting	1 (34) [1] [RCT]	Serious	Consistent*	Direct	Very imprecise	Undetected	Sparse	Very Low	NMA pooled, RR 0.97 (0.45, 2.11) Unable to estimate†	**Critical**
Filter termination due to clotting	1 (34) [2] [RCT]	Serious	N/A	Direct	Imprecise	Undetected	Sparse	Very low	RR 1.27 (0.88, 1.82) 1 more per 1000 (0 more to 3 more)	**Critical**
AEs: major bleeding	2 (234) [2] [RCT]	Serious	Consistent	Direct	Precise	Undetected	Large effect	High	**RR 0.45 (0.21 to 0.97)** 97 fewer per 1000 (130 fewer to 5 fewer)	**Critical**
AE: Stroke	0								No evidence	**Critical**
Discontinuation due to AE	2 (234) [2] [RCT]	Serious	Consistent	Direct	Precise	Undetected	Large effect	High	**RR 0.11 (0.03, 0.44)** 146 fewer per 1000 (159 fewer to 92 fewer)	**Critical**
Balance of Potential Benefits and Harms: Compared with LMWH, regional citrate anticoagulation greatly decreases the risk of major bleeding and discontinuation due to adverse events, without evidence of a difference in filter clotting.								**Certainty of Overall Evidence:** Low		

Abbreviations: AE = adverse events; RCT = randomized controlled trial; RR = risk ratio

* Single study consistent with NMA.
† Based on available data.

References

1. Zhou Z, Liu C, Yang Y, Wang F, Zhang L, Fu P. Anticoagulation options for continuous renal replacement therapy in critically ill patients: a systematic review and network meta-analysis of randomized controlled trials. Crit Care. 2023;27(1):222. Published 2023 Jun 7. doi:10.1186/s13054-023-04519-1
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Table S92. Regional citrate anticoagulation vs. regional unfractionated heparin in children

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Filter clotting	6 (145) [1] [NRCS]	Serious	Inconsistent	Direct	Imprecise	Undetected	None	Very low	RR 0.72 (0.38, 1.37) 110 fewer per 1000 filters (from 244 fewer to 145 more)	**Critical**
AE: Bleeding	1 (32) [1] [NRCS]	Serious	N/A	Direct	Imprecise	Undetected	Sparse	Very low	RR 0.80 (0.35, 1.85) 75 fewer per 1000 (from 244 fewer to 319 more)	**Critical**
AE: Systemic Bleeding	1 (130) [1] [NRCS]	Serious	N/A	Direct	Precise	Undetected	Sparse	Low	**RD −0.10 (−0.16, −0.04)** 97 fewer episodes per 1000 patients (from 157 fewer to 37 fewer)	**Critical**
AE: Stroke	0								No evidence	**Critical**
Heparin discontinuation due to AE (systemic bleeding and HIT)	1 (130) [1] [NRCS]	Serious	N/A	Direct	N/A	Undetected	Sparse	Very low	UFH 5/93 (5.4%) Citrate N/A	**Critical**
Balance of Potential Benefits and Harms: In children, regional citrate anticoagulation reduces the risk of systemic bleeding compared with regional UFH, with uncertain effects on other outcomes								**Certainty of Overall Evidence:** Very low		

Abbreviations: AE=Adverse events; AKI = Acute Kidney Injury; CRRT=Continuous renal replacement therapy; HIT=Heparin-induced thrombocytopenia; NRCS = nonrandomized controlled study; RR = risk ratio

References

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Table S93. Comprehensive care vs. usual care in adults post-AKI

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	4 (10789) [1-4] (1 RCT, 3 NRCS)	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	**RR 0.75 (0.70, 0.81)** 62 fewer per 1000 (from 75 fewer to 47 fewer)	**Critical**
MAKE	1 (78) [1] [RCT]	Serious	N/A	Direct	Imprecise	Undetected	Sparse	Very low	RR 1.12 (0.58, 2.19) 36 more per 1000 (from 167 fewer to 238 more)	**Critical**
CKD G5D	1 (9976) [3] [NRCS]	No serious	N/A	Direct	Precise	Undetected	Sparse, but large	Low	**HR 0.58 (0.54, 0.62)** 16 fewer per 1000 (from 17 fewer to 14 fewer)	**Critical**
New CKD	1 (41) [1] [RCT]	Serious	Consistent	Direct	Very imprecise	Undetected	Sparse	Very Low	RR 0.65 (0.17, 2.54) 75 fewer per 1000 (from 175 fewer to 325 more)	**Critical**
Recurrent AKI	1 (66) [1] [RCT]	Serious	Consistent	Direct	Very imprecise	Undetected	Sparse	Very Low	RR 1.0 (0.36, 2.78) 0 per 1000 (from 116 fewer to 324 more)	**Critical**
Health-related Quality of Life	1 (78) [1] [RCT]	Serious	Consistent	Direct	N/A	Undetected	Sparse	Very Low	Median difference −0.03	**Critical**
MACE	0								No evidence	Important
Hospital readmission	3 (833) [1-2,4] (1 RCT, 2 NRCS)	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	RR 0.94 (0.67, 1.33) 10 fewer per 1000 (from 54 fewer to 54 more)	Important
ED visits	0								No evidence	Important
Hospital visits	0								No evidence	Important
Balance of Potential Benefits and Harms: Comprehensive care post-AKI associated with reduced death and CKD requiring dialysis (CKD G5D), with uncertain effects on most other outcomes								**Certainty of Overall Evidence:** Moderate		

Abbreviations: ED = emergency department; AKI = acute kidney injury; CKD = chronic kidney disease; G = grade; HR = hazard ratio; MAKE = major adverse kidney event; N/A = not applicable; NRCS = non-randomized comparative study; RCT = randomized controlled trial; RR = risk ratio

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4. Ly H, Ortiz-Soriano V, Liu LJ, et al. Characteristics and Outcomes of Survivors of Critical Illness and Acute Kidney Injury Followed in a Pilot Acute Kidney Injury Clinic. Kidney Int Rep. 2021;6(12):3070-3073. Published 2021 Aug 27. doi:10.1016/j.ekir.2021.08.017

Table S94. Nephrologist follow-up vs. no nephrologist follow-up in post-AKI care in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Description of Findings	Importance of Outcome
Death	6 (16103) [1-2] [1 RCT, 5 NRCS]	Very serious	Consistent	Direct	Precise	Undetected	None	Low	**RR 0.84 (0.76, 0.93)** 57 fewer per 1000 (from 85 fewer to 25 fewer)	**Critical**
CKD G5D	5 (15400) [1-2] [1 RCT, 4 NRCS]	Very serious	Consistent	Direct	Precise	Undetected	None	Low	**RR 2.33 (2.12, 2.56)** 86 more per 1000 (from 73 more to 101 more)	**Critical**
CKD G3-G5	0								No evidence	**Critical**
Recurrent AKI	1 (169) [1] [NRCS]	Serious	Consistent	Direct	Imprecise	Undetected	Sparse	Low	RR 0.57 (0.33, 1.01) 197 fewer per 1000 (from 307 fewer to 5 more)	**Critical**
Health-related Quality of Life	0									**Critical**
MACE	1 (10716) [3] [NRCS]	Serious	N/A	Direct	Precise	Undetected	Sparse, but large	Low	**AdjHR 0.84 (0.73, 0.97)** 43 fewer per 1000 (from 72 fewer to 8 fewer)	Important
Hospital readmission	1 (10716) [3] [NRCS]	Serious	N/A	Direct	Imprecise	Undetected	Sparse, but large	Low	AdjHR 1.03 (0.93, 1.14) 19 more per 1000 (from 44 fewer to 89 more)	Important
ED visits	0								No evidence	Important
Hospital visits	0								No evidence	Important
Balance of Potential Benefits and Harms: Nephrologist follow-up associated with lower risk of death and possibly of recurrent AKI, but a higher risk of CKD requiring dialysis								**Certainty of Overall Evidence:** Low		

Abbreviations: AE = adverse event; AKI = acute kidney injury; CKD = chronic kidney disease; G = grade; HR = hazard ratio; MACE = major adverse cardiac event; N/A = not applicable; NRCS = non-randomized comparative study; RCT = randomized controlled trial, RR = risk ratio

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Table S95. Risk-based care vs. usual care in adults post-AKI

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Death	1 (155) [1] [RCT]	Serious	N/A	Direct	Imprecise	Undetected	Sparse	Very low	RR 1.44 (0.62, 3.33) 46 more per 1000 (from 40 fewer to 245 more)	**Critical**
CKD G5D (dialysis)	1 (155) [1]	Serious	N/A	Direct	Very imprecise	Undetected	Sparse	Very Low	RR 1.14 (0.25, 8.40) 4 more per 1000 (from 20 fewer to 192 more)	**Critical**
CKD G5	1 (155) [1] [RCT]	Serious	N/A	Direct	Imprecise	Undetected	Sparse	Very Low	RR 0.71 (0.35, 1.44) 57 fewer per 1000 (from 128 fewer to 87 more)	**Critical**
Recurrent AKI (with readmission)	1 (155) [1] [RCT]	Serious	N/A	Direct	Imprecise	Undetected	Sparse	Very Low	RR 1.60 (0.84, 3.05) 95 more per 1000 (from 31 fewer to 222 more)	**Critical**
Health-related Quality of Life	0								No evidence	**Critical**
MACE	0								No evidence	Important
Hospital readmission (with AKI or CKD)	1 (155) [1] [RCT]	Serious	N/A	Direct	Very Imprecise	Undetected	Sparse	Very Low	RR 0.96 (0.46, 2.01) 6 fewer per 1000 (from 121 fewer to 109 more)	Important
ED visits	0								No evidence	Important
Hospital visits	0								No evidence	Important
Balance of Potential Benefits and Harms: There is no difference in benefits and harms between risk-based care and usual care.								**Certainty of Overall Evidence:** Very low		

Abbreviations: ED = emergency department; AKI = acute kidney injury; CKD = chronic kidney disease; G = grade; MACE = major adverse cardiac event; N/A = not applicable; RCT = randomized controlled trial; RR = risk ratio

References

1. Pannu N, McBrien KA, Tan Z, et al. Advancing Community Care and Access to Follow-Up after Acute Kidney Injury Hospitalization: A Randomized Clinical Trial. J Am Soc Nephrol. 2025;36(3):441-450. doi:10.1681/ASN.0000000537

Table S96. Restart of RASi vs. no restart in adults post-AKI

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death, all cause	8 (81869) [1-8] [NRCS]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	**HR 0.83 (0.75, 0.91)** 66 fewer per 1000 (from 98 fewer to 34 fewer)	Critical
Recurrent AKI	4 (32655) [1,3-5] [NRCS]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	HR 1.02 (0.95, 1.08) 4 more per 1000 (from 8 fewer to 15 more)	Critical
MAKE[a]	1 (1145) [6] [NRCS]	Serious	N/A	Direct	Imprecise	Undetected	Sparse	Low	HR 1.09 (0.93, 1.28) 29 more per 1000 (from 26 fewer to 81 more)	Critical
CKD G3-G5 (CKD progression)	1 (9147) [3] [NRCS]	Serious	N/A	Direct	Precise	Undetected	Sparse, but large study	Low	HR 0.95 (0.87, 1.04) 13 fewer per 1000 (from 33 fewer to 11 more)	Critical
CKD G5D	2 (19110) [4,7] [NRCS]	Serious	Inconsistent	Direct	Imprecise	Undetected	None	Low	RR 0.92 (0.73, 1.17) 17 fewer per 1000 (from 57 fewer to 36 more)	Critical
Poor control of underlying condition:										Critical
Hypertension hospitalization	0								No evidence	
Heart failure hospitalization	2 (16912) [1,3] [NRCS]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	HR 0.98 (0.92, 1.04) 7 fewer per 1000 (from 29 fewer to 14 more)	
Myocardial infarction	1 (10165) [3] [NRCS]	Serious	N/A	Direct	Precise	Undetected	Sparse, but large study	Low	HR 0.94 (0.79, 1.11) 6 fewer per 1000 (from 18 fewer to 12 more)	
Stroke	2 (17034) [1,3] [NRCS]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	HR 0.97 (0.84, 1.13) 3 fewer per 1000 (from 13 fewer to 14 more)	
MACE[b]	2 (19790) [3-4] [NRCS]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	**HR 0.88 (0.85, 0.92)** 47 fewer per 1000 (from 60 fewer to 31 fewer)	
Hypervolemia	0								No evidence	
Quality of life	0								No evidence	Important
Discontinuation due to AE	0								No evidence	Important
SAE: Moderate or severe hyperkalemia	2 (18772) [3-4] [NRCS]	Serious	Inconsistent	Direct	Imprecise	Undetected	None	Low	HR 1.12 (0.87, 1.43) 24 more per 1000 (from 25 fewer to 89 more)	Important

Balance of Potential Benefits and Harms:
In patients with prior RASi use, RASi restart post-AKI reduces the likelihood of death, and MACE, with no evidence of effects on other outcomes.

Certainty of Overall Evidence:
Moderate

Abbreviations: RASi = Renin–angiotensin system inhibitors; ACEi = angiotensin converting enzyme inhibitors; ARB = angiotensin II receptor blockers; AE = adverse event; AKI = Acute Kidney Injury; CKD = chronic kidney disease; HR = hazard ratio (adjusted); MACE = major adverse cardiovascular events; MAKE = major adverse kidney event; N/A = not applicable; NRCS = non-randomized control study; SAE = serious adverse event

[a] MAKE defined as RRT initiation, sustained eGFR decline, or mortality.

[b] MACE defined as composite cardiovascular death, myocardial infarction, and stroke.

References

1. Bidulka P, Fu EL, Leyrat C, et al. Stopping renin-angiotensin system blockers after acute kidney injury and risk of adverse outcomes: parallel population-based cohort studies in English and Swedish routine care. *BMC Med*. 2020;18(1):195. Published 2020 Jul 29. doi:10.1186/s12916-020-01659-x
2. Gayat E, Hollinger A, Cariou A, et al. Impact of angiotensin-converting enzyme inhibitors or receptor blockers on post-ICU discharge outcome in patients with acute kidney injury. Intensive Care Med. 2018;44(5):598-605. doi:10.1007/s00134-018-5160-6
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7. Wu PC, Lin CF, Kao CC, Wu CJ, Chien LN. Long-Term Renal Outcomes of ACEI or ARB Use after Recovery from Dialysis-Requiring Acute Kidney Injury: A Population-Based Cohort Study. Int J Gerontol. 2025;19(2):109-114. doi:10.6890/IJGE.202504_19(2).0009
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Table S97. Post-AKI use of RASi (new start) vs. no RASi in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death, all cause	5 (31970) [1-5] [NRCS]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	**HR 0.87 (0.81, 0.95)** 44 fewer per 1000 From 65 fewer to 16 more	Critical
Recurrent AKI	1 (12095) [6] [NRCS]	No limitations	N/A	Direct	Highly imprecise	Undetected	Sparse, but large study	Low	OR 0.71 (0.45, 1.12) 18 fewer per 1000 (from 34 fewer to 7 more)[a]	Critical
MAKE[b]	1 (2144) [2]	Serious	N/A	Direct	Imprecise	Undetected	Sparse	Low	HR 1.20 (0.95, 1.51) 65 more per 1000 (from 19 fewer to 141 more)	Critical
CKD G3-G5 (CKD progression)	1 (5490) [5] [NRCS]	No limitations	N/A	Direct	Imprecise	Undetected	Sparse, but large study	Low	HR 1.09 (0.94, 1.27) 11 more per 1000 (from 8 fewer to 34 more)	Critical
CKD G5D	2 (12945) [3,5] [NRCS]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	HR 0.99 (0.90, 1.10) 1 fewer per 1000 (from 9 fewer to 9 more)	Critical
Poor control of underlying condition:										Critical
Hypertension hospitalization	0								No evidence	
Heart failure hospitalization	0								No evidence	
Myocardial infarction	0								No evidence	
Stroke	0								No evidence	
MACE	0								No evidence	
Hypervolemia	0								No evidence	
Quality of life	0								No evidence	Important
Discontinuation due to AE	0								No evidence	Important
SAE: Hyperkalemia	0								No evidence	Important
Balance of Potential Benefits and Harms: In RASi-naïve patients, RASi use post-AKI reduces the likelihood of death, with no evidence of effects on other outcomes.								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AE = adverse event; AKI = acute kidney injury; CKD = chronic kidney disease; HR = hazard ratio (adjusted); MACE = major adverse cardiovascular events; MAKE = major adverse kidney event; N/A = not applicable; NRSC = non-randomized control study; OR = odds ratio; RASi = renin–angiotensin system inhibitors; SAE = serious adverse event

[a] Calculation of absolute difference based on adjusted OR and assumed CR risk based on crude incidence rate per 100 person-years.
[b] MAKE defined as RRT initiation, sustained eGFR decline, or mortality

References

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Appendix D. Additional Evidence profiles developed as part of the evidence review

Table S98. Clinical decision support systems (alerts) in adults

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI	1 (1916) [1]	Not serious	N/A	Direct	Unclear	Undetected	Sparse, but large	Low	Sensitivity 0.89 Specificity 0.61	Critical
Severe AKI	0								none	Critical
AKI Stage 3	0								none	Critical
Mortality	0								none	Critical
RRT	0								none	Critical
Duration of AKI	0								none	Important
CKD G3-G5 (at ≥90 days)	0								none	Important
CKD G5D (dialysis) (at ≥90 days)	0								none	Important
Readmission (30-day)	1 (1916) [1]	Not serious	N/A	Direct	Unclear	Undetected	Sparse, but large	Low	Sensitivity 0.87 Specificity 0.76	(Not prioritized)
Balance of Potential Benefits and Harms: CDSS has moderate* sensitivity to predict AKI and 30-day readmission, with poor specificity† to predict AKI and moderate* specificity to predict readmission								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = Acute Kidney Injury; CKD = chronic kidney disease; G = grade; N/A = not applicable; RRT = renal replacement therapy

* *A priori* defined as 75-89% sensitivity or specificity
† *A priori* defined as 51-74% sensitivity or specificity

References

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Table S99. Kinetic eGFR difference in adults with AKI

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Incident AKI	1 (140) [1]	No serious	N/A	Direct	Precise	Undetected	Sparse	Very low	keGFR difference AUC 0.78 (0.73, 0.82)	**Critical**
	1 (140) [1]	No serious	N/A	Direct	Precise	Undetected	Sparse	Very low	keGFR ratio AUC 0.82 (0.78, 0.86) Sn* 66.4% Sp* 77.3% PPV* 62.8% NPV* 79.9%	
AKI Stage 3	0								No evidence	**Critical**
Mortality	0								No evidence	**Critical**
RRT	0								No evidence	**Critical**
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Balance of Potential Benefits and Harms: Kinetic eGFR difference and ratio have moderate accuracy to predict AKI incidence †								**Certainty of Overall Evidence:** Very low		

Abbreviations: AKI = Acute Kidney Injury; AUC = area under the curve; CKD = chronic kidney disease; eGFR = estimated glomerular filtration rate; keGFR = kinetic estimated glomerular filtration rate; N/A = not applicable; RRT = renal replacement therapy; Sn = sensitivity; Sp = specificity;

* Cutoff = 10%.
† *A priori*, test accuracy defined as moderate for AUC 0.76-0.85.

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Table S100. Serum NGAL in adults

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI	40 (9087) [1]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	Sn 0.76 (0.72-0.80) Sp 0.80 (0.76-0.83)	Critical
Severe AKI (stage 2-3)	10 (NR) [1]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	Sn 0.82 (0.65-0.92) Sp 0.72 (0.58-0.83)	Critical
AKI requiring RRT	6 (NR) [1]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	Sn 0.74 (0.61-0.84) Sp 0.81 (0.66-0.90)	Critical
Death	0								none	Critical
MAKE90	0								none	Critical
Duration of AKI	0								none	Important
CKD G3-G5	0								none	Important
CKD G5D (dialysis)	0								none	Important
Balance of Potential Benefits and Harms: In adults, sNGAL has low to moderate* sensitivity and specificity to predict AKI outcomes.								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = Acute Kidney Injury; CKD = chronic kidney disease; MAKE = major adverse kidney injury; sNGAL = serum neutrophil gelatinase-associated lipocalin; RRT = renal replacement therapy; Sn = sensitivity; Sp = specificity

* *A priori*, low sensitivity or specificity defined as 51-75% and moderate sensitivity or specificity defined as 76-89%

References

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Table S101. Serum NGAL in children

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI	10 (961) [1]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	AUC 0.74 (0.64-0.84) Sn 0.77 (0.62-0.87) Sp 0.77 (0.61-0.88	Critical
Severe AKI (stage 2-3)	3 (595) [1]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	AUC 0.61 (0.51-0.70)	Critical
AKI requiring RRT	0								none	Critical
Death	0								none	Critical
MAKE90	0								none	Critical
Duration of AKI	0								none	Important
CKD G3-G5	0								none	Important
CKD G5D (dialysis)	0								none	Important
Balance of Potential Benefits and Harms: In pediatrics, sNGAL has moderate* sensitivity and specificity to predict AKI incidence. sNGAL has low † accuracy to predict AKI incidence and stages.								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = Acute Kidney Injury; AUC = area under the curve; CKD = chronic kidney disease; G = grade; MAKE = major adverse kidney event; Sn = sensitivity; sNGAL = serum neutrophil gelatinase-associated lipocalin; Sp = specificity

* Arbitrarily defined as 76-89% sensitivity or specificity
† Arbitrarily defined as AUC 0.51-0.75

References

1. Meena J, Thomas CC, Kumar J, Mathew G, Bagga A. Biomarkers for prediction of acute kidney injury in pediatric patients: a systematic review and meta-analysis of diagnostic test accuracy studies. *Pediatr Nephrol*. 2023;38(10):3241-3251. doi:10.1007/s00467-023-05891-4

Table S102. Ultrasound (power Doppler) in adults who are critically ill or have postoperative conditions

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI	5 (752) [1]	Serious	Inconsistent	Direct	Sn: Imprecise* Sp, AUC: Precise	Undetected	None	Low	AUC: 0.86 (0.83 – 0.89) Sn: 0.64 (0.48 – 0.77) Sp: 0.90 (0.81 – 0.95)	Critical
Severe AKI (stage 3)	0								No evidence	Critical
AKI requiring RRT	0								No evidence	Critical
Death	0								No evidence	Critical
MAKE90	0								No evidence	Critical
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D (dialysis)	0								No evidence	Important
Balance of Potential Benefits and Harms: In adults, power Doppler has low sensitivity, but high specificity, with high overall accuracy to predict incident AKI†								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = Acute Kidney Injury; AUC = Area Under the Curve; CKD = Chronic Kidney Disease; MAKE90 = Major Adverse Kidney Events at 90 days; PDU = Power Doppler Ultrasound; RRT = renal replacement therapy; Sn = sensitivity; Sp = specificity

* Lower bound of 95% confidence interval <50%
† Thresholds for sensitivity and specificity were *a priori* defined as 0.51-0.75 low and ≥0.90 high, and thresholds for AUC as 0.86-1.00 as high

References

1. Wei Q, Zhu Y, Zhen W, et al. Performance of resistive index and semi-quantitative power doppler ultrasound score in predicting acute kidney injury: A meta-analysis of prospective studies. PLoS One. 2022;17(6):e0270623. Published 2022 Jun 28. doi:10.1371/journal.pone.0270623

Table S103. Ultrasound (renal resistive index) in adults who are critically ill or have postoperative conditions

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Incident AKI	23 (2400) [1]	Serious	Inconsistent*	Direct	Precise	Detected†	None	Low	AUC 0.83 (0.80 – 0.86) Sn: 0.76 (0.70 –0.81) Sp: 0.79 (0.72 – 0.85)	Critical
Severe AKI (stage 3)	0								No evidence	Critical
AKI requiring RRT	0								No evidence	Critical
Death	0								No evidence	Critical
MAKE90	0								No evidence	Critical
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Balance of Potential Benefits and Harms: In adults, renal resistive index has moderate sensitivity, specificity and accuracy to predict AKI incidence‡								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = Acute Kidney Injury; AUC = Area Under the Curve; CKD = chronic kidney disease; MAKE90 = Major Adverse Kidney Events at 90 days; RRT = renal replacement therapy; Sn = sensitivity; Sp = specificity

* Cutoff value for the test ranged between 0.659 and 0.850; I^2 was 80% for sensitivity and 94% for specificity.

† Per Deeks asymmetry test, P <0.001, such that larger studies tended to have smaller diagnostic odds ratios. This finding partly recapitulates the finding of inconsistency across studies; thus, the CoE was not further downgraded for publication bias.

‡ *A priori*, moderate sensitivity and specificity defined as 76-89% and moderate accuracy defined as AUC 0.76-0.85.

References

1. Wei Q, Zhu Y, Zhen W, et al. Performance of resistive index and semi-quantitative power doppler ultrasound score in predicting acute kidney injury: A meta-analysis of prospective studies. PLoS One. 2022;17(6):e0270623. Published 2022 Jun 28. doi:10.1371/journal.pone.0270623

Table S104. Ultrasound (renal pulsatility index) in children in the ICU

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI (at 72 hours)	2 (142) [1-2]	Serious	Inconsistent	Direct	Sn, Sp: Precise AUC: Imprecise*	Undetected	None	Low	AUC 0.84 (0.36 – 0.98) Sn range: 0.73 – 0.92 Sp range: 0.67 – 0.79	Critical
Severe AKI (stage 3)	0								No evidence	Critical
AKI requiring RRT	0								No evidence	Critical
Death	0								No evidence	Critical
MAKE90	0								No evidence	Critical
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Balance of Potential Benefits and Harms: In children, renal pulsatility index has variable test accuracy to predict incident AKI across studies.								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = Acute Kidney Injury; AUC = Area Under the Curve; CKD = Chronic Kidney Disease; ICU = Intensive Care Unit; MAKE90 = Major Adverse Kidney Events at 90 days; RRT = renal replacement therapy; Sn = sensitivity; Sp = specificity

* Range of AUC 95% CI >0.2 (larger than the ranges for low, moderate, and high accuracy).

References

1. De Souza FM, De Carvalho AV, Ferraz IS, et al. Acute kidney injury in children undergoing cardiac surgery: predictive value of kidney arterial Doppler-based variables. Pediatr Nephrol. 2024;39(7):2235-2243. doi:10.1007/s00467-024-06319-3
2. de Carvalho AV, Ferraz IS, de Souza FM, et al. Acute kidney injury in critically ill children: predictive value of renal arterial Doppler assessment. Pediatr Res. 2023;93(6):1694-1700. doi:10.1038/s41390-022-02296-1

Table S105. Ultrasound (renal resistive index) in children in the ICU

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI (at 72 hr)	3 (188) [1-3]	Serious	Inconsistent	Direct	Imprecise*	Undetected	Sparse	Low	AUC 0.86 (0.46 – 0.98) Sn range: 0.73 – 0.92 Sp range: 0.55 – 0.85	Critical
Severe AKI (stage 3)	0								No evidence	Critical
AKI requiring RRT	0								No evidence	Critical
Death	0								No evidence	Critical
MAKE90	0								No evidence	Critical
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Balance of Potential Benefits and Harms: In children, renal resistive index has variable test accuracy to predict incident AKI across studies.								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = Acute Kidney Injury; AUC = Area Under the Curve; CKD = Chronic Kidney Disease; ICU = Intensive Care Unit; MAKE90 = Major Adverse Kidney Events at 90 days; RRT = renal replacement therapy; Sn = sensitivity; Sp = specificity

* Range of AUC 95% CI >0.2 (larger than the ranges for low, moderate, and high accuracy).

References

1. Shen H, Na W, Li Y, Qu D. The clinical significance of renal resistance index (RRI) and renal oxygen saturation (RrSO2) in critically ill children with AKI: a prospective cohort study. BMC Pediatr. 2023;23(1):224. Published 2023 May 6. doi:10.1186/s12887-023-03941-2
2. De Souza FM, De Carvalho AV, Ferraz IS, et al. Acute kidney injury in children undergoing cardiac surgery: predictive value of kidney arterial Doppler-based variables. Pediatr Nephrol. 2024;39(7):2235-2243. doi:10.1007/s00467-024-06319-3
3. de Carvalho AV, Ferraz IS, de Souza FM, et al. Acute kidney injury in critically ill children: predictive value of renal arterial Doppler assessment. Pediatr Res. 2023;93(6):1694-1700. doi:10.1038/s41390-022-02296-1

Table S106. RAI and NGAL clinical decision pathway vs. control in children in the pediatric ICU

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death (prior to PICU D/C, among those w/CRRT)	1 (178) [NRCS]	Very serious	N/A	Indirect*	Precise	Undetected	Sparse	Very Low	**OR 0.46 (0.25, 0.85)** 189 fewer per 1000 (336 fewer to 43 fewer)	Critical
AKI, any stage	0								No evidence	Critical
AKI, stage 3	0								No evidence	Critical
AKI requiring RRT	0								No evidence	Critical
MAKE (at ≥90 days)	0								No evidence	Critical
Duration of AKI	0								No evidence	Important
CKD G3-G5 (at ≥90 days)	0								No evidence	Important
CKD G5D (dialysis) (at ≥90 days)	0								No evidence	Important
Balance of Potential Benefits and Harms: RAI and NGAL guided pathway associated with lower risk of death in children in the ICU who ultimately received CRRT. No other evidence.								Certainty of Overall Evidence: Very Low		

Abbreviations: AKI = Acute Kidney Injury; CKD = Chronic Kidney Injury; CRRT = continuous renal replacement therapy; MAKE: Major adverse kidney events; N/A = not applicable; NGAL = neutrophil gelatinase-associated lipocalin; NRCS = nonrandomized comparative study (pre-post); OR = odds ratio; PICU = pediatric intensive care unit; RAI = renal angina index

* Due to restriction in who was analyzed for death.

References

1. Goldstein SL, Krallman KA, Roy JP, Collins M, Chima RS, Basu RK, Chawla L, Fei L. Real-Time Acute Kidney Injury Risk Stratification-Biomarker Directed Fluid Management Improves Outcomes in Critically Ill Children and Young Adults. Kidney Int Rep. 2023 Sep 22;8(12):2690-2700. doi: 10.1016/j.ekir.2023.09.019. PMID: 38106571; PMCID: PMC10719644.

Table S107. Gelatin vs. albumin/crystalloids in adults who are critically ill or undergoing surgery

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Death	9 (1316) [1] [RCT]	No limitations	Consistent	Direct	Precise	Undetected	None	High	RR 1.08 (0.83, 1.42) 11 more per 1000 (from 23 fewer to 57 more)	Critical
AKI, any stage	2 (152) [1] [RCT]	Serious limitations	Not consistent	Direct	Imprecise	Undetected	None	Low	RR 1.70 (0.71, 4.07) 67 more per 1000 (from 28 fewer to 294 more)	Critical
AKI, stage 3	0								No evidence	Critical
AKI requiring RRT	0								No evidence	Critical
MAKE90	0								No evidence	Critical
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Serious adverse events	0								No evidence	Important
Balance of Potential Benefits and Harms: There is no evidence that the choice of gelatin vs. albumin or crystalloids affects the risk of death or incident AKI in adults, with no trial evidence for other outcomes								Certainty of Overall Evidence: Moderate		

Abbreviations: AKI = Acute Kidney Injury; CKD: Chronic Kidney Injury; MAKE90 = Major Adverse Kidney Injury in 90 days; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy; MAKE: Major adverse kidney events.

References

1. Moeller C, Fleischmann C, Thomas-Rueddel D, Vlasakov V, Rochwerg B, Theurer P, Gattinoni L, Reinhart K, Hartog CS. How safe is gelatin? A systematic review and meta-analysis of gelatin-containing plasma expanders vs crystalloids and albumin. J Crit Care. 2016 Oct;35:75-83. doi: 10.1016/j.jcrc.2016.04.011. Epub 2016 Apr 23. PMID: 27481739.

Table S108. Albumin vs. alternatives in adults with cirrhosis

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	42 (4554) [1] [RCT]	Serious limitations	Consistent	Direct	Precise	Undetected*	None	Moderate	**OR 0.81 (0.67, 0.98)** 37 fewer per 1000 (from 66 fewer to 4 fewer)	Critical
AKI, any stage ("renal impairment")†	29 (3103) [1] [RCT]	Serious limitations	Consistent	Indirect †	Precise	Undetected	None	Low	**OR 0.63 (0.45, 0.88)** 45 fewer per 1,000 (from 73 fewer to 15 fewer)	Critical
AKI, stage 3	0								No evidence	Critical
AKI requiring RRT	0								No evidence	Critical
MAKE90	0								No evidence	Critical
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Serious AE	0								No evidence	Important
Balance of Potential Benefits and Harms: Compared with a variety of other fluids, there is a lower risk of death and "renal impairment" in adults with cirrhosis given albumin								Certainty of Overall Evidence: Low		

Abbreviations: AE = adverse event; AKI = Acute Kidney Injury; CKD = Chronic Kidney Injury; MAKE: Major adverse kidney events; RCT = randomized controlled trial; OR = odds ratio; RRT = renal replacement therapy

* Bai et al. 2022 (PMID 36048318) reported an Egger's test P = 0.565.
† The outcome "renal impairment" is not defined beyond "the definitions proposed by each included study."

References

1. Bai Z, Wang L, Wang R, et al. Use of human albumin infusion in cirrhotic patients: a systematic review and meta-analysis of randomized controlled trials. *Hepatol Int*. 2022;16(6):1468-1483. doi:10.1007/s12072-022-10374-z

Table S109. Aggressive intravenous fluid resuscitation vs. nonaggressive fluid resuscitation adults with pancreatitis

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Publication Bias	Precision of the Estimate	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Description of Findings	Importance of Outcome
Mortality	11 (3034) [1] [3 RCT 8 NRCS]	Serious limitations	Inconsistent	Direct	Undetected*	Precise	None	Low	RR 1.28 (95% CI 0.85, 1.94) 23 more per 1000 (from 12 fewer to 76 more)	Critical
Incident AKI	6 (1606) [1] [1 RCT 5 NRCS]	Serious limitations	Consistent	Direct	Undetected	Precise	Large effect	High	**RR 2.01 (95% CI 1.56, 2.58)** 98 more per 1000 (from 54 more to 153 more)	Critical
RRT	0								No evidence	Critical
MAKE	0								No evidence	Critical
AKI Stage 3	0								No evidence	Important
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Serious AE: Respiratory Failure / Ventilatory Support Need	6 (1555) [1] [2 RCT 4 NRCS]	Serious limitations	Consistent	Direct	Undetected	Precise	None	Moderate	**RR 1.69 (95% CI 1.13, 2.50)** 146 more per 1000 (from 27 more to 317 more)	Critical
Balance of Potential Benefits and Harms: Aggressive IV fluid resuscitation increases the incidence of AKI and respiratory failure, but does not affect mortality								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AE = adverse event; AKI = Acute Kidney Injury; CKD = chronic kidney disease; CI = confidence interval; MAKE = major adverse kidney event; NRCS = nonrandomized comparative study; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

* Per Ding 2023 37523599, Egger P 0.15.

References

1. Ding X, Chen B. Effect of Aggressive Intravenous Fluid Resuscitation Versus Nonaggressive Fluid Resuscitation in the Treatment of Acute Pancreatitis: A Systematic Review and Meta-Analysis. *Pancreas*. 2023;52(2):e89-e100. doi:10.1097/MPA.0000000000002230

Table S110. Restrictive intravenous fluid resuscitation vs. standard fluid resuscitation in adults with traumatic hemorrhagic shock

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Publication Bias	Precision of the Estimate	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Mortality	24 (1473) [1] [20 RCT, 4 NRCS]	Some limitations	Consistent	Direct	Serious*	Precise	Large effect	Moderate	**RR:0.50 (95% CI 0.40, 0.61)*** 133 fewer per 1000 (from 160 fewer to 104 fewer)	Critical
Incident AKI	7 (1217) [1] [5 RCT, 2 NRCS]	Some limitations	Inconsistent	Direct	Undetected	Imprecise	None	Low	RR:1.19 (95% CI 0.65, 2.21) 11 more per 1000 (from 20 fewer to 69 more)	Critical
RRT	0								No evidence	Critical
MAKE	0								No evidence	Critical
AKI Stage 3	0								No evidence	Important
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Serious AE: ARDS	10 (1428) [1] [7 RCT, 3 NRCS]	Some limitations	Consistent	Direct	Undetected†	Precise	Large effect	High	**RR:0.44 (95% CI 0.33, 0.59)** 92 fewer per 1000 (from 111 fewer to 68 fewer)	Critical
Balance of Potential Benefits and Harms: Restricted fluid resuscitation reduces mortality and acute respiratory distress syndrome, but with no evidence of effect on incident AKI								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AE = adverse event; AKI = Acute Kidney Injury; ARDS = acute respiratory distress syndrome; CI = confidence interval; CKD = chronic kidney disease; MAKE = major adverse kidney event; NRCS = nonrandomized comparative study; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

* Evidence of publication bias per Owattanapanich 2018 PMID 30558650. Egger P<0.001. Using trim and fill method to adjust for the publication bias yielded similar but somewhat weaker effect: RR 0.65 (95% CI 0.52, 0.80).

† Egger P = 0.45

References

1. Owattanapanich N, Chittawatanarat K, Benyakorn T, Sirikun J. Risks and benefits of hypotensive resuscitation in patients with traumatic hemorrhagic shock: a meta-analysis. Scand J Trauma Resusc Emerg Med. 2018;26(1):107. Published 2018 Dec 17. doi:10.1186/s13049-018-0572-4

Table S111. Restrictive fluid resuscitation vs. standard fluid resuscitation in adults with sepsis

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Publication Bias	Precision of the Estimate	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Mortality	9 (3938) [1-2] [RCT]	Some limitations	Consistent	Direct	Undetected	Precise	None	Moderate	RR 0.94 (95% CI 0.83-1.06) 20 fewer per 1000 (from 56 fewer to 20 more)	Critical
Incident AKI	3 (1730) [1] [RCT]	Some limitations	Consistent	Direct	Undetected	Precise	None	Moderate	RR: 0.95 (95% CI 0.80–1.14) 11 fewer per 1000 (from 44 fewer to 31 more)	Critical
RRT	5 (1973) [1-2] [RCT]	Some limitations	Consistent	Direct	Undetected	Precise	None	Moderate	RR: 0.88 (95% CI 0.63–1.25) 7 fewer per 1000 (from 21 fewer to 14 more)	Critical
MAKE	0								No evidence	Critical
AKI Stage 3	0								No evidence	Important
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Serious AE	1 (1563) [2] [RCT]	No limitations	Consistent	Direct	Undetected	Imprecise	Sparse (but large)	Low	RD 2 events (−10–14) 20 more per 1000 (from 100 fewer to 140 more)	Critical
Balance of Potential Benefits and Harms: No evidence of a difference in mortality or AKI outcomes with restricted vs. standard fluid resuscitation in sepsis								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AE = adverse event; AKI = Acute Kidney Injury; CI = confidence interval; CKD = chronic kidney disease; MAKE = major adverse kidney event; RCT = randomized controlled trial; RD = risk difference; RR = risk ratio; RRT = renal replacement therapy

References

1. Reynolds PM, Stefanos S, MacLaren R. Restrictive resuscitation in patients with sepsis and mortality: A systematic review and meta-analysis with trial sequential analysis. Pharmacotherapy. 2023;43(2):104-114. doi:10.1002/phar.2764
2. Shapiro NI, Douglas IS, et al. Early Restrictive or Liberal Fluid Management for Sepsis-Induced Hypotension. N Engl J Med. 2023;388(6):499-510. doi:10.1056/NEJMoa2212663

Table S112. Oral vs. intravenous rehydration in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	2 (70) [1] [RCT]	Serious limitations	Inconsistent	Direct	Imprecise	Undetected	None	Very Low	OR 1.33 (0.33 to 5.36) 47 more per 1000 (from 96 fewer to 623 more)	Critical
AKI, incident	1 (24) [1] [RCT]	Serious limitations	N/A	Direct	Imprecise	Undetected	Sparse, rare event	Very Low	RD 0.01 (-0.21, 0.24) [no events in IV arm] 14 more per 1000 (from 209 fewer to 237 more)	Critical
AKI, stage 3	0								No evidence	Critical
AKI requiring RRT	0								No evidence	Critical
MAKE90	0								No evidence	Critical
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Serious adverse events	0								No evidence	Critical
Balance of Potential Benefits and Harms: There is insufficient evidence to determine the relative effect of oral vs. intravenous rehydration in adults (but no evidence of a difference)								**Certainty of Overall Evidence:** Very Low		

Abbreviation: AKI = acute kidney injury; CKD = chronic kidney disease; MAKE = major adverse kidney injury; N/A = not applicable; OR= odds ratio; RCT = randomized controlled trial; RD = risk difference; RRT = renal replacement therapy

References

1. Hsiao KH, Kalanzi J, Watson SB, et al. Oral/enteral fluid resuscitation in the initial management of major burns: A systematic review and meta-analysis of human and animal studies. Burns Open. 2024;8(4):None. doi:10.1016/j.burnso.2024.100364

Table S113. Goal-directed therapy vs. usual care in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Description of Findings	Importance of Outcome
Death	0								No evidence	Critical
AKI, incident	65 (9308) [1] [Post-op] [RCT] 4 (1369) [2] [Medical] [RCT]	No serious	Consistent	Direct	Precise	Undetected	None	High	*Post-op:* **OR 0.74 (0.62 to 0.87)** 24 fewer per 1000 (from 35 fewer to 12 fewer) *Medical:* **RR 0.70 (0.54, 0.90)** 315 fewer per 1000 (from 554 fewer to 77 fewer)	Critical
AKI, stage 3	0								No evidence	Critical
RRT	10 (5098) [2] [RCT]	Serious limitations	Inconsistent	Direct	Precise	Undetected	None	Low	OR 1.04 (0.86 to 1.26) 4 more per 1000 (from 13 fewer to 24 more)	Critical
MAKE90	0								No evidence	Critical
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Serious adverse events	0								No evidence	Critical
Balance of Potential Benefits and Harms: Goal-directed therapy reduces the incidence of AKI but no evidence of an effect on RRT								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = Acute renal injury; CKD = chronic kidney disease; ED = Emergency department; ICU = intensive care unit; MAKE = major adverse kidney injury; OR = Odd ratio; Post-op = post operation; RCT = RR = risk ratio; RRT = renal replacement therapy

References

1. Giglio M, Dalfino L, Puntillo F, Brienza N. Hemodynamic goal-directed therapy and postoperative kidney injury: an updated meta-analysis with trial sequential analysis. Crit Care. 2019 Jun 26;23(1):232. doi: 10.1186/s13054-019-2516-4. PMID: 31242941; PMCID: PMC6593609.
2. Zhao CC, Ye Y, Li ZQ, Wu XH, Zhao C, Hu ZJ. Effect of goal-directed fluid therapy on renal function in critically ill patients: a systematic review and meta-analysis. Ren Fail. 2022 Dec;44(1):777-789. doi: 10.1080/0886022X.2022.2072338. PMID: 35535511; PMCID: PMC9103701.

Table S114. Vasopressin receptor agonists vs. norepinephrine in adults

Outcome	# Studies (Patients) [RCT]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Mortality, postop [post-surgical]	5 (493) [1]	Serious	Consistent	Direct	Imprecise	Undetected	Rare events [most studies], incomplete reporting	Very low	adjHR 1.11 (0.62, 1.96)* 5 more per 1000 (from 87 fewer to 78 more)	Critical
[sepsis]	4 (606) [2]	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	RR 1.03 (0.88, 1.22) 13 more per 1000 (from 48 fewer to 84 more)	
Incident AKI [periop hypotension]	5 (468) [1]	Serious	Inconsistent	Direct	Precise	Undetected	None	Low	**RR 0.45 (0.26, 0.80)** 153 fewer per 1000 (from 208 fewer to 56 fewer)	Critical
RRT [sepsis]	2 (539) [2]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	**RR 0.76 (0.59, 0.99)** 82 fewer per 1000 (from 142 fewer to 4 fewer)	Critical
MAKE	0								No evidence	Critical
AKI Stage 3	0								No evidence	Important
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
CKD G5D	0								No evidence	Important
Serious AE [sepsis]	3 (1286) [2]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	RR 0.95 (0.79, 1.15) 7 fewer per 1000 (from 32 fewer to 23 more)	Critical
Balance of Potential Benefits and Harms: Vasopressin receptor agonists lower the risk of incident AKI and RRT, compared with norepinephrine, but do not affect short-term mortality, without evidence of difference in serious adverse events								**Certainty of Overall Evidence:** Low		

Abbreviations: AE = adverse event; AKI = Acute Kidney Injury; CKD = chronic kidney disease; HR = hazard ratio; MAKE = major adverse kidney injury; N/A = not applicable; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy.

* Based on single study that reported sufficient data. Absolute difference based on unadjusted risk difference.

References

1. Heybati K, Xie G, Ellythy L, et al. Outcomes of Vasopressin-Receptor Agonists Versus Norepinephrine in Adults With Perioperative Hypotension: A Systematic Review. J Cardiothorac Vasc Anesth. 2024;38(7):1577-1586. doi:10.1053/j.jvca.2024.03.014
2. Sedhai YR, Shrestha DB, Budhathoki P, et al. Vasopressin versus norepinephrine as the first-line vasopressor in septic shock: A systematic review and meta-analysis. J Clin Transl Res. 2022;8(3):185-199. Published 2022 May 25.

Table S115. Terlipressin vs. terlipressin + norepinephrine for hepatorenal syndrome in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	0								No evidence	**Critical**
Reversal of HRS	1 (60) [1] [RCT]	Serious	N/A	Direct	Precise	Undetected	Sparse	Low	**RR 0.65 (0.43, 0.98)** 268 fewer per 1000 (from 437 fewer to 15 fewer)	**Critical**
AKI requiring RRT	0								No evidence	**Critical**
MAKE (at ≥ 90days)	0								No evidence	**Critical**
Serious adverse events, treatment-related	1 (60) [1] [RCT]	Serious	N/A	Direct	Imprecise*	Undetected	Sparse, but large effect	Low	RR 2.75 (0.99, 7.68) 233 more per 1000 (from 1 fewer to 888 more)	
Serious adverse events: Arrythmia	1 (60) [1] [RCT]	Serious	N/A	Direct	Very imprecise	Undetected	Sparse	Very Low	RR 3.00 (0.33, 27.24) 66 more per 1000 (from 22 fewer to 866 more)	**Critical**
Discontinuation due to AE	0								No evidence	**Critical**
Duration of AKI	0								No evidence	Important
Quality of life	0								No evidence	Important
Balance of Potential Benefits and Harms: Terlipressin alone results in a lower HRS reversal rate compared with combination terlipressin and norepinephrine, but with an increase in overall serious treatment-related adverse events								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = Acute Kidney Injury; HRS = hepatorenal syndrome, RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

* Imprecise if consider the effect size to be a consistent with a clinical effect (P = 0.053).

References

1. Singh V, Jayachandran A, De A, Singh A, Chandel S, Sharma N. Combination of terlipressin and noradrenaline versus terlipressin in hepatorenal syndrome with early non-response to terlipressin infusion: A randomized trial. Indian J Gastroenterol. 2023;42(3):388-395. doi:10.1007/s12664-023-01356-6

Table S116. Intravenous bicarbonate vs. usual care in adults and children* who are critically ill with acidosis

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death, 28 d †	4 (1140)[1-3,6] [RCT]	Some limitations	Inconsistent	Direct	Precise	Undetected	None	Moderate	RR 0.88 (0.72,1.07) 46 fewer per 1000 (107 fewer to 27 more)	Critical
AKI Stage 3	2 (46)[5,6] [RCT]	No limitations	Consistent	Direct	Very imprecise	Undetected	None	Low	RR 0.89 (0.43, 1.82) 46 fewer per 1000 (from 342 fewer to 238 more)	Critical
RRT	4 (1071)[2,3,5,6] [RCT]	Some limitations	Consistent	Direct	Precise	Undetected	None	High	**RR 0.69 (0.60, 0.80)** **131 fewer per 1000** **(168 fewer to 84 fewer)**	Critical
MAKE90	1 (627)[3]	No limitations	N/A	Direct	Precise	Undetected	Sparse, but large	Low	RR 0.99 (0.91, 1.07) 8 fewer per 1000 (76 fewer to 59 more)	Critical
ICU LOS	4 (1071)[3,5-7] [RCT]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	MD 0.41 days (-0.68, 1.49)	Important
Hospital LOS	4 (730)[2,3,5,6] [RCT]	Some limitations	Consistent	Direct	Precise	Undetected	None	Moderate	MD 0.05 days (-2.49, 2.60)	Important
Serious AE	3 (1041)[2,3,5] [RCT]	Some limitations	Consistent	Direct	Imprecise	Undetected	Sparse	Very low	Severe electrolyte abnormality (1 study): RR 1.85 (0.41, 8.32) 142 more per 1000 (from 1220 fewer to 98 more) Infection (2 studies): RR 1.02 (0.80, 1.29) 6 more per 1000 (60 fewer to 87 more)	Critical
Discontinuation due to AE	0								No evidence	Important
Balance of Potential Benefits and Harms: Intravenous bicarbonate for acidosis reduces the risk of RRT with no evidence of effect on mortality, other kidney outcomes or harms.								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AE = adverse event; AKI = Acute Kidney Injury; MAKE = Major Adverse Kidney Injury; MD = mean difference; N/A = not applicable; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy.

* One study randomized 50 children.[7] Combined with other studies.
† Analyses at other time points were similar in effect size and were not statistically significant.

References

1. Fang ZX, Li YF, Zhou XQ, Zhang Z, Zhang JS, Xia HM, Xing GP, Shu WP, Shen L, Yin GQ. Effects of resuscitation with crystalloid fluids on cardiac function in patients with severe sepsis. BMC Infect Dis. 2008 Apr 17;8:50. doi: 10.1186/1471-2334-8-50. PMID: 18419825; PMCID: PMC2364628.
2. Jaber S, Paugam C, Futier E, Lefrant JY, Lasocki S, Lescot T, Pottecher J, Demoule A, Ferrandière M, Asehnoune K, Dellamonica J, Velly L, Abback PS, de Jong A, Brunot V, Belafia F, Roquilly A, Chanques G, Muller L, Constantin JM, Bertet H, Klouche K, Molinari N, Jung B; BICAR-ICU Study Group. Sodium bicarbonate therapy for patients with severe metabolic acidaemia in the intensive care unit (BICAR-ICU): a multicentre, open-label, randomised controlled, phase 3 trial. Lancet. 2018 Jul 7;392(10141):31-40. doi: 10.1016/S0140-6736(18)31080-8. Epub 2018 Jun 14. Erratum in: Lancet. 2018 Dec 8;392(10163):2440. doi: 10.1016/S0140-6736(18)33040-X. PMID: 29910040.
3. Jung B, Jabaudon M, De Jong A, Bitker L, Audard J, Klouche K, Sarton B, Guitton C, Lasocki S, Rieu B, Canet E, Jeantrelle C, Roquilly A, Mayaux J, Verdonk F, Pottecher J, Ferrandiere M, Riu B, Garcon P, Assefi M, Detouche P, Forel JM, Roger C, Bourenne J, Jacquier S, Bougon D, Rolle A, Corne P, Benchabane N, Richard JC, Asehnoune K, Chanques G, Reignier J, Belafia F, Fosset M, Huguet H, Futier E, Molinari N, Jaber S; BICARICU-2 Study Group. Sodium Bicarbonate for Severe Metabolic Acidemia and Acute Kidney Injury: The BICARICU-2 Randomized Clinical Trial. JAMA. 2025 Oct 29:e2520231. doi: 10.1001/jama.2025.20231. Epub ahead of print. PMID: 41159812; PMCID: PMC12573113.
4. Lo KB, Garvia V, Stempel JM, Ram P, Rangaswami J. Bicarbonate use and mortality outcome among critically ill patients with metabolic acidosis: A meta analysis. Heart Lung. 2020;49(2):167-174. doi:10.1016/j.hrtlng.2019.10.007
5. Schneider AG, Bellomo R, Reade M, et al. Safety evaluation of a trial of lipocalin-directed sodium bicarbonate infusion for renal protection in at-risk critically ill patients. Crit Care Resusc. 2013;15(2):126-133.
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Table S117. Corticosteroids vs. control in hemolysis, elevated liver enzymes and low platelets (HELLP) syndrome in pregnant women

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	4 (355) [RCT]	Serious limitations	Consistent	Direct	Very imprecise	Undetected	None	Very Low	RR 1.39 (0.42, 4.55) 10 more per 1000 (from 15 fewer to 89 more)	Critical
Resolution of HELLP syndrome	0									Critical
Incident AKI	4 (376) [RCT]	Serious limitations	Consistent	Direct	Imprecise	Undetected	None	Low	RR 0.75 (0.43, 1.30) 36 fewer per 1000 (from 82 fewer to 43 more)	Critical
AKI requiring RRT	0								No evidence	Critical
MAKE90	0								No evidence	Critical
CKD G5D	0								No evidence	Critical
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
Serious AE	0								No evidence	Critical
Discontinuation due to AE	0								No evidence	Critical
Balance of Potential Benefits and Harms: No evidence of beneficial effect of corticosteroids to prevent death or AKI in pregnant women with HELLP								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = Acute Kidney Injury; HELLP = Hemolysis, Elevated Liver enzymes, and Low Platelets; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy; MAKE = major adverse kidney event; AE = adverse event

References

1. Sun WJ, Hu J, Zhang Q, Shan JM. Administration of corticosteroid therapy for HELLP syndrome in pregnant women: evidences from seven randomized controlled trials. Hypertens Pregnancy. 2023;42(1):2276726. doi:10.1080/10641955.2023.2276726

Table S118. Haptoglobin vs. control in adults with hemolysis

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	**Description of Findings**	**Importance of Outcome**
Death	3 (97) [2 RCT, 1 NRCS]	Very serious	Consistent	Direct	Very imprecise	Undetected	None	Very Low	OR 1.41 (0.49, 4.06) 95 more per 1000 (from 118 fewer to 710 more)	Critical
Resolution of hemolysis	0								No evidence	Critical
Incident AKI	3 (571) [1 RCT, 2 NRCS]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	**OR 0.65 (0.44, 0.96)** 104 fewer per 1000 (from 167 fewer to 12 fewer)	Critical
AKI requiring RRT	0								No evidence	Critical
MAKE90	0								No evidence	Critical
CKD G5D	0								No evidence	Critical
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
Serious AE	0								No evidence	Critical
Discontinuation due to AE	0								No evidence	Critical
Balance of Potential Benefits and Harms: Haptoglobin lowers the risk of incident AKI with unclear/unknown effect on other outcomes.								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = Acute Kidney Injury; RCT = randomized controlled trial; OR = odds ratio; RRT = renal replacement therapy; MAKE = major adverse kidney event; CKD = chronic kidney disease; AE = adverse event

References

1. Baldetti L, Labanca R, Belletti A, et al. Haptoglobin Administration for Intravascular Hemolysis: A Systematic Review. Blood Purif. 2024;53(11-12):851-859. doi:10.1159/000539363

Table S119. Continuous renal replacement therapy vs. conventional therapy in adults with rhabdomyolysis

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	1 (43) [RCT]	Very serious	Consistent	Direct	Imprecise*	Undetected	Sparse	Very Low	RR 0.17 (0.02, 1.37) 208 fewer per 1000 (from 245 fewer to 93 more)	Critical
Resolution of rhabdomyolysis	0								No evidence	Critical
Incident AKI	0								No evidence	Critical
AKI requiring RRT	0								No evidence	Critical
MAKE90	0								No evidence	Critical
CKD G5D	0								No evidence	Critical
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
Serious AE: Arrhythmia (AE)	1 (36) [RCT]	Very serious	Consistent	Direct	Very imprecise	Undetected	Sparse	Very Low	RR 0.67 (0.21, 2.18) 105 fewer par 1000 (from 251 fewer to 375 more)	Critical
Serious AE: Shock	1 (36) [RCT]	Very serious	Consistent	Direct	Very imprecise	Undetected	Sparse	Very Low	RR 0.63 (0.14, 2.81) 84 fewer per 1000 (from 195 fewer to 411 more)	Critical
Serious AE: Gastrointestinal hemorrhage	1 (36) [RCT]	Very serious	Consistent	Direct	Very imprecise	Undetected	Sparse	Very Low	RR 0.98 (0.40, 2.40) 7 fewer per 1000 (from 218 fewer to 510 more)	Critical
Discontinuation due to AE	0								No evidence	Important
Balance of Potential Benefits and Harms: Imprecise suggestion that CRRT may reduce risk of death with no evidence about AKI and rare adverse events								**Certainty of Overall Evidence:** Very Low		

Abbreviations: AKI = Acute Kidney Injury; CRRT = continuous renal replacement therapy; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy; MAKE = major adverse kidney event; CKD = chronic kidney disease; AE = adverse event

* Imprecise under the assumption that there may be an effect

References

1. Zeng X, Zhang L, Wu T, Fu P. Continuous renal replacement therapy (CRRT) for rhabdomyolysis. Cochrane Database Syst Rev. 2014;2014(6):CD008566. Published 2014 Jun 15. doi:10.1002/14651858.CD008566.pub

Table S120. Febuxostat vs. allopurinol therapy in adults with tumor lysis syndrome

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	0								No evidence	Critical
Resolution of TLS	2 (428) [RCT]	Serious	Consistent	Direct	Very imprecise	Undetected	None	Very Low	OR 1.45 (0.42, 5.02) 431 more per 1000 (from 556 fewer to 3851 more)	Critical
Incident AKI	0								No evidence	Critical
AKI requiring RRT	0								No evidence	Critical
MAKE90	0								No evidence	Critical
CKD G5D	0								No evidence	Critical
Duration of AKI	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
Serious AE	0								No evidence	Critical
Discontinuation due to AE	0								No evidence	Important
Balance of Potential Benefits and Harms: Highly imprecise evidence								**Certainty of Overall Evidence:** Very Low		

Abbreviations: AKI = Acute Kidney Injury; RCT = randomized controlled trial; OR = odds ratio; RRT = renal replacement therapy; TLS = tumor lysis syndrome; MAKE = major adverse kidney event; CHD = chronic kidney disease; AE = adverse event

References

1. Bellos I, Kontzoglou K, Psyrri A, Pergialiotis V. Febuxostat administration for the prevention of tumour lysis syndrome: A meta-analysis. *J Clin Pharm Ther*. 2019;44(4):525-533. doi:10.1111/jcpt.12839

Table S121. Bartholomew score in adults undergoing coronary angiography or angioplasty

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI, any stage	5 (7,930) [1]	Some limitations	Inconsistent*	Direct	Imprecise †	Undetected	None	Moderate	AUC 0.75 (0.56-0.93) ‡	Critical
Duration of AKI	0								No evidence	Important
Balance of Potential Benefits and Harms: In adults, Bartholomew score has moderate accuracy to predict AKI incidence §								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = Acute Kidney Injury; AUC = Area Under the Curve.

* $I^2 = 97\%$
† 95% confidence interval for AUC spans a range >0.2.
‡ Pooled AUC from external validation studies. In the derivation study, AUC was 0.89 (0.74 – 0.99).
§ Thresholds for AUC were arbitrarily defined as ≤0.74 low, 0.75-0.89 moderate, ≥0.90 high.

References

1. Allen DW, Ma B, Leung KC, et al. Risk Prediction Models for Contrast-Induced Acute Kidney Injury Accompanying Cardiac Catheterization: Systematic Review and Meta-analysis. Can J Cardiol. 2017;33(6):724-736. doi:10.1016/j.cjca.2017.01.018

Table S122. Brown score in adults undergoing coronary angiography or angioplasty

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Incident AKI, any stage	1 (27,905) [1]	Some limitations	NA	Direct	Precise	Undetected	Single study	Low	AUC 0.70 (0.70-0.71)*	Critical
Duration of AKI	0								No evidence	Important
Balance of Potential Benefits and Harms: In adults, Brown score has low accuracy to predict AKI incidence †								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = Acute Kidney Injury; AUC = Area Under the Curve; NA = Not Applicable.

* AUC from an external validation study. In the derivation study, AUC was 0.84 (0.80– 0.89).
† Thresholds for AUC were arbitrarily defined as ≤0.74 low, 0.75-0.89 moderate, ≥0.90 high.

References

1. Allen DW, Ma B, Leung KC, et al. Risk Prediction Models for Contrast-Induced Acute Kidney Injury Accompanying Cardiac Catheterization: Systematic Review and Meta-analysis. Can J Cardiol. 2017;33(6):724-736. doi:10.1016/j.cjca.2017.01.018

Table S123. Ghani score in adults undergoing coronary angiography or angioplasty

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI, any stage	1 (805) [1]	Some limitations	NA	Direct	Imprecise*	Undetected	Single study	Low	AUC 0.65 (0.51-0.78) †	Critical
Duration of AKI	0								No evidence	Important
Balance of Potential Benefits and Harms: In adults, Ghani score has low accuracy to predict AKI incidence ‡								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = Acute Kidney Injury; AUC = Area Under the Curve; NA = Not Applicable.

* 95% confidence interval for AUC spans a range >0.2.
† AUC from an external validation study. In the derivation study, AUC was 0.61 (0.25– 0.97).
‡ Thresholds for AUC were arbitrarily defined as ≤0.74 low, 0.75-0.89 moderate, ≥0.90 high.

References

1. Allen DW, Ma B, Leung KC, et al. Risk Prediction Models for Contrast-Induced Acute Kidney Injury Accompanying Cardiac Catheterization: Systematic Review and Meta-analysis. Can J Cardiol. 2017;33(6):724-736. doi:10.1016/j.cjca.2017.01.018

Table S124. Marenzi score in adults undergoing coronary angiography or angioplasty

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Incident AKI, any stage	1 (891) [1]	Some limitations	NA	Direct	Precise	Undetected	Single study	Low	AUC 0.57 (0.51-0.62) †	Critical
Duration of AKI	0								No evidence	Important
Balance of Potential Benefits and Harms: In adults, Marenzi score has low accuracy to predict AKI incidence ‡								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = Acute Kidney Injury; AUC = Area Under the Curve; NA = Not Applicable.

* 95% confidence interval for AUC spans a range >0.2.
† AUC from an external validation study. In the derivation study, AUC was not reported.
‡ Thresholds for AUC were arbitrarily defined as ≤0.74 low, 0.75-0.89 moderate, ≥0.90 high.

References

1. Allen DW, Ma B, Leung KC, et al. Risk Prediction Models for Contrast-Induced Acute Kidney Injury Accompanying Cardiac Catheterization: Systematic Review and Meta-analysis. Can J Cardiol. 2017;33(6):724-736. doi:10.1016/j.cjca.2017.01.018

Table S125. Mehran score in adults undergoing coronary angiography or angioplasty

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI, any stage	14 (16,211) [1]	Some limitations	Inconsistent*	Direct	Precise	Undetected	None	Moderate	AUC 0.72 (0.63-0.80) †	Critical
Duration of AKI	0								No evidence	Important
Balance of Potential Benefits and Harms: In adults, Mehran score has low accuracy to predict AKI incidence ‡								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = Acute Kidney Injury; AUC = Area Under the Curve.

* $I^2 = 98\%$
† Pooled AUC from external validation studies. In the derivation study, AUC was 0.67 (0.62 – 0.72).
‡ Thresholds for AUC were arbitrarily defined as ≤0.74 low, 0.75-0.89 moderate, ≥0.90 high.

References

1. Allen DW, Ma B, Leung KC, et al. Risk Prediction Models for Contrast-Induced Acute Kidney Injury Accompanying Cardiac Catheterization: Systematic Review and Meta-analysis. Can J Cardiol. 2017;33(6):724-736. doi:10.1016/j.cjca.2017.01.018

Table S126. Tsai score in adults undergoing coronary angiography or angioplasty

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI, any stage	1 (11,041) [1]	Some limitations	NA	Direct	Precise	Undetected	Single study	Low	AUC 0.76 (0.72-0.80)*	Critical
Duration of AKI	0								No evidence	Important
Balance of Potential Benefits and Harms: In adults, Tsai score has moderate accuracy to predict AKI incidence †								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = Acute Kidney Injury; AUC = Area Under the Curve; NA = Not Applicable.

* AUC from an external validation study. In the derivation study, AUC was 0.71 (0.71– 0.72).
† Thresholds for AUC were arbitrarily defined as ≤0.74 low, 0.75-0.89 moderate, ≥0.90 high.

References

1. Allen DW, Ma B, Leung KC, et al. Risk Prediction Models for Contrast-Induced Acute Kidney Injury Accompanying Cardiac Catheterization: Systematic Review and Meta-analysis. Can J Cardiol. 2017;33(6):724-736. doi:10.1016/j.cjca.2017.01.018

Table S127. Tziakas score in adults undergoing coronary angiography or angioplasty

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI, any stage	2 (2,929) [1]	Some limitations	Consistent	Direct	Precise	Undetected	Sparse	Low	AUC 0.74 (0.71-0.77)*	Critical
Duration of AKI	0								No evidence	Important
Balance of Potential Benefits and Harms: In adults, Tziakas score has low accuracy to predict AKI incidence.**								**Certainty of Overall Evidence:** Low		

Abbreviations: AKI = Acute Kidney Injury; AUC = Area Under the Curve.

* Pooled AUC from external validation studies. In the derivation study, AUC was 0.86 (0.80 – 0.93).
** Thresholds for AUC were arbitrarily defined as ≤0.74 low, 0.75-0.89 moderate, ≥0.90 high.

References

1. Allen DW, Ma B, Leung KC, et al. Risk Prediction Models for Contrast-Induced Acute Kidney Injury Accompanying Cardiac Catheterization: Systematic Review and Meta-analysis. Can J Cardiol. 2017;33(6):724-736. doi:10.1016/j.cjca.2017.01.018

Table S128. Predication models, overall, in adults undergoing coronary angiography or angioplasty

Outcome	# Studies (Patients)	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI, any stage	64 (2,092,369)[1]	Some limitations	Inconsistent*	Direct	Precise	Undetected	None	Moderate	AUC 0.83 (0.82-0.84) Sn 0.74 (0.70-0.78) Sp 0.78 (0.75-0.82)	Critical
Duration of AKI	0								No evidence	Important
Balance of Potential Benefits and Harms: In adults, risk scores have moderate accuracy, low sensitivity and moderate specificity to predict AKI incidence. †								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AKI = acute kidney injury; AUC = area under the curve; Sn = sensitivity; Sp = specificity.

* Substantial heterogeneity was reported without I^2 estimate.
† Thresholds for AUC, sensitivity and specificity were arbitrarily defined as ≤0.74 low, 0.75-0.89 moderate, ≥0.90 high.

References

1. Feng Y, Jun M, Wang AY, et al. Predicting Contrast-Associated Acute Kidney Injury. JAMA Netw Open. 2025;8(3):e250107. Published 2025 Mar 3. doi:10.1001/jamanetworkopen.2025.0107

Table S129. Prophylactic intravenous isotonic 0.9% saline vs. oral hydration with intravascular contrast in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Incident AKI	6 (661)[1] [RCT]	Some serious	Consistent	Direct	Very imprecise	Undetected	No studies	Very Low	OR 0.91 (0.27–3.32)* 9 fewer per 1000 (from 77 fewer to 243 more)	Critical
Mortality	Not analyzed								No evidence	Critical
RRT	Not analyzed								No evidence	Critical
Duration of AKI	Not analyzed								No evidence	Important
Balance of Potential Benefits and Harms: No evidence regarding the effect of intravenous isotonic fluid versus oral hydration to prevent contrast-induced nephropathy.								**Certainty of Overall Evidence:** Very low		

Abbreviations: AKI = Acute Kidney Injury; N/A = not applicable; NMA = network meta-analysis; RR = relative risk; RRT = renal replacement therapy

* Based on network meta-analysis of 60 trials that evaluate prophylactic fluid for intravenous contrast.

References

1. Cai Q, Jing R, Zhang W, Tang Y, Li X, Liu T. Hydration Strategies for Preventing Contrast-Induced Acute Kidney Injury: A Systematic Review and Bayesian Network Meta-Analysis. J Interv Cardiol. 2020 Feb 11;2020:7292675. doi: 10.1155/2020/7292675. PMID: 32116474; PMCID: PMC7036123.

Table S130. Continuous renal replacement therapy vs. sustained low efficiency dialysis (SLED) in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Mortality, 90 d	5* (448) [1] [RCT]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	RR 1.06 (0.85 to 1.33) 30 more per 1000 (from 74 fewer to 164 more)	Critical
Time on RRT	NR* (NR) [1] [RCT]	Serious	Consistent	Direct	Very imprecise	Undetected	None	Very low	MD 0.57 days (-2.98 to 4.11)	Critical
CKD G5D	0								No evidence	Critical
Quality of life	0								No evidence	Critical
Serious AE	0								No evidence	Critical
Recovery of kidney function	4* (NR) [1] [RCT]	No serious	Inconsistent	Direct	Imprecise	Undetected	None	Low	RR 0.88 (0.65 to 1.19) Absolute undefined	Critical
Length of Hospital stay	3* (NR) [1] [RCT]	No serious	Consistent	Direct	Imprecise†	Undetected	None	Moderate	MD 6.58 days (-0.93 to 14.09)	Important
Length of ICU stay	NR* (NR) [1] [RCT]	No serious	Inconsistent	Direct	Imprecise	Undetected	None	Low	MD 1.78 days (-2.31 to 5.87)	Important
CKD G3-G5	0								No evidence	Important
Balance of Potential Benefits and Harms: No evidence of differences in death, AKI outcomes, or length of stay								**Certainty of Overall Evidence:** Low		

Abbreviations: AE = adverse event; AKI = acute kidney injury; CKD = chronic kidney disease; d = day; ICU = intensive care unit; MD = mean difference; NR = not reported; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement

* Based on network meta-analysis.
† For the conclusion of a possible effect (P = 0.086)

References

1. Ye Z, Wang Y, Ge L, et al. Comparing Renal Replacement Therapy Modalities in Critically Ill Patients With Acute Kidney Injury: A Systematic Review and Network Meta-Analysis. Crit Care Explor. 2021;3(5):e0399. Published 2021 May 12. doi:10.1097/CCE.0000000000000399

Table S131. Peritoneal dialysis vs. sustained low efficiency dialysis (SLED) in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death, all-cause	2* (NR) [1] [RCT]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	RR 0.91 (0.75 to 1.11) 57 fewer per 1000 (from 159 fewer to 70 more)	Critical
Time on RRT	NR* (NR) [1] [RCT]	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	MD -2.43 days (-6.02 to 1.16)	Critical
CKD G5D	0								No evidence	Critical
Quality of life	0								No evidence	Critical
Serious AE	0								No evidence	Critical
Recovery of kidney function	2* (NR) [1] [RCT]	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	RR 1.02 (0.68 to 1.51) 6 more per 1000 (from 95 fewer to 151 more)	Critical
Length of Hospital stay	0								No evidence	Important
Length of ICU stay	NR* (NR) [1] [RCT]	No serious	N/A	Indirect [No direct]	Precise	Undetected	Large effect	Low	**MD -8.22 days (-16.00 to -0.44)**	Important
CKD G3-G5	0								No evidence	Important
Balance of Potential Benefits and Harms: Peritoneal dialysis results in a large decrease in ICU length of stay with no evidence of an effect on death or AKI outcomes								**Certainty of Overall Evidence:** Moderate		

Abbreviations: AE = adverse event; AKI = acute kidney injury; CKD = chronic kidney disease; G = grade; ICU = intensive care unit; MD = mean difference; N/A = not applicable; NR = not reported; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

* Based on network meta-analysis.

References

1. Ye Z, Wang Y, Ge L, et al. Comparing Renal Replacement Therapy Modalities in Critically Ill Patients With Acute Kidney Injury: A Systematic Review and Network Meta-Analysis. Crit Care Explor. 2021;3(5):e0399. Published 2021 May 12. doi:10.1097/CCE.0000000000000399

Table S132. Intermittent hemodialysis vs. sustained low efficiency dialysis (SLED) in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Description of Findings	Importance of Outcome
Death, all-cause	NR* (NR) [1] [RCT]	Serious	Consistent	Indirect [No direct]	Imprecise	Undetected	None	Very low	RR 1.02 (0.79 to 1.31) Absolute undefined	Critical
Time on RRT	NR* (NR) [1] [RCT]	No serious	Consistent	Indirect [No direct]	Imprecise	Undetected	None	Very low	MD 1.44 days (-3.00 to 5.89)	Critical
CKD G5D	0								No evidence	Critical
Quality of life	0								No evidence	Critical
Serious AE	0								No evidence	Critical
Recovery of kidney function	NR* (NR) [1] [RCT]	Serious	Consistent	Indirect [No direct]	Precise		None	Very low	RR 0.77 (0.53 to 1.12) Absolute undefined	Critical
Length of Hospital stay	NR* (NR) [1] [RCT]	Serious	Consistent	Indirect [No direct]	Imprecise	Undetected	None	Very low	MD 4.11 days (-4.88 to 13.10)	Important
Length of ICU stay	NR* (NR) [1] [RCT]	Serious	Consistent	Indirect [No direct]	Imprecise	Undetected	None	Very low	MD 3.14 days (-2.40 to 8.68)	Important
CKD G3-G5	0								No evidence	
Balance of Potential Benefits and Harms: No evidence of differences in effects of SLED or intermittent HD on AKI, mortality, or length of stay outcomes								**Certainty of Overall Evidence:** Very Low		

Abbreviations: AE = adverse event; AKI = acute kidney injury; CKD = chronic kidney disease; ICU = intensive care unit; MD = mean difference; NR = not reported; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy; SLED = sustained low efficiency dialysis.

* Based on network meta-analysis.

References

1. Ye Z, Wang Y, Ge L, et al. Comparing Renal Replacement Therapy Modalities in Critically Ill Patients With Acute Kidney Injury: A Systematic Review and Network Meta-Analysis. Crit Care Explor. 2021;3(5):e0399. Published 2021 May 12. doi:10.1097/CCE.0000000000000399.

Table S133. Newcastle Infant Dialysis Ultrafiltration System vs. traditional RRT in children in the ICU

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Description of Findings	Importance of Outcome
Death, all-cause	1 (97) [1] [RCT]	Very serious	N/A	Direct	Precise	Undetected	Sparse, Large effect	Low	**RR 2.66 (1.20, 5.87)** 26 more 1000 (from 214 more to 628 more)	Critical
Time on RRT	0								No evidence	Critical
CKD G5D	0								No evidence	Critical
Quality of life	0								No evidence	Critical
Serious AE	1 (97) [1] [RCT]	Very serious	N/A	Direct	Precise	Undetected	Sparse	Very low	RR 0.93 (0.82, 1.06) 67 fewer per 1000 (from 171 fewer to 57 more)	Critical
Recovery of kidney function	0								No evidence	Critical
Length of Hospital stay	0								No evidence	Important
Length of ICU stay	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
Balance of Potential Benefits and Harms: The Newcastle infant dialysis ultrafiltration system reduces the risk of death, without evidence of increased serious adverse events								**Certainty of Overall Evidence:** Low		

Abbreviations: AE = adverse event; AKI = Acute Kidney Injury; CKD = chronic kidney disease; ICU = intensive care unit; N/A = not applicable; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

References

1. Lambert H, Hiu S, Coulthard MG, et al. The Infant KIdney Dialysis and Ultrafiltration (I-KID) Study: A Stepped-Wedge Cluster-Randomized Study in Infants, Comparing Peritoneal Dialysis, Continuous Venovenous Hemofiltration, and Newcastle Infant Dialysis Ultrafiltration System, a Novel Infant Hemodialysis Device. *Pediatr Crit Care Med*. 2023;24(7):604-613. doi:10.1097/PCC.0000000000003220

Table S134. Intensified vs. standard extended dialysis in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Description of Findings	Importance of Outcome
Death, 28 d	1 (156) [1] [RCT]	Some limitations	N/A	Direct	Imprecise	Undetected	Sparse	Very Low	RR 1.15 (0.79, 1.67) 58 more per 1000 (from 81 fewer to 259 more)	Critical
Time on RRT	0								No evidence	Critical
CKD G5D	0								No evidence	Critical
Quality of life	0								No evidence	Critical
Complete recovery	1 (156) [1] [RCT]	Some limitations	N/A	Direct	Imprecise	Undetected	Sparse	Very Low	RR 0.95 (0.69, 1.32) 31 fewer per 1000 (from 195 fewer to 202 more)	Important
Hospital LOS	0								No evidence	Important
ICU LOS	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
Serious AE	0								No evidence	Critical
Balance of Potential Benefits and Harms: No difference in death or complete recovery								**Certainty of Overall Evidence:** Very Low		

Abbreviations: AE = Adverse events; CKD = chronic kidney disease; d = day; G = grade; ICU = intensive care unit; LOS = length of stay; N/A = not applicable; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

References

1. Faulhaber-Walter R, Hafer C, Jahr N, et al. The Hannover Dialysis Outcome study: comparison of standard versus intensified extended dialysis for treatment of patients with acute kidney injury in the intensive care unit. Nephrol Dial Transplant. 2009;24(7):2179-2186. doi:10.1093/ndt/gfp035

Table S135. High-volume vs. conventional volume hemofiltration in children with sepsis

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Death	1 (76) [1] [RCT]	Some limitations	N/A	Direct	Imprecise*	Undetected	Sparse	Very Low	RR 0.31 (0.09, 1.03) † 182 fewer per 1000 (from 349 fewer to 15 fewer) †	Critical
Time on RRT	1 (76) [1] [RCT]	Some limitations	N/A	Direct	Precise	Undetected	Sparse	Very low	−1.2 d (−2.1, −0.3)	Critical
Complete recovery	0								No evidence	Important
CKD G5D	0								No evidence	Critical
Quality of life	0								No evidence	Critical
Length of hospital stay	0								No evidence	Important
Length of ICU stay	0								No evidence	Important
CKD G3-G5 (at ≥90 days)	0								No evidence	Important
Serious AE	1 (76) [1] [RCT]	Some limitations	N/A	Direct	Very imprecise	Undetected	No events	Very low	No events	Critical
Balance of Potential Benefits and Harms: High volume hemofiltration lowers the risk of death, time on RRT, with unclear effect on serious adverse events								**Certainty of Overall Evidence:** Very Low		

Abbreviations: AE= Adverse events; CKD = chronic kidney disease; G = grade; ICU = intensive care unit; N/A = not applicable; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy

* Downgraded for imprecision for the interpretation that high-volume hemofiltration decreases the risk of death.
† Not statistically significant analyzed as risk ratio, but statistically significant analyzed as risk difference.

References

1. Meng SQ, Yang WB, Liu JG, et al. Evaluation of the application of high volume hemofiltration in sepsis combined with acute kidney injury. Eur Rev Med Pharmacol Sci. 2018;22(3):715-720. doi:10.26355/eurrev_201802_14298

Table S136. Intensive vs. minimal standard dosing of peritoneal dialysis in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Description of Findings	Importance of Outcome
Death, 30 d	1 (75) [1] [RCT]	Some limitations	N/A	Direct	Imprecise	Undetected	Sparse	Very Low	RR 1.11 (0.80, 1.51) 70 more per 1000 (from 128 fewer to 326 more)	Critical
Time on RRT	0								No evidence	Critical
CKD G5D	1 (75) [1] [RCT]	Some limitations	N/A	Direct	Highly imprecise	Undetected	Sparse	Very Low	RR 0.64 (0.34, 1.18) 30 fewer per 1000 (from 55 fewer to 15 more)	Critical
Quality of life	0								No evidence	Critical
Complete recovery	0								No evidence	Important
Hospital LOS	1 (75) [1] [RCT]	Some limitations	N/A	Direct	Highly imprecise	Undetected	Sparse	Very Low	MD 0.17 (−6.37, 6.7)	Important
ICU LOS	0								No evidence	Important
CKD G3-G5	0								No evidence	Important
Serious AE	0								No evidence	Critical
SAE: Peritonitis	1 (75) [1] [RCT]	Some limitations	N/A	Direct	Highly imprecise	Undetected	Sparse	Very Low	RR 1.85 (0.50, 6.84) 71 more per 1000 (from 41 fewer to 487 more)	Critical
Balance of Potential Benefits and Harms: No difference in death or CKD 5D, or in harms								**Certainty of Overall Evidence:** Very Low		

Abbreviations: AE = Adverse events; CKD = chronic kidney disease; d = day; G = grade; ICU = intensive care unit; LOS = length of stay; MD = mean difference; N/A = not applicable; RCT = randomized controlled trial; RR = risk ratio; RRT = renal replacement therapy; SAE = serious adverse events

References

1. Parapiboon W, Jamratpan T. Intensive Versus Minimal Standard Dosage for Peritoneal Dialysis in Acute Kidney Injury: A Randomized Pilot Study. Perit Dial Int. 2017;37(5):523-528. doi:10.3747/pdi.2016.00260

Table S137. Nafamostat vs. no anticoagulation in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Description of Findings	Importance of Outcome
Filter clotting	1 (55) [1] [RCT]	Serious	Consistent [in NMA]	Direct	Precise	Undetected	Sparse	Low	**RR 0.63 (0.42, 0.95)** 221 fewer per 1000 filters (from 346 fewer to 30 fewer)	**Critical**
AE: Major bleeding	4 (195) [2]	Serious	Inconsistent	Direct	Very imprecise	Undetected	None	Very low	RR 1.03 (0.41, 2.57) 2 more per 1000 (from 44 fewer to 118 more)	**Critical**
Discontinuation due to AE	1 (73) [3] [RCT]	Serious	N/A	Direct	Very imprecise	Undetected	Sparse	Very low	RR 5.14 (0.26, 103.39) 56 more per 1000 (from 19 fewer to 130 more)	**Critical**
AE: Stroke	0								No evidence	**Critical**
Balance of Potential Benefits and Harms: Nafamostat reduces the risk of filter clotting with an unclear effect on adverse events								**Certainty of Overall Evidence:** Low		

Abbreviations: AE = Adverse events; CRRT = Continuous renal replacement therapy; ICU = Intensive care unit; MD = mean difference; NA = not applicable; RCT = randomized controlled trial; RD = risk difference; RR = risk ratio

References

1. Choi Tsujimoto H, Tsujimoto Y, Nakata Y, et al. Pharmacological interventions for preventing clotting of extracorporeal circuits during continuous renal replacement therapy. Cochrane Database Syst Rev. 2020;12(12):CD012467. Published 2020 Dec 14. doi:10.1002/14651858.CD012467.pub3
2. Lin Y, Shao Y, Liu Y, et al. Efficacy and safety of nafamostat mesilate anticoagulation in blood purification treatment of critically ill patients: a systematic review and meta-analysis. Ren Fail. 2022;44(1):1263-1279. doi:10.1080/0886022X.2022.2105233
3. Lee YK, Lee HW, Choi KH, Kim BS. Ability of nafamostat mesilate to prolong filter patency during continuous renal replacement therapy in patients at high risk of bleeding: a randomized controlled study. PLoS One. 2014 Oct 10;9(10):e108737. doi: 10.1371/journal.pone.0108737. PMID: 25302581

Table S138. Regional citrate anticoagulation vs no anticoagulation in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Publication Bias	Precision of the Estimate	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Filter clotting	2 (107) [1-2] [RCT]	Serious	Consistent	Direct	Undetected	Precise	None	Low	**RR 0.43 (0.26, 0.71)** 340 fewer episodes per 1000 patients (from 441 fewer to 173 fewer)	**Critical**
AE: Bleeding	3 (148) [1-2] [RCT]	Serious	Consistent	Direct	Undetected	Imprecise*	None	Low	RR 0.36 (0.11, 1.16) 71 fewer per 1000 (from 99 fewer to 18 more)	**Critical**
AE: Stroke	0								No evidence	**Critical**
Discontinuation due to AE	0								No evidence	**Critical**
Balance of Potential Benefits and Harms: Compared with no anticoagulation, regional citrate anticoagulation may decrease the risk of filter clotting and bleeding,								**Certainty of Overall Evidence:** Low		

Abbreviations: AE = Adverse events; CRRT = Continuous renal replacement therapy; ICU = Intensive care unit; MD = mean difference; NA = not applicable; RCT = randomized controlled trial; RR = risk ratio

* Imprecise from the perspective of there being a clinical effect (of similar magnitude of other outcomes)

References

1. Zhou Z, Liu C, Yang Y, Wang F, Zhang L, Fu P. Anticoagulation options for continuous renal replacement therapy in critically ill patients: a systematic review and network meta-analysis of randomized controlled trials. Crit Care. 2023;27(1):222. Published 2023 Jun 7. doi:10.1186/s13054-023-04519-1
2. Ratanarat R, Phairatwet P, Khansompop S, Naorungroj T. Customized Citrate Anticoagulation versus No Anticoagulant in Continuous Venovenous Hemofiltration in Critically Ill Patients with Acute Kidney Injury: A Prospective Randomized Controlled Trial. Blood Purif. 2023;52(5):455-463. doi:10.1159/000529076

Table S139. Systemic unfractionated heparin vs. low molecular weight heparin in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Filter clotting	0								No evidence	**Critical**
AEs: major bleeding	5 (153) [1] [RCT]	Serious	Inconsistent	Direct	Very imprecise	Undetected	None	Very Low	RR 0.58 (0.13, 2.58) 43 fewer per 1000 (from 90 fewer to 163 more)	**Critical**
Discontinuation due to AE	2 (77) [1] [RCT]	Serious	Inconsistent	Direct	Very imprecise	Undetected	None	Very Low	RR 0.29 (0.02, 3.53) 75 fewer per 1000 (from 103 fewer to 266 more)	**Critical**
AE: Stroke	0								No evidence	**Critical**
Balance of Potential Benefits and Harms: There is uncertainty about the difference in outcomes of systemic UFH vs. LMWH								**Certainty of Overall Evidence:** Very Low		

Abbreviations: AE = Adverse events; CRRT = Continuous renal replacement therapy; MD = mean difference; NA = not applicable; RCT = randomized controlled trial; RR = risk ratio

References

1. Tsujimoto H, Tsujimoto Y, Nakata Y, et al. Pharmacological interventions for preventing clotting of extracorporeal circuits during continuous renal replacement therapy. Cochrane Database Syst Rev. 2020;12(12):CD012467. Published 2020 Dec 14. doi:10.1002/14651858.CD012467.pub3

Table S140. Systemic unfractionated heparin vs. prostaglandin I2 in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Filter clotting	1 (21) [1] [RCT]	Serious	Consistent [in NMA]	Direct	Very imprecise	Undetected	Sparse	Very Low	RR 1.22 (0.27, 5.46) 44 more episodes per 1000 patients (from 146 fewer to 892 more)	**Critical**
AE: major bleeding	1 (27) [2] [RCT]	Serious	N/A	Direct	Very imprecise	Undetected	Sparse, rare events	Very Low	0 events in both arms	**Critical**
AE: Stroke	0								No evidence	**Critical**
Discontinuation due to AE	0								No evidence	**Critical**
Balance of Potential Benefits and Harms: There is uncertainty about the difference in outcomes of systemic unfractionated heparin and prostaglandin I2								**Certainty of Overall Evidence:** Very Low		

Abbreviations: AE = Adverse events; NA = not applicable; RCT = randomized controlled trial; RR = risk ratio

References

1. Zhou Z, Liu C, Yang Y, Wang F, Zhang L, Fu P. Anticoagulation options for continuous renal replacement therapy in critically ill patients: a systematic review and network meta-analysis of randomized controlled trials. Crit Care. 2023;27(1):222. Published 2023 Jun 7. doi:10.1186/s13054-023-04519-1
2. Tsujimoto H, Tsujimoto Y, Nakata Y, et al. Pharmacological interventions for preventing clotting of extracorporeal circuits during continuous renal replacement therapy. Cochrane Database Syst Rev. 2020;12(12):CD012467. Published 2020 Dec 14. doi:10.1002/14651858.CD012467.pub3

Table S141. Systemic unfractionated heparin vs. no anticoagulation in adults

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Filter clotting	0								No evidence	**Critical**
AEs: major bleeding	1 (20) [1] [RCT]	Serious	N/A	Direct	Very imprecise	Undetected	Sparse	Very Low	RR 0.50 (0.05, 4.67) 100 fewer per 1000 (from 190 fewer to 734 more)	**Critical**
AE: Stroke	0								No evidence	**Critical**
Discontinuation due to AE	0								No evidence	**Critical**
Balance of Potential Benefits and Harms: There is uncertainty about the difference in outcomes of systemic UFH vs. no pharmacologic intervention								**Certainty of Overall Evidence:** Very Low		

Abbreviations: AE = Adverse events; ICU = Intensive care unit; MD = mean difference; NA = not applicable; RCT = randomized controlled trial; RR = risk ratio

References

1. Tsujimoto H, Tsujimoto Y, Nakata Y, et al. Pharmacological interventions for preventing clotting of extracorporeal circuits during continuous renal replacement therapy. Cochrane Database Syst Rev. 2020;12(12):CD012467. Published 2020 Dec 14. doi:10.1002/14651858.CD012467.pub3

Table S142. NSAIDs vs. no NSAIDs in adults post-AKI

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Publication Bias	Precision of the Estimate	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death, 3 year	0								No evidence	Critical
Recurrent AKI episode	0								No evidence	Critical
Poor control of underlying condition	0								No evidence	Critical
MAKE90	0								No evidence	Important
CKD G3-G5 (de novo or progressive CKD)	1 (2461) [1] [NRCS]	Serious limitations	N/A	Direct	Undetected	Precise	Sparse, but large	Low	HR 0.91 (0.74, 1.13) Insufficient data to estimate absolute difference	Important
CKD G5D	0								No evidence	Important
Quality of life	0								No evidence	Important
Serious AE	0								No evidence	Critical
Discontinuation due to AE	0								No evidence	Critical
Balance of Potential Benefits and Harms: Using NSAIDs post-AKI is not associated with increased de novo or progressive CKD								Certainty of Overall Evidence: Very Low		

Abbreviations: AE = adverse event; AKI = Acute Kidney Injury; CKD: Chronic Kidney Injury; HR = hazard ratio; MAKE: Major adverse kidney events at 90 days; N/A = not applicable; NRCS = nonrandomized comparative study; NSAIDs = Non-steroidal Anti-inflammatory drugs

References

1. Schreier DJ, Rule AD, Kashani KB, et al. Nephrotoxin Exposure in the 3 Years following Hospital Discharge Predicts Development or Worsening of Chronic Kidney Disease among Acute Kidney Injury Survivors. *Am J Nephrol*. 2022;53(4):273-281. doi:10.1159/000522139

Table S143. RASi (combined restart, new start, or any use) vs. control in adults post-AKI

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death, all cause	11 (81869) [1-11] [NRCS]	Serious	Consistent[a]	Direct	Precise	Undetected	None	Moderate	**HR 0.82 (0.77, 0.88)** 67 fewer per 1000 (from 87 fewer to 44 fewer)	Critical
Recurrent AKI	7 (64,431) [1-4,6-7,12] [NRCS]	Serious	Inconsistent	Direct	Precise	Undetected	None	Low	HR 1.03 (0.95, 1.12) 3 more per 1000 (from 6 fewer to 14 more)	Critical
MAKE	1 (3289) [10] [NRCS]	Serious	N/A	Direct	Precise	Undetected	Sparse, but large	Low	adjHR 1.12 (0.98, 1.28) 40 more per 1000 (from 7 fewer to 86 more)	Critical
CKD G3-G5 (CKD progression)	3 (28,828) [3,6-7] [NRCS]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	**HR 0.93 (0.87, 0.99)** [Favors RAASi use post-AKI]] 14 fewer per 1000 (from 26 fewer to 2 fewer)	Critical
CKD G5D	3 (44146) [4,6,8] [NRCS]	Serious	Inconsistent	Direct	Precise	Undetected	None	Low	**HR 0.89 (0.80, 0.99)** 10 fewer per 1000 (from 18 fewer to 1 more)	Critical
Poor control of underlying condition:										Critical
Hypertension hospitalization	1 (18,912) [6] [NRCS]	No serious	N/A	Direct	Very imprecise	Undetected	Sparse, rare event	Very low	adjHR 1.06 (0.38, 2.95) 0 more per 1000 (from 0 fewer to 1 more)	
Heart failure hospitalization	4 (36,593) [1,3,6-7] [NRCS]	Serious	Inconsistent	Direct	Imprecise	Undetected	None	Very low	HR 1.02 (0.76, 1.37) 6 more per 1000 (from 78 fewer to 103 more)	
Myocardial infarction	1 (10,165) [3] [NRCS]	Serious	N/A	Direct	Precise	Undetected	Sparse, but large	Low	adjHR 1.06 (0.90, 1.26) 5 more per 1000 (from 9 fewer to 23 more)	
Stroke	2 (17,034) [1,3] [NRCS]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	HR 0.97 (0.84, 1.13) 3 fewer per 1000 (from 15 fewer to 12 more)	
MACE[a]	2 (19,790) [3-4]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	**HR 0.88 (0.85, 0.92)** [Favors RAASi use post-AKI] 46 fewer per 1000 (from 59 fewer to 30 fewer)	

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings		
								Certainty of Evidence	Description of Findings	Importance of Outcome
Hypervolemia	1 (18,912)[6] [NRCS]	Serious	N/A	Direct	Imprecise	Undetected	Sparse, rare event	Very low	adjHR 0.79 (0.38, 1.67) 0 more per 1000 (from 1 fewer to 1 more)	
Quality of life	0								No Evidence	Important
Discontinuation due to AE	0								No Evidence	Important
SAE: Moderate or severe hyperkalemia	3 (37,684)[3-4,6] [NRCS]	Serious	Inconsistent	Direct	Precise	Undetected	None	Low	HR 1.20 (0.94, 1.52) 27 more per 1000 (from 8 fewer to 68 more)	Critical
Balance of Potential Benefits and Harms: RASi use post-AKI reduces the likelihood of death, MACE, and CKD progression, with no evidence of effects on other outcomes.								Certainty of Overall Evidence: Moderate		

Abbreviations: AE = adverse event; AKI = Acute Kidney Injury; CKD = chronic kidney disease; G = grade (eg, G3, grade 3); HR = hazard ratio; MACE = major adverse cardiac event; MAKE = major adverse kidney event; N/A = not applicable; NRCS = non-randomized control study; RASi = Renin–angiotensin system inhibitors; SAE = serious adverse event

[a] MACE defined as composite cardiovascular death, myocardial infarction, and stroke.

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Table S144. Any RASi (any use) vs. control in adults post-AKI

Outcome	# Studies (Patients) [Design]	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Precision of the Estimate	Publication Bias	Sparseness (or Other Considerations)	Summary of Findings: Certainty of Evidence	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death, all cause	7 (93582) [1-7] [NRCS]	Serious	Consistent	Direct	Precise	Undetected	None	Moderate	**HR 0.80 (0.70, 0.92)** 75 fewer per 1000 (from 116 fewer to 29 fewer)	Critical
Recurrent AKI	2 (19681) [2-3] [NRCS]	Serious	Inconsistent	Direct	Imprecise	Undetected	None	Low	HR 1.06 (0.75, 1.50) 2 more per 1000 (from 10 fewer to 20 more)	Critical
MAKE[a]	1 (3289) [4] [NRCS]	Serious	N/A	Direct	Imprecise	Undetected	Sparse	Low	HR 1.12 (0.98, 1.28) 40 more per 1000 (from 7 fewer to 86 more)	Critical
CKD G3-G5 (CKD progression)	2 (19681) [2-3] [NRCS]	Serious	Consistent	Direct	Imprecise	Undetected	None	Low	HR 0.84 (0.62, 1.13) 23 fewer per 1000 (from 56 fewer to 18 more)	Critical
CKD G5D	2 (35838) [2,5] [NRCS]	Serious	Consistent	Direct	Precise	Undetected	None	Low	HR 0.95 (0.89, 1.01) 4 fewer per 1000 (from 10 fewer to 1 more)	Critical
Poor control of underlying condition:										Critical
Hypertension hospitalization	1 (18912) [2] [NRCS]	No limitations	N/A	Direct	Highly imprecise	Undetected	Sparse, but large study; very low probability event	Low	HR 1.06 (0.38, 2.95) 0 per 1000 (from 0 to 1 more)	
Heart failure hospitalization	2 (19681) [2-3] [NRCS]	Serious	Inconsistent	Direct	Highly imprecise	Undetected	None	Very low	HR 1.12 (0.50, 2.52) 1 more per 1000 (from 3 fewer to 10 more)	
Myocardial infarction	0								No evidence	
Stroke	0								No evidence	
MACE	0								No evidence	
Hypervolemia	1 (18912) [2] [NRCS]	No limitations	N/A	Direct	Highly imprecise	Undetected	Sparse, but large study	Low	HR 0.79 (0.38, 1.67) 0 per 1000 (from 1 fewer to 1 more)	
Quality of life	0								No evidence	Important
Discontinuation due to AE	0								No evidence	Important
SAE: Hyperkalemia hospitalization	1 (18912) [2]	No limitations	N/A	Direct	Imprecise	Undetected	Sparse, but large study	Low	**HR 1.56 (1.07, 2.27)** 3 more per 1000 (from 0 to 6 more)	Important

Balance of Potential Benefits and Harms:	Certainty of Overall Evidence:
Analyses of RASi use (any) post-AKI found reduced likelihood of death with *increased* risk of hyperkalemia hospitalization, with no evidence of effects on other outcomes.	Moderate

Abbreviations: AE = adverse event; AKI = acute kidney injury; CKD = chronic kidney disease; HR = hazard ratio (adjusted); MACE = major adverse cardiovascular events; MAKE = major adverse kidney event; N/A = not applicable; NRCS = non-randomized control study; RASi = renin–angiotensin system inhibitors

[a] MAKE defined as RRT initiation, sustained eGFR decline, or mortality.

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